

To: Users of *VESA DisplayPort Alt Mode on USB Type-C Standard*
(*DisplayPort Alt Mode*)

From: Video Electronics Standards Association

Date: December 31, 2020

Subject: *DisplayPort Alt Mode Standard*

This Standard has been revised from its original published version to remove use of offensive terms. The specific revisions that have been made include the following:

- The term “master” was globally changed to “Primary”
- The term “slave” was globally changed to “Secondary”

THIS PAGE INTENTIONALLY LEFT BLANK.

VESA DisplayPort Alt Mode on USB Type-C Standard (DisplayPort Alt Mode)

Version 2.0
12 March, 2020

vesa.org

Purpose

The purpose of this Standard is to define the use of *DisplayPort*[™] as an Alternate Mode for *USB Type-C*[®] connectors and cables.

Summary

This Standard defines use of the *DisplayPort* protocol with *USB Type-C* connectors and cables (also referred to as “*USB-C*[®] connectors and cables”). It defines *DisplayPort* protocol support between two devices that use USB-C-to-USB-C cable assemblies to support *USB-C* connectors. It defines cable assemblies that adapt the *USB-C* connector to *DisplayPort* connectors. It defines cable assemblies and adapters that adapt DisplayPort Alt Mode to other video protocols. It defines the simultaneous use of *USB 3.2* and the *DisplayPort* protocol on the *USB-C* connector.

This Standard defines the *USB-C* connector pin assignments and *DisplayPort* bit rates to be used in various scenarios when using the above cable assemblies and adapters. It defines how the *USB PD* Alternate Mode Discovery and Entry Processes are applied to *DisplayPort* on the *USB-C* connector.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Contents

Purpose	1
Summary	1
Preface	9
Intellectual Property	9
Trademarks	9
Patents	10
Support for this Standard	10
Acknowledgments	11
Revision History	13
Section 1	Introduction
	15
1.1	Purpose
	15
1.2	Scope
	15
1.3	Document Organization
	16
1.4	Terms, Acronyms, and Abbreviations
	17
1.5	Conventions
	23
1.5.1	Precedence
	23
1.5.2	Keywords
	23
1.5.3	Numbering
	23
1.6	Reference Documents
	24
Section 2	Overview
	26
2.1	Introduction to USB-C and Alternate Mode Operation
	26
2.2	USB-C Receptacles and Plugs
	28
2.3	USB-C Cables and Interconnectivity
	28
2.4	Multi-function Systems
	32
Section 3	Pin Assignments and Descriptions
	33
3.1	Pin Assignment Overview
	34
3.2	USB-C DP Pin Assignments
	35
3.2.1	Pin Assignment Support Mandates
	37
3.2.2	DisplayPort Support Mandates
	38
3.3	DP Source Device Pin Assignment C
	39
3.4	DP Source Device Pin Assignment D
	40
3.5	DP Source Device Pin Assignment E
	41
3.6	DP Sink Device Pin Assignment C
	43
3.7	DP Sink Device Pin Assignment D
	44
3.8	DP Sink Device Pin Assignment E
	45
3.9	AUX_CH and HPD
	47
3.9.1	AUX_CH
	47
3.9.2	HPD Conversion
	52
3.10	DP Connector CONFIG1 and CONFIG2 Pins
	61

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Section 4	USB-C Cables Used for DisplayPort	62
4.1	Active Cable Overview	63
4.2	Scenario 1 – USB-C Passive or Active Cable	63
4.2.1	Scenario 1 – Active USB-C-to-USB-C Cables	68
4.2.2	Scenario 1 – Active USB-C-to-USB-C Cable USB PD Mandates	69
4.3	Scenarios 2a and 2b – USB-C-to-DP/mDP Cable	72
4.3.1	Scenario 2a – USB-C DP Source Device-to-DP/mDP Sink Device	72
4.3.2	Scenario 2b – DP/mDP Source Device-to-USB-C DP Sink Device	76
4.3.3	Scenarios 2a and 2b – USB-C-to-DP/mDP Passive Main-Link Cable Electrical Properties	80
4.3.4	Scenarios 2a and 2b – USB-C-to-DP/mDP Cable Functionality	85
4.3.5	Scenarios 2a and 2b – Active USB-C-to-DP Cables	91
4.3.6	Scenarios 2a and 2b – USB-C-to-DP Cable Power	92
4.4	Scenarios 3a and 3c – USB-C DP Source Device-to-Protocol Converter	93
4.4.1	Scenarios 3a and 3c – HDMI Protocol Converters	94
4.4.2	Scenarios 3a and 3c – VGA Protocol Converters	94
4.4.3	Scenarios 3a and 3c – DVI Protocol Converters	94
4.5	Scenario 4a – USB-C DP Source Device-to-Multi-function Dock	95
4.6	Scenarios 4c and 4d – USB-C DP Source Device to Multi-function Cradle/Captive Cable Dock	100
4.7	Scenarios 5a and 5b – USB-C Plug-to-DP/mDP Active Receptacle Adapters	102
4.7.1	Scenario 5a – USB-C Plug DP Source-to-DP/mDP Receptacle Adapter	104
4.7.2	Scenario 5b – DP/mDP Sink Receptacle-to-USB-C DP Sink Adapter	110
4.7.3	Scenarios 5a and 5b – USB-C Plug-to-DP/mDP Active Receptacle Adapter Directionality	116
4.7.4	Scenarios 5a and 5b – Active Adapter USB Behavior	116
4.7.5	Scenarios 5a and 5b – Active Receptacle Adapter Power	116
4.8	Active Cable Power Consumption	117
4.8.1	Active Cable Sleep	117
4.8.2	Active Cable Suspend	117
4.9	Active Cable Legacy USB-C Support Mandates	118
4.10	Methods for Determining DP Directionality in an Active Cable	119
4.10.1	AUX_CH Pull-up/Pull-down Resistor Method for Detecting DP Direction	119
4.10.2	AUX Transaction Method for Detecting Direction	120

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Section 5	Discovery and USB PD	121
5.1	Connection Discovery	121
5.1.1	DisplayPort Alt Mode Entry	122
5.1.2	USB PD Data Role Policies	124
5.1.3	USB PD V_{CONN} Policies	125
5.1.4	DP Source Device on a USB-C Receptacle	127
5.1.5	DP Sink Device on a USB-C-powered USB-C Receptacle	127
5.1.6	DP Sink Device on an Unrestricted Externally Powered USB-C Receptacle	127
5.1.7	DP Source Device on a USB-C Plug	127
5.1.8	DP Sink Device on a USB-C Plug	127
5.1.9	USB-C-to-DP Receptacle Adapter, Cable Adapter, or Protocol Adapter	127
5.2	DisplayPort Alt Mode Discovery, Entry, and Exit	128
5.2.1	Structured VDM Header	128
5.2.2	DisplayPort Enter Mode Command	132
5.2.3	DisplayPort Status Update Command	133
5.2.4	DisplayPort Configure Command	135
5.2.5	USB PD Attention Command Request	138
5.2.6	DisplayPort Exit Mode Command	138
5.3	Active Cable Support	139
5.3.1	Active USB-C-to-USB-C Cable V_{CONN} Source	142
5.4	VDM Flows	143
Section 6	Miscellaneous	157
6.1	V_{CONN} and V_{BUS}	157
6.1.1	Active Cable Behavior	157
6.1.2	V_{CONN} Overcurrent Protection	157
6.2	Low-power Operation	158
6.2.1	Display Sleep	158
6.2.2	USB Suspend General Properties (Informative)	158
6.2.3	USB-C DFP_U	159
6.2.4	USB-C UFP_U that Supports a V_{CONN} -powered Adapter	160
6.3	Equivalent USB Functionality	160
6.4	USB BB URL	160
Appendix A	DisplayPort Alt Mode Examples (Informative)	161
A.1	DisplayPort Alt Mode Examples	161
A.2	USB-C USB Host-to-DP Video Adapter Example	161
A.2.1	DP Video Adapter Assumptions	162
A.2.2	Block Diagrams	163
A.2.3	Solution	166
A.3	USB Dock with DisplayPort Alt Mode Example	169
A.3.1	USB Dock with DisplayPort Alt Mode Assumptions	169
A.3.2	Block Diagram	170
A.3.3	Solution	171
Appendix B	Main Contributor History (Previous Versions)	174

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Tables

Table 1:	Patents	10
Table 2:	Main Contributors to DP AM v2.0	11
Table 3:	Revision History	13
Table 1-1:	Terms, Acronyms, and Abbreviations Referenced from USB-related Specifications (Informative)	17
Table 1-2:	Non-USB-related Terms, Acronyms, and Abbreviations Used in this Standard	21
Table 1-3:	Keywords	23
Table 1-4:	Numbering	23
Table 1-5:	Reference Documents	24
Table 2-1:	Inherent Support of DisplayPort Alt Mode by Full-Featured USB-C Standard Passive Cable Assemblies Specified by the USB-IF	28
Table 2-2:	Optional Support of DisplayPort Alt Mode by USB-C Active Cable Assemblies and Captive Cables Specified by the USB-IF	29
Table 2-3:	Support of DisplayPort Alt Mode by USB-C-enabled Cables and Adapters Defined by this Standard	30
Table 2-4:	Further Description of Cables and Adapters Listed in Table 2-1 through Table 2-3 (Informative)	31
Table 3-1:	DP Source and Sink Device Pin Assignment Summary	34
Table 3-2:	USB-C Receptacle DP Source Device Pin Assignment Summary – Normal Plug Orientation	35
Table 3-3:	USB-C Receptacle DP Source Device Pin Assignment Summary – Flipped Plug Orientation	35
Table 3-4:	USB-C Receptacle DP Sink Device Pin Assignment Summary – Normal Plug Orientation	36
Table 3-5:	USB-C Receptacle DP Sink Device Pin Assignment Summary – Flipped Plug Orientation	36
Table 3-6:	USB-C Receptacle DP Source Device Pin Assignment Mandates	37
Table 3-7:	USB-C Receptacle DP Sink Device Pin Assignment Mandates	37
Table 3-8:	USB-C Plug DP Source Device Pin Assignment Mandates (for DP Sink Device)	38
Table 3-9:	USB-C Plug DP Sink Device Pin Assignment Mandates (for DP Source Device)	38
Table 3-10:	DP Source Device Pin Assignment C – Normal Plug Orientation	39
Table 3-11:	DP Source Device Pin Assignment C – Flipped Plug Orientation	39
Table 3-12:	DP Source Device Pin Assignment D – Normal Plug Orientation	40
Table 3-13:	DP Source Device Pin Assignment D – Flipped Plug Orientation	40
Table 3-14:	DP Source Device Pin Assignment E – Normal Plug Orientation	41
Table 3-15:	DP Source Device Pin Assignment E – Flipped Plug Orientation	42
Table 3-16:	DP Sink Device Pin Assignment C – Normal Plug Orientation	43
Table 3-17:	DP Sink Device Pin Assignment C – Flipped Plug Orientation	43
Table 3-18:	DP Sink Device Pin Assignment D – Normal Plug Orientation	44
Table 3-19:	DP Sink Device Pin Assignment D – Flipped Plug Orientation	44

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 3-20:	DP Sink Device Pin Assignment E – Normal Plug Orientation.	45
Table 3-21:	DP Sink Device Pin Assignment E – Flipped Plug Orientation.	46
Table 3-22:	HPD-to-USB PD Converter Timing.	54
Table 3-23:	USB PD-to-HPD Converter Timing.	56
Table 4-1:	Active Cable USB PD Command Mandates	69
Table 4-2:	Active Cable DP Capabilities (VDO in the Responder USB PD Discover Modes VDM) .	70
Table 4-3:	Active Cable DisplayPort Status Update	70
Table 4-4:	Active Cable DisplayPort Configurations	71
Table 4-5:	Scenario 2a – USB-C DP Source Device-to-DP/mDP Sink Device Use, Normal Plug Orientation	75
Table 4-6:	Scenario 2b – DP/mDP Source Device-to-USB-C DP Sink Device Use, Normal Plug Orientation	79
Table 4-7:	Nyquist Frequencies and Their ILFitAtNq Mandate	80
Table 4-8:	DP Connector AUX_CH Signal States	86
Table 4-9:	DP Connector HPD and AUX_CH Signal States when Disconnected –or– in Adapter DP Source Device Configuration.	89
Table 4-10:	DP Connector HPD and AUX_CH Signal States in Adapter DP Sink Device Configuration.	90
Table 4-11:	USB-C Cable DP Use (Multi-function Cradle/Captive Cable Dock)	101
Table 4-12:	Scenario 5a – USB-C Plug DP Source-to-DP/mDP Receptacle Adapter Use	109
Table 4-13:	Scenario 5b – DP/mDP Sink Receptacle-to-USB-C DP Sink Adapter Use.	115
Table 5-1:	USB-C and USB PD Power and Data Roles	123
Table 5-2:	USB PD Data Role Policies	124
Table 5-3:	USB PD V _{CONN} Policies	125
Table 5-4:	Structured VDM Header.	129
Table 5-5:	DP Capabilities (VDO in the Responder USB PD Discover Modes VDM)	131
Table 5-6:	DisplayPort Status Update	134
Table 5-7:	DFP_U Pin Assignment Selection Mandates	136
Table 5-8:	DisplayPort Configurations	137
Table B-1:	Main Contributor History (Previous Versions)	174

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figures

Figure 2-1:	USB-C Receptacle Interface (Front View) (Informative)	27
Figure 2-2:	USB-C Plug Interface (Front View) (Informative)	27
Figure 3-1:	AUX_CH Connections Using USB-C-to-USB-C Cables	48
Figure 3-2:	AUX_CH Connections When Using a USB-C-to-DP Adapter – Example 1	49
Figure 3-3:	AUX_CH Connections When Using a USB-C-to-DP Adapter – Example 2	50
Figure 3-4:	AUX_CH Connections When Using a USB-C-to-DP Protocol Converter	51
Figure 3-5:	HPD Conversion Architecture	52
Figure 3-6:	HPD-to-USB PD Converter Connection State Machine	55
Figure 3-7:	USB PD-to-HPD Converter Connection State Machine	57
Figure 3-8:	HPD Communication When Using a USB-C-to-USB-C Cable	60
Figure 3-9:	HPD Communication When Using a USB-C-to-DP Adapter – Example 1	60
Figure 3-10:	HPD Communication When Using a USB-C-to-DP Adapter – Example 2	61
Figure 3-11:	HPD Communication When Using a USB-C-to-Protocol Converter	61
Figure 4-1:	Scenario 1 – USB-C Passive or Active Cable	63
Figure 4-2:	Scenario 1 Cable Wiring – Normal Plug Orientation	64
Figure 4-3:	Scenario 1 Cable Wiring – DP Source Device Flipped Plug Orientation	65
Figure 4-4:	Scenario 1 Cable Wiring – DP Sink Device Flipped Plug Orientation	66
Figure 4-5:	Scenario 1 Cable Wiring – DP Source and Sink Device Flipped Plug Orientation	67
Figure 4-6:	Scenario 2a – USB-C DP Source Device-to-DP/mDP Sink Device	72
Figure 4-7:	Scenario 2a Cable Wiring – Normal Plug Orientation on a Full-size DP Connector	73
Figure 4-8:	Scenario 2a Cable Wiring – DP Source Device Flipped Plug Orientation on a Full-size DP Connector	74
Figure 4-9:	Scenario 2b – DP/mDP Source Device-to-USB-C DP Sink Device	76
Figure 4-10:	Scenario 2b Cable Wiring – Normal Plug Orientation on a Full-size DP Connector	77
Figure 4-11:	Scenario 2b Cable Wiring – DP Sink Device Flipped Plug Orientation on a Full-size DP Connector	78
Figure 4-12:	ILFitAtNq for UHBR10	81
Figure 4-13:	ILFitAtNq for HBR3	81
Figure 4-14:	ILFitAtNq for HBR2	82
Figure 4-15:	ILFitAtNq for HBR	82
Figure 4-16:	IMR as a Function of ILFitAtNq at UHBR10 and HBR3	83
Figure 4-17:	IRL as a Function of ILFitAtNq at UHBR10 and HBR3	84
Figure 4-18:	USB-C-to-DP Adapter Additional USB PD	86
Figure 4-19:	Scenario 3a – USB-C DP Source Device Protocol Converter Cable	93
Figure 4-20:	Scenario 3c – USB-C DP Source Device Protocol Converter Adapter or Dongle	93
Figure 4-21:	Scenario 4a – USB-C DP Source Device-to-Multi-function Dock	95
Figure 4-22:	Scenario 4a Multi-function Cable Wiring – Normal Plug Orientation	96
Figure 4-23:	Scenario 4a Multi-function Cable Wiring – DP Source Device Flipped Plug Orientation	97
Figure 4-24:	Scenario 4a Multi-function Cable Wiring – DP Sink Device Flipped Plug Orientation	98

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 4-25:	Scenario 4a Multi-function Cable Wiring – DP Source and Sink Device Flipped Plug Orientation.	99
Figure 4-26:	Scenario 4c – USB-C DP Source Device Connected to a Multi-function Cradle Dock . . .	100
Figure 4-27:	Scenario 4d – USB-C DP Source Device Connected to a Multi-function Captive Cable Dock	100
Figure 4-28:	USB-C Plug-to-DP/mDP Active Receptacle Adapter (Main-Link AC-coupling Mandates)	103
Figure 4-29:	Scenario 5a – USB-C Plug DP Source-to-DP/mDP Receptacle Adapter	104
Figure 4-30:	Scenario 5a Full-size DP Adapter Wiring – Normal Plug Orientation	105
Figure 4-31:	Scenario 5a Full-size DP Adapter Wiring – Flipped Plug Orientation	106
Figure 4-32:	Scenario 5a mDP Adapter Wiring – Normal Plug Orientation	107
Figure 4-33:	Scenario 5a mDP Adapter Wiring – Flipped Plug Orientation	108
Figure 4-34:	Scenario 5b – DP/mDP Sink Receptacle-to-USB-C DP Sink Adapter	110
Figure 4-35:	Scenario 5b Full-size DP Adapter Wiring – Normal Plug Orientation	111
Figure 4-36:	Scenario 5b Full-size DP Adapter Wiring – Flipped Plug Orientation	112
Figure 4-37:	Scenario 5b mDP Adapter Wiring – Normal Plug Orientation	113
Figure 4-38:	Scenario 5b mDP Adapter Wiring – Flipped Plug Orientation	114
Figure 5-1:	Connect Initial Actions.	144
Figure 5-2:	DFP_U DisplayPort Alt Mode Discovery and Entry Processes.	146
Figure 5-3:	DFP_U DisplayPort Configuration	147
Figure 5-4:	DFP_U DisplayPort Operation and Reconfiguration.	148
Figure 5-5:	DFP_U Reconfigure to USB Configuration.	149
Figure 5-6:	DFP_U Exit Mode Process.	150
Figure 5-7:	UFP_U DisplayPort Alt Mode Discovery and Entry Processes.	151
Figure 5-8:	UFP_U USB Configuration Wait State	153
Figure 5-9:	UFP_U DisplayPort Configuration	154
Figure 5-10:	UFP_U DP Operation.	155
Figure 5-11:	UFP_U Exit Mode Process.	156
Figure 5-12:	USB PD Hard Reset and Disconnect.	156
Figure A-1:	USB Host System Block Diagram	163
Figure A-2:	DP Video Adapter Cable Block Diagram.	164
Figure A-3:	Example Block Diagram of USB Dock with DisplayPort Alt Mode.	170

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Preface

Intellectual Property

Copyright © 2014 – 2020 [Video Electronics Standards Association](#). All rights reserved.

Excerpts from [USB4™](#) are Copyright © 2007 – 2019, USB4 Promoter Group (Apple, Inc.; HP, Inc.; Intel Corporation; Microsoft Corporation; Renesas Corporation; STMicroelectronics, Inc.; and Texas Instruments), and are reproduced with permission.

Excerpts from [USB-C®](#) are Copyright © 2014 – 2019, USB4 Promoter Group (Apple, Inc.; HP, Inc.; Intel Corporation; Microsoft Corporation; Renesas; STMicroelectronics, Inc.; and Texas Instruments), and are reproduced with permission.

Excerpts from [USB PD](#) are Copyright © 2010 – 2019, USB4 Promoter Group (Apple, Inc.; HP, Inc.; Intel Corporation; Microsoft Corporation; Renesas; STMicroelectronics, Inc.; and Texas Instruments), and are reproduced with permission.

While every precaution has been taken in the preparation of this Standard, the [Video Electronics Standards Association](#) and its contributors assume no responsibility for errors or omissions, and make no warranties, expressed or implied, of functionality or suitability for any purpose.

Trademarks

[DisplayPort](#) is a trademark and VESA is a registered trademark of the [Video Electronics Standards Association](#).

[HDMI](#) and High-Definition Multimedia Interface are trademarks or registered trademarks of [HDMI Licensing Administrator, Inc.](#), in the United States and other countries.

[USB4](#) is a trademark and [USB Type-C](#) and [USB-C](#) are registered trademarks of the [USB Implementers Forum, Inc.](#)

All other trademarks, registered trademarks, or servicemarks used within this Standard are the property of their respective owners.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Patents

VESA® draws attention to the fact that compliance with this Standard might involve the use of a patent or other intellectual property right (collectively, “IPR”). VESA takes no position concerning the evidence, validity, and/or scope of this IPR.

The following holders of this IPR have assured VESA that they are willing to license the IPR on Reasonable and Non-Discriminatory (RAND) terms. The statement of the holder of this IPR is registered with VESA.

Table 1: Patents

Holder Name	Contact Information	Claims Known
Apple, Inc. 1 Infinite Loop MS 169-3IPL Cupertino, CA 95014 408-783-5093	BJ Watrous Senior Standards Counsel (bjwatrous@apple.com)	—

Attention is drawn to the possibility that some of the elements of this VESA Standard might be the subject of IPR other than that identified above. VESA shall not be held responsible for identifying any or all such IPR, and has made no inquiry into the possible existence of any such IPR.

THIS STANDARD IS BEING OFFERED WITHOUT ANY WARRANTY WHATSOEVER, AND IN PARTICULAR, ANY WARRANTY OF NON-INFRINGEMENT IS EXPRESSLY DISCLAIMED. ANY IMPLEMENTATION OF THIS STANDARD SHALL BE MADE ENTIRELY AT THE IMPLEMENTER’S OWN RISK, AND NEITHER VESA, NOR ANY OF ITS MEMBERS OR SUBMITTERS, SHALL HAVE ANY LIABILITY WHATSOEVER TO ANY IMPLEMENTER OR THIRD PARTY FOR ANY DAMAGES OF ANY NATURE WHATSOEVER DIRECTLY OR INDIRECTLY ARISING FROM THE IMPLEMENTATION OF THIS STANDARD.

Support for this Standard

Clarifications and application notes to support this Standard may have been written. To obtain the latest Standard and any supporting documentation, contact VESA.

If you have a product that incorporates DisplayPort on a USB Type-C connector, ask the company that manufactured your product for assistance. If you are a manufacturer, VESA can assist you with any clarification you might need.

Submit all comments or reported errors to support@vesa.org.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Acknowledgments

This Standard would not have been possible without the efforts of VESA's DP on USB Type-C® Connector Subgroup. In particular, [Table 2](#) lists the individuals and their companies that contributed significant time and knowledge to this version of the Standard.

Table 2: Main Contributors to DP AM v2.0

Company	Name	Contribution
Advanced-Connectek, Inc.	Steve Sedio	Technical Content
Advanced Micro Devices, Inc.	Dennis Au	Technical Content
	Jim Hunkins	Technical Content
Allion Labs, Inc.	John Lin	Technical Content
	Brian Shih	Technical Content
Amphenol AssembleTech	Jesse Jaramillo	Technical Content
Analogix Semiconductor	Greg Stewart	Technical Content
Apple, Inc.	Sree Anantharaman	Technical Content
	Brian Baek	Technical Content
	Bill Cornelius	Technical Content
	Christine Krause	Document Editor, Technical Content
	Alan Kobayashi	DisplayPort Task Group Chair
	Scott Jackson	Technical Content
	Jennifer Tsai	Technical Content
Avatar Tech Pubs	Trish McDermott	Technical Writer
BizLink Technology, Inc.	Morphy Hsieh	Technical Content
Chrontel, Inc.	David Tsai	Technical Content
Dai-Ichi Seiko (formerly I-PEX)	Alan Kinningham	Technical Content
Diodes	Tat Choi	Technical Content
DisplayLink Corporation	Tony Lambord	Technical Content
EIZO Corp.	Yoshinobu Bamba	Technical Content
Ellisys Corporation	Abel Astley	Subgroup Co-leader, Technical Content
	Rick Bogart	Technical Content
Foxconn Interconnect Technology, Ltd.	Glenn Moore	Technical Content
Fresco Logic, Inc.	Tom Burton	Technical Content
GoPro, Inc.	Bill Orner	Technical Content
Intel Corporation	Mike Hamann	Technical Content
	Robert "Jamie" Johnston	Technical Content
	Ziv Kabiry	Technical Content
	Christine Krause	Technical Content
	Sara Shachor	Technical Content
	Gal Yedidia	Technical Content

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 2: Main Contributors to DP AM v2.0 (Continued)

Company	Name	Contribution
INVECAS, Inc.	Gyudong Kim	Technical Content
JAE Electronics, Inc.	Mark Saubert	Technical Content
Keysight Technologies	Brian Fetz	Technical Content
LUXSHARE-ICT, Inc.	Scott Shuey	Technical Content
Microchip Technology	Mark Bohm	Technical Content
Mitas Electronics	Greg Young	Technical Content
NVIDIA Corporation	David Stears	Technical Content
	Bill Simms	Technical Content
NXP Semiconductors (formerly Philips Semiconductor)	Vijendra Kuroodi	Technical Content
Parade Technologies, Ltd.	Craig Wiley	Subgroup Leader, Technical Content
Qualcomm, Inc.	Alan Chan	Technical Content
	Chris Wiesner	Technical Content
Realtek Semiconductor Corporation	Jay Lin	Technical Content
Siliconworks Co., Ltd.	Ben Huh	Technical Content
Synaptics, Inc.	Jeff Lukanc	Technical Content
	Mark Miller	Technical Content
Synopsys	Moren Christiansen	Technical Content
Tektronix, Inc.	Yogesh Pai	Technical Content
	Medhu Acharya	Technical Content
Teledyne LeCroy	Mike Micheletti	Technical Content
Texas Instruments	Mike Campbell	Technical Content
	Yoon Lee	Technical Content
	Michael Koltun	Technical Content
UL	Paul Vanderlann	Technical Content
	Eric Wall	Technical Content
VESA – CPM	Jim Choate	Technical Content
VIA Technologies, Inc.	Jay Tseng	Technical Content
VTM Group	Bob Crepps	Technical Content

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Revision History

Table 3: Revision History

Date	Version	Description
March 12, 2020	2.0	- Updated to comply with DP v2.0 and current USB-related Specifications - Changed some acronyms and abbreviations, as listed in Table 1-2, to remove ambiguity with those used in some of the referenced USB-related Specifications - Updated passive USB-C-to-DP cable electrical parameters to include UHBR10 - Updated Section 5 and Section 6 to provide further clarity and align with updated referenced Specifications - Added changes made to <i>DP AM v1.0b</i> through ratification of the following SCRs: <i>DP_Alt_Mode_1.0b_IRL_IMR_INEXT_SCR_D1</i> <i>DP_Alt_Mode_Spec_C_to_DP_cable_adapter_SCR_draft1</i> <i>DP_Alt_Mode_V1.0b_SCR Connected State</i> <i>Type-C Alt Mode Add receptacle adapter</i> <i>Type-C DP Alt Mode Cplug DRD SCR</i> <i>USB Type-C DP Alt Mode Active Cable</i> - Replaced “DFP_D” and “UFP_D” with “DP Source device” and “DP Sink device,” respectively - Updated reference documents in Table 1-5, and hyperlinked external document references to their Table 1-5 entries - Applied minor editorial improvements
November 3, 2017	1.0b	- Added changes made to <i>DP AM v1.0a</i> through ratification of the following SCRs: <i>Main link AC capacitor value guidance</i> <i>USB-C to DP adapter cable reversibility clarification</i> <i>Object Position Note Correction</i> <i>USB Type-C DisplayPort Improved Interoperability</i> <i>Update External Spec References</i> <i>DP Alt Mode Attention Pacing</i> <i>Deprecate DockPort</i> – Removes DockPort support, including DFP_D Pin Assignment F, from this Standard <i>Clarify Pinout E</i> - In all but the document title, shortened the name of the mode defined by this Standard to “DisplayPort Alt Mode” - Added relaxation of DP_PWR mandates - Updated to reflect the change to USB 3.2 to allow optional AC capacitors on the USB SuperSpeed RX system design - Removed reference to GEN2_BR - Applied minor editorial improvements

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 3: Revision History (Continued)

Date	Version	Description
August 5, 2015	1.0a	Added proposed changes from the following <i>DP AM v1.0</i> SCRs: • <i>DisplayPort Alt Mode Editorial Corrections</i> • <i>DisplayPort Alt Mode Plug Requirement Corrections & Protocol clarifications</i> • <i>DisplayPort Alt Mode VCONN Low Power</i> • <i>DisplayPort Alt Mode Billboard URL</i> • <i>DP Alt Mode VGA Adaptor Requirement Reduction to 1 Lane at HBR and HBR2</i> • <i>DisplayPort Alt Mode Capabilities Clarification</i> • <i>DisplayPort Alt Mode Flipped Plug Orientation Additions and Revisions</i>
September 22, 2014	1.0	Initial release of the Standard.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

1 Introduction

USB-C[®] envisages support for Alternate Modes that use *USB Type-C*[®] (*USB-C*) connectors and cables, in which various *USB-C* connector pins may be reconfigured to support interfaces that are beyond the scope of *USB-C*.

USB-C (and *USB PD*, which it leverages) provides a Discovery Process for discovering the support of Alternate Modes in connected *USB* devices, including specifying the method for changing modes, whereby an Alternate Mode is used on the *USB-C* connector. *USB-C* provides mandates by which Alternate Modes shall abide.

This Standard defines the use of *DisplayPort*[™] as an Alternate Mode for the *USB-C* connector (*DisplayPort Alt Mode*).

1.1 Purpose

The purpose of this Standard is to enable an ecosystem in which a *USB-C* connector may be used to support *DisplayPort* and other video protocols (e.g., *HDMI*[®] and VGA) in a way that is fully interoperable with existing *DisplayPort* and other video protocol products, with an emphasis on a small number of cable types to minimize user confusion.

This Standard defines:

- Use of *DisplayPort Alt Mode* on the *USB-C*-to-*USB-C* cable assembly
- Cable assemblies that adapt the *USB-C* connector to *DisplayPort* (*DP*) connectors
- Cable assemblies and adapters for *DisplayPort Alt Mode* to other video protocols
- Simultaneous use of *USB* SuperSpeed and *DisplayPort* for docking applications

Additionally, this Standard defines pin assignments and *DP* bit rates that are to be used in various scenarios using the cable assemblies and adapters listed above.

This Standard also describes how the *USB Power Delivery* (*USB PD*) Discovery Process is applied to *DisplayPort Alt Mode*.

1.2 Scope

This Standard is intended to complement *USB-C* and *DP Standard*.

Normative information is provided to allow interoperability of components designed to this Standard. Informative material, when provided, may illustrate possible design implementations.

Section 1.6 provides a complete list of referenced Specifications and Standards. Nothing in this Standard is intended to change the normative mandates of the referenced *USB*-related Specifications. Nothing in this Standard is intended to change the normative mandates of *DP Standard*, except where explicitly stated. Extracts are copied from the referenced Specifications/Standards to assist with clarity of description; however, all such material is informative and the referenced Specifications/Standards should be consulted for the normative mandates. If the mandates stated within this Standard and similar mandates in the referenced Specifications/Standards conflict, the referenced Specifications/Standards shall prevail, except where explicitly stated for *DP Standard*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

1.3 Document Organization

This Standard is organized into the following sections that define the overall architecture and structure of DisplayPort Alt Mode on the *USB-C* interface:

- [Section 1 – Introduction](#)

This section provides an overview of this Standard. This section also includes a list of acronyms and glossary of terms that are used within this Standard and references to related Specifications/Standards, particularly *USB-C*.

- [Section 2 – Overview](#)

This section provides a brief overview on how this Standard leverages *USB-C*, and Alternate Mode support within *USB-C*, to support *DisplayPort* across *USB-C* connectors and cables. Other cable and connector adapter types that are defined within this Standard are also described.

- [Section 3 – Pin Assignments and Descriptions](#)

This section describes the various pin assignments on the *USB-C* connector when transmitting and receiving *DP* Main-Link data. It also describes *DP* AUX_CH and HPD mandates.

- [Section 4 – USB-C Cables Used for DisplayPort](#)

This section describes the cable assemblies and adapters used for *DisplayPort* through a *USB-C* connector.

- [Section 5 – Discovery and USB PD](#)

This section describes the mechanisms for discovery of *DisplayPort* support on a *USB-C* connector and the negotiation procedures for entering and exiting DisplayPort Alt Mode and establishing DisplayPort Configuration.

- [Section 6 – Miscellaneous](#)

This section describes *DP* Source and Sink device power management, low-power operation, and equivalent *USB* functionality.

- [Appendix A](#)

This appendix provides two examples of using DisplayPort Alt Mode.

- [Appendix B](#)

This appendix lists the contributors to past releases of this Standard.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

1.4 Terms, Acronyms, and Abbreviations

Table 1-1 lists terms, acronyms, and abbreviations that are defined in the USB-related Specifications referenced in Table 1-5, and are provided here for reference.

Table 1-1: Terms, Acronyms, and Abbreviations Referenced from USB-related Specifications (Informative)

Term	Definition
Accept Message	Accept Message. <i>USB PD</i> Message that is used to indicate that the requested action is accepted.
active cable	Cable that incorporates signal conditioning logic on the high-speed data path. An active cable is electronically marked (see electronically marked cable) to differentiate it from a passive cable , and has signal conditioning characteristics that may be controlled by the DFP_U .
Alternate Mode or Alt Mode	Operation defined by a vendor or standards organization that is associated with an SVID assigned by the USB Implementers Forum, Inc. (USB-IF; see usb.org/members). Entry to and exit from an Alternate Mode is controlled by the <i>USB PD</i> Structured VDM Enter Mode and Exit Mode commands. Note: <i>The Alternate Mode defined by this Standard is DisplayPort Alt Mode.</i>
AMA	Alternate Mode Adapter.
Attention Command	Attention Command. <i>USB PD</i> command request that is used by the UFP_U to request service from the DFP_U .
captive cable	Cable that is terminated on one end with a USB Type-C plug and has a vendor-specific means of connection (i.e., hardwired or custom detachable) on the opposite end.
CC	Configuration Channel. Used in the discovery, configuration, and management of connections across a USB Type-C cable.
Contract (Explicit Power Contract)	<i>USB PD</i> agreement on both power level and direction that is reached between a Port Pair . May be explicitly negotiated between the Port Pair (Explicit), –or– an implicit power level that is defined by the current state (Implicit). In this Standard, refers to a <i>USB PD</i> Explicit Power Contract unless specifically stated otherwise.
DFP	Downstream-Facing Port. Specifically associated with the flow of data in a USB connection. Typically, the ports on a USB host or the ports on a hub to which devices are connected. In its initial state, the DFP sources V_{BUS} and V_{CONN} , and supports data. A charge-only DFP only sources V_{BUS} (see DFP_U in Table 1-2).
direct-attach	Indicates that a USB host's DFP_U is connected directly, either with or without a cable (e.g., thumb drive), to the device's UFP_U . (<i>USB-C</i>)
Discover Identity Command	Discover Identity Command. <i>USB PD</i> command request from an Initiator to identify its Port Partner and for an Initiator (V_{CONN} Source) to identify the Responder (cable plug).
Discover Modes Command	Discover Modes Command. <i>USB PD</i> command from an Initiator to discover the Alternate Modes that its Port Partner supports and for an Initiator (V_{CONN} Source) to identify the Responder (cable plug).
Discover SVIDs Command	Discover SVIDs Command. <i>USB PD</i> command and request set that are used by an Initiator to discover the USB SID or the Vendor ID that its Port Partner supports, and for an Initiator (V_{CONN} Source) to identify the Responder (cable plug).
dock	General term used to describe products that provide multi-function capability and/or port expanding-type functionality when connected to a USB Type-C DP Source device (e.g., Multi-function cradle dock).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 1-1: Terms, Acronyms, and Abbreviations Referenced from USB-related Specifications (Informative) (Continued)

Term	Definition
DP_SID	See SID .
DR	Data Role.
DRD	Dual-Role-Data. USB Type-C acronym that refers to a USB port that is capable of operating as either a DFP or UFP (USB host or device, respectively). The role that the port initially takes is determined by the port's power role at attachment. A Source port takes on the data role of a DFP. A Sink port takes on the data role of a UFP. The port's data role may be changed dynamically using a USB PD DR_Swap Message .
DRP	Dual-Role-Power. USB Type-C acronym that refers to a USB data port that is capable of operating as either a power source or power sink. The role that the port offers may be fixed to either a power source or power sink, or may alternate between the two port states. The port's power role may be changed dynamically using a USB PD PR_Swap Message .
DR_Swap Message	DR_Swap Message. USB PD Message that is used to exchange DFP and UFP operation between Port Partners that both utilize USB Type-C connectors while maintaining the direction of power flow over V_{BUS} .
electronically marked cable	USB Type-C cable that uses USB PD to provide the cable's characteristics.
Enter Mode Process	Enter Mode Process. USB PD command sequence that uses Structured VDMs that result in the Port Partners entering a Mode.
Exit Mode Process	Exit Mode Process. USB PD command sequence that uses Structured VDMs that result in the Port Partners exiting a Mode.
Gen 1	Adjective used to refer to USB SuperSpeed 5.0-Gbps link rate.
Gen 2	Adjective used to refer to USB SuperSpeed or USB4™ 10-Gbps link rate.
Gen 3	Adjective used to refer to USB4 20-Gbps link rate.
GoodCRC Message	GoodCRC Message. USB PD Message that is sent by the receiver to acknowledge that the previous Message was correctly received (i.e., had a good CRC).
Hard Reset	Hard Reset . USB PD ordered set that is used to reset the USB PD Phy when an error event occurs.
Initiator	Port that initiates a Vendor Defined Message (VDM). Independent of the port's USB PD role (e.g., Provider, Consumer, Provider/Consumer, or Consumer/Provider).
Not_Supported Message	Not_Supported Message. USB PD Message that is used by the Responder to indicate that the Message received from the Initiator is not supported.
OCP	Over-Current Protection.
passive cable	Cable that does not incorporate any electronics to condition the data path signals. May or may not be electronically marked.
Phy	Physical layer.
Port Pair	Two attached PD-capable ports. (USB PD)
Port Partner	When a USB PD Contract (Explicit Power Contract) is negotiated between a Port Pair connected by a USB cable, these ports are known as Port Partners. (USB PD)
PR_Swap Message	PR_Swap Message. USB PD Message that may be transmitted by either Port Partner to request an exchange of V_{BUS} Source.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 1-1: Terms, Acronyms, and Abbreviations Referenced from USB-related Specifications (Informative) (Continued)

Term	Definition
Rd	Pull-down resistor on the <i>USB Type-C CC</i> wire that is used to indicate that the port is a <i>UFP</i> .
RDO	Request Data Object.
Reject Message	Reject Message. <i>USB PD</i> Message that is used by the <i>Responder</i> to indicate that the Message received from the <i>Initiator</i> is rejected and no action will be taken.
Responder	Port that responds to the Initiator of a Vendor Defined Message (<i>VDM</i>). Independent of the port's <i>USB PD</i> role (e.g., Provider, Consumer, Provider/Consumer, or Consumer/Provider).
Responder ACK response	<i>USB PD</i> response that is the normal return and shall be sent to indicate that the Command request was received and handled normally.
Responder BUSY response	<i>USB PD</i> response that is sent in the response to a <i>VDM</i> when the <i>Responder</i> is unable to immediately respond to the Command request, but the Command request may be retried.
Responder NAK response	<i>USB PD</i> response that is returned when the Command request: - Has an invalid parameter (e.g. invalid <i>SVID</i> or Mode) - May not be acted on because the configuration is not correct (e.g., a Mode that has a dependency on another Mode or a request to exit a Mode that is not Active) - Is an Unrecognized Message - Leaves the handling of “Responder NAK” up to the <i>Initiator</i>
Rp	Pull-up resistor on the <i>USB Type-C CC</i> wire that is used to indicate that the port is a <i>DFP</i> .
RX	<i>USB SuperSpeed</i> and <i>USB4</i> receiver differential pair (possibly qualified to indicate positive or negative side, and/or channel).
SBU	Sideband Use. (<i>USB-C</i>)
SID	Standard ID. Unique 16-bit unsigned value assigned by the USB-IF to identify an industry standard. The <i>USB</i> SID associated with <i>DisplayPort</i> is DP_SID.
SOP	Start of Packet. K-code marker used to delineate the start of a packet. Three start of packet sequences are defined – SOP, SOP', and SOP'', with SOP* used when referring to all three (in place of SOP/SOP'/SOP'').
SVID	Generic term for either an <i>SID</i> or <i>VID</i> . Used by <i>USB PD</i> Structured <i>VDMs</i> when requesting <i>SIDs</i> and <i>VIDs</i> from a <i>USB</i> device.
TX	<i>USB SuperSpeed</i> and <i>USB4</i> transmitter differential pair (possibly qualified to indicate positive or negative side, and/or channel).
UFP	Upstream-Facing Port. Specifically associated with the flow of data in a <i>USB</i> connection. The port on a device or a hub that connects to a <i>USB</i> host or the <i>DFP</i> of a hub. In its initial state, the UFP sinks V _{BUS} and supports data (see <i>UFP_U</i> in Table 1-2).
USB AV	<i>USB</i> Audio/Video. (<i>USB AV</i>)
USB BB	<i>USB</i> Billboard. (<i>USB BB</i>)
USB-C	Generic reference to a <i>USB-C</i> plug, <i>USB-C</i> receptacle, –or– <i>USB-C</i> cable assembly. (<i>USB-C</i>) Used interchangeably to represent <i>USB Type-C</i> (primarily outside the definitions provided within this table, which reflect the definitions as they appear in USB-related Specifications).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 1-1: Terms, Acronyms, and Abbreviations Referenced from USB-related Specifications (Informative) (Continued)

Term	Definition
USB Full-Featured Type-C cable	Full-Featured <i>USB-C</i> cable. USB Type-C-to-USB Type-C cable that supports <i>USB 2.0</i> and <i>USB 3.2</i> and/or <i>USB4</i> data operation. Includes <i>SBU</i> wires.
USB Full-Featured Type-C plug	Full-Featured <i>USB-C</i> plug. <i>USB Type-C</i> plug specifically designed to implement the <i>USB Full-Featured Type-C cable</i> .
USB PD	USB Power Delivery. (<i>USB PD</i>)
USB SuperSpeed	<i>USB SuperSpeed</i> . Adjective referring to the architectural layer portions of a device defined in <i>USB 3.2</i> when in x1 operation.
USB Type-C plug	<i>USB-C</i> plug. Plug that conforms to <i>USB Type-C</i> mechanical and electrical mandates.
USB Type-C receptacle	<i>USB-C</i> receptacle. Receptacle that conforms to <i>USB Type-C</i> mechanical and electrical mandates.
VCONN_Swap Message	VCONN_Swap Message. <i>USB PD</i> Message that may be sent by either <i>Port Partner</i> to request an exchange of V_{CONN} Source.
V_{CONN} -powered accessory	Accessory that is powered from V_{CONN} to operate in an <i>Alternate Mode</i> .
VDM	Vendor Defined Message. Provided to allow vendors to exchange information beyond that defined by <i>USB PD</i> .
VDO	Vendor Data Object. Used to transmit vendor-specific information as part of a VDM .
VID	Vendor ID. Unique 16-bit value assigned by the USB-IF to identify a vendor.
Wait Message	Wait Message. <i>USB PD</i> Message that is a valid response to a request, <i>USB PD DR_Swap Message</i>, <i>USB PD PR_Swap Message</i>, or <i>USB PD VCONN_Swap Message</i> that is sent: - To signal the <i>DP</i> Sink device that the <i>DP</i> Source device is unable to meet the request at this time - By the recipient of a <i>USB PD DR_Swap Message</i> to indicate that the recipient is unable to do a Data Role Swap at this time - By the recipient of a <i>USB PD PR_Swap Message</i> to indicate that the recipient is unable to do a Power Role Swap at this time - By the recipient of a <i>USB PD VCONN_Swap Message</i> that is not currently the V_{CONN} Source to indicate that the recipient is unable to do a V_{CONN} Swap at this time

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 1-2 lists terms, acronyms, and abbreviations used in this Standard that are **not** included in the USB-related Specifications. Uppercase is applied in the definition to indicate the letters used in acronyms and abbreviations.

Table 1-2: Non-USB-related Terms, Acronyms, and Abbreviations Used in this Standard

Term	Definition
adapter	Product with a <i>USB-C</i> plug at one end and a non- <i>USB-C</i> plug or receptacle at the other end. May be implemented as a <i>cable assembly</i> or may integrate the connectors into a single unit.
ARC	Audio Return Channel.
AUX	Used in the context of a transaction on the <i>DP auxiliary channel</i> .
AUX_CH	Auxiliary CHannel. Used in reference to the <i>DP auxiliary channel</i> physical connection.
auxiliary channel	<i>AUX_CH</i> . Physical property (pins and signal traces) over which AUX transaction traffic is routed. Half-duplex, bidirectional channel between a <i>DPTX</i> and <i>DPRX</i> . Consists of one differential pair transporting data. An upstream <i>DP</i> device is the Primary that initiates an AUX transaction. A downstream <i>DP</i> device is the Secondary that responds to the AUX transaction initiated by the Primary.
cable assembly	Product implemented as a cable with a plug at both ends. May include active electronics for identification and/or signal conditioning.
CEC	Consumer Electronics Channel.
DFP_U	Downstream-Facing Port, referred to as “ <i>DFP</i> ” in <i>USB PD</i> . Typically the ports on a <i>USB</i> host, –or– the ports on a hub, to which devices are connected.
DisplayPort or DP	<i>DisplayPort</i> or <i>DP</i> . Connector or <i>DisplayPort</i> port presented through a connector defined in <i>DP Standard</i> .
DisplayPort Configuration	<i>USB-C</i> interface configuration in which certain signals are associated with <i>DisplayPort</i> functionality, as defined in this Standard.
DisplayPort Alt Mode	DisplayPort Mode, as defined by <i>DP Standard</i> . Alternate Mode in which, when entered, the <i>USB-C</i> interface behaves according to this Standard.
DisplayPort Configure Command	<i>DisplayPort Configure</i> Command. Command that is used to configure the pin assignment for <i>DisplayPort</i> operation (see Section 5.2.4 for further details).
DisplayPort Status Update Command	<i>DisplayPort Status Update</i> Command. Command that is used to communicate the state of the <i>DP Source device/DP Sink device</i> connection (see Section 5.2.3 for further details).
DP Branch device	<i>DP Branch device</i> . Device containing a <i>DPRX</i> and either a <i>DPTX</i> or transmitter of another AV transport interface (“legacy interface”).
DPRX	<i>DisplayPort</i> Receiver.
DP Sink device	<i>DP Sink device</i> . Contains the <i>DPRX</i> and at least one stream sink. Leaf device of <i>DP</i> topology. Refers to both a <i>DP Sink device</i> –or– <i>DP Sink device</i> on a <i>DP Branch device</i> .
DP Source device	<i>DP Source device</i> . Contains the <i>DPTX</i> and at least one stream source. Root device of <i>DP</i> topology. Refers to both a <i>DP Source device</i> –or– <i>DP Source device</i> on a <i>DP Branch device</i> .
DPTX	<i>DisplayPort</i> Transmitter.
GPU	Graphics Processor Unit.
HBR	<i>DisplayPort</i> High Bit Rate (2.7Gbps/lane). (<i>DP Standard</i>)
HBR2	<i>DisplayPort</i> High Bit Rate 2 (5.4Gbps/lane). (<i>DP Standard</i>)
HBR3	<i>DisplayPort</i> High Bit Rate 3 (8.1Gbps/lane). (<i>DP Standard</i>)

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 1-2: Non-USB-related Terms, Acronyms, and Abbreviations Used in this Standard (Continued)

Term	Definition
HPD event	HPD_Low, HPD_High, –or– IRQ_HPDP event.
IFEXT	Integrated Far-End CrossTalk.
IL	Insertion Loss.
ILFit	Insertion Loss Fit.
ILFitAtNq	Insertion Loss Fit At Nyquist frequencies.
IMR	Integrated Multi-Reflection.
INEXT	Integrated Near-End CrossTalk.
IRL	Integrated Return Loss.
LTPR	LT-Tunable PHY Repeater. Link training-tunable component that is composed of the DPTX and DPRX PHY circuits. (<i>DP Standard</i>)
mDP	mini DisplayPort. (<i>DP Standard</i>)
multi-function	USB-C connector configuration that supports USB SuperSpeed x1 at the same time as DisplayPort x2.
PHY	PHYSical layer.
protocol converter	Adapter that converts from a DP to non-DP audio-video protocol.
RBR	DisplayPort Reduced Bit Rate (1.62Gbps/lane). (<i>DP Standard</i>)
UFP_U	Upstream-Facing Port, referred to as “UFP” in USB PD . Port on a device or hub that connects to a USB host –or– the DFP_U of a hub.
UHBR10	DisplayPort Ultra-High Bit Rate 10 (10Gbps/lane). (<i>DP v2.0 and higher</i>)
UHBR13.5	DisplayPort Ultra-High Bit Rate 13.5 (13.5Gbps/lane). (<i>DP v2.0 and higher</i>)
UHBR20	DisplayPort Ultra-High Bit Rate 20 (20Gbps/lane). (<i>DP v2.0 and higher</i>)
UI	Unit Interval.
USB-C DP Sink device	USB-C DP Sink device. USB-C connector configured as a DP Sink device.
USB-C DP Source device	USB-C DP Source device. USB-C connector configured as a DP Source device.
USB Configuration	USB-C interface configuration in which all signals have their default USB functionality, as specified in USB-C .
USB Mode	Mode that is in use when DisplayPort Alt Mode is not in use.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

1.5 Conventions

1.5.1 Precedence

If there is a conflict between text, figures, and tables, the precedence shall be tables, figures, and then text.

1.5.2 Keywords

Table 1-3 lists keywords that differentiate between the levels of mandates and **options** within this Standard.

Table 1-3: Keywords

Keyword	Definition
informative	Information that discusses and clarifies mandates and features.
may	Indicates a choice that does not have an implied preference.
N/A	Indicates that a field or value is not applicable, does not have a defined value, and shall not be checked or used by the recipient.
normative	Features that are mandated by this Standard.
optional	Features that are not mandated by this Standard. However, if an optional feature is implemented, the feature shall be implemented as defined by this Standard (optional normative).
RESERVED	Indicates RESERVED bits, bytes, words, fields, and code values that are set aside for future standardization. Their use and interpretation may be specified by future extensions to this Standard and, unless otherwise stated, shall not be used or adapted by vendor implementation. A RESERVED bit, byte, word, or field shall be cleared to 0 by the transmitter and ignored by the receiver. RESERVED field values shall not be transmitted by the transmitter and, if received, shall be ignored by the receiver.
shall	Indicates a mandate. Designers shall implement all such mandates to ensure interoperability with other compliant devices.
should	Indicates flexibility of choice with a preferred alternative. Equivalent to the phrases “is recommended” and “it is recommended that.”

1.5.3 Numbering

Table 1-4 lists the different types of numbering that are used within this Standard.

Table 1-4: Numbering

Number Type	Definition
Binary	Numbers that are immediately followed by a lowercase “b” (e.g., 01b). Also used for individual register bit values (i.e., 0, 1).
Decimal	Numbers that are not immediately followed by either a “b” or “h” (e.g., 5).
Hexadecimal	Numbers that are immediately followed by a lowercase “h” (e.g., 10h).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

1.6 Reference Documents

Table 1-5 lists the various reference documents used and/or referenced within this Standard. Users of this Standard are advised to ensure that they have the latest versions/revisions of reference documents.

Table 1-5: Reference Documents

Document	Version/ Revision ^a	Publication Date	Referenced As
<i>High-Bandwidth Digital Content Protection System, Amendment for DisplayPort^b</i>	Version 1.3 Revision 1.1	January 15, 2010	<i>HDCP for DP r1.3</i>
<i>High-Bandwidth Digital Content Protection System, Mapping HDCP to DisplayPort^b</i>	Revision 2.2	December 21, 2012	<i>HDCP to DP r2.2</i>
	Revision 2.3	January 22, 2019	<i>HDCP to DP r2.3</i>
<i>High-Bandwidth Digital Content Protection System (applicable to HDMI)^b</i>	Revision 1.4	July 8, 2009	<i>HDCP to HDMI r1.4</i>
<i>High-Bandwidth Digital Content Protection System, Mapping HDCP to HDMI^b</i>	Revision 2.2	February 13, 2013	<i>HDCP to HDMI r2.2</i>
	Revision 2.3	February 28, 2018	<i>HDCP to HDMI r2.3</i>
<i>High-Definition Multimedia Interface Specification^c</i>	Version 2.1	November 28, 2017	<i>HDMI v2.1</i> <i>HDMI</i>
<i>Universal Serial Bus 2.0 Specification^d</i>	Release 2.0	April 27, 2000	<i>USB 2.0</i> <i>USB</i>
<i>Universal Serial Bus 3.2 Specification^d</i>	Release 3.2	September 22, 2017	<i>USB 3.2</i> <i>USB</i>
<i>Universal Serial Bus 4 (USB4TM) Specification^d</i>	Version 1.0	August 29, 2019	<i>USB4</i> <i>USB</i>
<i>Universal Serial Bus Device Class Definition for Audio/Video Devices – AV Device Class Overview & AVFunction Definition^d</i>	Release 1.0	December 07, 2011	<i>USB AV</i>
<i>Universal Serial Bus Device Class Definition for Billboard Devices^d</i>	Revision 1.21 ^g	September 8, 2016	<i>USB BB</i>
<i>Universal Serial Bus Power Delivery Specification^d</i>	Revision 2.0, Version 1.3 ^{e g}	January 12, 2017	<i>USB PD 2.0</i> <i>USB PD 3.0</i>
	Revision 3.0, Version 2.0 ^{e g}	August 29, 2019	<i>USB PD</i>
<i>Universal Serial Bus Type-C Cable and Connector Specification^d</i>	Release 2.0 ^g	August 29, 2019	<i>USB-C</i> <i>USB Type-C</i>
<i>VESA DisplayPort (DP) Standard^f</i>	Version 2.0 ^g	June 26, 2019	<i>DP v2.0</i> <i>DisplayPort</i> <i>DP</i> <i>DP Standard</i>

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 1-5: Reference Documents (Continued)

Document	Version/ Revision ^a	Publication Date	Referenced As
<i>VESA DisplayPort Dual-Mode (DP++) Standard^f</i>	Version 1.1	February 8, 2013	<i>DP++ DP++ Standard</i>
<i>VESA Glossary of Terms and Acronyms Specification^f</i>	Current	Current	
<i>VESA Intellectual Property Rights (IPR) Policy^f</i>	200D	March 27, 2017	

- a. All references include subsequently published errata, specification change notices or engineering change notices, etc.
- b. Published by Digital Content Protection (DCP). See digital-cp.com.
- c. Published by HDMI Licensing Administrator, Inc. See hdmi.org.
- d. Published by USB Implementers Forum, Inc. See usb.org/developers/docs.
- e. Implementations of this Standard may be compliant with the relevant mandates specified in **USB PD 2.0** or those specified in **USB PD 3.0**. **USB PD 3.0** should be followed because it provides significant improvements in **USB PD** bus arbitration. **USB PD 3.0** also mandates fallback to **USB PD 3.0** for cases in which the Port Partner does not support **USB PD 3.0**. A future version of this Standard may mandate use of **USB PD 3.0** to take advantage of **USB PD** bus arbitration improvements.
- f. See vesa.org/vesa-member-downloads/.
- g. This version of the referenced Specification/Standard is correct at the time of publication of this Standard. In the event that a later version of the referenced Specification/Standard is published, reference should be made to the latest published version.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

2 Overview

This section provides a brief overview on how this Standard leverages *USB-C*, and Alternate Mode support within *USB-C*, to support *DisplayPort* across *USB-C* connectors and cables. Other cable and connector adapter types that are defined within this Standard are also described.

2.1 Introduction to USB-C and Alternate Mode Operation

USB-C is an open standard that is published by USB Implementers Forum, Inc. (USB-IF), as indicated in Table 1-5. *USB-C* describes interoperable *USB-C* plugs, receptacles, and cables that include support for up to four high-speed differential signal pairs, each of which is used to transport high-speed data. This *USB-C* connectivity ecosystem also includes support for low-speed sideband signals, including a data pair that is dedicated for *USB 2.0*, and support for power delivery. These lower-speed signals are also used to discover Alternate Mode capability and configuration.

In addition to providing support for *USB 2.0*, *USB 3.2*, *USB4*, *USB PD*, and other Specifications published by the USB-IF, *USB-C* also defines Alternate Mode operation that allows the use of *USB-C* plugs, receptacles, and cables by other interface Standards. This Standard uses Alternate Mode operation to transport *DP* across *USB-C* plugs, receptacles, and cables.

As described in this Standard, DisplayPort Alt Mode may use all four of the *USB-C* high-speed differential pairs to implement four *DP* Main-Link lanes from a *DP* Source device to a *DP* Sink device, –or– from any downstream-facing *DP* device to any upstream-facing *DP* device. This means that DisplayPort Alt Mode, using *USB-C* connectors and cabling, may have the same performance capability as native *DP* connectors and cabling.

In addition to supporting four *DP* Main-Link lanes, this Standard also describes configurations that support up to two *DP* Main-Link lanes along with two high-speed differential pairs dedicated for a single *USB 3.2* bidirectional lane. This provides simultaneous support of DisplayPort Alt Mode with high-speed *USB 3.2* data across the same *USB-C* connector and cable, which is useful for external hardware such as docking stations.

USB4 includes the ability to tunnel *DisplayPort*, *USB 3.x*, and PCI Express; however, support for *USB4* may **not** be combined with DisplayPort Alt Mode due to conflicts in the sideband signaling between the two protocols. *USB4* hosts and hubs support DisplayPort Alt Mode and may negotiate to enter DisplayPort Alt Mode with downstream *USB PD*-capable devices that support DisplayPort Alt Mode, –or– may alternately tunnel *DisplayPort* over *USB4* if the downstream device supports *USB4*.

Within *USB-C*, use of the sideband and communication channel signals that are provided within the *USB-C* interface is also described for Alternate Mode operation. This Standard includes use of these signals to accommodate *DP* AUX_CH, HPD, and power, as needed, for protocol adapters and active cables.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 2-1 illustrates the *USB-C* receptacle, as specified in *USB-C*. Figure 2-2 illustrates the comprehensive functional signal plan for the *USB-C* plug. Both figures are provided here for reference. (See *USB-C* for further details.)

A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
GND	TXp1	TXn1	V _{BUS}	CC1	D+	D-	SBU1	V _{BUS}	RXn2	RXp2	GND
GND	RXp1	RXn1	V _{BUS}	SBU2	D-	D+	CC2	V _{BUS}	TXn2	TXp2	GND
B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1

Figure 2-1: USB-C Receptacle Interface (Front View) (Informative)

A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1
GND	RXp2	RXn2	V _{BUS}	SBU1	D-	D+	CC	V _{BUS}	TXn1	TXp1	GND
GND	TXp2	TXn2	V _{BUS}	V _{CONN}			SBU2	V _{BUS}	RXn1	RXp1	GND
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12

Figure 2-2: USB-C Plug Interface (Front View) (Informative)

Notable differences between the *USB-C* receptacle illustrated in Figure 2-1 and *USB-C* plug illustrated in Figure 2-2 are as follows:

- Only one CC pin is connected through the cable; the “unused” receptacle CC pin is reconfigured as V_{CONN}, which provides a power source that is available for electronics that may be incorporated into *USB-C* plugs, cable assemblies, –or– adapters).
- Only one D+/D- pair instead of two, which is used for *USB 2.0* data, is connected through the cable. Two pairs are implemented in the receptacle to enable flip-ability of the plug and cable.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

2.2 USB-C Receptacles and Plugs

For *USB-C* interfaces that support DisplayPort Alt Mode, this Standard uses the *USB-C* receptacles and plugs as specified in *USB-C*.

For native *DP* interfaces, such as the plug used on a USB-C-to-DP conversion cable (see [Section 4.3](#)) or the receptacle used on a USB-C plug-to-DP active receptacle adapter (see [Section 4.7](#)), this Standard uses the *DP* plugs and receptacles as defined in *DP Standard*. This includes full-size *DP* and mini DP (mDP) connectors.

Protocol adapters (see [Section 4.4](#)), *HDMI*, VGA, or DVI plugs and receptacles are defined by the corresponding Specification/Standard.

2.3 USB-C Cables and Interconnectivity

DisplayPort Alt Mode is supported by the *USB-C* cables defined by the USB-IF in *USB-C*, and by the USB-C-to-DP/mDP cables defined in this Standard. Each combination of *DP* and *USB* signaling is associated with a specific pin assignment (see [Section 3.1](#)). A *DP* Source or Sink device may support fewer than the maximum permitted number of lanes, as defined in [Section 3.2.1](#).

Note: *Pin Assignment C is mandatory for DP Source and Sink devices (DP Source devices – see [Table 3-10](#) and [Table 3-11](#); DP Sink devices – see [Table 3-16](#) and [Table 3-17](#)) that support DisplayPort Alt Mode receptacles (see [Section 3.2.1](#)) – this provides a guarantee of interoperability on passive USB-C cables.*

[Table 2-1](#) lists the types of USB-C-to-USB-C passive cables that are specified by *USB-C* and support DisplayPort Alt Mode. These cables are guaranteed to support DisplayPort Alt Mode by design and certification, are **not** certified by VESA and shall **not** use any VESA logo.

Table 2-1: Inherent Support of DisplayPort Alt Mode by Full-Featured USB-C Standard Passive Cable Assemblies Specified by the USB-IF

Reference Number in Table 2-4	Cable Type	DisplayPort Alt Mode Directionality and Main-Link Lane-count Capability ^a	Minimum DP Bit-rate Capability ^a	Certification and Logo Applicability
1	Full-Featured <i>USB-C USB 3.2</i> Gen 1 (5Gbps) Cables	Bidirectional 4 Lanes	UHBR10	<i>USB</i> certification and logo only
2	Full-Featured <i>USB-C USB 3.2</i> and <i>USB4</i> Gen 2 (10Gbps) Cables			No VESA certification or logo
3	Full-Featured <i>USB-C USB4</i> Gen 3 (20Gbps) Cables			

a. DisplayPort Alt Mode directionality, lane-count, and bit-rate support is guaranteed by design when implemented as per the applicable USB-related Specification and certified as the denoted cable type. DP bit rates from RBR up to the listed bit rate are supported.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 2-2 lists the types of USB-C-to-USB-C active cables that are specified by *USB-C* and may optionally support DisplayPort Alt Mode. Such cables that are enabled to support DisplayPort Alt Mode may be certified by VESA with the *DisplayPort* logo and *TBD* bit-rate designator.

Table 2-2: Optional Support of DisplayPort Alt Mode by USB-C Active Cable Assemblies and Captive Cables Specified by the USB-IF

Reference Number in Table 2-4	Cable Type	DisplayPort Alt Mode Directionality and Main-Link Lane-count Capability Mandates ^a	Minimum DP Bit-rate Capability Options for VESA Certification ^b	Certification and Logo Applicability
4	<i>USB-C</i> Active Cables, Certified for <i>USB 3.2</i> Gen 1 (5Gbps) or Gen 2 (10Gbps) Operation, and Enabled with DisplayPort Alt Mode Support	Bidirectional 4 Lanes	HBR3 UHBR10 UBHR13.5 ^c UHBR20 ^c	<i>USB</i> certification and logo with a bit-rate capability designator
5	<i>USB-C</i> Active Cables, Certified for <i>USB4</i> Gen 2 (10Gbps) or Gen 3 (20Gbps) Operation, and Enabled with DisplayPort Alt Mode Support			VESA certification and <i>DisplayPort</i> logo with a bit-rate capability designator
6	Captive Cable with <i>USB-C</i> connector	Device-specific	Device-specific	<i>USB</i> and <i>DisplayPort</i> logos, as applicable

- a. Configuration that shall be supported for VESA certification.
- b. *DP* bit-rate capability levels supported through bit-rate compliance testing and certification. Bit rates from RBR up to the listed bit rates are supported.
- c. UHBR13.5 and UHBR20 electrical interface and compliance test specifications will be defined by future VESA Specifications.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 2-3 lists the types of *USB-C*-enabled cables to *DP* adapters defined by this Standard that support DisplayPort Alt Mode. These cables and adapters may be certified by VESA and may be marked with the *DisplayPort* logo and *TBD* bit-rate designator.

Table 2-3: Support of DisplayPort Alt Mode by USB-C-enabled Cables and Adapters Defined by this Standard

Reference Number in Table 2-4	Cable Type	DisplayPort Alt Mode Directionality and Main-Link Lane-count Capability Mandates ^a	Minimum DP Bit-rate Capability Options for VESA Certification ^b	Certification and Logo Applicability
7	USB-C-to-DP Passive Cable	Bidirectional 4 Lanes	UHBR10 UBHR13.5 ^c UHBR20 ^c	<i>USB</i> compliance testing may be performed for <i>USB BB</i> and <i>USB PD</i> functions; however, a <i>USB</i> logo is not applicable to these products
8	USB-C-to-DP Active Cables	Unidirectional ^d or Bidirectional 4 Lanes	HBR3 UHBR10 UBHR13.5 ^c	
9	USB-C Plug-to-DP Active Receptacle Adapters	Bidirectional 4 Lanes	UHBR20 ^c	
10	DisplayPort Alt Mode Protocol Adapter Cable with <i>HDMI</i> , VGA, or DVI Plug	Unidirectional Lane count is device-specific	Protocol-specific	VESA certification and <i>DisplayPort</i> logo with a bit-rate capability designator
11	DisplayPort Alt Mode Protocol Adapter with <i>HDMI</i> , VGA, or DVI Receptacle			

- a. Configuration that shall be supported for VESA certification.
- b. Minimum *DP* bit-rate capability levels supported through bit-rate compliance testing and certification. Bit rates from RBR up to the listed bit rates are supported.
- c. UHBR13.5 and UHBR20 electrical interface and compliance test specifications will be defined by future VESA Specifications.
- d. Unidirectional USB-C-to-DP active cables shall be marked in a way that indicates the correct connection orientation.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 2-4 provides further description of the devices listed in Table 2-1 through Table 2-3, by reference number.

**Table 2-4: Further Description of Cables and Adapters
Listed in Table 2-1 through Table 2-3 (Informative)**

Reference Number	Cable Type	Defining Specification(s)/Standard(s)
	General Description and Capabilities	
1, 2, 3	Full-Featured <i>USB-C</i> Cable Assemblies (Passive)	<i>USB BB</i> ; <i>USB-C</i> ; <i>USB PD</i> ; Section 4.2
	Full-Featured <i>USB-C</i> cables that are not identified as “active” in the <i>USB PD Discover Identity</i> Command response. Passive cables that are certified for <i>USB 3.2</i> Gen 1/Gen 2 and <i>USB4</i> Gen 2/Gen 3 operation are defined to support DisplayPort Alt Mode and <i>DP</i> bit rates up to UHBR10 and up to four lanes of operation. Bidirectional operation is supported.	
4, 5	<i>USB-C</i> Active Cable Assemblies	<i>USB BB</i> ; <i>USB-C</i> ; <i>USB PD</i> ; Section 4.2
	<i>USB-C</i> active cables include a <i>USB-C</i> plug at both ends and identified as “active” in the <i>USB PD Discover Identity</i> Command response. Such cables are not defined within <i>USB-C</i> to support DisplayPort Alt Mode (DisplayPort Alt Mode support is beyond the scope of <i>USB-C</i>). However, active cables may be designed to support DisplayPort Alt Mode up to a particular <i>DP</i> bit rate, and up to a particular number of lanes. Bidirectionality shall be supported. Cables that support DisplayPort Alt Mode shall advertise this by way of <i>USB PD</i> and the DP_SID.	
6	Captive Cables with a <i>USB-C</i> Connector	<i>USB BB</i> ; <i>USB-C</i> ; <i>USB PD</i> ; Section 4.6
	Captive cable assembly, as defined by <i>USB-C</i> , is a cable assembly that is terminated on one end with a <i>USB-C</i> plug and has a vendor-specific connect means (i.e., hardwired or custom detachable) on the opposite end. Such cables are not defined to support DisplayPort Alt Mode; therefore, such support is beyond the scope of <i>USB-C</i> . Such cables may be designed to support DisplayPort Alt Mode.	
7	USB-C-to-DP Conversion Cables (Passive)	<i>USB BB</i> ; <i>USB PD</i> ; Section 4.3
	USB-C-to-DP conversion cables that are not identified as “active” in the <i>USB PD Discover Identity</i> Command and that have a <i>USB-C</i> plug at one end and a full-size <i>DP</i> plug or mDP plug at the other end. Such cables are VESA certified to the DP8K performance level and support <i>DP</i> bit rates up to UHBR20, and up to four lanes of bidirectional operation.	
8	Active USB-C-to-DP Conversion Cables	<i>USB BB</i> ; <i>USB PD</i> ; Section 4.1, Section 4.2, Section 4.8, Section 4.9, and Section 4.10
	Active USB-C-to-DP conversion cables have a <i>USB-C</i> plug at one end and a full-size <i>DP</i> plug or mDP plug at the other end and identified as “active” in the <i>USB PD Discover Identity</i> Command response. Such cables are VESA certified to the DP8K performance level and support <i>DP</i> bit rates up to UHBR20, and up to four lanes of unidirectional or bidirectional operation. These cables report DisplayPort Alt Mode support by way of <i>USB PD</i> and the DP_SID.	
9	Active USB-C Plug-to-DP Receptacle Adapters	<i>USB BB</i> ; <i>USB PD</i> ; Section 4.7
	Active USB-C plug-to-DP receptacle adapters have a <i>USB-C</i> plug at one end and a full-size <i>DP</i> receptacle or mDP receptacle at the other end. Such adapters are VESA certified to the DP8K performance level and support <i>DP</i> bit rates up to UHBR20, and up to four lanes of bidirectional operation.	

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 2-4: Further Description of Cables and Adapters
Listed in Table 2-1 through Table 2-3 (Informative) (Continued)

Reference Number	Cable Type	Defining Specification(s)/Standard(s)
	General Description and Capabilities	
10	DisplayPort Alt Mode Protocol Adapter Cables with an <i>HDMI</i> , VGA, or DVI Plug	<i>HDMI v2.1</i> ; <i>USB BB</i> ; <i>USB PD</i> ; Section 4.4.1 through Section 4.4.3
	Protocol adapter cable with <i>HDMI</i> , VGA, or DVI has a <i>USB-C</i> plug at one end and an <i>HDMI</i> , VGA, or DVI plug at the other end. <i>DP</i> bit-rate and lane-count support varies according to the supported output protocol. The <i>USB-C</i> connector end plugs into the receptacle of a downstream-facing DisplayPort Alt Mode-enabled device and provides video output that is converted to <i>HDMI</i> , VGA, or DVI from the appropriate plug type.	
11	DisplayPort Alt Mode Protocol Adapters with an <i>HDMI</i> , VGA, or DVI Receptacle	<i>HDMI v2.1</i> ; <i>USB BB</i> ; <i>USB PD</i> ; Section 4.4.1 through Section 4.4.3
	Protocol adapter with <i>HDMI</i> , VGA, or DVI has a <i>USB-C</i> plug at one end and an <i>HDMI</i> , VGA, or DVI receptacle at the other end. <i>DP</i> bit-rate and lane-count support varies according to the supported output protocol. The <i>USB-C</i> connector end plugs into the receptacle of a downstream-facing DisplayPort Alt Mode-enabled device and provides video output that is converted to <i>HDMI</i> , VGA, or DVI from the appropriate receptacle type.	

2.4 Multi-function Systems

A *USB-C* interface may support one of the following protocols:

- *USB* SuperSpeed, as specified in *USB 3.2*, and/or *USB4*
- *DisplayPort*, as defined in *DP Standard*
- Both *USB* SuperSpeed x1 and *DP* x2 simultaneously, on separate *USB* SuperSpeed data signal pairs

Multi-function systems may include *USB* hosts, docks, and peripheral *USB-C* devices that support DisplayPort Alt Mode. Multi-function systems also include dongle-like devices that provide both protocol conversation as well as a *USB* port, in addition to the *USB-C* plug that connects a DisplayPort Alt Mode-enabled device with the DPTX.

Some systems with multi-function support, such as docks and displays with *USB* functionality, may need the ability to change between various USB/DisplayPort Configurations, depending on how the system is being implemented. It may also be necessary to pre-establish a particular configuration when the system is first connected, although the configuration may need to change again later. To support this, a mechanism is provided for the system to indicate its preferred configuration to its Port Partner (see Section 5.2.3).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3 Pin Assignments and Descriptions

This section describes the various pin assignments on the *USB-C* connector when transmitting and receiving *DP* Main-Link data. It also describes *DP* AUX_CH and HPD mandates.

All *USB-C* pin assignments defined in this Standard define the pin assignments on the *USB-C* receptacle and are provided for when the plug is inserted in normal and inverted (“flipped”) orientations. The corresponding plug’s pin assignments are those specified for the receptacle when the plug is inserted in the receptacle in normal orientation. The plug pin assignment does **not** change with orientation. The system with the *USB-C* receptacle shall provide the appropriate pin functions on the receptacle to match the plug pin assignments when the plug is inserted in the receptacle in normal or flipped orientation. The system design may implement support for plug rotation by multiplexers in the system design or by pin function adaptation within the *DP* Source or Sink device’s silicon design. The *USB-C* plugs on adapters and USB-C-to-DP cables always use the normal orientation pin assignment. Pin A5 is always CC and pin B5 is always V_{CONN} .

A *DP* Source device refers to both a *DP* Source device –or– *DP* Source device on a *DP* Branch device. A *DP* Sink device refers to both a *DP* Sink device –or– *DP* Sink device on a *DP* Branch device. A *USB-C* receptacle that supports *DP* Source device functionality shall support the *DP* Source device pin assignments defined in [Section 3.2.1](#). Likewise, a *USB-C* receptacle that supports *DP* Sink devices shall support the *DP* Sink device pin assignments defined in [Section 3.2.1](#).

A *USB-C* plug that is intended to plug directly into a receptacle-based *DP* Source device (e.g., a plug that behaves as a *DP* Sink device –or– *DP* Sink device on a *DP* Branch device) shall support one or more *DP* Source device pin assignments. Likewise, a *USB-C* plug that is intended to plug directly into a receptacle-based *DP* Sink device (e.g., a plug that behaves as a *DP* Source device –or– *DP* Source device on a *DP* Branch device) shall support one or more *DP* Sink device pin assignments.

Note that port directionality advertised for a *USB-C* plug is opposite to the directionality of the *DisplayPort* functionality implemented behind the plug. For example, a USB-C-to-HDMI or USB-C-to-VGA protocol converter will implement *DisplayPort* upstream functionality (it is a DPRX); however, the *USB-C* plug on the protocol converter will advertise a *DP* Source device pin assignment because it is intended to plug into a *USB-C DP* Source device receptacle.

Pin assignments are advertised and configured as defined in [Section 4](#) and [Section 5](#).

All associations of *USB-C* functions with specific pins on the *USB-C* connector are informative. Reference should be made to *USB-C* for normative *USB-C* pin functionality.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.1 Pin Assignment Overview

Note: Pin assignments A, B, and F have been deprecated.

Three possible pin assignments are defined for **USB-C** when the **USB-C** connector of a video Source device is configured for DisplayPort Alt Mode – C, D, or E. Three possible pin assignments are also defined for **USB-C** when the **USB-C** connector of a video Sink device is configured for DisplayPort Alt Mode – C, D, or E. Both ends of a USB-C-to-USB-C cable shall use pin assignments with the same letter. Application of the various pin assignments within various scenarios is provided in [Section 4](#).

Pin Assignments C and D are intended for use with USB-C-to-USB-C cables and with adapters from **USB-C** to other video standards. Pin Assignment E is intended for use with adapters from USB-C-to-DP plugs or receptacles.

DP protocol is carried when using Pin Assignments C, D, and E.

[Table 3-1](#) summarizes the **DP** Source and Sink device pin assignments.

Table 3-1: DP Source and Sink Device Pin Assignment Summary

Device Type	Pin Assignment	Description
DP Source Device	C, E	DP Source device Pin Assignments C and E are identical. - USB-C-to-USB-C cable assembly has crossovers between TX1 and RX1, and TX2 and RX2, as specified in USB-C. DP cable assembly has crossovers defined in Section 4.3.
	D	Multi-function pin assignment used for simultaneously carrying one lane of USB SuperSpeed and two DP lanes.
DP Sink Device	C	Derived from the DP Source device Pin Assignment C, taking into account the pin mapping of A2 – A3 to B11 – B10, A11 – A10 to B2 – B3 and SBU1 to SBU2 crossovers in the USB-C-to-USB-C cable assembly.
	D	Derived from the DP Source device Pin Assignment D taking into account the pin mapping of A2 – A3 to B11 – B10, A11 – A10 to B2 – B3 and SBU1 to SBU2 crossovers in the USB-C-to-USB-C cable assembly.
	E	Derived from DP Source device Pin Assignment E, taking into account the mandate that the USB-C-to-DP/mDP cable assembly is bidirectional. In effect, this exchanges ML_Lane_0, ML_Lane_1, ML_Lane_2, and ML_Lane_3 with ML_Lane_3, ML_Lane_2, ML_Lane_1, and ML_Lane_0, respectively.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.2 USB-C DP Pin Assignments

Note: *TX changes to TX1 or TX2, based on whether the plug orientation is normal or flipped. respectively. RX changes to RX1 or RX2, based on whether the plug orientation is normal or flipped. respectively.*

Table 3-2 summarizes the *DP* Source device pin assignments when the attached plug is in normal orientation. Table 3-3 summarizes the *DP* Source device pin assignments when the attached plug is in flipped orientation.

Table 3-2: USB-C Receptacle DP Source Device Pin Assignment Summary – Normal Plug Orientation

Receptacle Pin Number	Pin Assignment and Cable Type ^a		
	C	D ^b	E
	USB-C-to-USB-C or protocol converter	USB-C-to-USB-C or protocol converter	USB-C-to-DP
A11 – A10	ML_Lane_0	ML_Lane_0	ML_Lane_0
A2 – A3	ML_Lane_2	TX	ML_Lane_2
B11 – B10	ML_Lane_3	RX	ML_Lane_3
B2 – B3	ML_Lane_1	ML_Lane_1	ML_Lane_1
A8	AUX_CH_P	AUX_CH_P	AUX_CH_P
B8	AUX_CH_N	AUX_CH_N	AUX_CH_N

- a. See Table 3-10, Table 3-12, and Table 3-14 for Pin Assignments C, D, and E, respectively.
b. Pin Assignment D is **not** supported by USB4.

Table 3-3: USB-C Receptacle DP Source Device Pin Assignment Summary – Flipped Plug Orientation

Receptacle Pin Number	Pin Assignment and Cable Type ^a		
	C	D ^b	E
	USB-C-to-USB-C or protocol converter	USB-C-to-USB-C or protocol converter	USB-C-to-DP
A11 – A10	ML_Lane_3	RX	ML_Lane_3
A2 – A3	ML_Lane_1	ML_Lane_1	ML_Lane_1
B11 – B10	ML_Lane_0	ML_Lane_0	ML_Lane_0
B2 – B3	ML_Lane_2	TX	ML_Lane_2
A8	AUX_CH_N	AUX_CH_N	AUX_CH_N
B8	AUX_CH_P	AUX_CH_P	AUX_CH_P

- a. See Table 3-11, Table 3-13, and Table 3-15 for Pin Assignments C, D, and E, respectively.
b. Pin Assignment D is **not** supported by USB4.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 3-4 summarizes the *DP* Sink device pin assignments when the attached plug is in normal orientation. Table 3-5 summarizes the *DP* Sink device pin assignments when the attached plug is in flipped orientation.

Table 3-4: USB-C Receptacle DP Sink Device Pin Assignment Summary – Normal Plug Orientation

Receptacle Pin Number	Pin Assignment and Cable Type ^a		
	C	D ^b	E
	USB-C-to-USB-C or protocol converter	USB-C-to-USB-C or protocol converter	DP-to-USB-C
A11 – A10	ML_Lane_1	ML_Lane_1	ML_Lane_3 ^c
A2 – A3	ML_Lane_3	TX	ML_Lane_1 ^c
B11 – B10	ML_Lane_2	RX	ML_Lane_0 ^c
B2 – B3	ML_Lane_0	ML_Lane_0	ML_Lane_2 ^c
A8	AUX_CH_N	AUX_CH_N	AUX_CH_P ^d
B8	AUX_CH_P	AUX_CH_P	AUX_CH_N ^d

- a. See Table 3-16, Table 3-18, and Table 3-20 for Pin Assignments C, D, and E, respectively.
b. Pin Assignment D is **not** supported by USB4.
c. The polarity on each lane is inverted when Pin Assignment E is used at the *DP* Sink device receptacle.
d. AUX_CH_P and AUX_CH_N are inverted with respect to the other pin assignments when Pin Assignment E is used at the *DP* Sink device receptacle.

Table 3-5: USB-C Receptacle DP Sink Device Pin Assignment Summary – Flipped Plug Orientation

Receptacle Pin Number	Pin Assignment and Cable Type ^a		
	C	D ^b	E
	USB-C-to-USB-C or protocol converter	USB-C-to-USB-C or protocol converter	DP-to-USB-C
A11 – A10	ML_Lane_2	RX	ML_Lane_0 ^c
A2 – A3	ML_Lane_0	ML_Lane_0	ML_Lane_2 ^c
B11 – B10	ML_Lane_1	ML_Lane_1	ML_Lane_3 ^c
B2 – B3	ML_Lane_3	TX	ML_Lane_1 ^c
A8	AUX_CH_P	AUX_CH_P	AUX_CH_N ^d
B8	AUX_CH_N	AUX_CH_N	AUX_CH_P ^d

- a. See Table 3-17, Table 3-19, and Table 3-21 for Pin Assignments C, D, and E, respectively.
b. Pin Assignment D is **not** supported by USB4.
c. The polarity on each lane is inverted when Pin Assignment E is used at the *DP* Sink device receptacle.
d. AUX_CH_P and AUX_CH_N are inverted with respect to the other pin assignments when Pin Assignment E is used at the *DP* Sink device receptacle.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Pins A2, A3, B2, and B3 shall be AC-coupled in receptacle-based systems between the port orientation switch, –or– *DP* Source or Sink device (when plug rotation support is provided within the *DP* silicon), and the *USB-C* connector. Pins A10, A11, B10, and B11 may be DC-connected in receptacle-based systems. These mandates match the *USB* SuperSpeed operation specifications for using these pins. Adapter cables and plug-based systems shall include AC capacitors on pins A10, A11, B10, and B11 (see [Figure 4-28](#)).

When configuring channels, take care to ensure that the channels may be configured to carry either *USB* SuperSpeed or *DP* signals. In particular, capacitors should be chosen to be suitable for both protocols. A 220-nF $\pm 20\%$ AC-coupling capacitor should be used on the channels shared with *USB* SuperSpeed TX channels in the system design. The *USB* SuperSpeed RX channel AC-coupling capacitors, if implemented, shall be as specified in [USB 3.2](#). Adapters should use a 220-nF $\pm 20\%$ AC-coupling capacitor on the channels associated with the *USB* SuperSpeed RX pins (A10, A11, B10, and B11).

3.2.1 Pin Assignment Support Mandates

[Table 3-6](#) and [Table 3-7](#) list the mandatory pin assignments for a *USB-C* receptacle that supports *DisplayPort*. The mandatory pin assignments depend on whether the port also implements a *USB* host or device (including a Sink with Accessory Support, as specified in *USB-C*).

Table 3-6: USB-C Receptacle DP Source Device Pin Assignment Mandates

	Pin Assignment and Cable Type		
	C	D	E
	USB-C-to-USB-C or protocol converter	USB-C-to-USB-C or protocol converter	USB-C-to-DP
USB Host	Mandatory	Optional	Mandatory
USB Device	Mandatory	Optional	Mandatory

Table 3-7: USB-C Receptacle DP Sink Device Pin Assignment Mandates

	Pin Assignment and Cable Type		
	C	D	E
	USB-C-to-USB-C	USB-C-to-USB-C	USB-C-to-DP
USB Host	Mandatory	Optional	Mandatory
USB Device	Mandatory	Optional	Mandatory

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 3-8 lists the mandates for a **DP** Sink device that is implemented with a **USB-C** plug and is therefore intended to plug into a **DP** Source device receptacle. The **USB-C** plug on the **DP** Sink device therefore supports one or more of the **DP** Source device receptacle pin assignments.

Table 3-8: USB-C Plug DP Source Device Pin Assignment Mandates (for DP Sink Device)

	Pin Assignment		
	C	D	E
USB Host	Mandatory if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Mandatory if C is not supported; otherwise, optional
USB Device	Mandatory if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Mandatory if C is not supported; otherwise, optional

Table 3-9 lists the mandates for a **DP** Source device that is implemented with a **USB-C** plug and is therefore intended to plug into a **DP** Sink device receptacle. The **USB-C** plug on the **DP** Source device therefore supports one or more of the **DP** Sink device receptacle pin assignments.

Table 3-9: USB-C Plug DP Sink Device Pin Assignment Mandates (for DP Source Device)

	Pin Assignment ^a		
	C	D	E
USB Host	Mandatory if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Mandatory if C is not supported; otherwise, optional
USB Device	Mandatory if E is not supported; otherwise, optional	Optional if C is supported; otherwise, prohibited	Mandatory if C is not supported; otherwise, optional

a. See Table 3-16 through Table 3-21 for the pin assignments.

3.2.2 DisplayPort Support Mandates

Systems that support a **DP** Source device on a **USB-C** receptacle shall support at least HBR.

Systems that support a **DP** Source or Sink device on a **USB-C** receptacle may support one, two, or four **DP** lanes, according to the system's functionality.

Multi-function cradle docks that support **DisplayPort** on a **USB-C** plug that are designed to complement a specific product or product family and are restricted by mechanical constraints, may support one, two, or four **DP** lanes, according to the complementary system's functionality.

Notes: See Section 4.4 for the **HDMI**, **VGA**, or **DVI** protocol converter lane count and bandwidth mandates.

The number of **DP** lanes used for a particular connection shall follow **DP Standard** mandates.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.3 DP Source Device Pin Assignment C

Table 3-10 and Table 3-11 define use and configuration of the *USB-C* connector pins when the pins are reconfigured to support *DisplayPort* in normal and flipped plug orientations, respectively. *DP* electrical mandates for Pin Assignment C shall comply with *DP Standard* for bit rates up to UHBR10, HBR3, HBR2, HBR, or RBR, as supported by the system. The protocol shall be as defined in *DP Standard*. Pin Assignment C is appropriate only for USB-C-to-USB-C passive and active cables and *DP* Sink devices with *USB-C* plugs.

Table 3-10: DP Source Device Pin Assignment C – Normal Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 2	ML_Lane_2_P	B11	Positive half of Lane 3	ML_Lane_3_P
A3	Negative half of Lane 2	ML_Lane_2_N	B10	Negative half of Lane 3	ML_Lane_3_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_N
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_P	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 0	ML_Lane_0_N	B3	Negative half of Lane 1	ML_Lane_1_N
A11	Positive half of Lane 0	ML_Lane_0_P	B2	Positive half of Lane 1	ML_Lane_1_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

Table 3-11: DP Source Device Pin Assignment C – Flipped Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML_Lane_1_P	B11	Positive half of Lane 0	ML_Lane_0_P
A3	Negative half of Lane 1	ML_Lane_1_N	B10	Negative half of Lane 0	ML_Lane_0_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_P
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_N	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 3	ML_Lane_3_N	B3	Negative half of Lane 2	ML_Lane_2_N
A11	Positive half of Lane 3	ML_Lane_3_P	B2	Positive half of Lane 2	ML_Lane_2_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.4 DP Source Device Pin Assignment D

Table 3-12 and Table 3-13 define use and configuration of the *USB-C* connector pins when the pins are reconfigured to support *DisplayPort* combined with *USB* SuperSpeed (multi-function) in normal and flipped plug orientations, respectively. *DP* electrical mandates for Pin Assignment D shall comply with *DP Standard* for bit rates up to UHBR10, HBR3, HBR2, HBR, or RBR, as supported by the system. The protocol shall be as defined in *DP Standard*. Pin Assignment D is appropriate only for USB-C-to-USB-C passive and active cables and *DP* Sink devices with *USB-C* plugs. This pin assignment is **not** supported by *USB4*. *USB 3.2* shall be used when combined with DisplayPort Alt Mode.

Table 3-12: DP Source Device Pin Assignment D – Normal Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2		(<i>USB-C</i>)	B11		(<i>USB-C</i>)
A3		(<i>USB-C</i>)	B10		(<i>USB-C</i>)
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_N
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_P	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 0	ML_Lane_0_N	B3	Negative half of Lane 1	ML_Lane_1_N
A11	Positive half of Lane 0	ML_Lane_0_P	B2	Positive half of Lane 1	ML_Lane_1_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

Table 3-13: DP Source Device Pin Assignment D – Flipped Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML_Lane_1_P	B11	Positive half of Lane 0	ML_Lane_0_P
A3	Negative half of Lane 1	ML_Lane_1_N	B10	Negative half of Lane 0	ML_Lane_0_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_P
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_N	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10		(<i>USB-C</i>)	B3		(<i>USB-C</i>)
A11		(<i>USB-C</i>)	B2		(<i>USB-C</i>)
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.5 DP Source Device Pin Assignment E

Table 3-14 and Table 3-15 define use and configuration of the *USB-C* connector pins when the pins are reconfigured to support *DisplayPort* in normal and flipped plug orientations, respectively. *DP* electrical mandates for Pin Assignment E shall comply with *DP Standard* for bit rates up to UHBR10, HBR3, HBR2, HBR, or RBR, as supported by the system, except that the reference cable model shall be for the USB-C-to-DP cable defined in Section 4.3.3. Pin Assignment E is appropriate only for receptacles that accept plugs on USB-C-to-DP passive and active cables and the plugs on such cables.

Note: *The reference cable has lower loss than the HBR reference cable in DP Standard to permit higher losses in the system design due to the need for signal multiplexing.*

Table 3-14: DP Source Device Pin Assignment E – Normal Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 2	ML_Lane_2_P	B11	Positive half of Lane 3	ML_Lane_3_P
A3	Negative half of Lane 2	ML_Lane_2_N	B10	Negative half of Lane 3	ML_Lane_3_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_N
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_P	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 0	ML_Lane_0_N	B3	Negative half of Lane 1	ML_Lane_1_N
A11	Positive half of Lane 0	ML_Lane_0_P	B2	Positive half of Lane 1	ML_Lane_1_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 3-15: DP Source Device Pin Assignment E – Flipped Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 1	ML_Lane_1_P	B11	Positive half of Lane 0	ML_Lane_0_P
A3	Negative half of Lane 1	ML_Lane_1_N	B10	Negative half of Lane 0	ML_Lane_0_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_P
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_N	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 3	ML_Lane_3_N	B3	Negative half of Lane 2	ML_Lane_2_N
A11	Positive half of Lane 3	ML_Lane_3_P	B2	Positive half of Lane 2	ML_Lane_2_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.6 DP Sink Device Pin Assignment C

Table 3-16 and Table 3-17 define use and configuration of the *USB-C* connector pins when the pins are reconfigured to support *DisplayPort* in normal and flipped plug orientations, respectively. *DP* electrical mandates for Pin Assignment C shall comply with *DP Standard* for bit rates up to UHBR10, HBR3, HBR2, HBR, or RBR, as supported by the system. The protocol shall be as defined in *DP Standard*. Pin Assignment C is appropriate only for USB-C-to-USB-C passive and active cables and *DP* Source devices with *USB-C* plugs.

Table 3-16: DP Sink Device Pin Assignment C – Normal Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 3	ML_Lane_3_P	B11	Positive half of Lane 2	ML_Lane_2_P
A3	Negative half of Lane 3	ML_Lane_3_N	B10	Negative half of Lane 2	ML_Lane_2_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_P
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_N	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 1	ML_Lane_1_N	B3	Negative half of Lane 0	ML_Lane_0_N
A11	Positive half of Lane 1	ML_Lane_1_P	B2	Positive half of Lane 0	ML_Lane_0_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

Table 3-17: DP Sink Device Pin Assignment C – Flipped Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 0	ML_Lane_0_P	B11	Positive half of Lane 1	ML_Lane_1_P
A3	Negative half of Lane 0	ML_Lane_0_N	B10	Negative half of Lane 1	ML_Lane_1_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_N
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_P	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 2	ML_Lane_2_N	B3	Negative half of Lane 3	ML_Lane_3_N
A11	Positive half of Lane 2	ML_Lane_2_P	B2	Positive half of Lane 3	ML_Lane_3_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.7 DP Sink Device Pin Assignment D

Table 3-18 and Table 3-19 define use and configuration of the *USB-C* connector pins when the pins are reconfigured to support *DisplayPort* combined with *USB* SuperSpeed (multi-function) in normal and flipped plug orientations, respectively. *DP* electrical mandates for Pin Assignment D shall comply with *DP Standard* for bit rates up to UHBR10, HBR3, HBR2, HBR, or RBR, as supported by the system. The protocol shall be as defined in *DP Standard*. Pin Assignment D is appropriate only for USB-C-to-USB-C passive and active cables and *DP* Source devices with *USB-C* plugs. This pin assignment is **not** supported by *USB4*. *USB 3.2* shall be used when combined with DisplayPort Alt Mode.

Table 3-18: DP Sink Device Pin Assignment D – Normal Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2		(<i>USB-C</i>)	B11		(<i>USB-C</i>)
A3		(<i>USB-C</i>)	B10		(<i>USB-C</i>)
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_P
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_N	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Negative half of Lane 1	ML_Lane_1_N	B3	Negative half of Lane 0	ML_Lane_0_N
A11	Positive half of Lane 1	ML_Lane_1_P	B2	Positive half of Lane 0	ML_Lane_0_P
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

Table 3-19: DP Sink Device Pin Assignment D – Flipped Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Positive half of Lane 0	ML_Lane_0_P	B11	Positive half of Lane 1	ML_Lane_1_P
A3	Negative half of Lane 0	ML_Lane_0_N	B10	Negative half of Lane 1	ML_Lane_1_N
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_N
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_P	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10		(<i>USB-C</i>)	B3		(<i>USB-C</i>)
A11		(<i>USB-C</i>)	B2		(<i>USB-C</i>)
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.8 DP Sink Device Pin Assignment E

Table 3-20 and Table 3-21 define use and configuration of the *USB-C* connector pins when the pins are reconfigured to support *DisplayPort* in normal and flipped plug orientations, respectively. *DP* electrical mandates for Pin Assignment E shall comply with *DP Standard* for bit rates up to UHBR10, HBR3, HBR2, HBR, or RBR, as supported by the system, except that the reference cable model shall be for the USB-C-to-DP cable defined in Section 4.3.3. Pin Assignment E is appropriate only for receptacles that accept plugs on DP-to-USB-C passive and active cables and for plugs on such cables.

Notes: *The reference cable has lower loss than the HBR reference cable in DP Standard to permit higher losses in the system design due to the need for signal multiplexing.*

Inverted use of pins A8 and B8 (AUX_CH_N and AUX_CH_P, respectively) and inverted polarity of the Main-Link lanes on pins A2 – A3, A10 – A11, B11 – B10, and B3 – B2 compared to the DP Sink device Pin Assignments C and D.

Table 3-20: DP Sink Device Pin Assignment E – Normal Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Negative half of Lane 1	ML_Lane_1_N	B11	Negative half of Lane 0	ML_Lane_0_N
A3	Positive half of Lane 1	ML_Lane_1_P	B10	Positive half of Lane 0	ML_Lane_0_P
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_N
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_P	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Positive half of Lane 3	ML_Lane_3_P	B3	Positive half of Lane 2	ML_Lane_2_P
A11	Negative half of Lane 3	ML_Lane_3_N	B2	Negative half of Lane 2	ML_Lane_2_N
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 3-21: DP Sink Device Pin Assignment E – Flipped Plug Orientation

Pin	Description	DP Function ^a	Pin	Description	DP Function ^a
A1	Ground return	GND	B12	Ground return	GND
A2	Negative half of Lane 2	ML_Lane_2_N	B11	Negative half of Lane 3	ML_Lane_3_N
A3	Positive half of Lane 2	ML_Lane_2_P	B10	Positive half of Lane 3	ML_Lane_3_P
A4		(<i>USB-C</i>)	B9		(<i>USB-C</i>)
A5		(<i>USB-C</i>)	B8		AUX_CH_P
A6		(<i>USB-C</i>)	B7		(<i>USB-C</i>)
A7		(<i>USB-C</i>)	B6		(<i>USB-C</i>)
A8		AUX_CH_N	B5		(<i>USB-C</i>)
A9		(<i>USB-C</i>)	B4		(<i>USB-C</i>)
A10	Positive half of Lane 0	ML_Lane_0_P	B3	Positive half of Lane 1	ML_Lane_1_P
A11	Negative half of Lane 0	ML_Lane_0_N	B2	Negative half of Lane 1	ML_Lane_1_N
A12	Ground return	GND	B1	Ground return	GND

a. “(*USB-C*)” indicates no change from *USB-C*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9 AUX_CH and HPD

AUX_CH and HPD implementations are designed to support the scenarios defined in [Section 4](#), which may use one of the following:

- USB-C-to-USB-C cable assembly,
- USB-C-to-DP cable assembly, –or–
- USB-C-to-other video protocol converter

See the following sections for USB-C-to-DP adapter behavior when:

- **Detecting a DP Source or Sink device** – [Section 4.3.4.2](#)
- **Detecting an active cable DP Source or Sink device** – [Section 4.1](#) and [Section 4.8](#)

Equivalent implementations to those illustrated within this section are permissible.

3.9.1 AUX_CH

AUX_CH shall be supported on the [USB-C](#) connector when operating in DisplayPort Alt Mode using differential 100- Ω -terminated signaling on the receptacle pin A8 and B8 connection. AC-coupling capacitors shall be placed between the AUX_CH_P and AUX_CH_N transceiver and [USB-C](#) receptacle. The AUX_CH differential signal levels shall be as defined in [DP Standard](#), with the exception that the EYE diagrams are measured for 1UI after a differential signal transition.

Note: *Receptacle pins A8 and B8 remain disconnected within the system until the connection is changed to DisplayPort Alt Mode and a configuration is selected. Likewise, plug pins A8 and B8 on adapter cables remain disconnected until the connection is changed to DisplayPort Alt Mode and a configuration is selected.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.1.1 AUX_CH Schematics

The resistor tolerance shown in Figure 3-1, Figure 3-2, and Figure 3-3 is $\pm 5\%$. The capacitor tolerance shown in Figure 3-1, Figure 3-2, and Figure 3-3 is $\pm 20\%$, except for the capacitors on *DP* systems in which the minimum and maximum values are shown.

Figure 3-1 illustrates the system design of a *DP* Source device on a *USB-C* connector connected to a *DP* Sink device on a *USB-C* connector by way of a *USB-C*-to-*USB-C* cable. The 2-M Ω pull-down resistors on SBU1 and SBU2 are representative of the leakage of ESD and EMI/RFI components, including termination to ensure that there are no floating nodes, and are intended to show compliance with zSBU Termination in *USB-C*.

Note: These schematics do not show the ESD protection and EMI/RFI components typically included in the system designs.

The plug orientation switch or AUX_CH polarity inversion logic in the DPTX or DPRX is controlled by the *USB-C* state machine which determines the cable plug connection orientation at both cable ends.

Note: The 3.3-V levels in the adapters are derived from V_{CONN} because not all *DP* Sink devices provide *DP_PWR*.

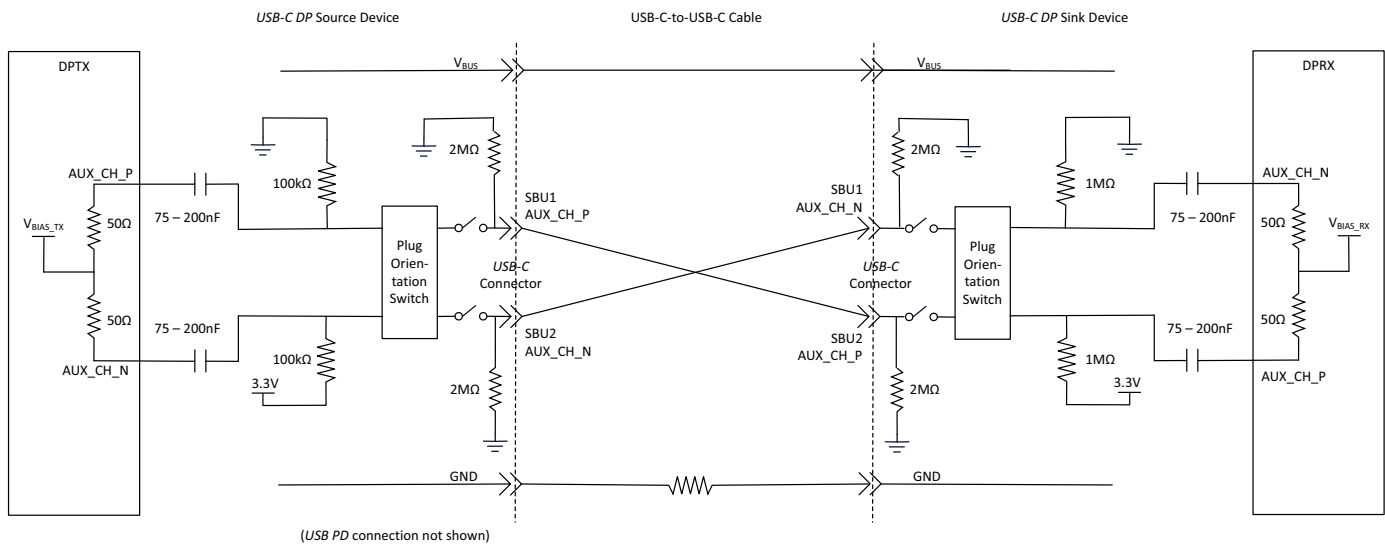


Figure 3-1: AUX_CH Connections Using USB-C-to-USB-C Cables

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 3-2 illustrates the system design of a *DP* Source device on a *USB-C* connector connected to a *DP* Sink device on a *DP* connector by way of a *USB-C*-to-*DP* adapter. Note that the SBU isolation switches are open; however, when a *DP* Sink device connection is discovered, the AUX_CH pull-up/pull-down resistors that emulate a *DP* Source device on the AUX_CH single-ended connections are engaged by closing the SBU isolation switches.

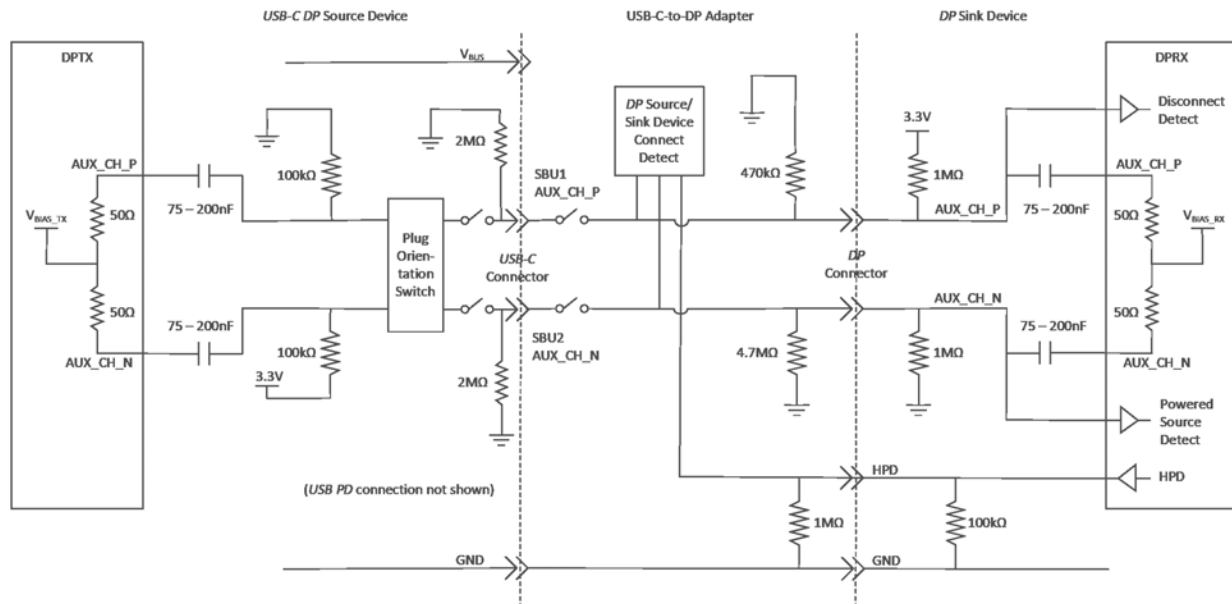


Figure 3-2: AUX_CH Connections When Using a *USB-C*-to-*DP* Adapter – Example 1

When designing *DP* Source/Sink Device Connect Detect system logic to detect the weak (nominal 1MΩ) pull-up resistor on AUX_CH_P in the *DP* Sink device, it is important to note that system leakage may be present in the connected *DP* Sink device due to its ESD and EMI components. Use of a 470-kΩ resistor represents a compromise between providing an adequate pull-down resistor to overcome these losses and the need for a relatively low connection-detection threshold. Actual values will depend on the specific system design. The weak (4.7MΩ) resistor on AUX_CH_N represents circuitry that is needed to avoid a floating conductor when the adapter is not connected to a *DP* Sink device without compromising the threshold in the *DP* Sink device that is used to detect the adapter's pull-up resistor.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 3-3 illustrates the system design of a *DP* Source device on a *DP* connector connected to a *DP* Sink device on a *USB-C* connector by way of a USB-C-to-DP adapter. Note that this adapter is the same adapter that is used in the previous example; however, *DP* Main-Link data is flowing in the opposite direction.

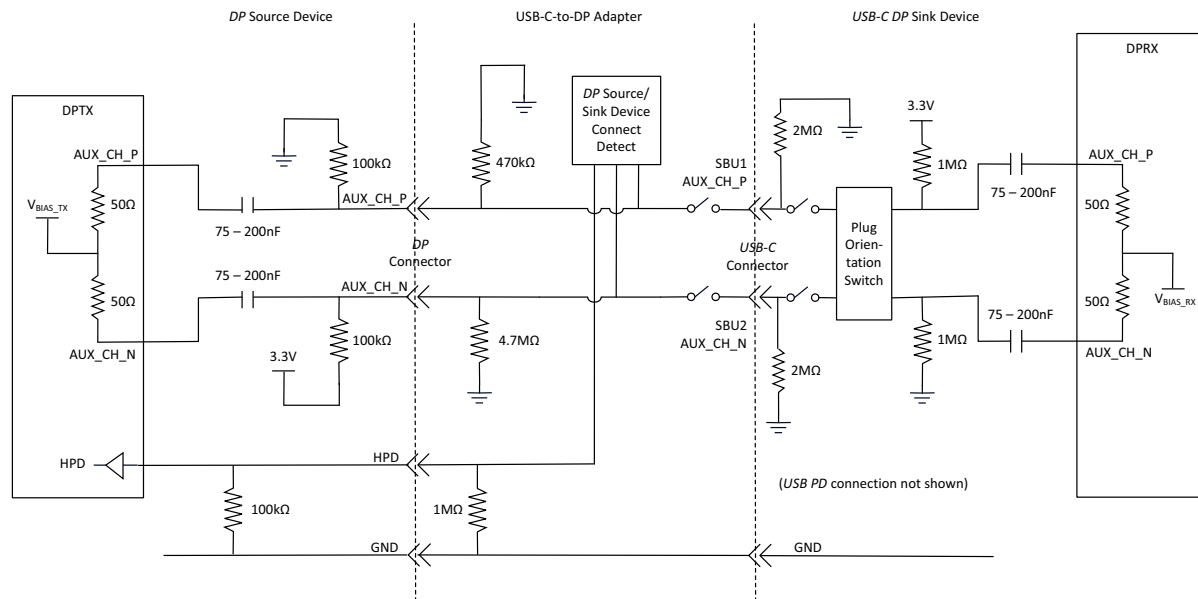


Figure 3-3: AUX_CH Connections When Using a USB-C-to-DP Adapter – Example 2

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.1.2 Protocol Converter AUX_CH Mandates

If the protocol converter is powered when the adapter is **not** in DisplayPort Alt Mode, the following mandates apply:

- Protocol adapter that is capable of detecting an upstream **DP** Source device disconnect by way of the **DP** AUX_CH common-mode voltage shall use the **USB-C** mechanism (instead of the AUX_CH method) to determine whether a disconnect has occurred. The adapter shall **not** drive HPD low when it detects an AUX_CH disconnect.
- Protocol adapter powered from **USB-C** V_{BUS} shall use the **USB-C** connection detection mechanism.
- Protocol adapter that does not use **USB-C** V_{BUS} for power shall **not** power on until the adapter enters DisplayPort Alt Mode.
- Protocol adapter that detects an upstream **DP** Source device is connected by way of the **DP** AUX_CH common-mode voltage mechanism shall **not** indicate a connection by driving HPD high until after DisplayPort Alt Mode is negotiated.

Figure 3-4 illustrates the system design of a **DP** Source device on a **USB-C** connector connected to a protocol converter. In this design, the protocol converter is used to convert **DP** protocol to a non-DP audio-video protocol.

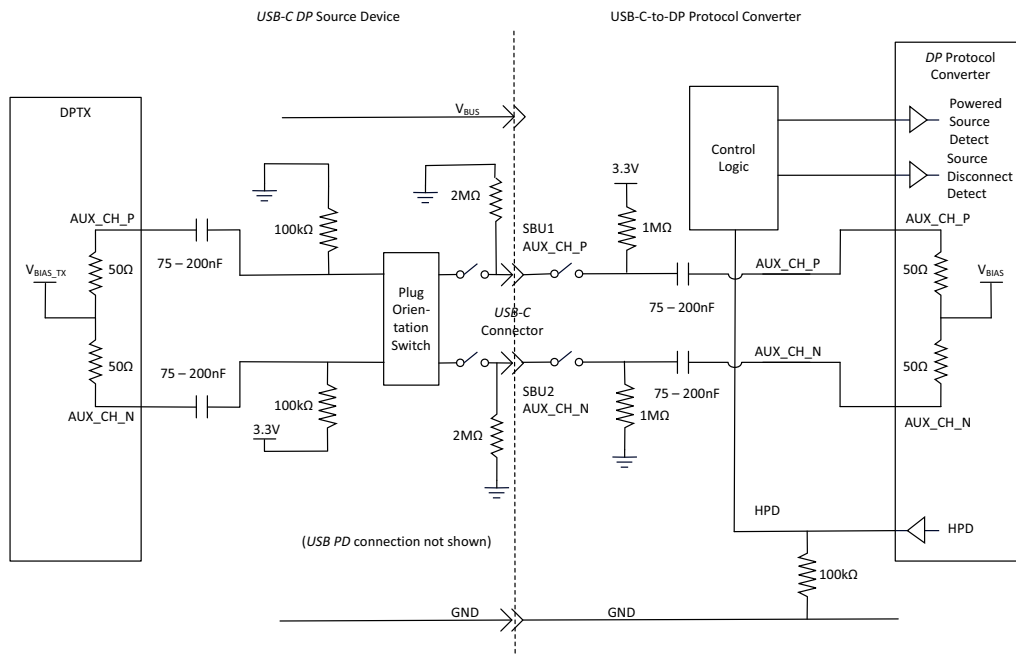


Figure 3-4: AUX_CH Connections When Using a USB-C-to-DP Protocol Converter

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.2 HPD Conversion

HPD shall be supported on the *USB-C* connector when operating in DisplayPort Alt Mode by way of *USB PD* Messages that carry *DisplayPort* status information. Two types of *USB PD* Messages may carry HPD – *DisplayPort Status Update* and *Attention* (see [Section 5.2.3](#) and [Section 5.2.5](#), respectively).

[Figure 3-5](#) illustrates the architecture for transmitting the HPD state over a *USB PD* connection, and converting between HPD (as defined in *DP Standard*) and *USB PD* messaging. The architecture defines HPD-to-USB PD and USB PD-to-HPD converters to perform the necessary conversions. The converters may be integrated within a system, –or– have their HPD driver or detector connected to a *USB PD* connector. See [Section 3.9.2.4](#) for detailed schematics of the various possible applications of this architecture.

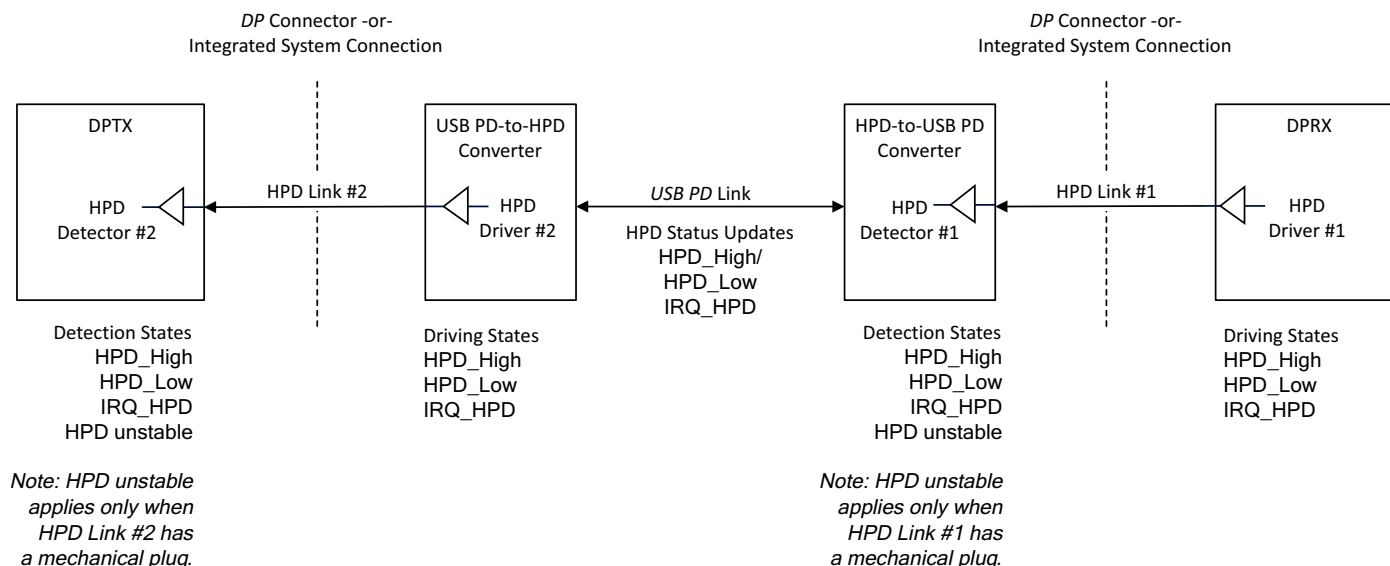


Figure 3-5: HPD Conversion Architecture

Two status bits are defined to communicate the state of HPD:

- **HPD State** – Indicates whether HPD’s logical state is high or low (respectively indicated as HPD_High or HPD_Low). When communicating the HPD state over *USB PD*, HPD’s logical state is considered as remaining high while receiving an IRQ_HP, and low while HPD is being de-bounced on a new mechanical connection. HPD’s logical state transitions from high to low when a low level on the HPD link is detected for 2ms (i.e., longer than the maximum IRQ_HP pulse detection time). HPD’s logical state is unchanged during glitches (as defined in *DP Standard*) on the HPD link.
- **IRQ_HP** – Indicates when a high-to-low transition on HPD followed by a low-to-high transition is detected between 250us and 2ms later, as defined in *DP Standard*.

[Table 5-6](#) defines the location for **HPD State** and **IRQ_HP** in the *USB PD DisplayPort Status Update* VDO.

A *USB PD* Message that contains the appropriate settings of these status bits is transmitted as soon as possible when an HPD logical status change is detected.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

DFP_Us and UFP_Us shall transmit updates to **HPD State** and **IRQ_HP** by way of the following commands:

- **DFP_U (including a Sink with Accessory Support, as specified in USB-C) – DisplayPort Status Update** Command (see [Section 5.2.3](#))
- **UFP_U – USB PD Attention** Command request (see [Section 5.2.5](#)), –or– as the response to a **DisplayPort Status Update** Command (see [Section 5.2.3](#))

A **USB PD** engine on a **USB-C** receptacle or plug may do the following:

- **DP Source device** – Incorporate an HPD driver, and then convert the **USB PD** Messages that carry the DisplayPort Status updates received across the **USB-C** connector into HPD signals as defined in [DP Standard](#) (USB PD-to-HPD converter), –or– provide an integrated implementation for communicating the HPD state
- **DP Sink device** – Incorporate an HPD detector that monitors an HPD signal's state as defined in [DP Standard](#), and then converts the received HPD signal to **USB PD** Messages that carry the DisplayPort Status updates across the **USB-C** connector (an HPD-to-USB PD converter), –or– provide an integrated implementation for generating the **USB PD** Messages

The HPD drivers and/or detectors in these converters shall comply with the HPD specifications defined in [DP Standard](#), [Section 3.3](#).

A USB-C-to-DP adapter shall:

- Implement a converter that is able to convert between the HPD signal and **USB PD** Messages in either direction
- Meet the timing mandates defined in [Section 3.9.2.1](#) and [Section 3.9.2.2](#), depending on the adapter's orientation

A USB-C-to-non-DP audio-video protocol converter may implement the USB PD-to-HPD converter (as defined in this section) or directly process the **DisplayPort Status Update** or **USB PD Attention** Commands.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.2.1 HPD-to-USB PD Timing

Per [USB PD](#), Alternate Mode Commands and Messages are to be transmitted only while the appropriate Alternate Mode is enabled. When DisplayPort Alt Mode is **not** enabled, an HPD-to-USB PD converter shall **not** transmit any **DisplayPort Status Update** or **USB PD Attention** Commands. When DisplayPort Alt Mode is enabled, the HPD-to-USB PD converter shall transmit the current HPD state as part of the initial **DisplayPort Status Update** Command exchange. The HPD detector may monitor the HPD state outside the time that DisplayPort Alt Mode is enabled so that the detector may perform HPD de-bouncing and track the HPD signal's current state.

[Table 3-22](#) defines the HPD-to-USB PD timing for passive adapter cables, active receptacle adapters, active adapter cables, and [USB-C DP](#) Sink devices. Active adapters shall be ready to respond to AUX transactions when forwarding [USB PD DisplayPort Status Update](#) or **Attention** Commands that have the [HPD State](#) bit set to 1. See [Section 3.9.2.3](#) for Message queuing mandates.

Table 3-22: HPD-to-USB PD Converter Timing

Parameter	Minimum Value (ms)	Maximum Value (ms)	Note
IRQ_HPDPulse/Glitch Detection Threshold	0.25		As defined in DP Standard , Table 3-4. <i>DP Standard</i> shall take precedence.
IRQ_HPDPulse/Hot Unplug Event Detection Threshold	2		As defined in DP Standard , Table 3-4. <i>DP Standard</i> shall take precedence.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 3-6 illustrates the HPD-to-USB PD converter connection state diagram. The process flow illustrated in Figure 3-6 assumes that the HPD_High, HPD_Low, and IRQ_HPDP signals are queued as per Section 3.9.2.3.1.

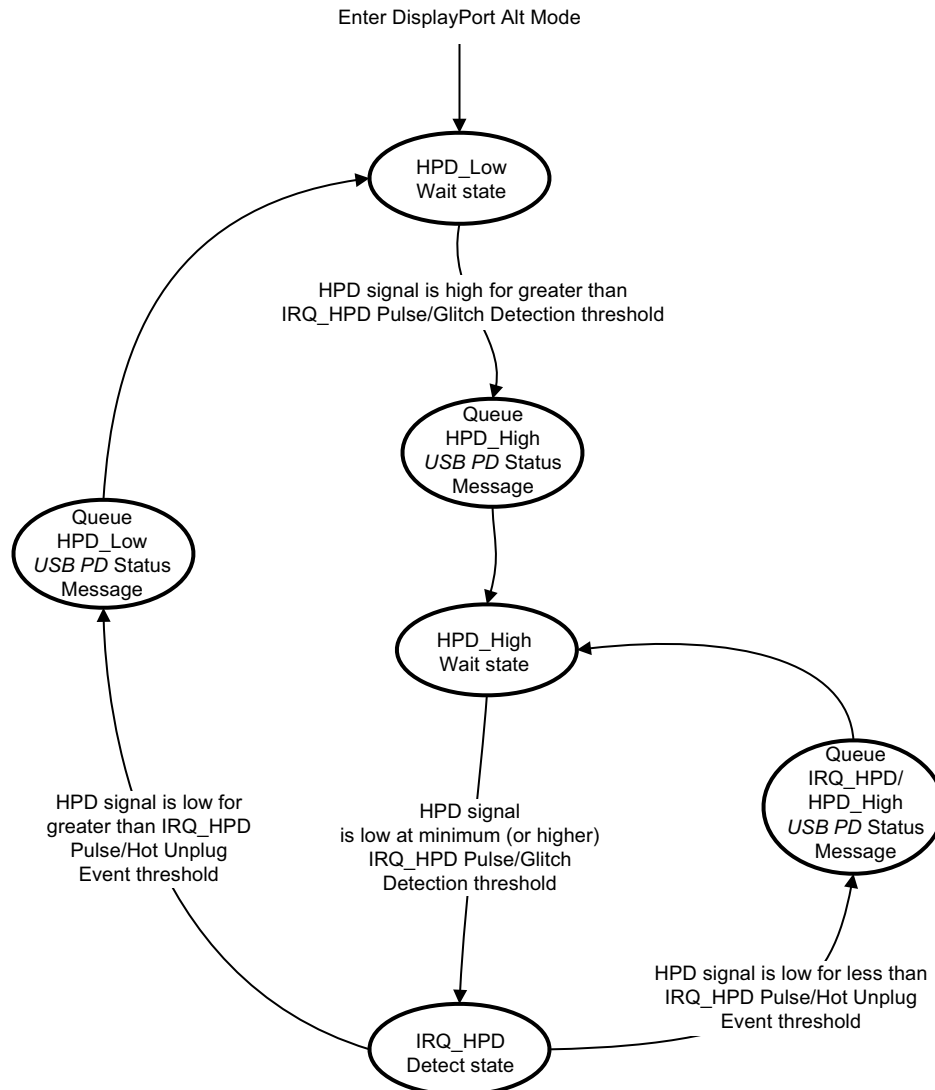


Figure 3-6: HPD-to-USB PD Converter Connection State Machine

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.2.2 USB PD-to-HPD Timing

A USB PD-to-HPD converter shall drive a low level on its HPD driver whenever DisplayPort Configuration on the *USB-C* interface is **not** enabled. The USB PD-to-HPD converter shall only indicate the HPD signal's current state after the SBU isolation switches have been closed to connect the AUX_CH. After DisplayPort Configuration is disabled, the USB PD-to-HPD converter shall drive a low level on its HPD driver before opening the SBU isolation switches to disconnect the AUX_CH.

[Table 3-22](#) defines the USB PD-to-HPD timing. Active receptacle adapters and active adapter cables shall drive the HPD signal high only when ready to respond to AUX transactions.

DP Source devices shall **not** drive the HPD signal high until HPD_WAIT time after DisplayPort Configuration is complete. If the HPD_High Message is received during the HPD_WAIT time, the USB PD-to-HPD converter should queue the Message until the timer expires, and then drive the HPD signal high. If the HPD_High Message is received after the HPD_WAIT timer times out, the HPD signal should be driven high with no delay. The HPD_WAIT timer should be reset on a V_{CONN} loss (DFP_U only), –or– after a new DisplayPort Configuration completion (DFP_U or UFP_U). See [Section 3.9.2.3](#) for Message queuing mandates.

Table 3-23: USB PD-to-HPD Converter Timing

Parameter	Minimum Value (ms)	Maximum Value (ms)	Note
IRQ_HPDPulse Width	0.5	1.0	As defined in <i>DP Standard</i> , Table 3-4. <i>DP Standard</i> shall take precedence.
HPD_Low Duration	3.0		As defined in Section 3.9.2.3.2 .
IRQ_HPDPMinimum Spacing	2.0		As defined in <i>DP Standard</i> , Table 3-4. <i>DP Standard</i> shall take precedence.
HPD_WAIT Timer	0	150	Applies to passive adapter cable, active receptacle adapter, and active adapter cable.
	50	150	Applies only to a <i>DP</i> Source device.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 3-7 illustrates the USB PD-to-HPD converter connection state diagram. The process flow illustrated in Figure 3-7 assumes that the *USB PD HPD State* and *IRQ_HPDP* Messages are queued as per Section 3.9.2.3.1.

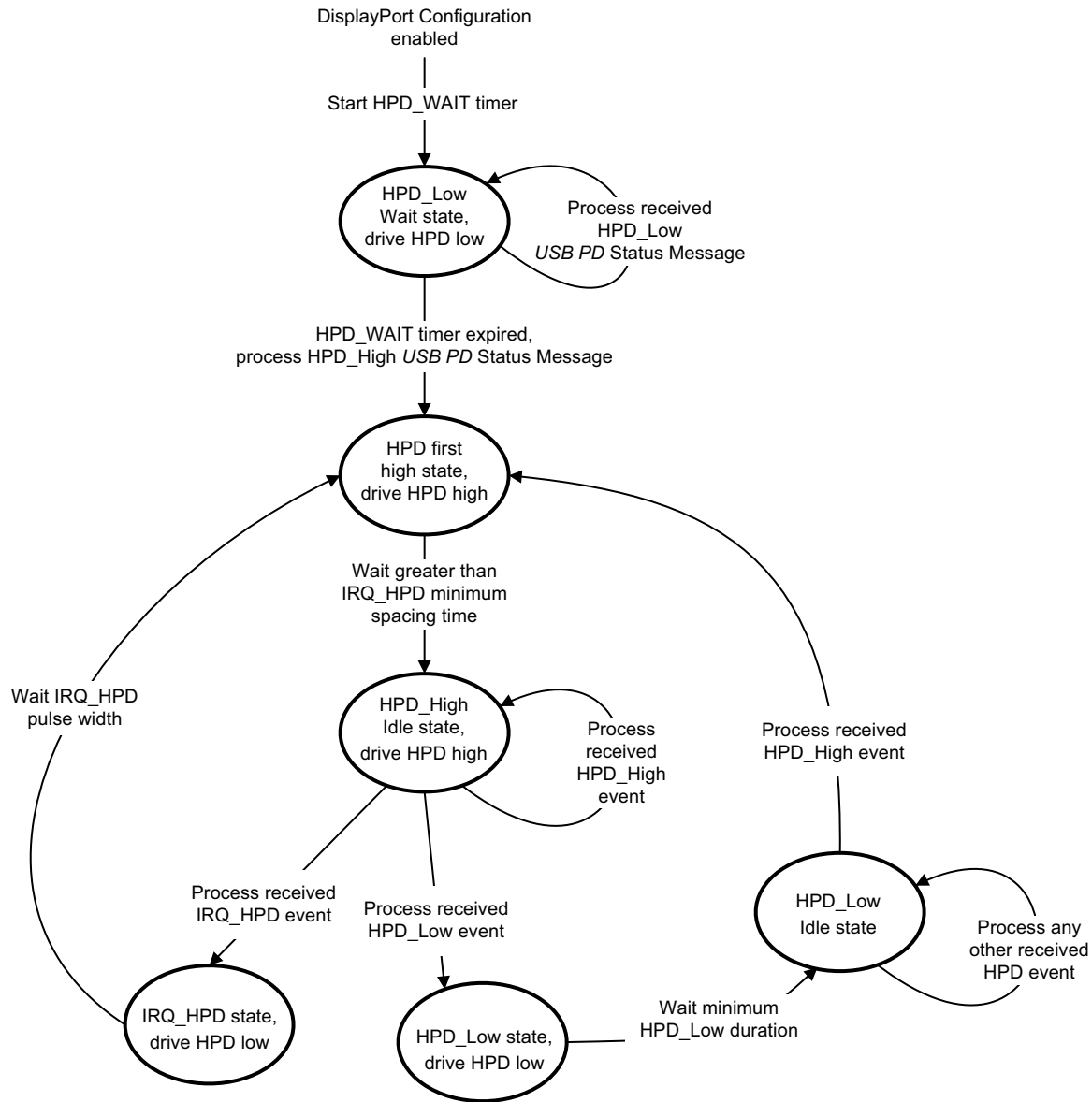


Figure 3-7: USB PD-to-HPD Converter Connection State Machine

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.2.3 Message Queuing

It is possible that other *USB PD* traffic will cause the HPD-to-USB PD converter to delay transmission of HPD's current state, and may cause the arrival timing of event indications to **not** correspond to the original HPD timing. Accordingly, the converters shall observe the following mandates:

- [HPD-to-USB PD Converter Message Queuing Mandates](#)
- [USB PD-to-HPD Converter Message Queuing Mandates](#)

Both are defined in the following subsections.

3.9.2.3.1 HPD-to-USB PD Converter Message Queuing Mandates

- If an IRQ_HPDP is detected while a previous IRQ_HPDP is in the queue, the converter shall transmit **DisplayPort Status Update** or *USB PD Attention* Commands for both IRQ_HPDP events. Any IRQ_HPDP events detected while two IRQ_HPDP events are in the queue may be discarded.
- If HPD_Low is detected while any previous HPD events are in the queue, the previous HPD events may be discarded. An HPD event can be an HPD_Low, HPD_High, –or– IRQ_HPDP event.
- If HPD_High is detected while a previous HPD_Low is in the queue, the converter shall transmit **DisplayPort Status Update** or *USB PD Attention* Commands for both HPD events.
- If an IRQ_HPDP is detected while a previous HPD_High is in the queue, both shall be transmitted, either as one **DisplayPort Status Update** or *USB PD Attention* Command (recommended) –or– separate **DisplayPort Status Update** or *USB PD Attention* Commands in the order detected.
- While DisplayPort Alt Mode is disabled, HPD events are **not** queued.
- When DisplayPort Alt Mode changes from enabled to disabled, any entries in the queue are discarded.
- When DisplayPort Alt Mode is enabled, the current HPD state is transmitted as part of the initial DisplayPort Status exchange.

Note: *The worst-case status buffering that is needed in the HPD-to-USB PD converter is the sequence HPD_Low, HPD_High, IRQ_HPDP, IRQ_HPDP.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.2.3.2 USB PD-to-HPD Converter Message Queuing Mandates

- If a single Message represents both a transition to HPD_High and an IRQ_HPD, the converter shall ensure that HPD is taken high for a minimum of 2ms before generating the IRQ_HPD.
- After generating a low-to-high transition on HPD, the converter shall ensure that HPD is maintained high for a minimum of 2ms, deferring taking HPD low on a subsequent HPD_Low or a concurrent or subsequent IRQ_HPD.
- After generating a high-to-low transition on HPD in response to the receipt of HPD_Low, the converter shall ensure that HPD is maintained low for a minimum of 3ms (i.e., the 2-ms IRQ pulse detect time plus a 1-ms margin), deferring taking HPD high after receiving a subsequent HPD_High event during this time.
- If a converter receives HPD_Low, any intervening HPD_High and/or IRQ_HPD in the queue shall be discarded.
- When DisplayPort Configuration is **not** selected and the converter is driving its HPD output low, the converter shall track HPD's current state, ready for appropriate indication when DisplayPort Configuration is subsequently selected.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

3.9.2.4 HPD Schematics

Figure 3-8 illustrates the schematic for an implementation of a USB PD-to-HPD converter in a *DP* Source device, an HPD-to-USB PD converter in a *DP* Sink device, and their connection by way of a USB-C-to-USB-C cable assembly. As previously indicated, either one or both of the *DP* Source and Sink devices may integrate the *USB PD* engine with the DPTX or DPRX (as appropriate), rather than using an explicit HPD connection.

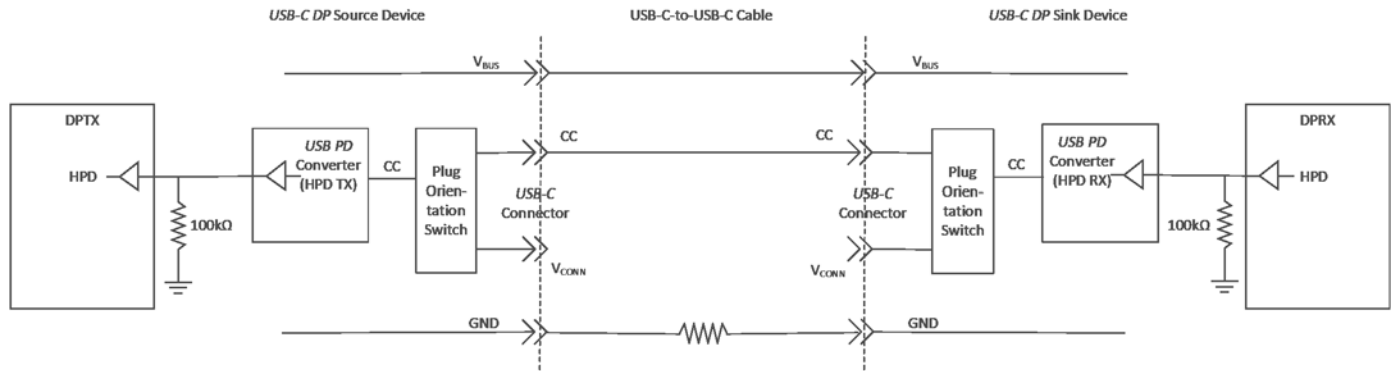


Figure 3-8: HPD Communication When Using a USB-C-to-USB-C Cable

Figure 3-9 illustrates the schematic for an implementation of a USB PD-to-HPD converter in a *DP* Source device connected to a *DP* Sink device on the *DP* connector by way of an HPD-to-USB PD converter in a USB-C-to-DP adapter. The HPD transmitter in the HPD-to-USB PD converter is disabled when used in this direction.

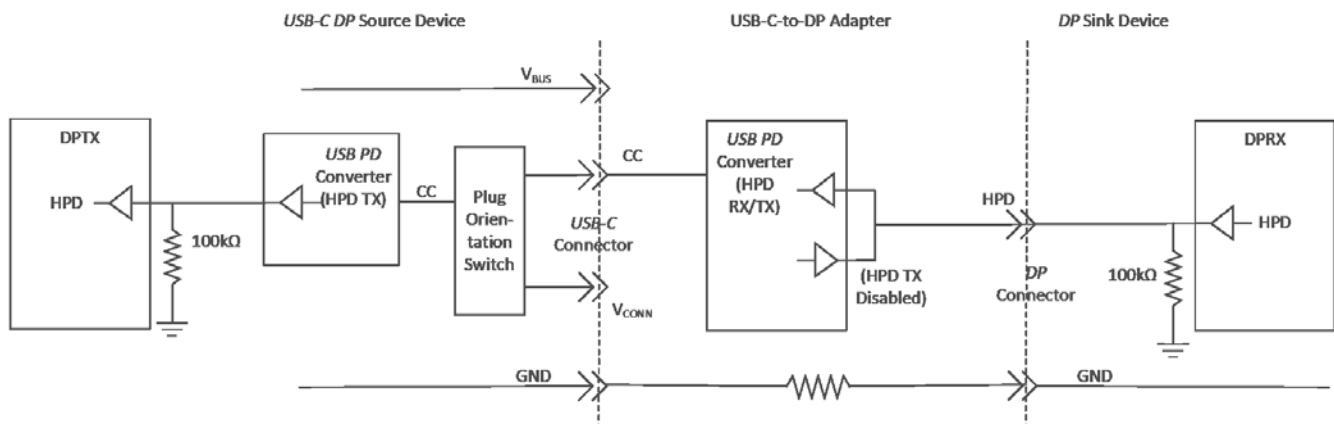


Figure 3-9: HPD Communication When Using a USB-C-to-DP Adapter – Example 1

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 3-10 illustrates the schematic for an implementation of an HPD-to-USB PD converter in a *DP* Sink device and its connection to a *DP* Source device on the *DP* connector by way of a USB PD-to-HPD converter in a DP-to-USB-C adapter. The HPD transmitter in the HPD-to-USB PD converter is enabled when used in this direction.

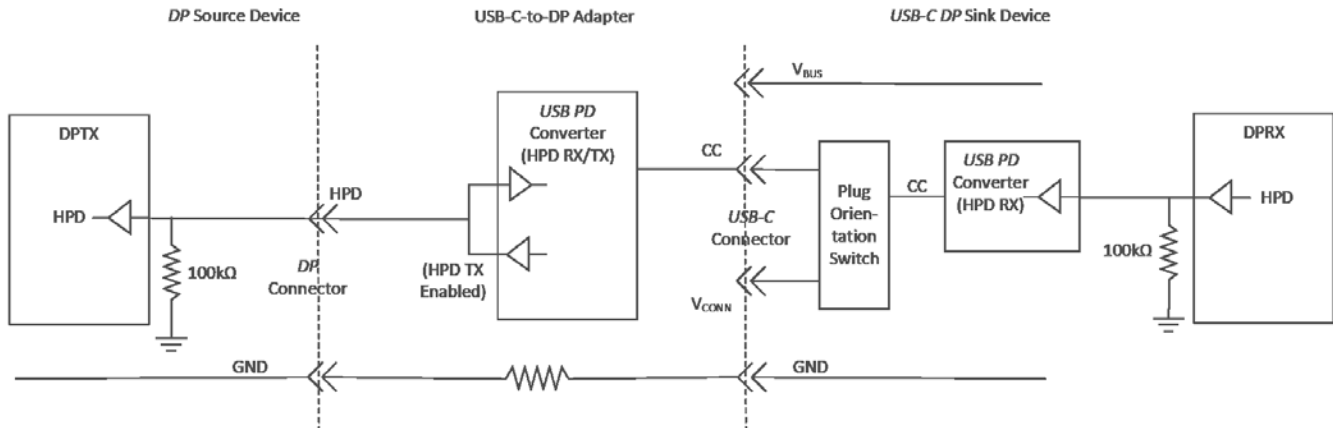


Figure 3-10: HPD Communication When Using a USB-C-to-DP Adapter – Example 2

Figure 3-11 illustrates the schematic for an implementation of a USB PD-to-HPD converter in a *DP* Source device and an HPD-to-USB PD converter in a protocol converter. In this design, the protocol converter is used to convert *DP* protocol to a non-DP audio-video protocol.

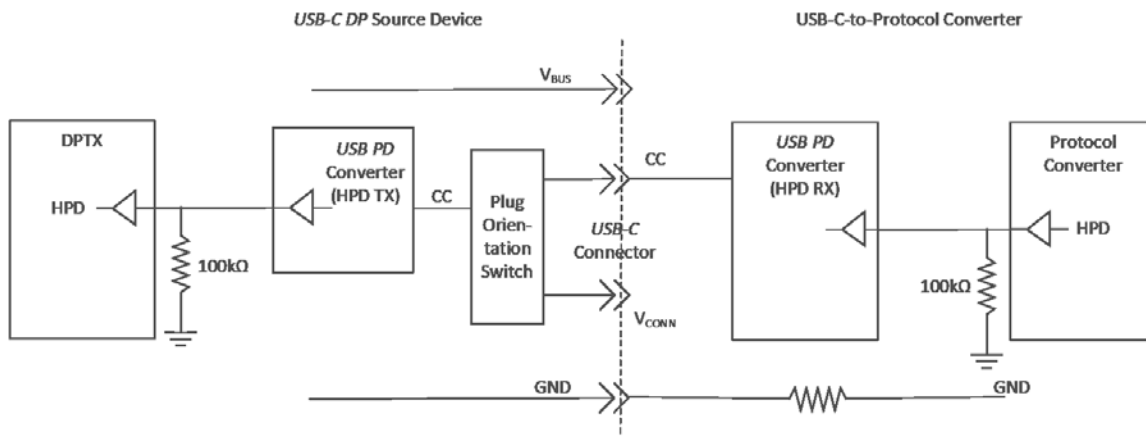


Figure 3-11: HPD Communication When Using a USB-C-to-Protocol Converter

3.10 DP Connector CONFIG1 and CONFIG2 Pins

The *DP* plug's CONFIG1 and CONFIG2 pins in USB-C-to-DP adapters shall each be pulled low to GND with a 1-MΩ pull-down resistor.

Note: *DP++* adapters are **not** supported on the USB-C-to-DP adapter's *DP* connector.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4 USB-C Cables Used for DisplayPort

Note: *Scenario 3b does not exist. Scenario 4b has been deprecated.*

This section describes the cable assemblies and adapters used for *DisplayPort* through a *USB-C* connector.

All associations of *USB-C* functions with specific pins on the *USB-C* connector are informative. Reference *should* be made to *USB-C* for normative *USB* pin functionality. The *USB-C* receptacle and plug mechanical and electrical properties are also specified in *USB-C*.

The following scenarios are described:

- [Scenario 1 – USB-C Passive or Active Cable](#)
- [Scenario 2a – USB-C DP Source Device-to-DP/mDP Sink Device](#)
- [Scenario 2b – DP/mDP Source Device-to-USB-C DP Sink Device](#)
- [Scenarios 3a and 3c – USB-C DP Source Device-to-Protocol Converter](#)
- [Scenario 4a – USB-C DP Source Device-to-Multi-function Dock](#)
- [Scenarios 4c and 4d – USB-C DP Source Device to Multi-function Cradle/Captive Cable Dock](#)
- [Scenario 5a – USB-C Plug DP Source-to-DP/mDP Receptacle Adapter](#)
- [Scenario 5b – DP/mDP Sink Receptacle-to-USB-C DP Sink Adapter](#)

The following topics are also discussed:

- [Active Cable Overview](#)
- [Active Cable Power Consumption](#)
- [Active Cable Legacy USB-C Support Mandates](#)
- [Methods for Determining DP Directionality in an Active Cable](#)

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.1 Active Cable Overview

Active cables are cables that incorporate signal conditioning logic on the high-speed path. This Standard defines the following active cable assemblies:

- USB-C connector-to-USB-C active cable assembly
- USB-C connector-to-DP/mDP active cable assembly

Active cables shall support the mandates defined in [Section 4.8](#) and [Section 4.9](#).

[USB-C](#) and [DP](#) active cables shall support the mandates defined in [Table 2-2](#) and [Table 2-3](#).

The active cable's data transfer functions shall continue uninterrupted in the event of a [USB PD VCONN_Swap](#) Message being performed between the two Port Partners.

Active cables may incorporate the methods defined in [Section 4.10](#) to determine [DisplayPort](#) directionality.

4.2 Scenario 1 – USB-C Passive or Active Cable

[Figure 4-1](#) illustrates an example of a Full-Featured [USB-C](#) passive or active cable assembly (as specified in [USB-C](#)) used for [DisplayPort](#). Bit rates are as per [DP Standard](#). Multi-function (simultaneous [DisplayPort](#) and [USB SuperSpeed](#)) is supported. Active cables shall follow the mandates defined in this subsection and [Section 4.8](#).

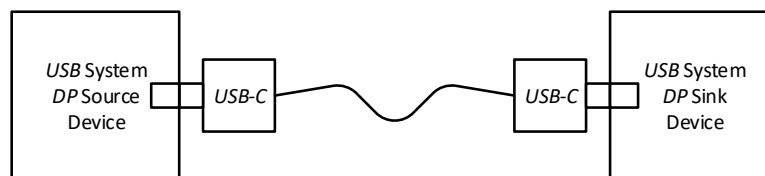


Figure 4-1: Scenario 1 – USB-C Passive or Active Cable

This scenario uses [DP](#) Source device Pin Assignment C at the [DP](#) Source device and [DP](#) Sink device Pin Assignment C at the [DP](#) Sink device (see [Table 3-10](#) and [Table 3-16](#), respectively). The following figures illustrate how [DisplayPort](#) uses the Full-Featured [USB-C](#) passive cable assembly's signal connections:

- [Figure 4-2](#) illustrates when the plugs at both ends are inserted in normal orientation
- [Figure 4-3](#) illustrates when the [DP](#) Source device plug is flipped and the [DP](#) Sink device plug is inserted in normal orientation
- [Figure 4-4](#) illustrates when the [DP](#) Source device plug is inserted in normal orientation and the [DP](#) Sink device plug is flipped
- [Figure 4-5](#) illustrates when the plugs at both ends are flipped

Note: *The figures omit the [USB PD Identification Circuitry](#) that is incorporated in the cable assembly.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

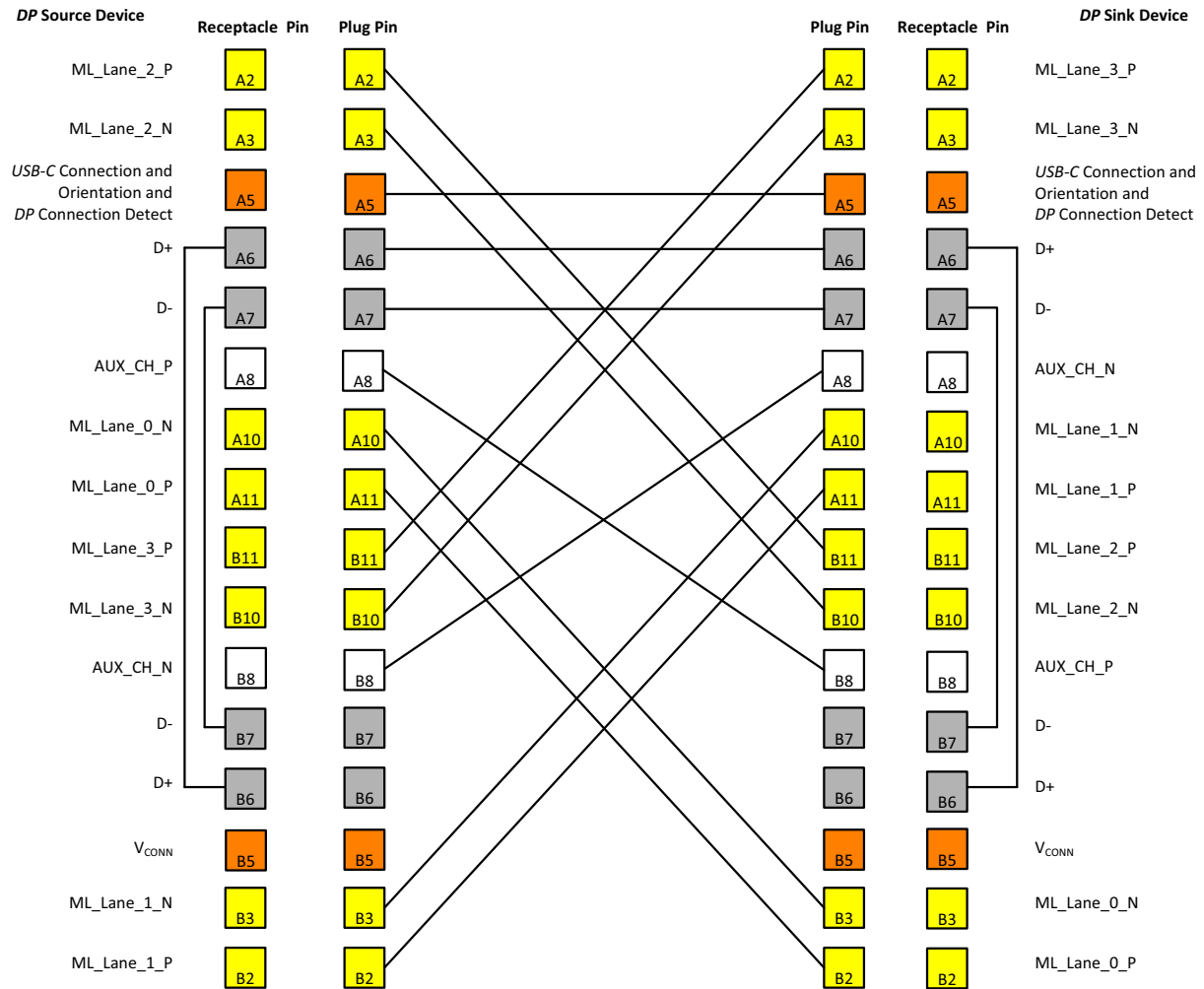


Figure 4-2: Scenario 1 Cable Wiring – Normal Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

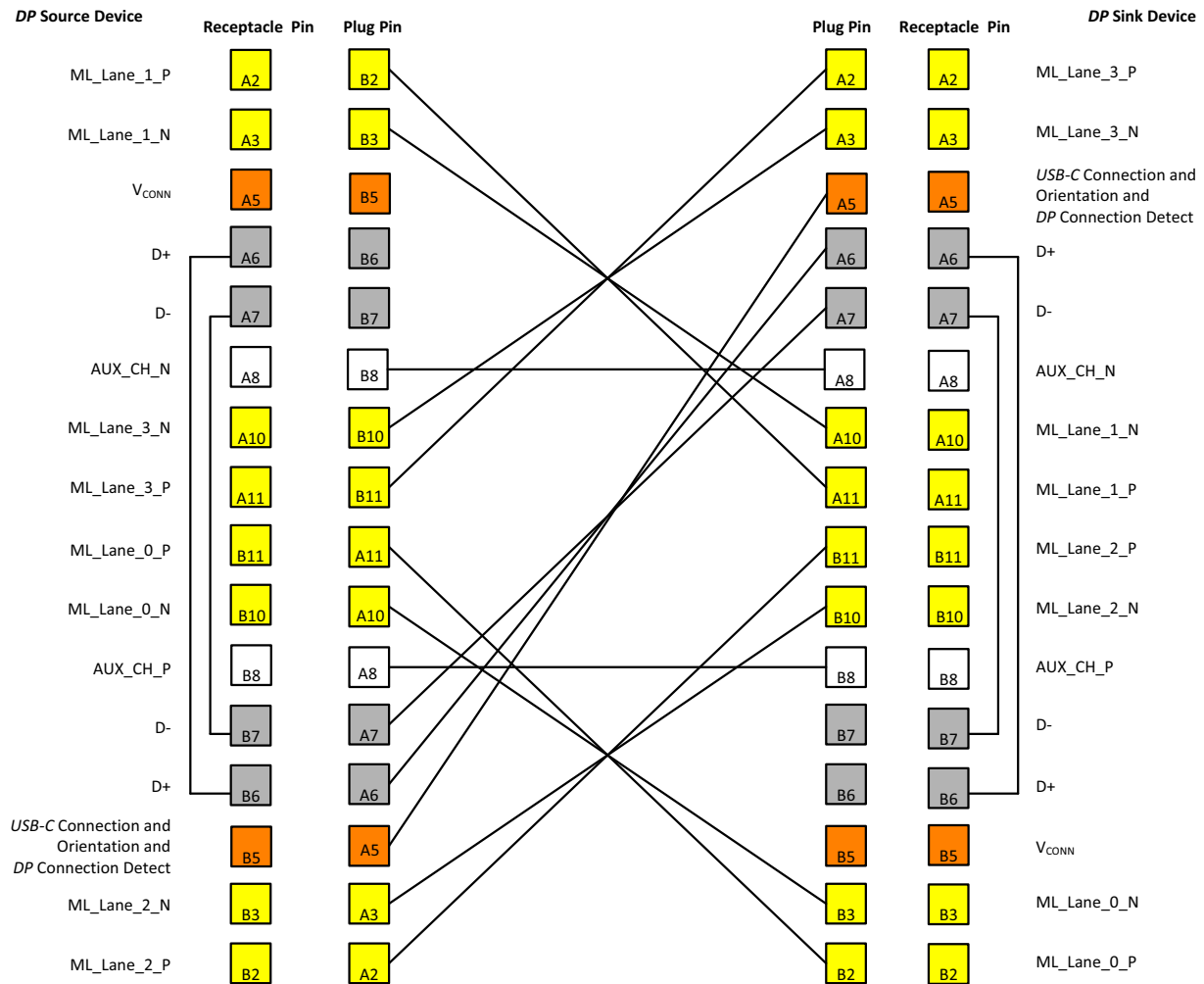


Figure 4-3: Scenario 1 Cable Wiring – DP Source Device Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

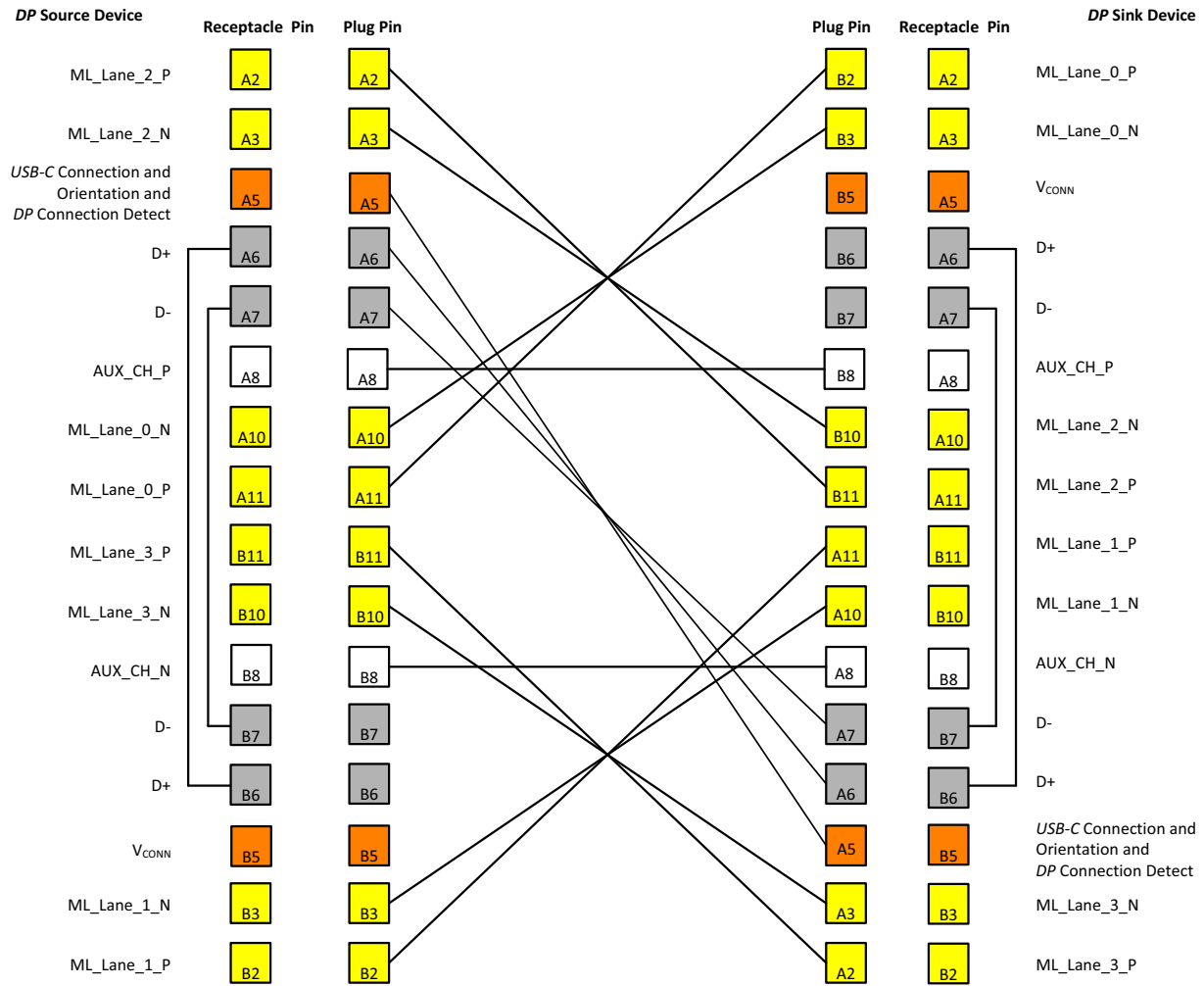


Figure 4-4: Scenario 1 Cable Wiring – DP Sink Device Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

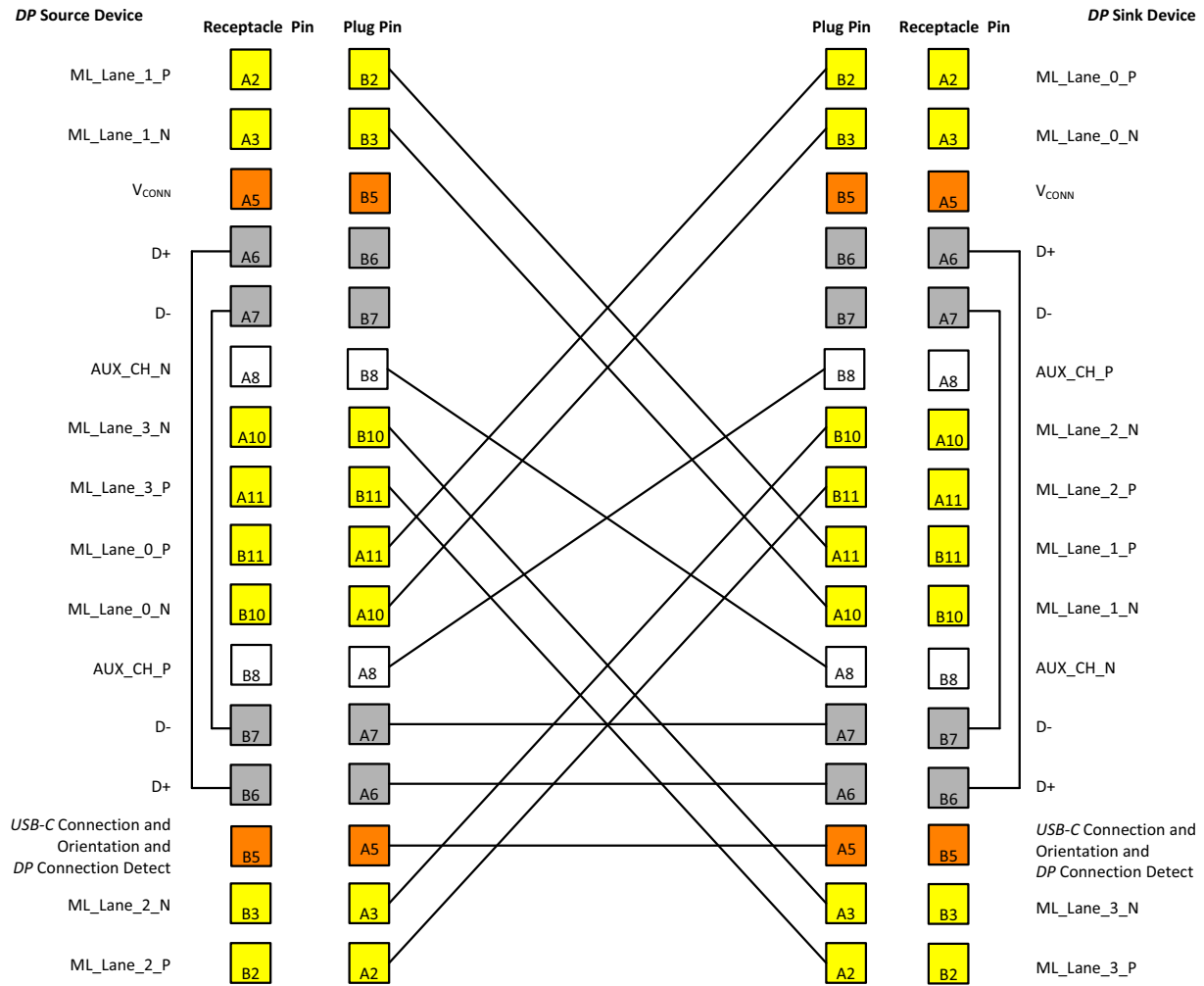


Figure 4-5: Scenario 1 Cable Wiring – DP Source and Sink Device Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.2.1 Scenario 1 – Active USB-C-to-USB-C Cables

An active USB-C-to-USB-C cable shall support at least a single *USB* SuperSpeed lane of *USB 3.2* (i.e., one *USB* path in each direction). An active cable shall support the mandates defined in *Table 2-2* and meet *USB-C* requirements. Active cables shall also follow the mandates defined in *Section 4.8* and *Section 4.9*.

4.2.1.1 Active USB-C-to-USB-C Cable Connections

The signal connections for a USB-C connector-to-USB-C connector active cable assembly shall be as defined in *Figure 4-2* through *Figure 4-5*. Both *DP* (4-, 2-, and 1-lane) and multi-function (one or two *DP* lanes and one *USB* SuperSpeed lane) shall be supported.

Note: *To support this, active cable plugs shall support Pin Assignments C and D for DP Source and Sink devices.*

4.2.1.2 Active USB-C-to-USB-C Cable Configured for USB Operation

An active USB-C-to-USB-C cable shall comply with *USB-C* power mandates when configured for *USB*.

4.2.1.3 Active USB-C-to-USB-C Configuration Timing

An active cable shall be ready to respond to AUX transactions 40ms after the *DisplayPort Configure* Command completes. Active cables shall behave as if HPD is high when receiving a *DisplayPort Configure* Command.

4.2.1.4 Active USB-C-to-USB-C Directionality Support

An active USB-C-to-USB-C cable shall support *DisplayPort* in either direction. The method by which the cable determines the signal direction is implementation-specific. *Section 4.10* describes methods that may be used for determining signal direction.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.2.2 Scenario 1 – Active USB-C-to-USB-C Cable USB PD Mandates

The active cable should behave as if the **DisplayPort Status Update** VDO has a high HPD state after receiving a **DisplayPort Configure** Command request addressed to SOP'/SOP", –or– after snooping a **DisplayPort Configure** Command request addressed to SOP per [Section 4.9](#).

An active USB-C-to-USB-C cable shall respond to commands as defined in [Table 4-1](#). **USB PD** and **USB-C** requirements shall take precedence.

Table 4-1: Active Cable USB PD Command Mandates^a

Command Type	SOP' Behavior	SOP" Behavior
USB PD Discover SVIDs	Normative	Ignore
USB PD Discover Modes	Normative	Ignore
Enter Mode	Normative	Normative (if SOP" supported)
Exit Mode	Normative	Normative (if SOP" supported)
DisplayPort Status Update	Normative	Normative (if SOP" supported)
DisplayPort Configure	Normative	Normative (if SOP" supported)
USB PD Attention	Ignore	Ignore

a. A response of "Normative" indicates that the Message type shall be supported.

An active cable shall support the DP-specific **DisplayPort Status Update** and **DisplayPort Configure** Commands in conjunction with the DP_SID. From the DFP_U's perspective, an active cable is configured to have the same **DP** role (**DP** Source or Sink device) as the UFP_U. The plugs at both active cable ends select the pin assignments accordingly. For example, if the DFP_U is also a **DP** Source device, the DFP_U shall issue a **DisplayPort Configure** Command request addressed to SOP' (and SOP", if present) to configure the cable as a **DP** Sink device (see [Section 4.2.1.4](#) for further details).

After entering DisplayPort Alt Mode, the active cable should connect the PHY Repeater's AUX_CH to provide the maximum time that is needed to prepare a response to AUX transactions that are received from the **DP** Source device. The active cable should assume that HPD is high when a **DisplayPort Configure** Command request is addressed to SOP'/SOP" is issued, –or– when snooping the **DisplayPort Configure** Command request addressed to SOP.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

When transmitting the Responder **USB PD Discover Modes** VDM, the active cable shall indicate its **DP** capabilities in the VDO, as defined in Table 4-2. Table 4-3 lists the active cable **DisplayPort Status Update** as transmitted in the VDO. Table 4-4 lists the active cable DisplayPort Configurations.

Table 4-2: Active Cable DP Capabilities (VDO in the Responder USB PD Discover Modes VDM)

Bit(s)	Description	Values
1:0	RESERVED	RESERVED (always 00b).
5:2	Signaling for Transport of DisplayPort Protocol ^a	XXX1b = Supports DP bit rates and electrical settings (shall always be set apart from diagnostic purposes). XX1Xb = RESERVED. X1XXb = RESERVED. 1XXXb = RESERVED.
7:6	RESERVED	RESERVED (always 00b).
15:8	DP Source Device Pin Assignments Supported	0Ch = Pin Assignments C and D are supported. All other values are RESERVED.
23:16	DP Sink Device Pin Assignments Supported	0Ch = Pin Assignments C and D are supported (USB-C-to-USB-C cable). All other values are RESERVED.
31:24	RESERVED	RESERVED (always 00h).

a. “X” value indicates “Don’t Care.”

Table 4-3: Active Cable DisplayPort Status Update

Bit(s)	Description	Values
2:0	RESERVED	RESERVED (always 00b).
3	Enabled	0 = Active cable DP functionality is disabled. 1 = Active cable DP functionality is enabled and operational.
31:4	RESERVED	RESERVED (always 0000000h).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-4: Active Cable DisplayPort Configurations

Bit(s)	Description	Values
1:0	Select Configuration	00b = Set configuration for <i>USB</i> . ^a 01b = Set configuration for active cable as a <i>DP</i> Source device (UFP_U is a <i>DP</i> Source device). ^b 10b = Set configuration for active cable as a <i>DP</i> Sink device (UFP_U is a <i>DP</i> Sink device). ^b 11b = RESERVED.
5:2	Signaling for Transport of DisplayPort Protocol	0h = Bit rate is unspecified (used only when the <i>Select Configuration</i> field is programmed for USB Configuration). 1h = Select <i>DP</i> bit rates and electrical settings. All other values are RESERVED.
7:6	RESERVED	RESERVED (always 00b).
15:8	Configure Active Cable Pin Assignment	00h = De-select pin assignment. 04h = Select Pin Assignment C. ^c 08h = Select Pin Assignment D. ^d All other values are RESERVED.
31:16	RESERVED	RESERVED (always 0000h).

a. Selects USB Configuration.

b. Selects DisplayPort Configuration.

c. See Table 3-10 and Table 3-11 for DP Source device Pin Assignment C.

See Table 3-16 and Table 3-17 for DP Sink device Pin Assignment C.

d. See Table 3-12 and Table 3-13 for DP Source device Pin Assignment D.

See Table 3-18 and Table 3-19 for DP Sink device Pin Assignment D.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.3 Scenarios 2a and 2b – USB-C-to-DP/mDP Cable

USB-C-to-DP passive cables shall support both Scenario 2a (USB-C DP Source device-to-DP/mDP Sink device) and Scenario 2b (DP/mDP Source device-to-USB-C DP Sink device). USB-C-to-DP active cables may support both Scenario 2a and 2b. USB-C-to-DP active cables that do **not** support both Scenario 2a and 2b should be clearly marked to indicate the *DP* directionality supported.

4.3.1 Scenario 2a – USB-C DP Source Device-to-DP/mDP Sink Device

Figure 4-6 illustrates a USB-C plug-to-DP/mDP plug passive or active cable assembly. Bit rates are as per *DP Standard*.

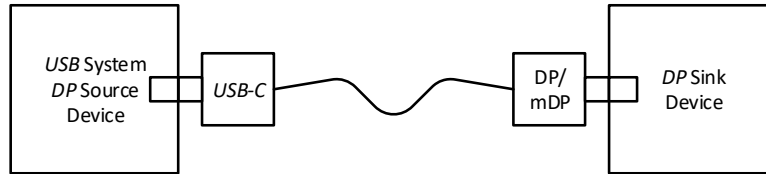


Figure 4-6: Scenario 2a – USB-C DP Source Device-to-DP/mDP Sink Device

This scenario uses *DP* Source device Pin Assignment E at the *DP* Source device (see Table 3-14 and Table 3-15). The pin numbering on the *DP* connector in Figure 4-7 and Figure 4-8 is that for a full-sized *DP* connector (see Table 4-5 for the mDP connector pin numbering). The following figures illustrate how *DisplayPort* uses the USB-C DP Source device-to-DP Sink device cable assembly's signal connections:

- Figure 4-7 illustrates this use when the *USB-C* plug is inserted in normal orientation
- Figure 4-8 illustrates this use when the *USB-C* plug is inserted in flipped orientation

Note: The figures omit the pull-down resistors on CONFIG1 and CONFIG2.

The cable assembly shall contain logic to implement *USB PD* and support the Discovery Process (see [Section 5](#)). The cable assembly contains a *USB BB* device, which the *USB PD* logic shall connect to D+/D- in the event that changing to use DisplayPort Alt Mode is unsuccessful. The cable assembly's connections behind pins A10, A11, B10, and B11 on the *USB-C* plug shall contain series AC capacitors. The capacitor shall be sufficiently large so that the series combination of the *USB 3.2* capacitance (75 to 265nF) and the cable assembly capacitance meets the *DP Standard* mandate. A 220nF $\pm 20\%$ capacitor should be used. Although *DP Standard* mandates these pins to be AC-coupled to the *DP* Source device, the pins may be DC-connected in the system behind the *USB-C* receptacle. The series AC capacitors on these pins are also required by *USB-C*.

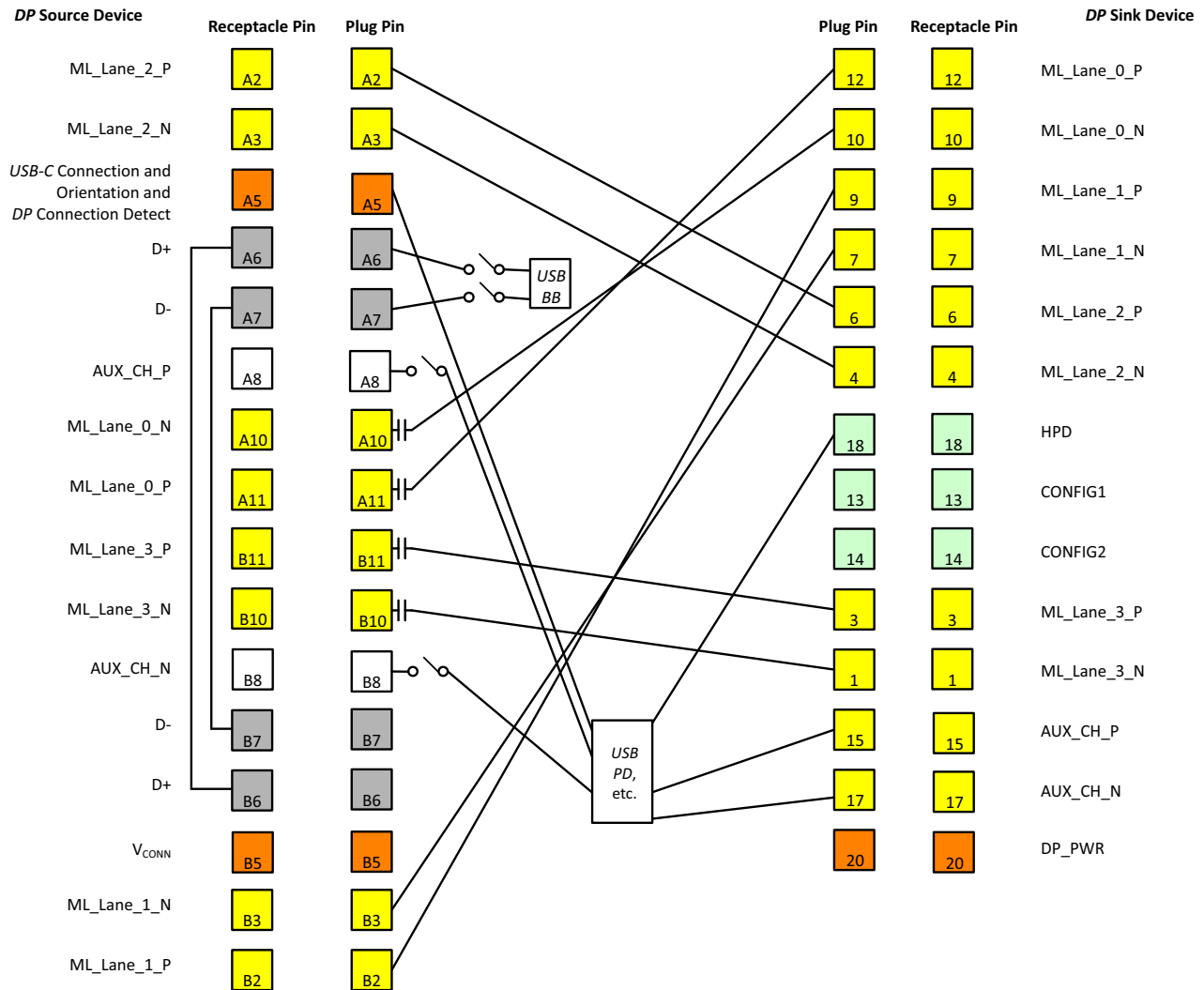


Figure 4-7: Scenario 2a Cable Wiring – Normal Plug Orientation on a Full-size DP Connector

Note: In [Figure 4-7](#), the *DP* pin assignments assume that the *DP* Source device is at the *USB-C* system.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

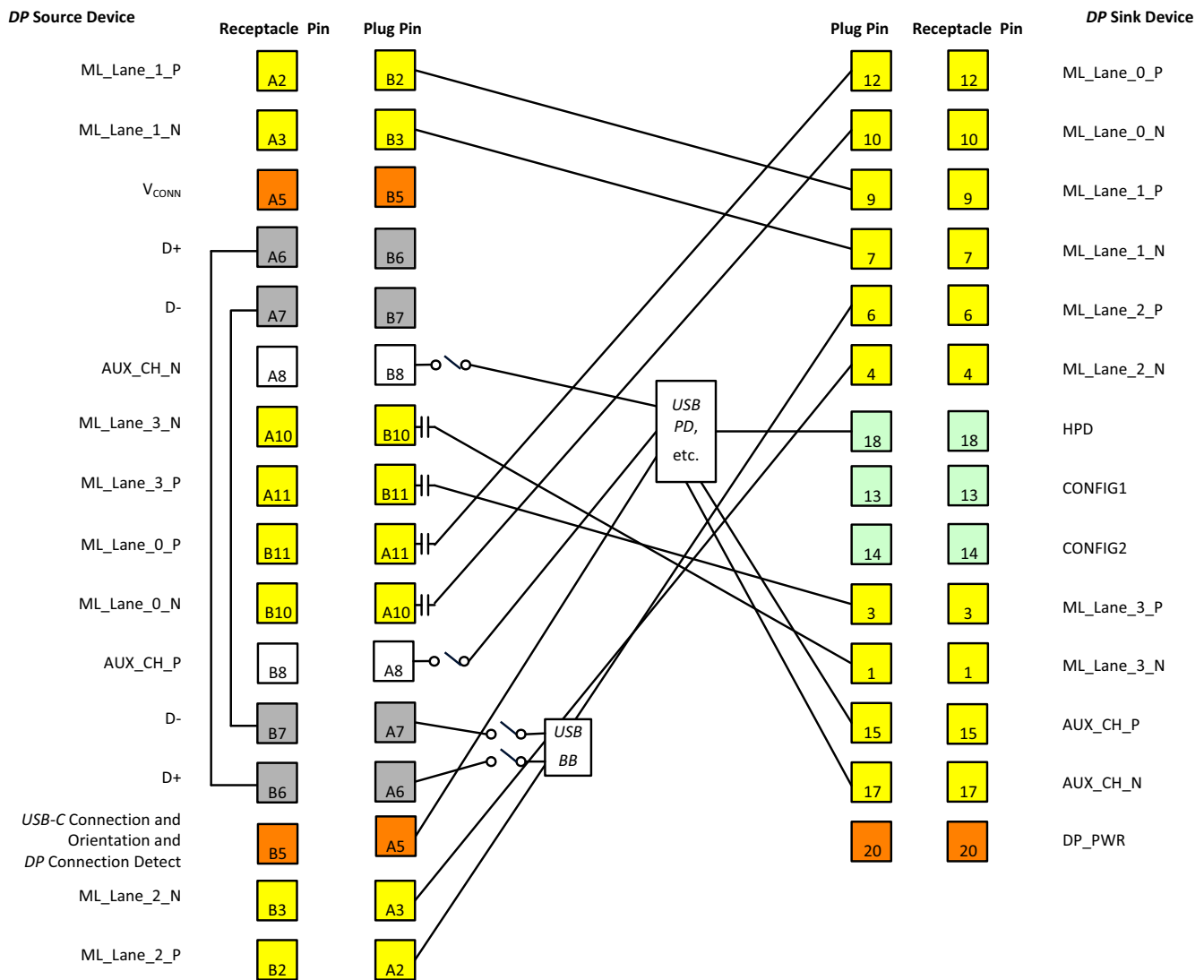


Figure 4-8: Scenario 2a Cable Wiring – DP Source Device Flipped Plug Orientation on a Full-size DP Connector

Note: In Figure 4-8, the DP pin assignments assume that the DP Source device is at the USB-C system.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-5 lists how *DisplayPort* uses the USB-C DP Source device-to-DP/mDP Sink device cable assembly's signal connections.

**Table 4-5: Scenario 2a – USB-C DP Source Device-to-DP/mDP Sink Device Use,
Normal Plug Orientation**

USB-C Plug #1 DP Source Device		Wire		DP/mDP Plug DP or mDP Sink Device		
Pin	DP Signal Name	Wire Number	Signal Name	DP Pin	mDP Pin	DP Signal Name
A1, B1, A12, B12	GND	1 [16]	GND_PWRrt1 [GND_PWRrt2]	2, 5, 8, 11, 16, 19	1, 7, 8, 13, 14, 19	GND
A4, B4, A9, B9	V _{BUS}	2	Powers <i>USB BB</i>			
A5	CC	3	Cable logic (<i>USB PD</i>)			
A6	D+	4	Cable logic (<i>USB BB</i>)			
A7	D-	5	Cable logic (<i>USB BB</i>)			
			1M to GND	13	4	CONFIG1
			1M to GND	14	6	CONFIG2
A2	ML_Lane_2_P	6	SDPp1	6	11	ML_Lane_2_P
A3	ML_Lane_2_N	7	SDPn1	4	9	ML_Lane_2_N
B11	ML_Lane_3_P	8	SDPp2	3	5	ML_Lane_3_P
B10	ML_Lane_3_N	9	SDPn2	1	3	ML_Lane_3_N
B2	ML_Lane_1_P	10	SDPp3	9	17	ML_Lane_1_P
B3	ML_Lane_1_N	11	SDPn3	7	15	ML_Lane_1_N
A11	ML_Lane_0_P	12	SDPp4	12	12	ML_Lane_0_P
A10	ML_Lane_0_N	13	SDPn4	10	10	ML_Lane_0_N
A8	AUX_CH_P	14	Cable logic	15	16	AUX_CH_P
B8	AUX_CH_N	15	Cable logic	17	18	AUX_CH_N
			Cable logic	18	2	HPD
B5	V _{CONN}	16	Powers Cable logic			
			No Connect	20	20	DP_PWR
Shell	Shield	Braid	Shield	Shell	Shell	Shield

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.3.2

Scenario 2b – DP/mDP Source Device-to-USB-C DP Sink Device

Figure 4-9 illustrates a DP/mDP plug-to-USB-C plug passive or active cable assembly. Bit rates are as per *DP Standard*.

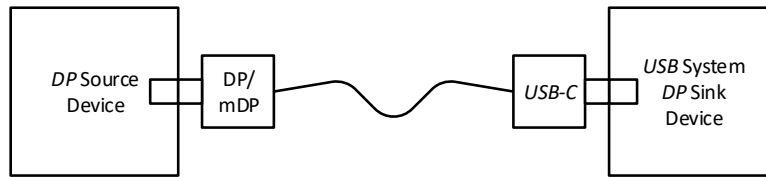


Figure 4-9: Scenario 2b – DP/mDP Source Device-to-USB-C DP Sink Device

This scenario uses *DP* Sink device Pin Assignment E (see Table 3-20 and Table 3-21). The cable assembly is the same cable assembly that is used in Scenario 2a, but with *DisplayPort* running in the opposite direction. The pin numbering on the *DP* connector in Figure 4-10 and Figure 4-11 is that for a full-sized *DP* connector (see Table 4-6 for the mDP connector pin numbering). The following figures illustrate how *DisplayPort* uses the DP Source device-to-USB-C DP Sink device cable assembly's signal connections:

- Figure 4-10 illustrates when the *USB-C* plug is inserted in normal orientation
- Figure 4-11 illustrates when the *USB-C* plug is inserted in flipped orientation

Pins A10, A11, B10, and B11 may be DC-connected in the system behind the *USB-C* receptacle; thus, the capacitors on these pins provide the optional AC-coupling permitted by *DP Standard*.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

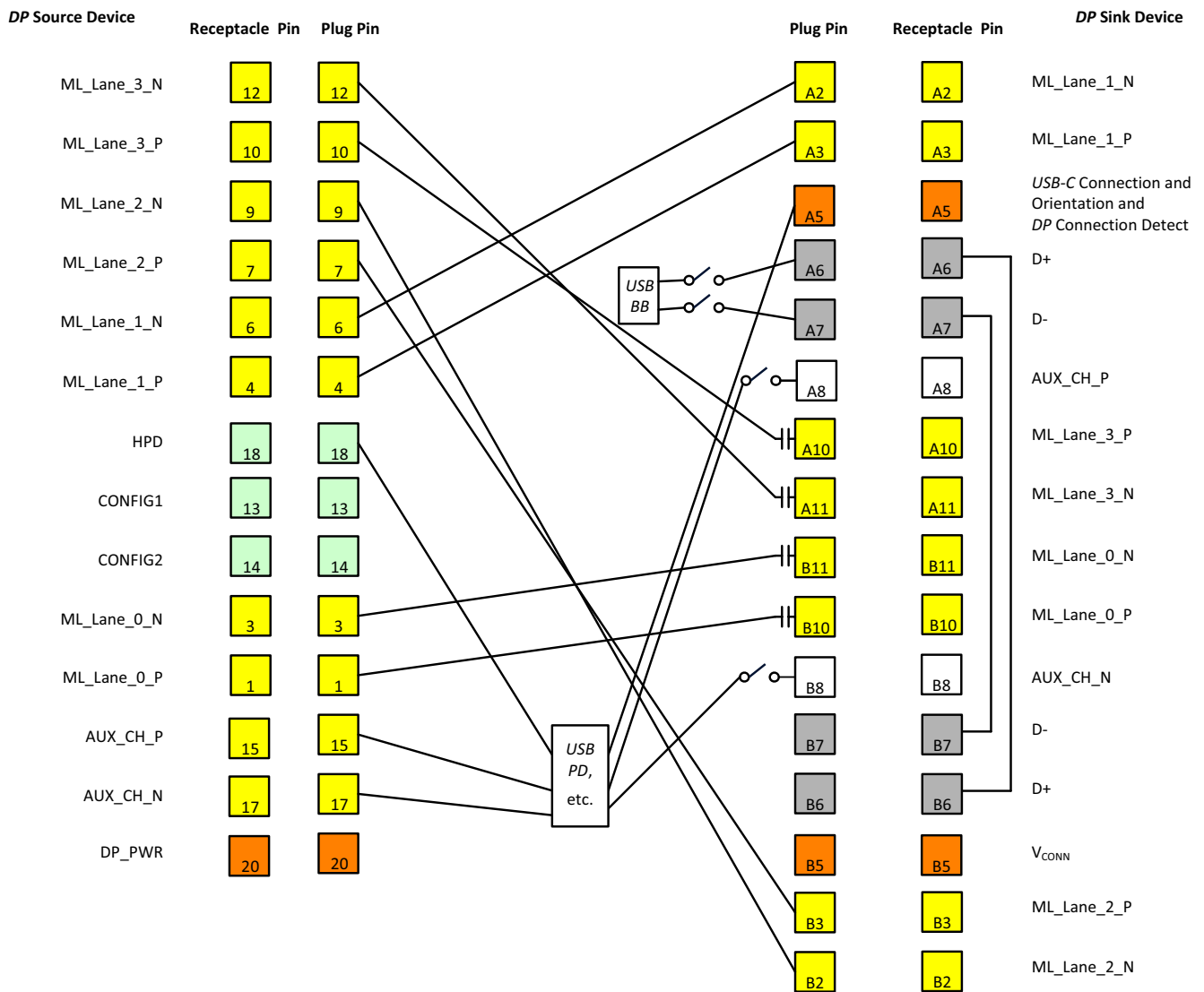


Figure 4-10: Scenario 2b Cable Wiring – Normal Plug Orientation on a Full-size DP Connector

Note: In Figure 4-10, the DP pin assignments assume that the DP Sink device is at the USB-C system.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

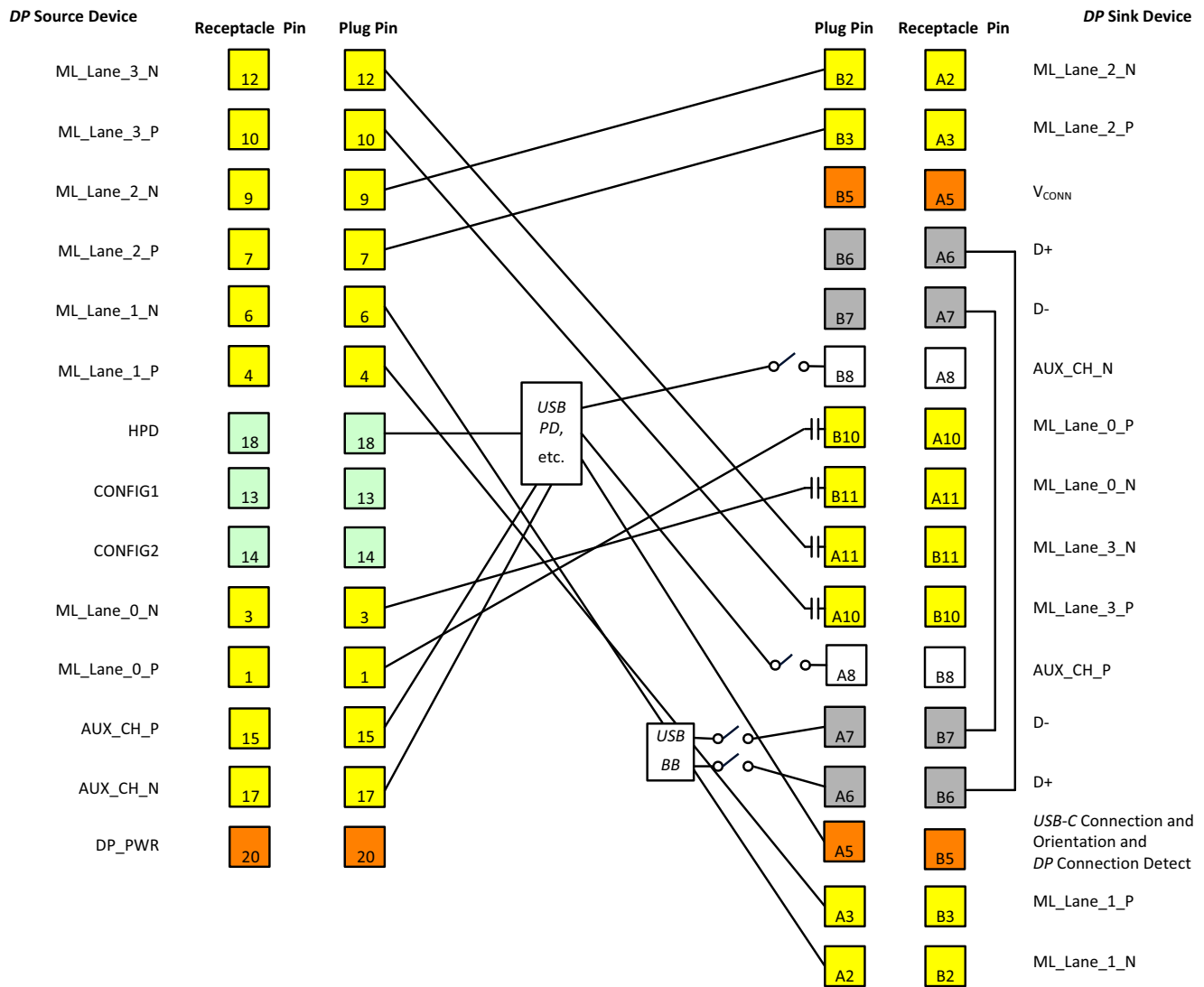


Figure 4-11: Scenario 2b Cable Wiring – DP Sink Device Flipped Plug Orientation on a Full-size DP Connector

Note: In *Figure 4-11*, the DP pin assignments assume that the DP Sink device is at the USB-C system.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-6 lists how *DisplayPort* uses the DP/mDP Source device-to-USB-C DP Sink device cable assembly's signal connections.

**Table 4-6: Scenario 2b – DP/mDP Source Device-to-USB-C DP Sink Device Use,
Normal Plug Orientation**

DP/mDP Plug DP or mDP Source Device			Wire		USB-C Plug DP Sink Device	
DP Signal Name	DP Pin	mDP Pin	Signal Name	Wire Number	Pin	DP Signal Name
GND	2, 5, 8, 11, 16, 19	1, 7, 8, 13, 14, 19	GND_PWRrt1 [GND_PWRrt2]	1 [16]	A1, B1, A12, B12	GND
			Powers <i>USB BB</i>	2	A4, B4, A9, B9	V _{BUS}
			Cable logic (<i>USB PD</i>)	3	A5	CC
			Cable logic (<i>USB BB</i>)	4	A6	D+
			Cable logic (<i>USB BB</i>)	5	A7	D-
CONFIG1	13	4	1M to GND			
CONFIG2	14	6	1M to GND			
ML_Lane_1_N	6	11	SDPp1	6	A2	ML_Lane_1_N
ML_Lane_1_P	4	9	SDPn1	7	A3	ML_Lane_1_P
ML_Lane_0_N	3	5	SDPp2	8	B11	ML_Lane_0_N
ML_Lane_0_P	1	3	SDPn2	9	B10	ML_Lane_0_P
ML_Lane_2_N	9	17	SDPp3	10	B2	ML_Lane_2_N
ML_Lane_2_P	7	15	SDPn3	11	B3	ML_Lane_2_P
ML_Lane_3_N	12	12	SDPp4	12	A11	ML_Lane_3_N
ML_Lane_3_P	10	10	SDPn4	13	A10	ML_Lane_3_P
AUX_CH_P	15	16	Cable logic	14	A8	AUX_CH_P
AUX_CH_N	17	18	Cable logic	15	B8	AUX_CH_N
HPD	18	2	Cable logic			
			Powers Cable logic	16	B5	V _{CONN}
DP_PWR	20	20	No Connect			
Shield	Shell	Shell	Shield	Braid	Shell	Shield

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.3.3 Scenarios 2a and 2b – USB-C-to-DP/mDP Passive Main-Link Cable Electrical Properties

The electrical specification of the passive cable assembly is based on the specification methodology for USB-C-to-USB-C passive cable assemblies (see [USB-C, Section 3.7.3](#)). Insertion Loss and Return Loss (IL and RL, respectively) shall be tested relative to an 85-Ω differential environment at the [USB-C](#) connector and relative to a 100-Ω differential environment at the [DP](#) connector. The appropriate parameters in the referenced methodology shall be adjusted accordingly. With respect to f_{max} and T_b in [USB-C, Section 3.7.3.2.2](#), the values to be used shall be $f_{max} = 12.15\text{GHz}$ and $T_b = 123.4\text{ps}$.

Note: Using the methodology specified in [USB-C](#), the measurement results for the [USB-C](#) end of a cable assembly do not include the [USB-C](#) receptacle, but do include the [DP](#) receptacle at the [DisplayPort](#) end.

4.3.3.1 Intra-pair Skew (Informative)

The intra-pair skew should be limited to 30ps. This is designed to support 2-m cables at 15ps/m.

4.3.3.2 Insertion Loss Fit at Nyquist Frequencies

[Table 4-7](#) lists the Nyquist frequencies for UHBR10, HBR3, HBR2, and HBR and their Insertion Loss Fit at Nyquist Frequency (ILFitAtNq) mandates.

Table 4-7: Nyquist Frequencies and Their ILFitAtNq Mandate

Bit Rate	Nyquist Frequency (GHz)	ILFitAtNq Mandate (dB)
UHBR10	5	≥ -13.3
HBR3	4.05	≥ -11.0
HBR2	2.7	≥ -8.5
HBR	1.35	≥ -5.5

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 4-12 through Figure 4-15 illustrate the Insertion Loss and the Insertion Loss Fit curves for an example cable assembly at UHBR10, HBR3, HBR2, and HBR, respectively. Each graph identifies the Insertion Loss and the Insertion Loss Fit for the Nyquist frequency corresponding to one of the four *DP* bit rates, also providing the ILFitAtNq limit in each case.

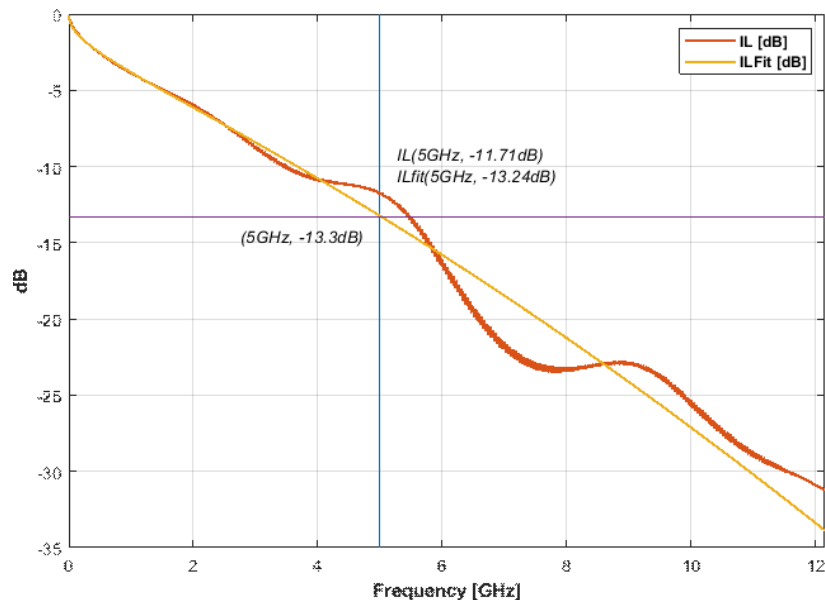


Figure 4-12: ILFitAtNq for UHBR10

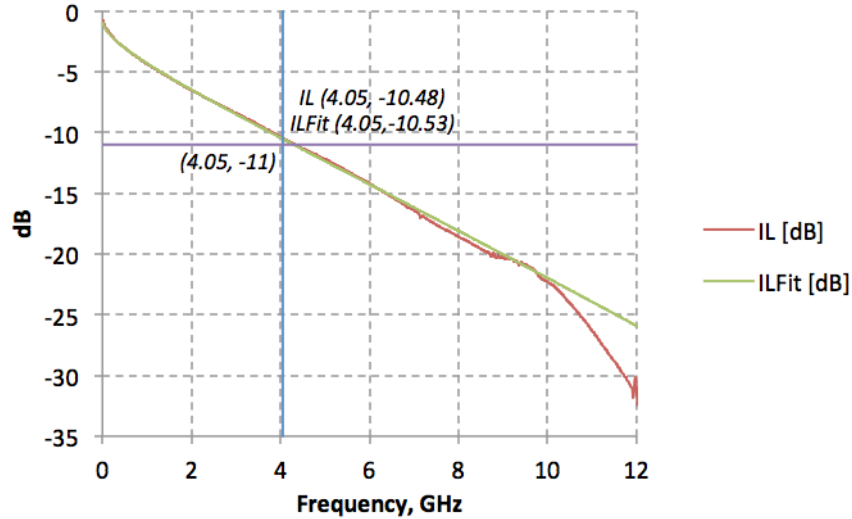


Figure 4-13: ILFitAtNq for HBR3

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

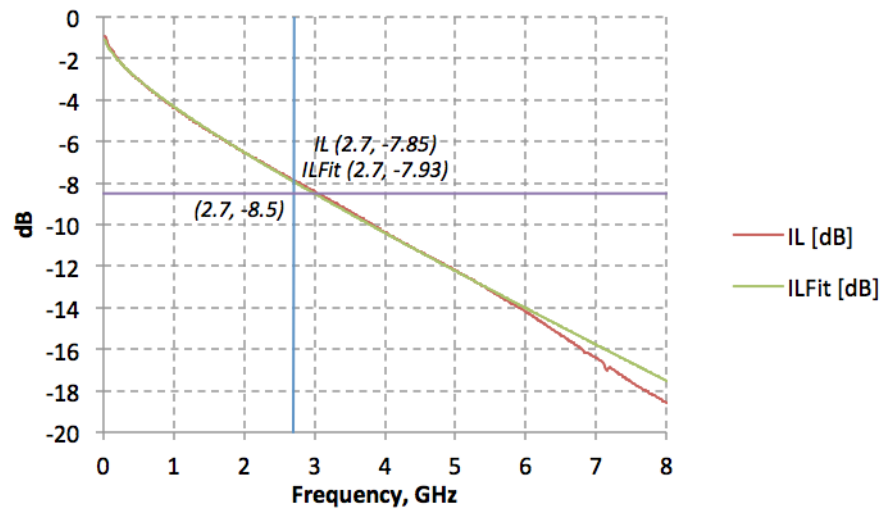


Figure 4-14: ILFitAtNq for HBR2

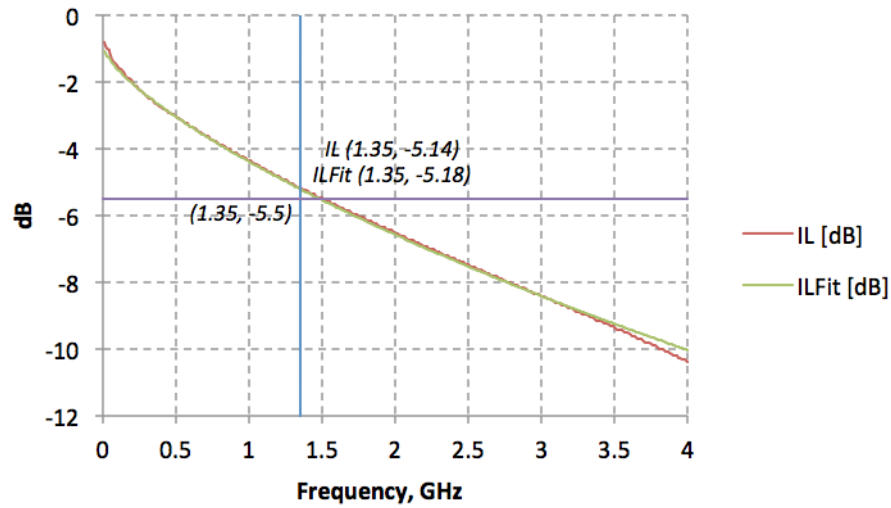


Figure 4-15: ILFitAtNq for HBR

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.3.3.3 Integrated Multi-reflection

For the Input Pulse Spectrum (used in the calculation of integrated multi-reflection (IMR)), the UHBR10, HBR3, HBR2, and HBR *DP* bit rates correspond to a UI of 100.0, 123.5, 185.2, and 370.4ps, respectively. The rise time is 0.4UI.

IMR has dependency on ILFitAtNq. More IMR may be tolerated when ILFitAtNq decreases.

The IMR limit is specified as a function of ILFitAtNq:

$$IMR \leq 0.0579 \times ILFitAtNq^2 + 1.64 \times ILFitAtNq - 23.23$$

Figure 4-16 illustrates the compensation curve of IMR as a function of ILFitAtNq at UHBR10 and HBR3.

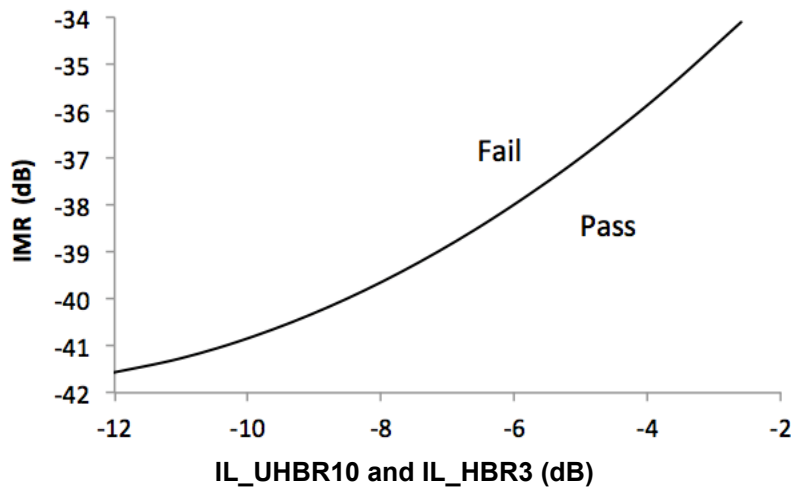


Figure 4-16: IMR as a Function of ILFitAtNq at UHBR10 and HBR3

4.3.3.4 Power-summed Integrated Crosstalk between DP Lanes

The power-sum limits for integrated crosstalk between the *DP* lanes shall comply with the following limits:

- $IFEXT \leq -35\text{dB}$ (up to 12GHz)
- $INEXT \leq -35\text{dB}$ (up to 12GHz) when measured through the *USB-C* connector
- $INEXT \leq -23\text{dB}$ (up to 12GHz) when measured through the *DP* connector

4.3.3.5 Integrated Return Loss

IRL has a strong dependency on $ILFitAtNq$, and its limit is specified as a function of $ILFitAtNq$:

$$IRL \leq 0.0336 \times ILFitAtNq^2 + 1.28 \times ILFitAtNq - 8.87$$

Figure 4-17 illustrates the compensation curve of IRL as a function of $ILFitAtNq$ at UHBR10 and HBR3.

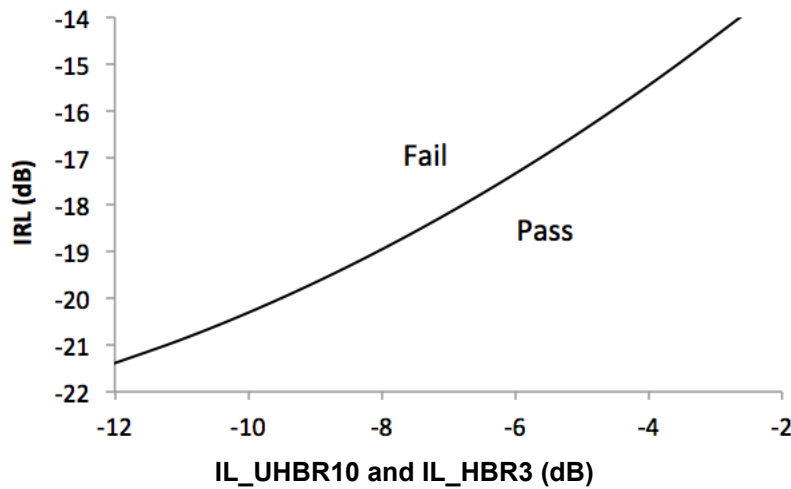


Figure 4-17: IRL as a Function of $ILFitAtNq$ at UHBR10 and HBR3

4.3.3.6 Differential-to-Common Mode Conversion

SCD12/SCD21 shall be less than or equal to -17dB from 100MHz to 8.1GHz.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.3.4 Scenarios 2a and 2b – USB-C-to-DP/mDP Cable Functionality

The cable assembly includes logic to perform the following functions:

- [USB PD](#) Support
- DP Source/Sink Device Detection and HPD Communication

Both are defined in the following subsections.

4.3.4.1 USB PD Support

The logic supports the following:

- [USB PD](#) protocol in the role of a UFP_U
- Negotiates a [USB PD](#) Explicit Power Contract
- Responds to [USB PD](#) Commands for the discovery of DisplayPort Alt Mode capability, entry and exit by the connected DFP_U using structured Vendor Defined Messages (VDMs) as specified in [USB PD](#) and [USB-C](#)

The logic reports whether the [DP](#) connector is:

- Disconnected,
- Connected to a [DP](#) Source device, –or–
- Connected to a [DP](#) Sink device

4.3.4.2 DP Source/Sink Device Detection and HPD Communication

The cable assembly shall contain logic to detect a connected [DP](#) Source or Sink device, and use [USB PD](#) Messages to report the following [DP](#) connector status:

- Disconnected,
- Connected to a [DP](#) Source device, –or–
- Connected to a [DP](#) Sink device

The [Section 3.9.2.1](#) and [Section 3.9.2.2](#) timing mandates shall be met. The cable assembly shall use [USB PD](#) Messages to report a change of state between these three states and support HPD-to-USB PD and USB PD-to-HPD conversion.

The following description outlines a method by which connection detection and HPD conversion may be performed. Any implementation with equivalent behavior may be used.

[DP Standard](#) defines pull-up and pull-down resistors, located between the AC-coupling capacitors, for the AUX_CH_P and AUX_CH_N signal pair at both the [DP](#) Source and Sink devices (R_{AUX_SRC} and R_{AUX_SNK} , respectively; see [Table 4-8](#) and [Figure 4-18](#)). This implementation allows a [DP](#) Sink device to determine when a [DP](#) Source device has been connected.

An unconnected USB-C-to-DP adapter will have an undefined voltage at AUX_CH_P and AUX_CH_N due to the AC-coupling capacitors (C_{AUX} in [Figure 4-18](#)). R_{AUXP_ADAPT} shall be connected to AUX_CH_P and R_{AUXN_ADAPT} shall be connected to AUX_CH_N to avoid an undefined logic level on AUX_CH_P and AUX_CH_N when the USB-C-to-DP adapter is disconnected (see [Table 4-8](#) and [Figure 4-18](#)) and to avoid an unknown logic state.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-8: DP Connector AUX_CH Signal States

Signal	DP Source Device	DP Sink Device	USB-C-to-DP Adapter	Logic Value When Connected
AUX_CH_P	$R_{AUX_SRC}^a$ pull-down resistor to GND.	$R_{AUX_SNK}^a$ pull-up resistor to 3.3V.	R_{AUXP_ADAPT} pull-down resistor to GND. Value is $470k\Omega \pm 20\%$.	0
AUX_CH_N	$R_{AUX_SRC}^a$ pull-up resistor to 3.3V.	$R_{AUX_SNK}^a$ pull-down resistor to GND.	R_{AUXN_ADAPT} pull-down resistor to GND. Value is $4.7M\Omega \pm 20\%$.	1

a. See [DP Standard](#), Table 3-5, for R_{AUX_SRC} and R_{AUX_SNK} values.

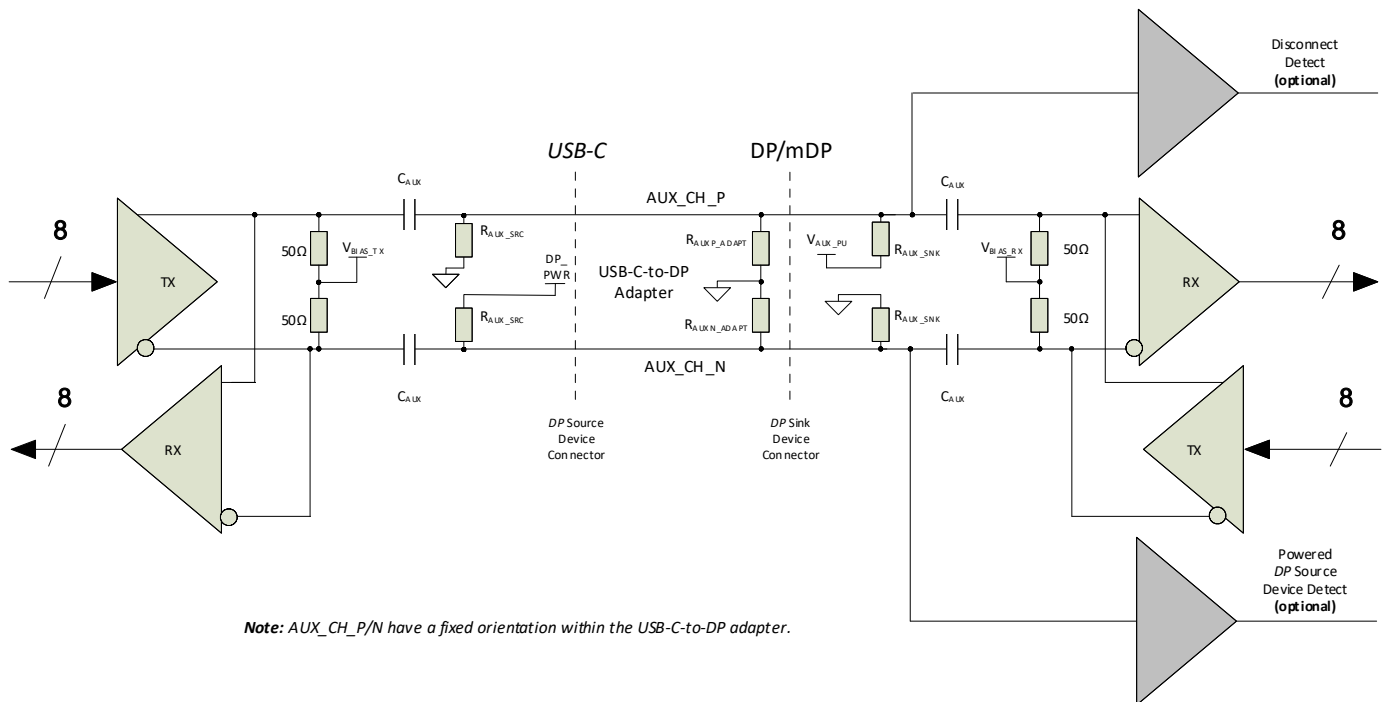


Figure 4-18: USB-C-to-DP Adapter Additional USB PD

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

After filtering to generate AUX_CH logic states while ignoring actual AUX_CH traffic and filtering HPD to avoid an IRQ_HPD being detected as a disconnect, the *DP* connector state may be determined.

Initially, the USB-C-to-DP adapter places its HPD driver in a high-impedance state.

A powered *DP* Source device shall always have R_{AUX_SRC} on AUX_CH_N. Thus, given that the adapter's SBU2 isolation switch is open, a logic 1 on AUX_CH_N indicates a connected *DP* Source device. If the adapter detects a connected *DP* Source device, the adapter uses *USB PD* messaging to report the device's presence to the *USB-C* system, and then waits for a **DisplayPort Configure** Command from the *USB-C* system to change to *DP* Sink Device Configuration. After receiving this command, the adapter closes the SBU isolation switches, thus connecting the AUX_CH_P and AUX_CH_N signals between the *DP* Sink device on the *USB-C* connector and *DP* Source device on the *DP* connector. The *DP* Sink device on the *USB-C* connector shall use *USB PD* Messages to provide HPD status updates to the adapter. After receiving the Messages, the adapter shall control the HPD signal on the *DP* connector accordingly. Because the *DP* Sink device on the *USB-C* connector has only a weak pull-up resistor on AUX_CH_P (R_{AUX_SNK}) and a weak pull-down resistor on AUX_CH_N (R_{AUX_SNK}), the filtered states on these connections do **not** change. If the *DP* Source device is subsequently disconnected from the *DP* connector, the following occurs:

- AUX_CH_N (filtered) state shall revert to a low state
- AUX_CH_P (filtered) state shall revert to a high state

The adapter shall use *USB PD* Messages to report the disconnected state to the *DP* Sink device, and then the *DP* Sink device will signal to isolate the SBU connections.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A powered *DP* Sink device shall provide the weak pull-up resistor on AUX_CH_P (R_{AUX_SRC}). Some displays do **not** provide this pull-up resistor; instead, those displays assert HPD without *DP* Source device detection. Thus, given that the adapter's SBU1 isolation switch is open, either a logic 1 (using a threshold appropriate for $R_{AUX_P_ADAPT}$) on AUX_CH_P or HPD indicates a connected *DP* Sink device. If the adapter detects a connected *DP* Sink device, the adapter uses *USB PD* Messages to report the device's presence to the *USB-C* system, and then waits for a **DisplayPort Configure** Command from the *USB-C* system to change to *DP* Source Device Configuration. After receiving this command, the adapter closes the SBU isolation switches, thus connecting the AUX_CH_P and AUX_CH_N on the *DP* connector to the *DP* Sink device. If the *DP* Sink device is waiting to detect the *DP* Source device before asserting HPD, the *DP* Sink device shall detect this when the SBU isolation switches are closed. The adapter uses *USB PD* Messages to provide HPD status updates to the *DP* Source device on the *USB-C* connector. When the SBU isolation switches are closed, the following occurs:

- R_{AUX_SRC} in the *DP* Source device at the *USB-C* receptacle shall cause the AUX_CH_P (filtered) state to transition to logic 0
- R_{AUX_SRC} in the *DP* Source device shall cause the AUX_CH_N (filtered) state to transition to logic 1

If the *DP* Sink device is subsequently disconnected from the *DP* connector, the HPD (filtered) state shall transition to logic 0. The adapter shall use *USB PD* Messages to report the disconnected state to the *DP* Source device, and then the *DP* Source device will signal to isolate the SBU connections.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-9 enumerates the possible states of HPD, AUX_CH_P, and AUX_CH_N when there is no connection on the *DP* connector or when the adapter is operating in *DP* Source Device Configuration (i.e., connected to a *DP* Sink device at the *DP* connector).

**Table 4-9: DP Connector HPD and AUX_CH Signal States when Disconnected
–or– in Adapter DP Source Device Configuration**

HPD	AUX_CH_P	AUX_CH_N	Interpretation
0	0	0	Nothing is connected, or a powered-down device is connected. Adapter may report that an external <i>DP</i> Sink device is connected. HPD_Low shall be asserted. When connected to a <i>DP</i> Sink device under marginal conditions, the weak R _{AUX_SNK} pull-up resistor may result in a V _{AUX_CH_P} of approximately 0.7V. Some implementations, however, may not be able to detect the approximate 0.7V voltage level as being high, in which case the implementation may optionally report a connection to the <i>DP</i> Sink device when the HPD, V _{AUX_CH_P} , and V _{AUX_CH_N} signals are all low. The DUT shall indicate <i>USB PD</i> HPD_Low until HPD_High is detected on the <i>DP</i> connector.
0	0	1	External <i>DP</i> Source device is connected.
0	1	0	External <i>DP</i> Sink device is connected with R _{AUX_SNK} on AUX_CH_P. No <i>DP</i> Source device is connected. <i>DP</i> Sink device presence is reported. AUX_CH_P is high. Actual logical level of this high may be anywhere between 0.7 and 3.6V. When connected to a <i>DP</i> Sink device under marginal conditions, the weak pull-up value of R _{AUX_SNK} may result in a V _{AUX_CH_P} of approximately 0.7V.
0	1	1	Invalid.
1	0	0	External <i>DP</i> Sink device is connected (i.e., one that does not implement the R _{AUX_SRC} resistors, but directly asserts HPD).
1	0	1	<i>DP</i> Sink device is connected. <i>DP</i> Source device is also connected. The adapter has already closed the SBU isolation switches, thereby connecting the R _{AUX_SRC} resistors from the <i>DP</i> Source device. This state is maintained until the <i>DP</i> Sink device is disconnected.
1	1	0	Invalid (possible transitory state between 0 1 0 and 1 0 1 if the <i>DP</i> Sink device detects a connection resulting from the adapter's weak R _{AUXP_ADAPT} pull-down resistor).
1	1	1	Invalid (this state may occur if the connected device is a <i>DP</i> Source device-side <i>DP++</i> adapter with Display Data Channel (DDC) pull-up resistors, as defined in <i>DP++ Standard</i>).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

After the adapter is operating in *DP* Sink Device Configuration (i.e., connected to a *DP* Source device on the *DP* connector) and converting HPD from the *USB PD* Messages received on the *USB-C* connector and driving HPD to the *DP* connector, the information provided in [Table 4-9](#) simplifies to that listed in [Table 4-10](#).

**Table 4-10: DP Connector HPD and AUX_CH Signal States in Adapter
DP Sink Device Configuration**

HPD	AUX_CH_P	AUX_CH_N	Interpretation
1	0	0	Powered-down device is connected.
1	0	1	External <i>DP</i> Source device is connected and operational.
1	1	0	Transitional state. When the previously connected <i>DP</i> Source device is disconnected, the adapter should change out of <i>DP</i> Sink Device Configuration and report the new status to the <i>USB-C DP</i> Sink device, which in turn should disable DisplayPort Configuration, thereby taking HPD to a logic low state and isolating the SBU connection.
1	1	1	Invalid.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.3.5 Scenarios 2a and 2b – Active USB-C-to-DP Cables

Active cables shall follow the mandates defined in [Section 4.9](#) and [Section 4.10](#).

4.3.5.1 Active USB-C-to-DP Cable Directionality Support

An active USB-C-to-DP cable may do the following:

- Support *DisplayPort* in both directions, as defined in [Table 2-3](#)
- Use the mechanisms described in [Section 4.10](#) to determine *DisplayPort* directionality

4.3.5.2 Signal Connections for Active USB-C-to-DP/mDP Cables

The signal connections for a USB-C connector-to-DP/mDP connector active cable assembly shall be as defined in [Section 4.3.1](#) and [Section 4.3.2](#). Thus, the active cable shall support *DP* Source and Sink device Pin Assignment E (**DP Source devices** – see [Table 3-14](#) and [Table 3-15](#); **DP Sink devices** – see [Table 3-20](#) and [Table 3-21](#)).

4.3.5.3 Active USB-C-to-DP Cable Support for USB PD

An active USB-C-to-DP cable shall support *USB PD Discover Identity* SOP. The cable shall behave as a USB-C-to-DP passive cable, as defined elsewhere in this Standard, except for the properties described in this subsection.

Active USB-C-to-DP cable behavior includes the following:

- Present as Rd/Ra on the *USB-C* connector so that the cable always resolves to UFP_U when the cable is first connected
- Powered from V_{CONN}
- Present *USB BB* on the *USB-C* connector's *USB 2.0* pins under appropriate circumstances
- Use the method described in [Section 4.10](#) to determine the *DP* directionality
- Shall **not** set the V_{BUS} required bit in the *USB PD Discover Identity* Command response (bit 3 of the AMA VDO; see *USB PD*) to 1 to allow for operation from V_{CONN} when connected to a Sink with Accessory Support
- Shall meet the [Table 3-21](#) mandates for an active adapter cable with the *USB-C* plug connected to the *DP* Source device
- Shall meet the [Table 3-22](#) mandates for an active adapter cable with the *USB-C* plug connected to the *DP* Sink device
- Active adapter cable shall behave as if it received an HPD high when the cable receives a *DisplayPort Configure* Command

After entering DisplayPort Alt Mode, the active cable should connect the PHY Repeater's AUX_CH to provide the maximum time that is needed to prepare a response to AUX transactions that are received from the *DP* Source device. The active cable should assume that HPD is high when a *DisplayPort Configure* Command request is addressed to SOP/SOP" is issued, –or– when snooping the *DisplayPort Configure* Command request addressed to SOP.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.3.6 Scenarios 2a and 2b – USB-C-to-DP Cable Power

A passive or active USB-C-to-DP cable shall operate at any voltage between 3.0 and 5.5V (as presented at the adapter's *USB-C* plug, there may be an IR drop on a V_{CONN} wire within the adapter's cable), and the adapter shall not consume in excess of 1.5W. **Optional** operation down to 2.7V ensures backward compatibility with earlier versions of *DP AM* and provides margin for voltage drop across V_{CONN} power switches, connectors, cables, and adapters. *USB PD* communications may be maintained if the V_{CONN} voltage falls below 3.0V. Power consumption shall not exceed 70mW and cable functionality may be disabled if V_{CONN} voltage falls below 3.0V. The cable shall report whether it continues to operate but disables *DP* functionality (see [Table 4-3](#)).

During the Discovery Process and whenever DisplayPort Alt Mode is **not** entered, the passive or active USB-C-to-DP cable shall comply with *USB-C* V_{CONN} power consumption mandates.

All active cables that support DisplayPort Alt Mode shall meet the power requirements defined in [Section 4.8](#).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.4

Scenarios 3a and 3c – USB-C DP Source Device-to-Protocol Converter

Figure 4-19 and Figure 4-20 illustrate configurations of a protocol converter that is designed to plug into a *USB-C DP* Source device. Figure 4-19 illustrates a protocol converter cable. Figure 4-20 illustrates a protocol converter adapter, or dongle.

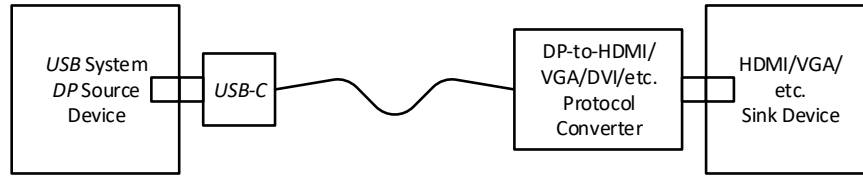


Figure 4-19: Scenario 3a – USB-C DP Source Device Protocol Converter Cable

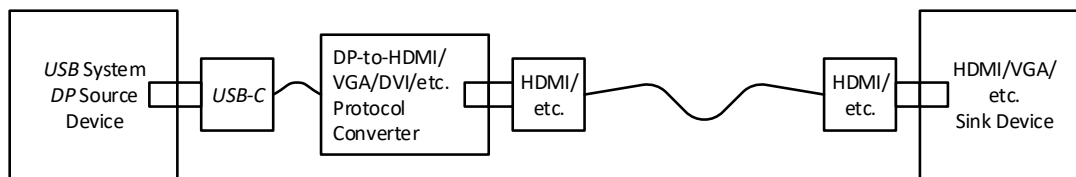


Figure 4-20: Scenario 3c – USB-C DP Source Device Protocol Converter Adapter or Dongle

These scenarios use *DP* Source device Pin Assignment C (see Table 3-10 and Table 3-11). The implementation of AUX_CH and HPD in the protocol converter is similar to that used in a *USB-C DP* Sink device (see Section 3.9).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.4.1 Scenarios 3a and 3c – HDMI Protocol Converters

HDMI protocol converters shall comply with the mandates for DP-to-HDMI protocol converter adapters defined in *DP* lane-count and link-rate mandates for *HDMI* protocol converters are device-specific (see Table 2-3).

In addition, the DETAILED_CAP_INFO_AVAILABLE bit in the DOWN_STREAM_PORT_PRESENT register (DPCD Address 00005h, bit 4) shall be set to 1 (i.e., the *DP* Source device Receiver Capability fields in registers starting at DPCD Address 00080h shall use the longer 4-byte format).

4.4.1.1 Audio Return Channel

The *HDMI* protocol converter may include a *USB 2.0* bridge device to convert the Audio Return Channel (ARC) to the *USB* Audio Device Class (audio input). For *HDMI* protocol converters and *DP* Source devices, CEC support is a prerequisite for using an ARC. ARC is supported only on *DP* Source devices that are also *USB* hosts.

4.4.2 Scenarios 3a and 3c – VGA Protocol Converters

VGA protocol converters shall support the following minimum resolutions for reduced and full blanking:

- **24-bpp reduced blanking** – 1920x1200 at 60Hz
- **24-bpp full blanking** – 1600x1200 at 60Hz

VGA protocol converters shall support two HBR lanes, but are **not** mandated to support bandwidths in excess of two HBR lanes.

4.4.3 Scenarios 3a and 3c – DVI Protocol Converters

Single-link DVI protocol converters shall support a minimum resolution of 1920x1200 at 60Hz, 24bpp. DVI protocol converters shall support two HBR lanes, but are **not** mandated to support bandwidths in excess of two HBR lanes.

4.5

Scenario 4a – USB-C DP Source Device-to-Multi-function Dock

Figure 4-21 illustrates a *USB-C DP* Source device connected to a multi-function dock.

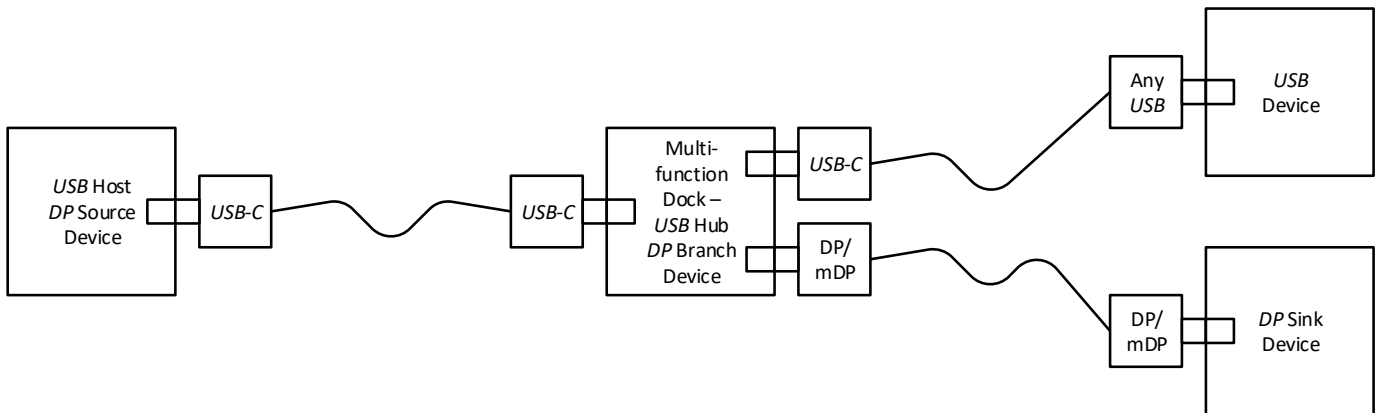


Figure 4-21: Scenario 4a – USB-C DP Source Device-to-Multi-function Dock

A multi-function dock includes a *USB* hub and a *DP* Branch device. This scenario uses *DP* Source device Pin Assignment D at the *DP* Source device (see Table 3-12 and Table 3-13) and *DP* Sink device Pin Assignment D at the *DP* Sink device (see Table 3-18 and Table 3-19). However, if the multi-function dock is able to be configured to apply all four *DP* lanes to the *DP* connector with *USB 2.0*, Pin Assignment C may be used instead (**DP Source devices** – see Table 3-10 and Table 3-11; **DP Sink devices** – see Table 3-16 and Table 3-17). The following figures illustrate how *DisplayPort* uses the Full-Featured *USB-C* cable assembly's signal connections:

- Figure 4-22 illustrates when the plugs at both ends are inserted in normal orientation
- Figure 4-23 illustrates when the *DP* Source device plug is flipped and the *DP* Sink device plug is inserted in normal orientation
- Figure 4-24 illustrates when the *DP* Source device plug is inserted in normal orientation and the *DP* Sink device plug is flipped
- Figure 4-25 illustrates when the plugs at both ends are flipped

Note: The figures omit the *USB PD* identification circuitry that is incorporated in the cable assembly.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

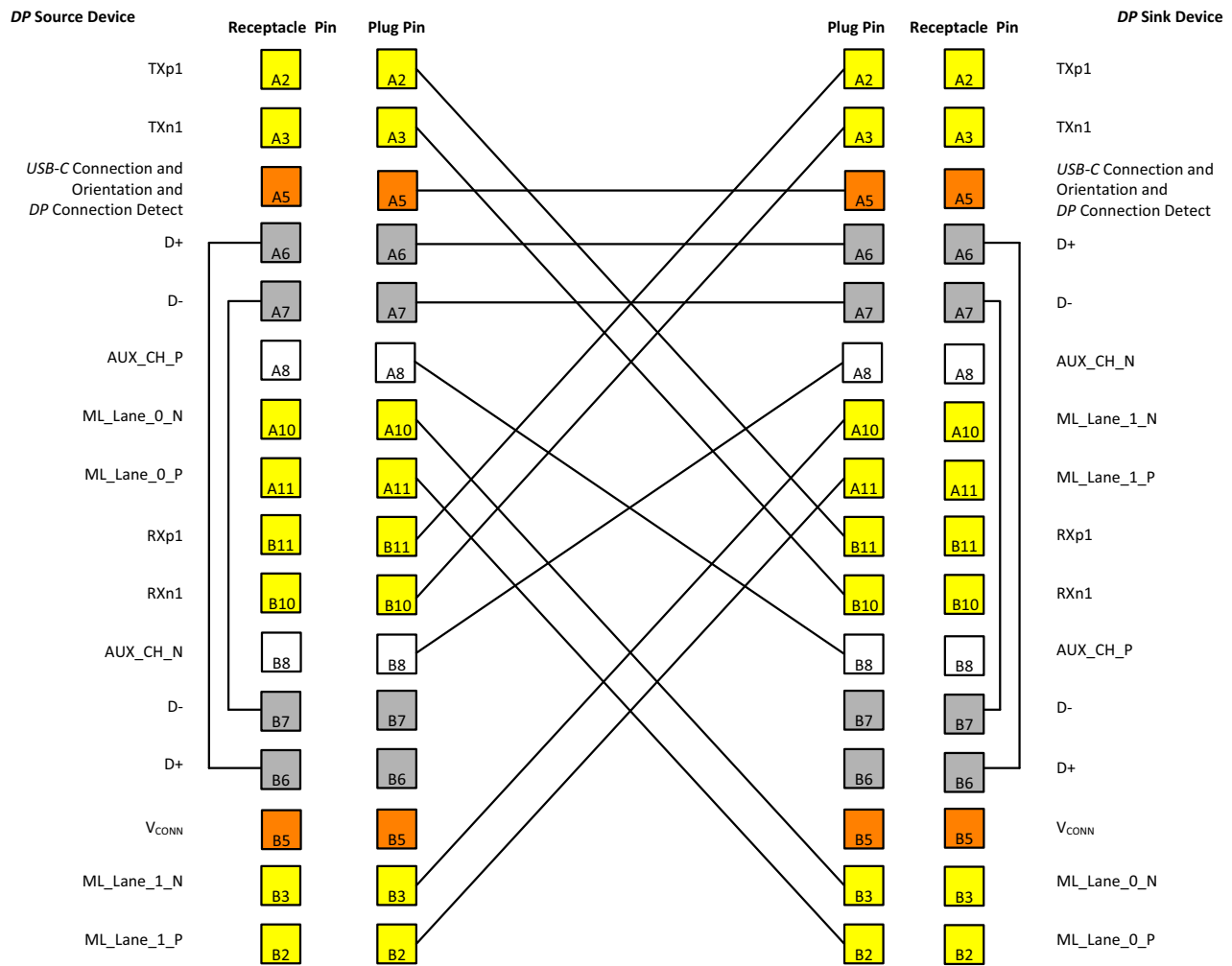


Figure 4-22: Scenario 4a Multi-function Cable Wiring – Normal Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

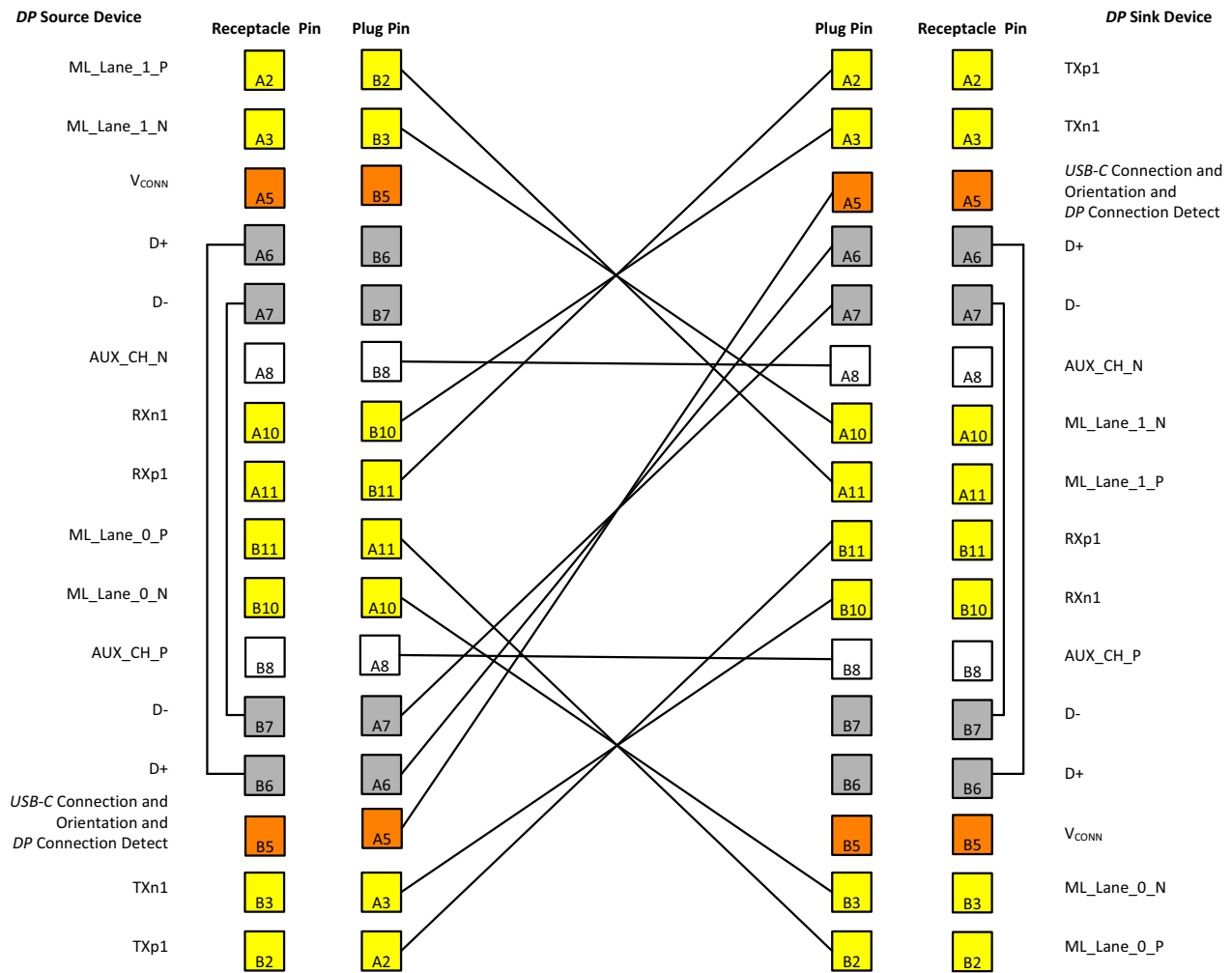


Figure 4-23: Scenario 4a Multi-function Cable Wiring – DP Source Device Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

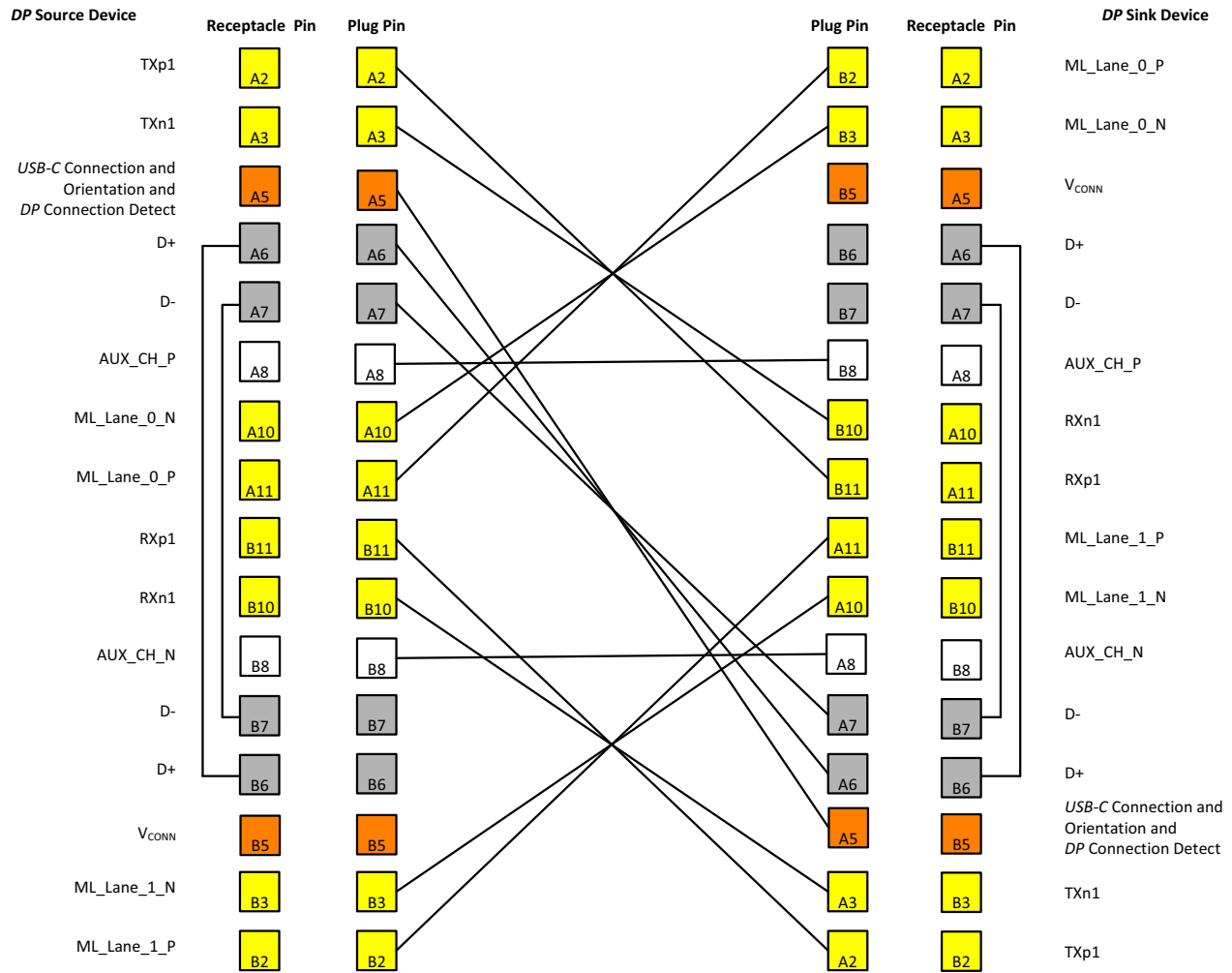


Figure 4-24: Scenario 4a Multi-function Cable Wiring – DP Sink Device Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

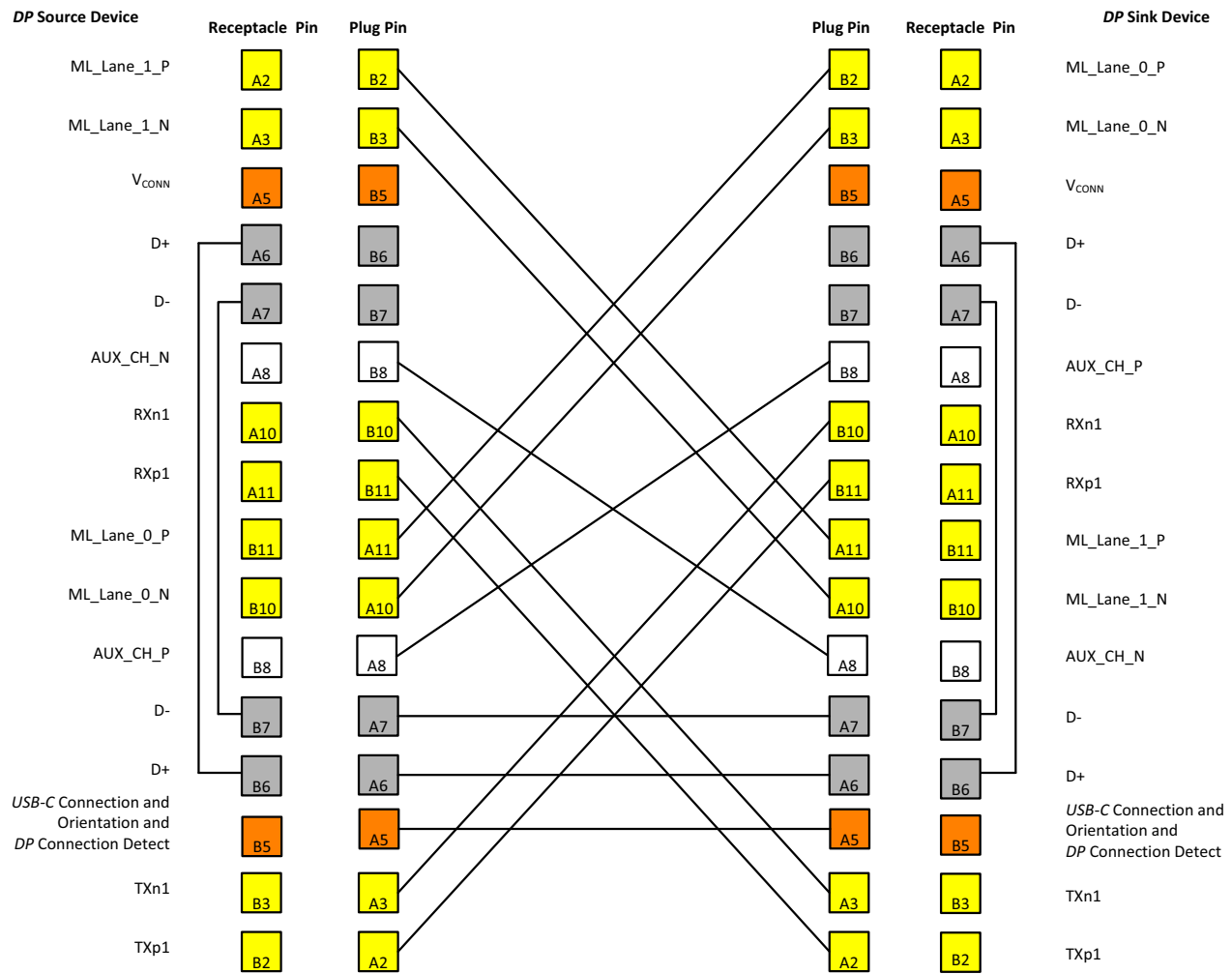


Figure 4-25: Scenario 4a Multi-function Cable Wiring – DP Source and Sink Device Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.6

Scenarios 4c and 4d – USB-C DP Source Device to Multi-function Cradle/Captive Cable Dock

Figure 4-26 illustrates a *USB-C DP* Source device connected to a multi-function cradle dock. Figure 4-27 illustrates a *USB-C DP* Source device connected to a multi-function captive cable dock.

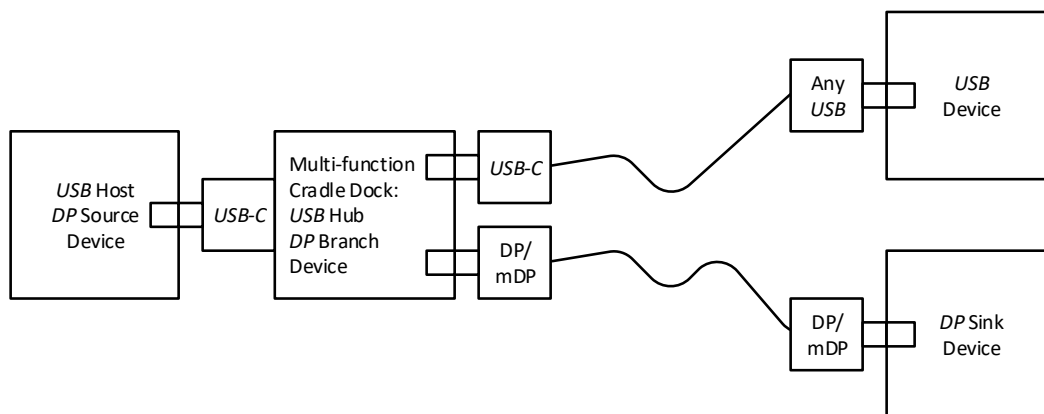


Figure 4-26: Scenario 4c – USB-C DP Source Device Connected to a Multi-function Cradle Dock

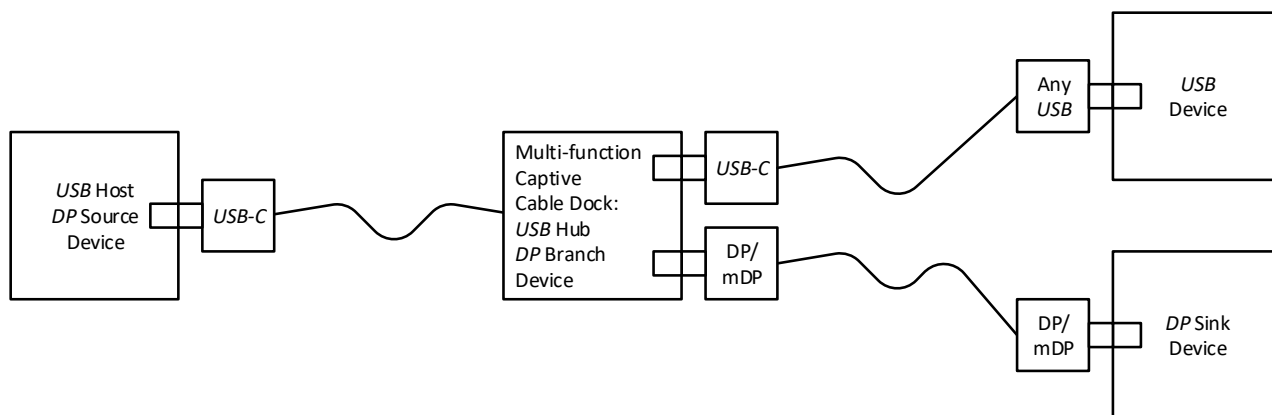


Figure 4-27: Scenario 4d – USB-C DP Source Device Connected to a Multi-function Captive Cable Dock

These scenarios use *DP* Source device Pin Assignment D at the *DP* Source device (see Table 3-12 and Table 3-13) and *DP* Sink device Pin Assignment D at the *DP* Sink device (see Table 3-18 and Table 3-19). However, if the multi-function dock is able to be configured to apply all four *DP* lanes to the *DP* connector, Pin Assignment C may be used instead (**DP Source devices** – see Table 3-10 and Table 3-11; **DP Sink devices** – see Table 3-16 and Table 3-17). Table 4-11 defines the *USB-C* plug's multi-function cradle/captive cable dock pin assignments.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-11: USB-C Cable DP Use (Multi-function Cradle/Captive Cable Dock)

USB-C Plug DP Source Device Pin Assignment D	
Pin	Signal Name
A1, B1, A12, B12	GND
A4, B4, A9, B9	V _{BUS}
A5	CC
A6	D+
A7	D-
A2	TXp1
A3	TXn1
B11	RXp1
B10	RXn1
B2	ML_Lane_1_P
B3	ML_Lane_1_N
A11	ML_Lane_0_P
A10	ML_Lane_0_N
A8	AUX_CH_P
B8	AUX_CH_N
B5	V _{CONN}
Shell	Shield

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.7

Scenarios 5a and 5b – USB-C Plug-to-DP/mDP Active Receptacle Adapters

USB-C plug-to-DP/mDP active receptacle adapters are intended to adapt the following:

- **Scenario 5a** – *DP* Source device that is connected to a *USB-C* plug to a DP/mDP Source device that is connected to a full-size *DP* or mDP receptacle
- **Scenario 5b** – DP/mDP Sink device that is connected to a *USB-C* plug to a *DP* Sink device that is connected to a full-size *DP* or mDP receptacle

The adapter shall support:

- Both Scenario 5a and Scenario 5b (i.e., the adapter shall be bidirectional). Direction determination may be accomplished as defined in [Section 4.10](#), with the adapter presenting the AUX_CH common-mode termination as a *DP* Sink device on the *DP* or mDP receptacle.
- Four *DP* lanes supporting RBR, HBR, HBR2, HBR3, and **optionally** higher bit rates (see [Table 2-3](#))

The adapter shall contain the following:

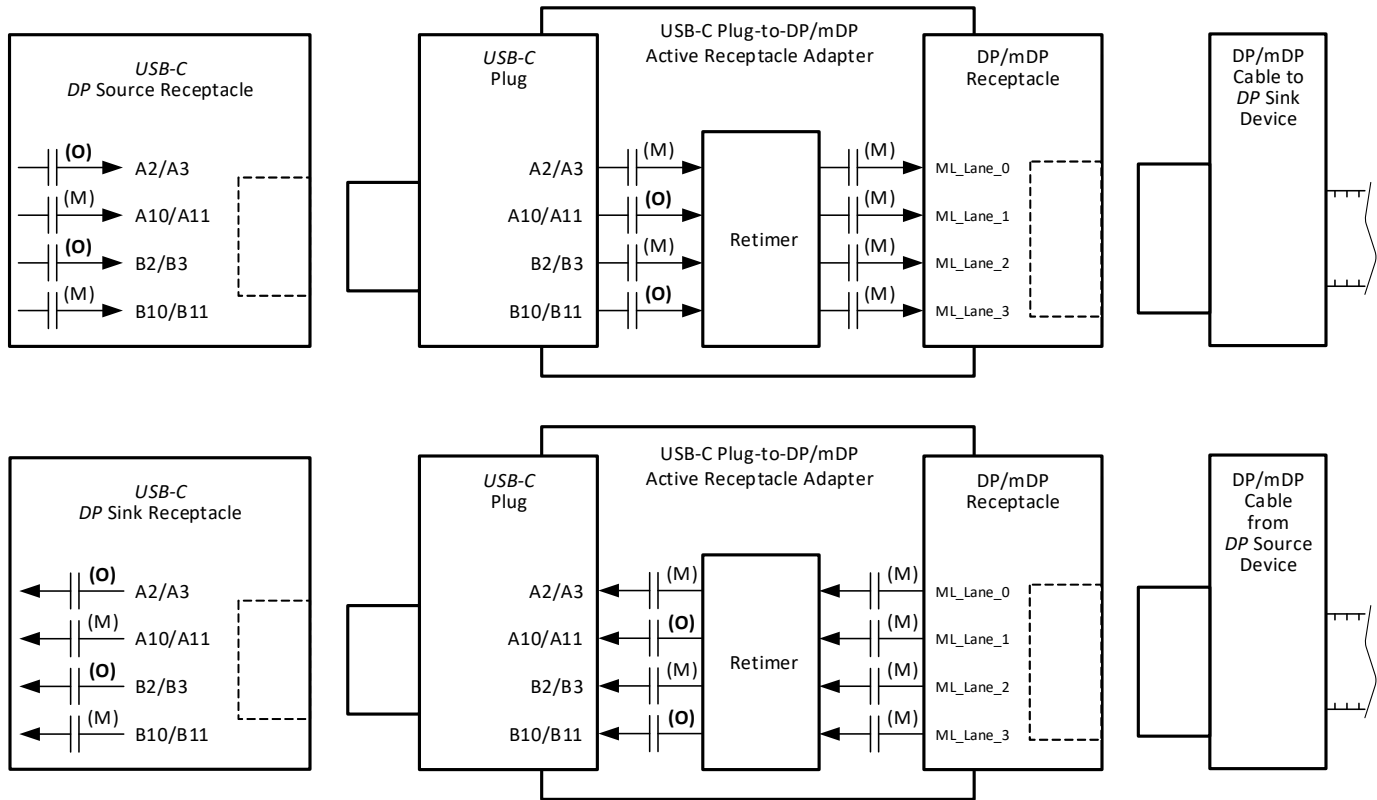
- LTTPR retimer, to present the following:
 - Compliant *DP* Source device signal on its DP/mDP receptacle whenever the adapter is receiving a compliant *DP* Source device signal on its *USB-C* plug
 - Compliant *DP* Sink device signal on its *USB-C* plug whenever the adapter is receiving a compliant *DP* Sink device signal on its DP/mDP receptacle

Note: *Passive DP/mDP receptacle adapters are **not** compliant with this Standard.*

- Logic to implement *USB PD* and support the Discovery Process (see [Section 4.3.4.2](#) and [Section 5](#))
- *USB PD* logic shall connect a *USB BB* device to D+/D- in the event that the logic is unable to negotiate sufficient V_{BUS} power to support the DP/mDP active cable

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

The adapter shall AC-couple **USB-C** pins A10, A11, B10, and B11 to the retimer with $220\text{nF} \pm 20\%$ capacitors. The adapter may AC-couple **USB-C** pins A2, A3, B2, and B3 to the retimer. The adapter's retimer shall be AC-coupled to the full-size **DP** or mDP receptacle, as defined for C_{TX} in **DP Standard**. Figure 4-28 illustrates these retimer configurations.



Notes:

- AC-coupling capacitors denoted with (M) are mandatory. AC-coupling capacitors denoted with (O) are **optional**.
- Both of the above configurations shall be supported.

**Figure 4-28: USB-C Plug-to-DP/mDP Active Receptacle Adapter
(Main-Link AC-coupling Mandates)**

The adapter shall present AUX_CH pull-up/pull-down resistors on the full-size **DP** or mDP receptacle as if it were a **DP** Source device.

The adapter shall meet the [Section 4.7.3](#) power mandates. The adapter shall be powered from V_{CONN} and correctly function with a passive DP/mDP cable without V_{BUS} being present. The adapter may not function with a DP/mDP active cable if the V_{BUS} power is insufficient.

The adapter shall behave as a USB-C-to-DP/mDP cable (see [Section 4.3](#) and [Section 5](#)) from the perspective of DisplayPort Alt Mode discovery and **USB PD**.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.7.1

Scenario 5a – USB-C Plug DP Source-to-DP/mDP Receptacle Adapter

Figure 4-29 illustrates a USB-C plug-to-DP/mDP active receptacle adapter assembly that adapts a *DP* Source device that is connected to a *USB-C* connector to a DP/mDP Sink device presented on a full-size *DP* or mDP connector.

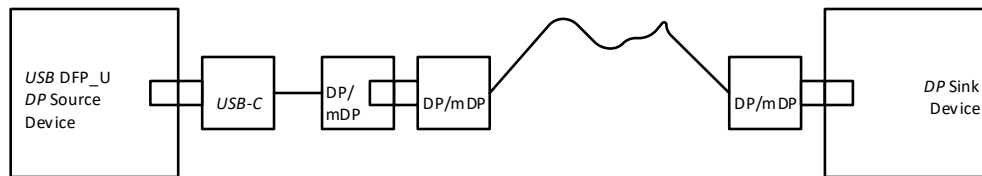


Figure 4-29: Scenario 5a – USB-C Plug DP Source-to-DP/mDP Receptacle Adapter

The *USB-C* plug adapter input is defined as TP2 or TP2_CTLE at its input, per *DP Standard*, Section 3.5.4.

The DP/mDP receptacle output shall meet the *DP Standard* mandates for a DP/mDP Main-Link transmitter (TP2; see *DP Standard*, Section 3.5.4).

The HPD-to-USB PD converter shall meet the Section 3.9.2.1 timing mandates.

Notes: *The adapter will need to incorporate an LTTTPR retimer on each of the four supported lanes.*

*The connections within the adapter remain the same regardless of the adapter *USB-C* plug's orientation within the *DP* Source device connector.*

The connections within the adapter remain the same regardless of whether the adapter is used in Scenario 5a or 5b.

This scenario uses *DP* Source device Pin Assignment E on the DP/mDP receptacle (see Table 3-14 and Table 3-15). The following figures illustrate how *DisplayPort* uses the USB-C plug DP Source-to-DP/mDP receptacle adapter's signal connections:

- Figure 4-30 illustrates this use for a full-size *DP* receptacle when the *USB-C* plug is inserted in normal orientation
- Figure 4-31 illustrates this use for a full-size *DP* receptacle when the *USB-C* plug is inserted in flipped orientation
- Figure 4-32 illustrates this use for an mDP receptacle when the *USB-C* plug is inserted in normal orientation
- Figure 4-33 illustrates this use for an mDP receptacle when the *USB-C* plug is inserted in flipped orientation

Note: *The figures omit the pull-down resistors on CONFIG1 and CONFIG2. Additionally, the bidirectional buffers are illustrative of LTTTPR retimers that are incorporated in a design and are **not** intended to imply a particular implementation.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

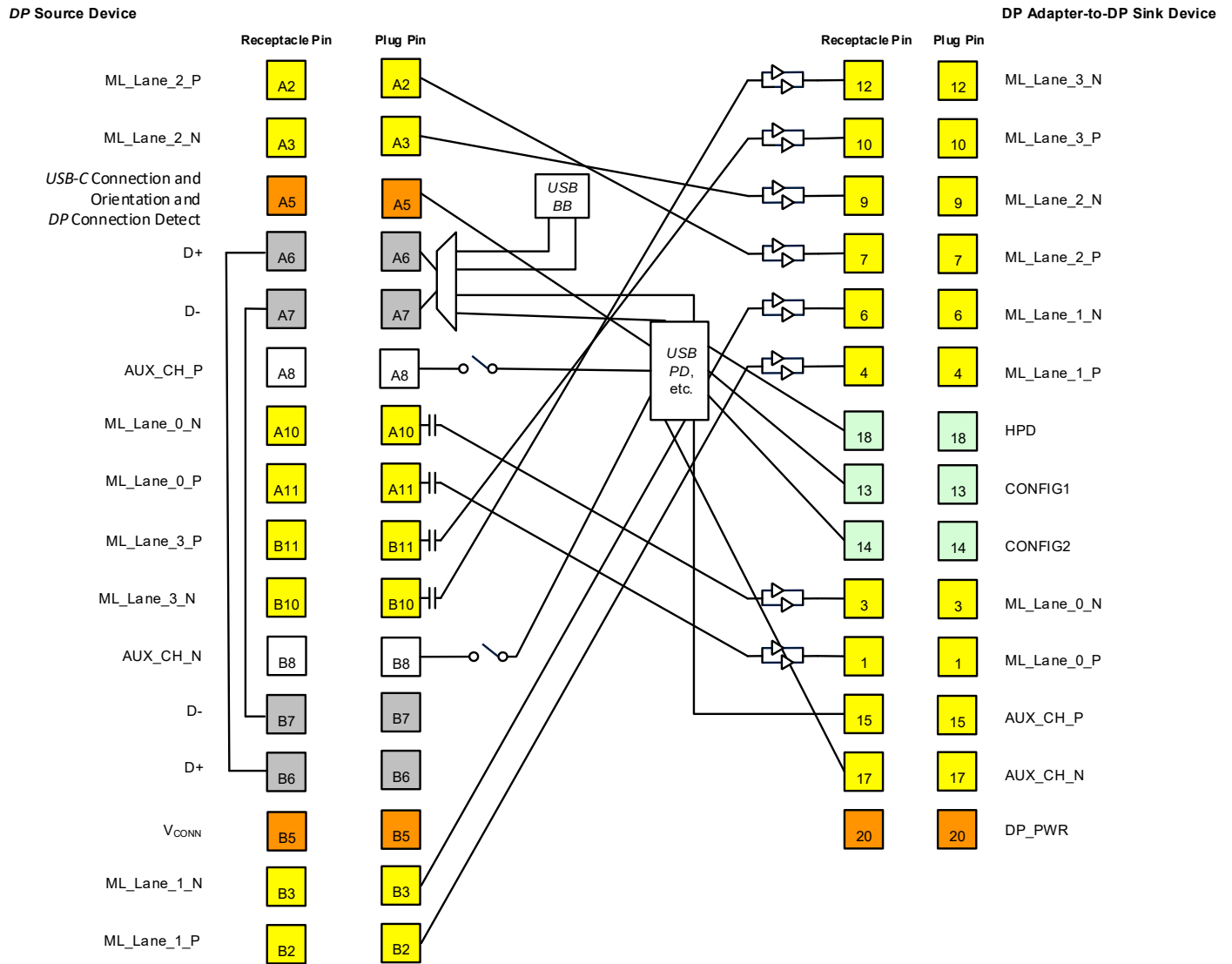


Figure 4-30: Scenario 5a Full-size DP Adapter Wiring – Normal Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

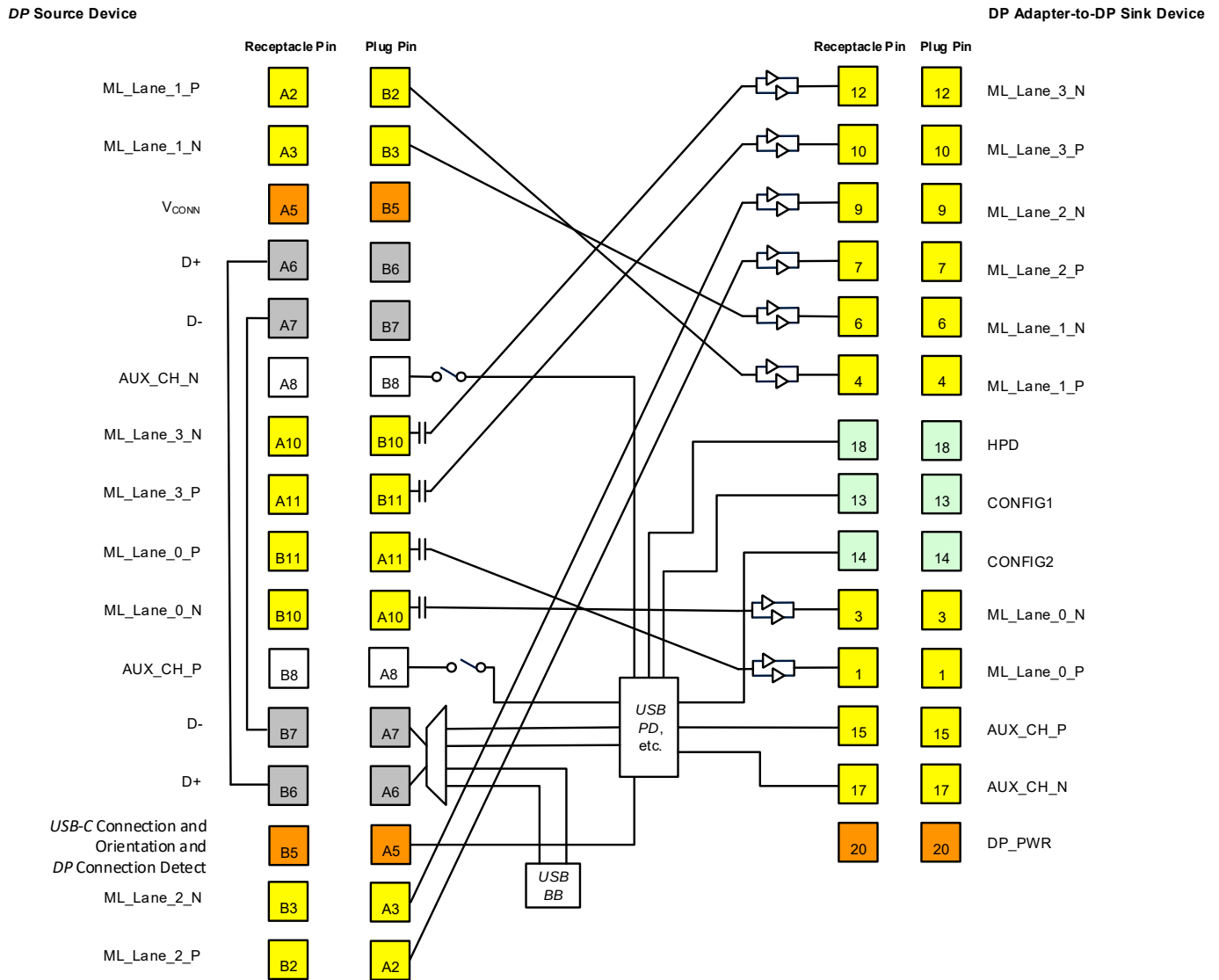


Figure 4-31: Scenario 5a Full-size DP Adapter Wiring – Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

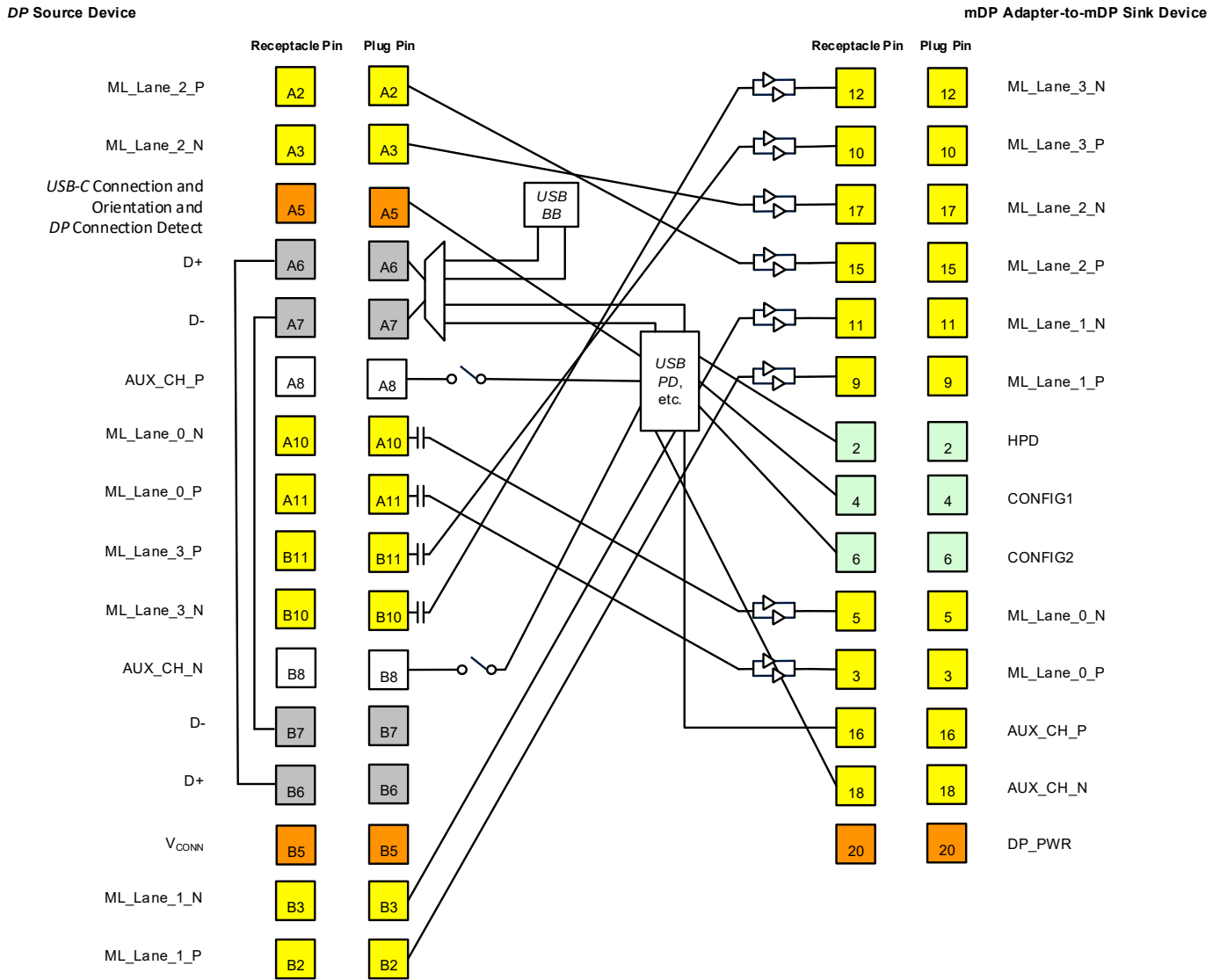


Figure 4-32: Scenario 5a mDP Adapter Wiring – Normal Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

DP Source Device

mDP Adapter-to-mDP Sink Device

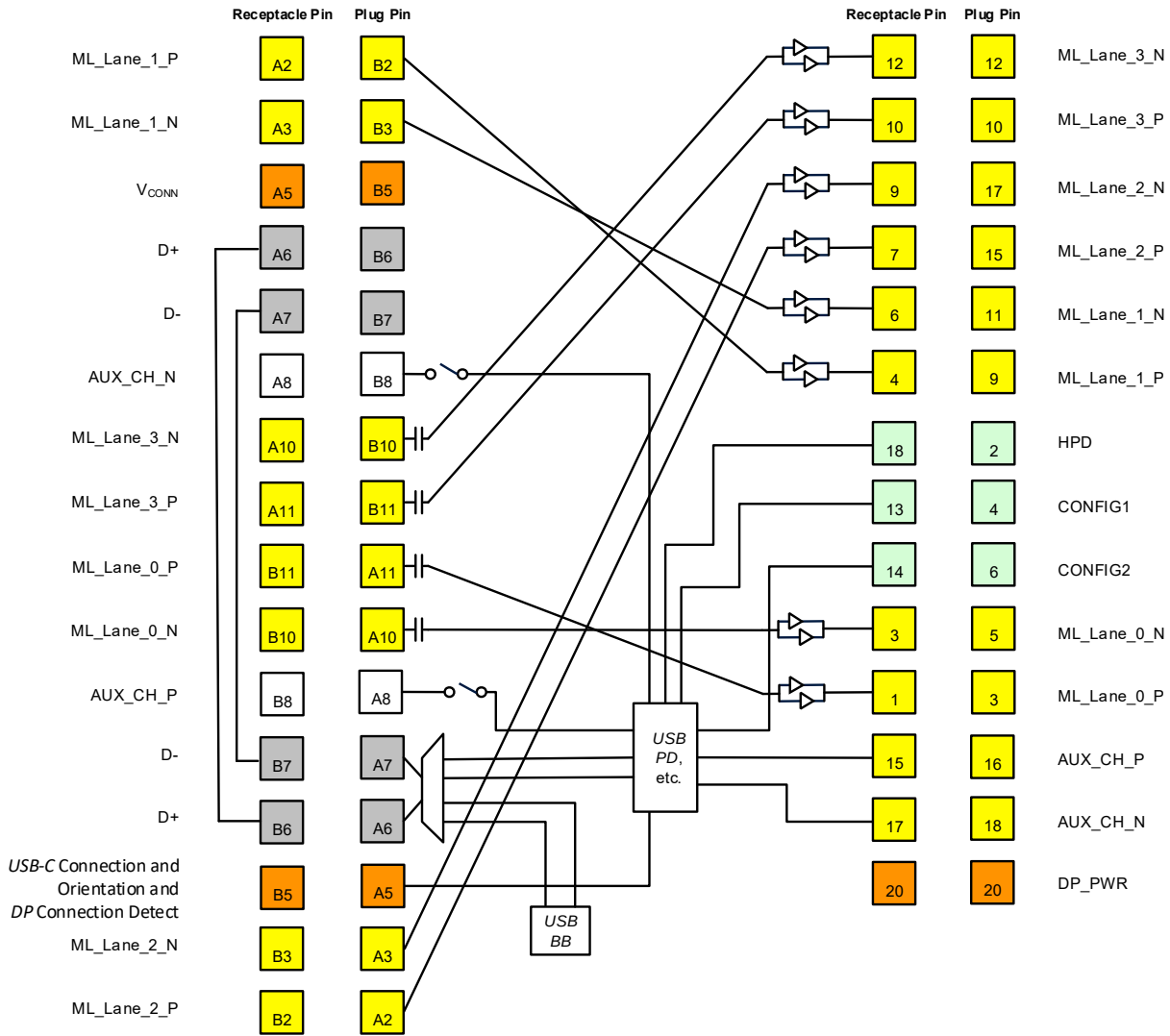


Figure 4-33: Scenario 5a mDP Adapter Wiring – Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-12 lists how *DisplayPort* uses the USB-C plug DP Source device-to-DP receptacle DP/mDP Source device adapter's signal connections.

Table 4-12: Scenario 5a – USB-C Plug DP Source-to-DP/mDP Receptacle Adapter Use

USB-C Plug DP Source Device		Wire		DP/mDP Receptacle DP/mDP Source Device		
Pin	DP Signal Name	Wire Number	Signal Name	DP Pin	mDP Pin	DP Signal Name
A1, B1, A12, B12	GND	1 [16]	GND_PWRrt1 [GND_PWRrt2]	2, 5, 8, 11, 16, 19	1, 7, 8, 13, 14, 19	GND
A4, B4, A9, B9	V _{BUS}	2	Powers <i>USB BB</i>			
A5	CC	3	Adapter logic (<i>USB PD</i>)			
A6	D+	4	Adapter logic (<i>USB BB</i>)			
A7	D-	5	Adapter logic (<i>USB BB</i>)			
			1M to GND	13	4	CONFIG1 ^a
			1M to GND	14	6	CONFIG2 ^a
A2	ML_Lane_2_P	6	SDPp1	6	11	ML_Lane_1_N
A3	ML_Lane_2_N	7	SDPn1	4	9	ML_Lane_1_P
B11	ML_Lane_3_P	8	SDPp2	3	5	ML_Lane_0_N
B10	ML_Lane_3_N	9	SDPn2	1	3	ML_Lane_0_P
B2	ML_Lane_1_P	10	SDPp3	9	17	ML_Lane_2_N
B3	ML_Lane_1_N	11	SDPn3	7	15	ML_Lane_2_P
A11	ML_Lane_0_P	12	SDPp4	12	12	ML_Lane_3_N
A10	ML_Lane_0_N	13	SDPn4	10	10	ML_Lane_3_P
A8 ^b	AUX_CH_P	14	Adapter logic	15	16	AUX_CH_P
B8 ^b	AUX_CH_N	15	Adapter logic	17	18	AUX_CH_N
			Adapter logic	18	2	HPD
B5	V _{CONN}	16	Powers Adapter logic			
			No Connect	20	20	DP_PWR ^c
Shell	Shield	Braid	Shield	Shell	Shell	Shield

a. CONFIG1 and CONFIG2 shall include 1-MΩ pull-down resistors.

b. A8/B8 shall include the AUX_CH DP Source device pull-up/pull-down resistors.

c. DP_PWR shall **not** be provided unless a short is detected between CONFIG1 and CONFIG2.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.7.2

Scenario 5b – DP/mDP Sink Receptacle-to-USB-C DP Sink Adapter

Figure 4-34 illustrates a USB-C plug-to-DP Sink device active receptacle adapter assembly that adapts a *USB-C DP* Sink device to a *DP* Source device that uses a full-size *DP* or mDP connector, driving a *DP* cable with full-size *DP* and/or mDP connectors.

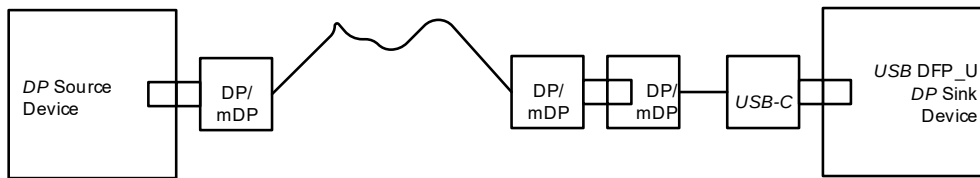


Figure 4-34: Scenario 5b – DP/mDP Sink Receptacle-to-USB-C DP Sink Adapter

The adapter's *DP* Sink input on the DP/mDP receptacle shall meet the *DP Standard* mandates for a DP/mDP Main-Link receiver (TP3_EQ; see *DP Standard*, Section 3.5.4).

The *USB-C* plug *DP* Sink output shall meet the *DP Standard*, Table 5-12, TX EYE opening and TX drive setting monotonicity test specifications defined for C1A, C2A, and C3A (TP3_EQ) (assuming that the input signal is one that a *DP* Sink device shall accept).

The USB PD-to-HPD converter shall meet the Section 3.9.2.2 timing mandates.

Notes: *The adapter will need to incorporate an LTTPR retimer on each of the four supported lanes.*

The connections within the adapter remain the same regardless of the adapter USB-C plug's orientation within the DP Sink device connector.

The connections within the adapter remain the same regardless of whether the adapter is used in Scenario 5a or 5b.

This scenario uses *DP* Sink device Pin Assignment E on the *USB-C DP* Sink device (see Table 3-20 and Table 3-21). The following figures illustrate how *DisplayPort* uses the DP/mDP Sink receptacle-to-USB-C DP Sink adapter's signal connections:

- Figure 4-35 illustrates this use for a full-size *DP* receptacle when the *USB-C* plug is inserted in normal orientation
- Figure 4-36 illustrates this use for a full-size *DP* receptacle when the *USB-C* plug is inserted in flipped orientation
- Figure 4-37 illustrates this use for an mDP receptacle when the *USB-C* plug is inserted in normal orientation
- Figure 4-38 illustrates this use for an mDP receptacle when the *USB-C* plug is inserted in flipped orientation

Note: *The figures omit the pull-down resistors on CONFIG1 and CONFIG2. Additionally, the bidirectional buffers are illustrative of LTTPR retimers that are incorporated in a design and are **not** intended to imply a particular implementation.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

DP Adapter-to-DP Sink Device

DP Source Device

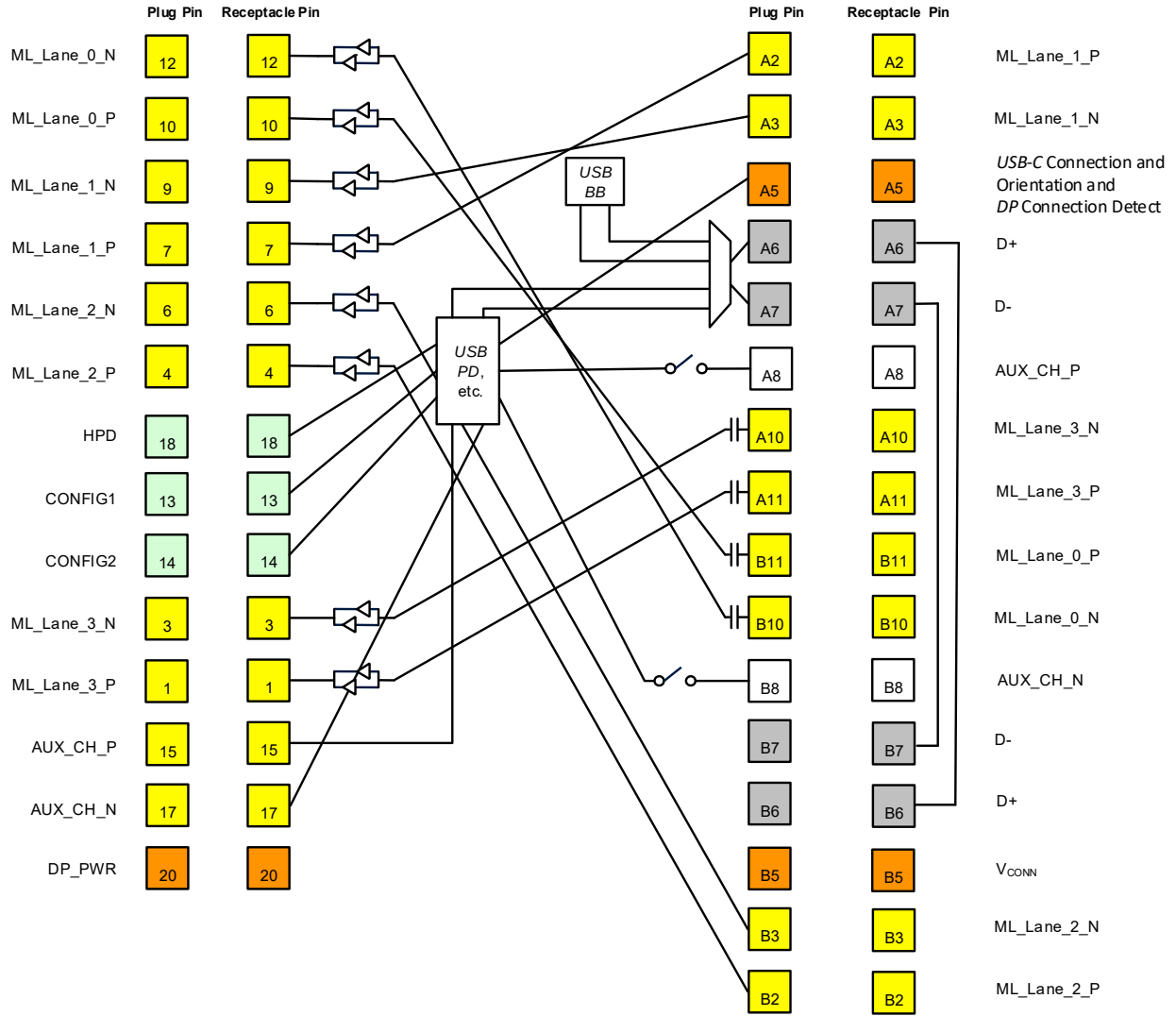


Figure 4-35: Scenario 5b Full-size DP Adapter Wiring – Normal Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

DP Adapter-to-DP Sink Device

DP Source Device

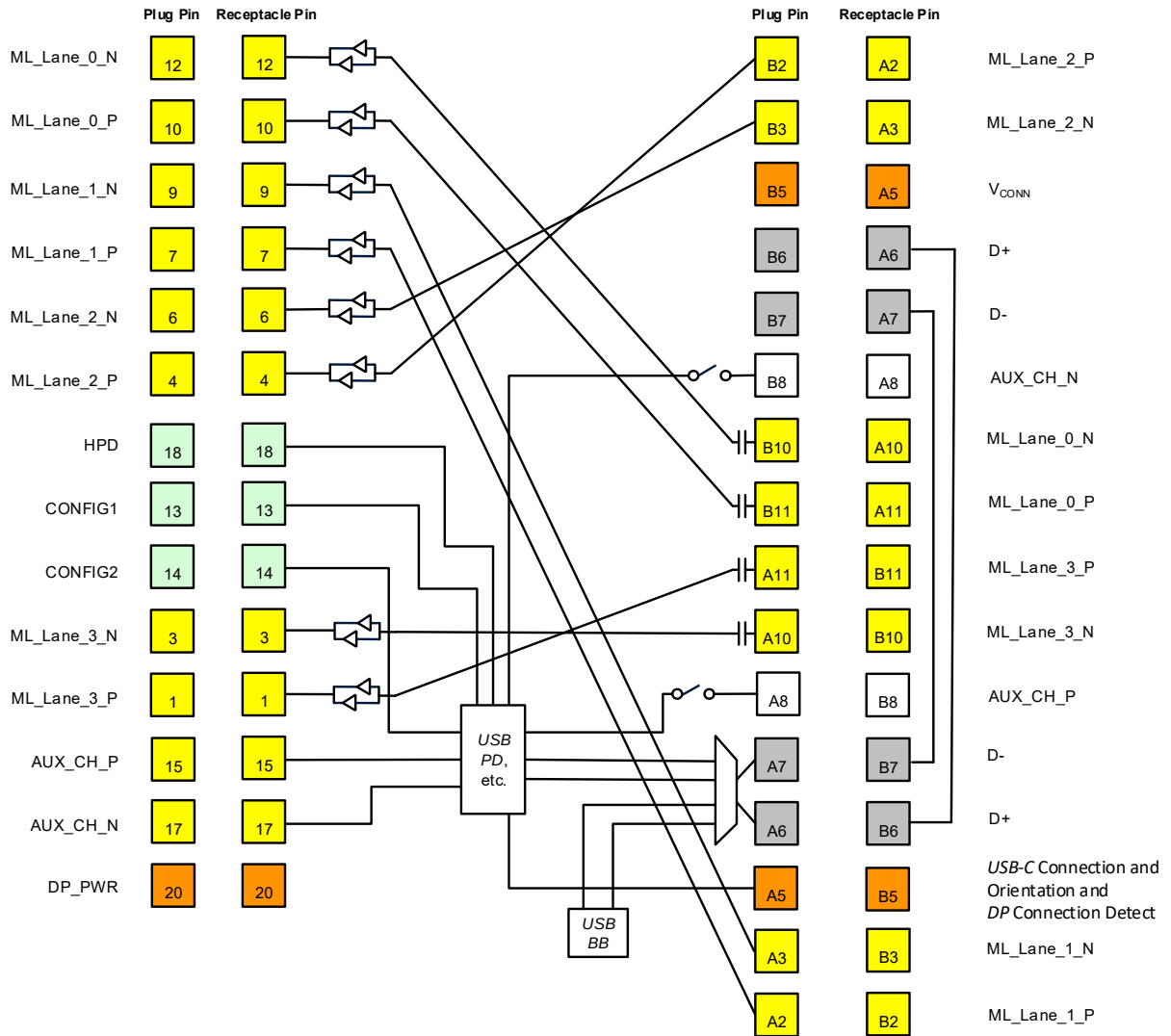


Figure 4-36: Scenario 5b Full-size DP Adapter Wiring – Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

mDP Adapter-to-mDP Sink Device

DP Source Device

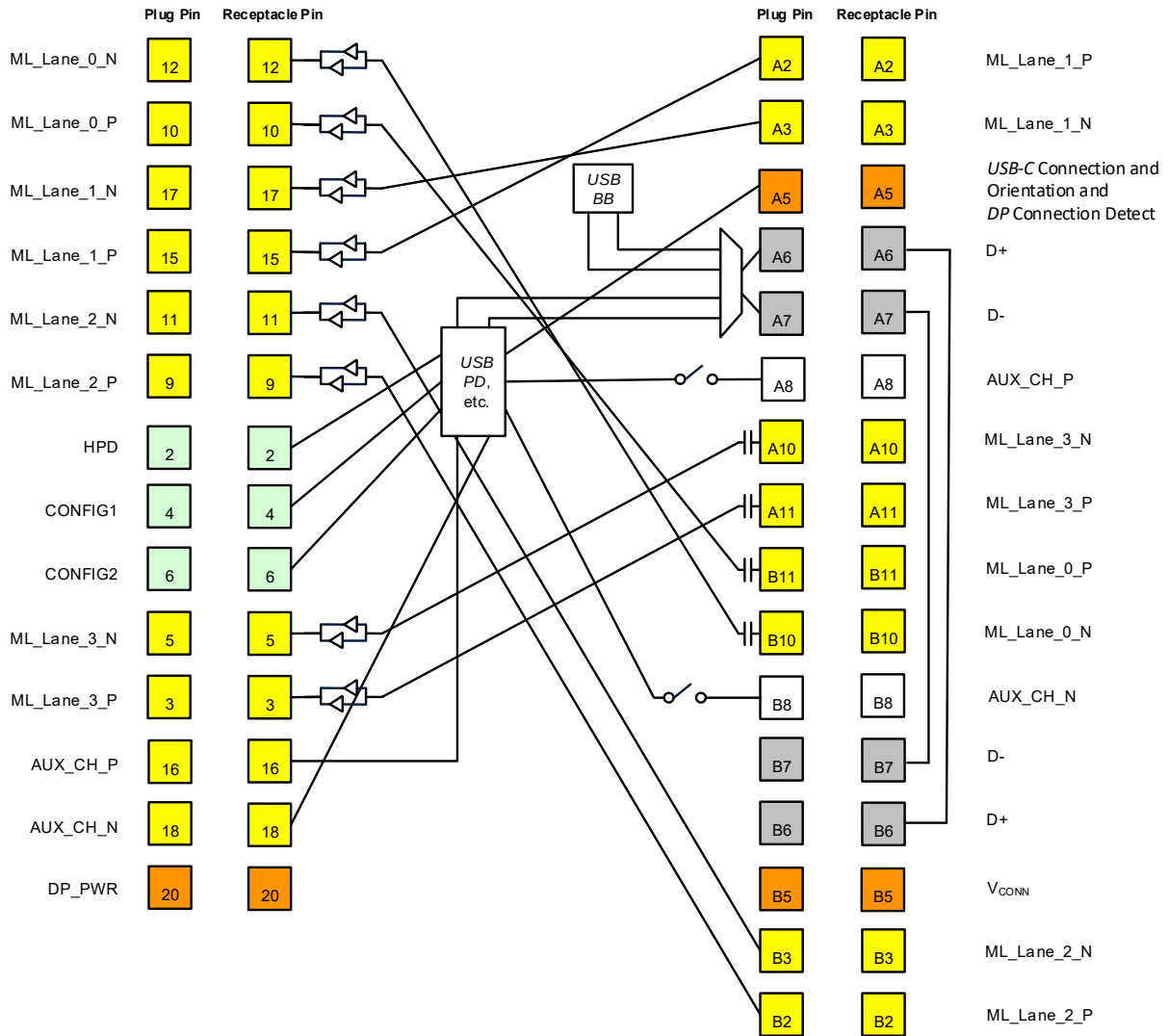


Figure 4-37: Scenario 5b mDP Adapter Wiring – Normal Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

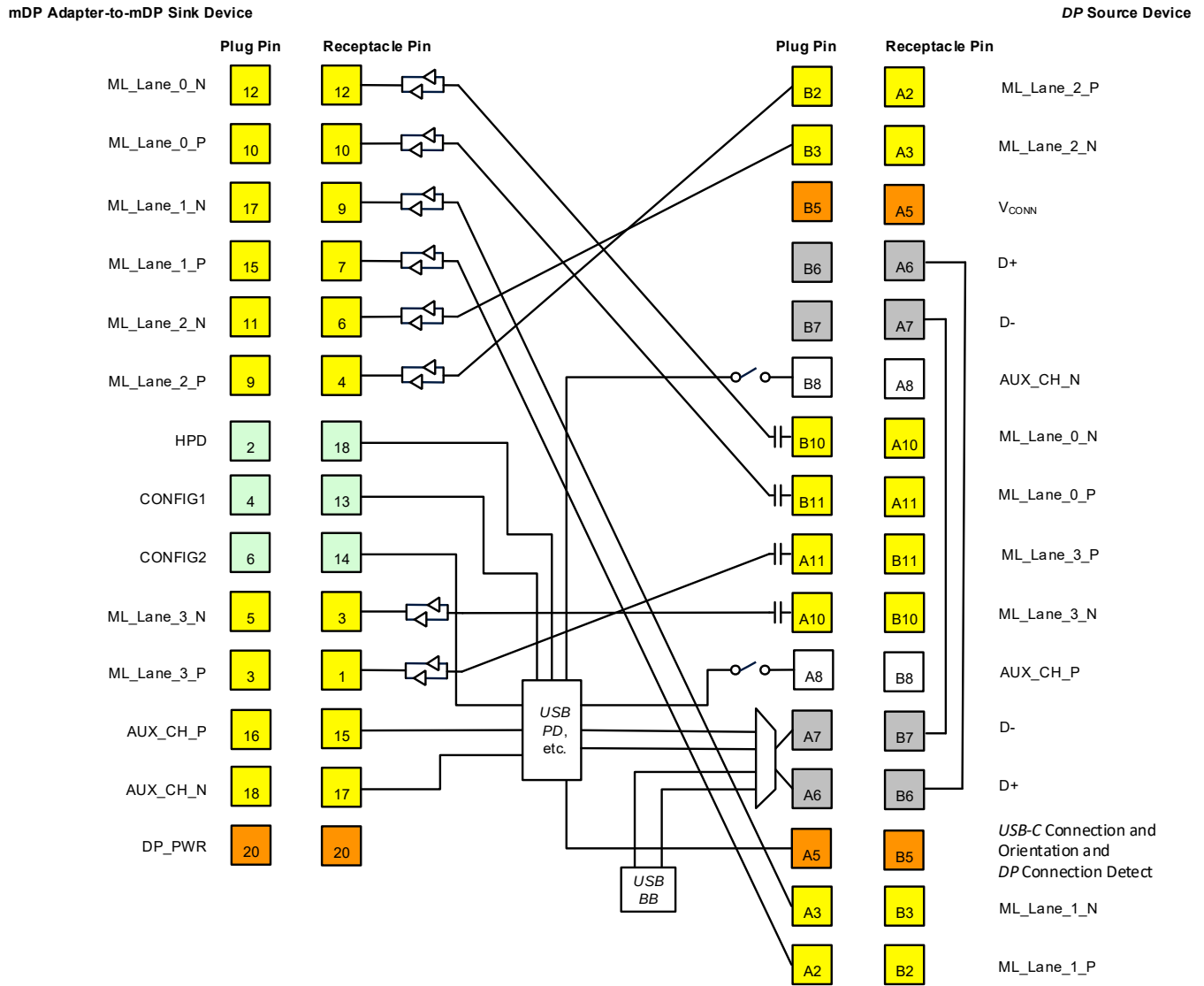


Figure 4-38: Scenario 5b mDP Adapter Wiring – Flipped Plug Orientation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 4-13 lists how *DisplayPort* uses the DP/mDP receptacle DP/mDP Sink device-to-USB-C plug DP Sink device adapter's signal connections.

Table 4-13: Scenario 5b – DP/mDP Sink Receptacle-to-USB-C DP Sink Adapter Use

DP/mDP Receptacle DP/mDP Sink Device			Wire		USB-C Plug DP Sink Device	
DP Pin	mDP Pin	DP Signal Name	Wire Number	Signal Name	Pin	DP Signal Name
2, 5, 8, 11, 16, 19	1, 7, 8, 13, 14, 19	GND	1 [16]	GND_PWRrt1 [GND_PWRrt2]	A1, B1, A12, B12	GND
			2	Powers <i>USB BB</i>	A4, B4, A9, B9	V _{BUS}
			3	Adapter logic (<i>USB PD</i>)	A5	CC
			4	Adapter logic (<i>USB BB</i>)	A6	D+
			5	Adapter logic (<i>USB BB</i>)	A7	D-
13	4	CONFIG1 ^a		1M to GND		
14	6	CONFIG2 ^a		1M to GND		
6	11	ML_Lane_2_P	6	SDPp1	A2	ML_Lane_1_N
4	9	ML_Lane_2_N	7	SDPn1	A3	ML_Lane_1_P
3	5	ML_Lane_3_P	8	SDPp2	B11	ML_Lane_0_N
1	3	ML_Lane_3_N	9	SDPn2	B10	ML_Lane_0_P
9	17	ML_Lane_1_P	10	SDPp3	B2	ML_Lane_2_N
7	15	ML_Lane_1_N	11	SDPn3	B3	ML_Lane_2_P
12	12	ML_Lane_0_P	12	SDPp4	A11	ML_Lane_3_N
10	10	ML_Lane_0_N	13	SDPn4	A10	ML_Lane_3_P
15	16	AUX_CH_P	14	Adapter logic	A8 ^b	AUX_CH_P
17	18	AUX_CH_N	15	Adapter logic	B8 ^b	AUX_CH_N
18	2	HPD		Adapter logic		
			16	Powers Adapter logic	B5	V _{CONN}
20	20	DP_PWR ^c		No Connect		
Shell	Shell	Shield	Braid	Shield	Shell	Shield

- a. CONFIG1 and CONFIG2 shall include 1-M Ω pull-down resistors.
 b. A8/B8 shall include the AUX_CH DP Sink device pull-up/pull-down resistors.
 c. DP_PWR shall **not** be provided unless a short is detected between CONFIG1 and CONFIG2.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.7.3 Scenarios 5a and 5b – USB-C Plug-to-DP/mDP Active Receptacle Adapter Directionality

The active receptacle adapter presents the AUX_CH terminations as a *DP* Source device that faces the *USB-C* plug and *DP* receptacle when entering DisplayPort Alt Mode. The adapter determines the *DP* Source and Sink device directions, and then configures its AUX_CH receivers to be ready to respond to the AUX transaction. The adapter shall transmit HPD_High only when its AUX_CH receivers are ready to respond to AUX transactions.

4.7.4 Scenarios 5a and 5b – Active Adapter USB Behavior

A UFP_U with a *USB-C* plug that contains *USB* and/or other functionality (e.g., the UFP_U is a Multi-function captive cable dock) shall power its *USB* and/or other functionality from V_{BUS} and **not** from V_{CONN} . If V_{BUS} is unavailable, the *USB* or other functionality may be disabled. DisplayPort Alt Mode functionality shall be supported, as long as V_{CONN} is available. An active adapter with *USB* and/or other functionality should advertise the needed operating power in the *USB PD* Sink RDO.

An active receptacle adapter shall comply with *DP Standard* for SLEEP power state entry/exit by snooping the AUX_CH traffic that is exchanged between the *DP* Source and Sink devices. The active receptacle adapter shall consume less than 70mW in the SLEEP power state.

4.7.5 Scenarios 5a and 5b – Active Receptacle Adapter Power

An active receptacle adapter shall operate at any voltage between 3.0 and 5.5V (as presented at the adapter's *USB-C* plug; there may be an IR drop on a V_{CONN} wire within the adapter's cable), and the adapter shall **not** consume in excess of 1.5W. **Optional** operation down to 2.7V ensures backward compatibility with earlier versions of *DP AM* and provides margin for voltage drop across V_{CONN} power switches, connectors, and adapters. *USB PD* communications may be maintained if the V_{CONN} voltage falls below 3.0V. Power consumption shall not exceed 70mW and *DP* adapter functionality may be disabled if V_{CONN} voltage falls below 2.7V. The adapter shall report if it continues to operate but disables *DP* functionality (see Table 5-6).

During the Discovery Process and whenever DisplayPort Alt Mode is **not** entered, the active receptacle adapter shall comply with *USB-C* V_{CONN} power consumption mandates.

Active receptacle adapters that support *DisplayPort* on a *USB-C* plug shall operate as V_{CONN} -powered accessories. These adapters are **not** mandated to provide power on V_{BUS} , and do **not** depend on the presence of power on V_{BUS} (except to power the presentation of a *USB BB* class, when needed).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.8 Active Cable Power Consumption

Active cable power consumption is defined as follows:

- Active cable shall function on V_{CONN} within the range of 3.0 to 5.5V, and **not** depend on V_{BUS} presence
- Active cable shall consume no more than 1.5W from V_{CONN}
- Active cable may be partially or fully self-powered (e.g., the cable end that is not receiving V_{CONN} may be self-powered)

4.8.1 Active Cable Sleep

Active cable sleep is defined as follows:

- Active cable shall comply with *DP Standard* for Sleep entry/exit by snooping the AUX_CH traffic that is exchanged between the *DP* Source and Sink devices
- Active cable shall meet the *DP Standard* power consumption mandate when configured for four *DP* lanes
- Active cable shall meet the *DP Standard* power consumption mandate when configured for Multi-function Mode with the *DP* and *USB* lanes in U3 Power Mode
- Sleep power is measured at least 500ms after V_{CONN} is applied and *USB PD* traffic has **not** been present for at least 500ms

4.8.2 Active Cable Suspend

Active cables shall program the Cable Termination Type field in the *USB PD Discover Identity* Command response to indicate that V_{CONN} is needed in the USB Suspend state.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.9

Active Cable Legacy USB-C Support Mandates

To support operation from legacy DFP_Us that do **not** generate the *DisplayPort* Messages that are defined in this section, an active cable shall snoop the *USB PD* DisplayPort Alt Mode Messages that are transmitted from the DFP_U to the UFP_U (addressed to SOP), as follows:

- If the active cable observes an **Enter Mode** Command request with the DP_SID addressed to SOP without having previously received an **Enter Mode** Command request addressed to SOP', the active cable shall obey the **Enter Mode** Command request as if the request is addressed to SOP', but without responding (including not issuing a *USB PD GoodCRC* Message)
- Active cable shall obey any Configure and **Exit Mode** Command requests that are addressed to SOP without having previously received similar requests addressed to SOP as if they were addressed to SOP', but without responding (including not issuing a *USB PD GoodCRC* Message)
- USB-C-to-USB-C active cable shall behave as if it has received an HPD high when the cable receives a **DisplayPort Configure** Command

A command that is addressed to SOP' shall take precedence over the same command addressed to SOP, regardless of the order in which the commands are received.

Note: *This snooping mandate does **not** support the normal robustness of *USB PD* communications; thus, this mandate is treated as a best-effort means of supporting legacy DFP_U systems.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.10 Methods for Determining DP Directionality in an Active Cable

4.10.1 AUX_CH Pull-up/Pull-down Resistor Method for Detecting DP Direction

The method defined in this section may be used for direction detection for USB-C-to-USB-C active cables and USB-C-to-DP cables. This method may be used after the DFP_U uses a **DisplayPort Configure** Command to configure the UFP_U as a **DP** Source or Sink device. The descriptions that follow define the cable's normal orientation. Flipped orientation is left to the user.

Both **USB-C** cable plugs should initially configure their AUX_CH_P and AUX_CH_N with the appropriate **DP** Sink device pin assignment, but present as a **DP** Source device for the AUX_CH common-mode termination (100-k Ω pull-down resistor on SBU2/AUX_CH_P, and 100-k Ω pull-up resistor on SBU1/AUX_CH_N).

If the cable plug is connected to a **DP** Source device, SBU1 and SBU2 shall be connected as follows:

- SBU1 shall be connected to AUX_CH_P in the **DP** Source device, which will have a 100-k Ω pull-down resistor
- SBU2 shall be connected to AUX_CH_N in the **DP** Source device, which will have a 100-k Ω pull-up resistor

As a result of the common-mode termination on AUX_CH_P and AUX_CH_N on the SBU pins, AUX_CH_P and AUX_CH_N will be pulled to approximately mid-rail by the resulting resistor divider.

If the cable plug is connected to a **DP** Sink device, AUX_CH_P and AUX_CH_N will be pulled to the appropriate levels as a result of the 10:1 resistor divider, as follows:

- Cable plug that detects that it is connected to a **DP** Source device shall configure its signal direction from the local plug to the remote plug, and may reconfigure its AUX_CH common-mode resistor pulls as a **DP** Sink device. The plug shall reconfigure itself for AUX_CH purposes as a **DP** Source device.
- Cable plug that detects that it is connected to a **DP** Sink device shall configure its signal direction from the remote plug to the local plug. The plug shall already be configured with the **DP** Sink device pin assignment for AUX_CH purposes.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

4.10.2

AUX Transaction Method for Detecting Direction

The method defined in this section may be used for direction detection for USB-C-to-USB-C active cables and USB-C-to-DP cables. This method may be used in active cables that terminate AUX_CH_P and AUX_CH_N from both plugs, such as for active cables that incorporate LTTPR functionality. This may be done after the DFP_U uses a **DisplayPort Configure** Command to configure the UFP_U as a **DP** Source or Sink device. The descriptions that follow define the cable's normal orientation. Flipped orientation is left to the user.

Both **USB-C** cable plugs should initially configure their AUX_CH_P and AUX_CH_N with the appropriate **DP** Sink device pin assignment, but present as a **DP** Source device for the AUX_CH common-mode termination (100-k Ω pull-down resistor on SBU2/AUX_CH_P, and 100-k Ω pull-up resistor on SBU1/AUX_CH_N), with no AUX_CH connection between the **USB-C** plugs.

If the cable plug is connected to a **DP** Source device, SBU1 and SBU2 shall be connected as follows:

- SBU1 shall be connected to AUX_CH_P in the **DP** Source device, which will have a 100-k Ω pull-down resistor
- SBU2 shall be connected to AUX_CH_N in the **DP** Source device, which will have a 100-k Ω pull-up resistor

As a result of the common-mode termination on AUX_CH_P and AUX_CH_N on the SBU pins, AUX_CH_P and AUX_CH_N will be pulled to approximately mid-rail by the resulting resistor divider.

If the cable plug is connected to a **DP** Sink device, AUX_CH_P and AUX_CH_N will be pulled to the appropriate levels as a result of the 10:1 resistor divider.

A DFP_U that is a **DP** Sink device shall transmit a **DisplayPort Status Update** Command with HPD State High after transmitting a **DisplayPort Configure** Command to configure the UFP_U as a **DP** Source device. A UFP_U that is a **DP** Sink device shall transmit a **USB PD Attention** Command request with a **DisplayPort Status Update** VDO with HPD State High **USB PD** Message after receiving a **DisplayPort Configure** Command to configure the UFP_U as a **DP** Sink device. The **DP** Source device shall transmit an AUX transaction after receiving the **DisplayPort Status Update** VDO with HPD State High. The plug that receives the first AUX transaction should configure itself for AUX_CH purposes as being connected to a **DP** Source device (AUX_CH_P and AUX_CH_N are inverted relative to the initial cable configuration). The plug that detects that it is connected to a **DP** Sink device should configure the signal direction from the remote plug to the local plug. The active cable is already configured with the **DP** Sink device pin assignment for AUX_CH purposes.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5 Discovery and USB PD

This section describes the mechanisms for discovery of *DisplayPort* support on a *USB-C* connector and the negotiation procedures for entering and exiting DisplayPort Alt Mode and establishing DisplayPort Configuration.

All descriptions of *USB PD* apart from DisplayPort Alt Mode-specific commands are informative. Reference should be made to *USB PD* for normative *USB PD* functionality.

5.1 Connection Discovery

If one or both of the connected ports are DRPs, as specified in *USB-C*, the DRP(s) resolve into one being a Source/ V_{CONN} Source, as specified in *USB-C*, and one being a Sink/UFP on first connection. [Table 5-1](#) defines the USB-C and DisplayPort Configurations.

This Standard does **not** mandate when power roles may be swapped or when the *USB PD* Explicit Power Contract may be renegotiated, beyond the mandates specified in *USB PD*.

A receptacle or plug that resolves as a DFP (possibly after a *USB PD DR_Swap* Message) is described as a DFP_U. A port that resolves as a UFP (possibly after a *USB PD DR_Swap* Message) is described as a UFP_U. The DFP_U or UFP_U role is static while DisplayPort Alt Mode is active.

If a *USB PD DR_Swap* Message is issued during DisplayPort Alt Mode discovery, the following shall occur:

- Current DFP, as specified in *USB-C*, shall abandon the Discovery Process, and
- New DFP, as specified in *USB-C*, shall initiate a new Discovery Process

The new DFP is described as a DFP_U. The new UFP is described as a UFP_U.

Note: *USB PD mandates that if there are any Active Modes between the port partners, a USB PD DR_Swap Message shall not be issued and a USB PD Hard Reset shall be performed if a USB PD DR_Swap Message is received.*

A receptacle that resolves as a DFP or UFP (possibly after a *USB PD DR_Swap* Message) shall follow the behavior in [Section 5.1.7](#) or [Section 5.1.8](#).

A plug that resolves as a DFP (possibly after a *USB PD DR_Swap* Message) shall follow the behavior defined in [Section 5.1.7](#) or [Section 5.1.8](#).

A Sink with Accessory Support, as specified in *USB-C*, that has detected a V_{CONN} -powered accessory is described as a DFP_U. The DFP_U shall be responsible for DisplayPort Alt Mode negotiation.

Note: *A Sink with Accessory Support, as specified in USB-C, is not mandated to operate as a USB host when the UFP uses Rp to pull up CC1/CC2 and discovers a DP-capable UFP.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.1.1 DisplayPort Alt Mode Entry

The DFP_U shall be in control of DisplayPort Alt Mode negotiation. Where possible, the *DP* Source device should also be the DFP_U (i.e., the *DP* Source device is in control of DisplayPort Alt Mode negotiation). However, there are some configurations in which the *DP* Sink device behaves as the DFP_U and controls the DisplayPort Alt Mode negotiation.

- Port shall follow *USB-C* for providing V_{CONN}
- Port that powers an active cable shall follow the mandates defined in [Section 5.3.1](#)
- **Port is a DFP_U and in DisplayPort Alt Mode** – Port shall exit the mode after receiving a *USB PD Attention* Command request with the [Exit DisplayPort Alt Mode Request](#) bit (bit 6) set to 1 in the *DisplayPort Status Update* (including first reconfiguring to USB Configuration)
- **Port is a DFP_U in DisplayPort Alt Mode and DisplayPort Configuration** – Port shall change to USB Configuration by issuing an appropriate *DisplayPort Configure* Command to both the UFP_U and cable after receiving a *USB PD Attention* Command request with the [DisplayPort/USB Configuration Request](#) bit (bit 5) set to 1 in the *DisplayPort Status Update*
- **Port is a DFP_U in DisplayPort Alt Mode and USB Configuration** – Port shall change to an appropriate DisplayPort Configuration by issuing an appropriate *DisplayPort Configure* Command to both the UFP_U and cable after receiving a *USB PD Attention* Command request with the [DisplayPort/USB Configuration Request](#) bit cleared to 0

A DisplayPort Alt Mode device shall implement one of the device types defined in [Table 5-1](#). The device-type Data Role policies and V_{CONN} policy options are outlined in [Section 5.1.2](#) and [Section 5.1.3](#), respectively. Details for each device type are defined in [Section 5.1.4](#) through [Section 5.1.9](#).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 5-1: USB-C and USB PD Power and Data Roles

Device Type				Device Configuration		
#	DP Device Type	USB-C Power Role	Toggles ^a	Supported USB PD Data Roles	USB PD Data Role Policy (before DisplayPort Alt Mode Entry) ^b	USB PD V _{CONN} Policy ^c
1	DP Source device with USB-C receptacle ^d	Source	No	DFP_U	DFP_U Only	Potential V _{CONN} Source
2				DFP_U and UFP_U	Prefer DFP_U	
3		Sink with Accessory Support ^e	Yes	DFP_U and UFP_U	Prefer DFP_U	
4				DFP_U and UFP_U	Prefer DFP_U	
5	DP Sink device with USB-C receptacle (USB-C-powered) ^f	Sink	No	UFP_U	UFP_U Only	Never V _{CONN} Source
6				DFP_U and UFP_U	Prefer UFP_U	Potential V _{CONN} Source
7		DRP Toggling (should use Try.SNK)	Yes	DFP_U and UFP_U	Prefer UFP_U	Potential V _{CONN} Source
8	DP Sink device with USB-C receptacle (unrestricted powered) ^g	DRP only	Yes	DFP_U and UFP_U	Prefer UFP_U	Potential V _{CONN} Source
9	DP Source device with USB-C plug ^h	DRP Toggling ⁱ	Yes	DFP_U and UFP_U	Prefer DFP_U	Potential Captive V _{CONN} Source
10	DP Sink device with USB-C plug ^j	Sink ^k	No	UFP_U	UFP_U Only	Never V _{CONN} Source
11				DFP_U and UFP_U	No Preference	Never V _{CONN} Source
12		DRP Toggling (should use Try.SNK) ⁱ	Yes	DFP_U and UFP_U	Prefer UFP_U	Potential Captive V _{CONN} Source
13	USB-C-to-DP receptacle adapter, cable adapter, –or– protocol adapter (USB-C powered) ^l	V _{CONN} -powered accessory (Rd/Ra)	No	UFP_U	UFP_U Only	Never V _{CONN} Source
14				DFP_U and UFP_U	Prefer UFP_U	

- a. See [USB-C](#) for toggle definition.
- b. See [Section 5.1.2](#) for policy definitions.
- c. See [Section 5.1.3](#) for policy definitions.
- d. See [Section 5.1.4](#).
- e. See [USB-C](#) for behavior definition.
- f. See [Section 5.1.5](#).
- g. See [Section 5.1.6](#).
- h. See [Section 5.1.7](#).
- i. Captive-cable DP Source or Sink device shall present Ra during the unattached.SNK state if V_{CONN} is needed.
- j. See [Section 5.1.8](#)
- k. Sink with a captive cable shall present either Rd/Zopen or Rd/Ra on the CC pins, as specified in [USB-C](#).
- l. See [Section 5.1.9](#).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.1.2 USB PD Data Role Policies

Table 5-2 lists the *USB PD* Data Role policy definitions.

Table 5-2: USB PD Data Role Policies

Policy	Definition
Prefer DFP_U	- Issue a <i>USB PD DR_Swap</i> Message to become a DFP_U USB Mode response – Transmit a <i>USB PD Accept</i> Message DisplayPort Alt Mode response – Issue a <i>USB PD Hard Reset</i> - Issue a <i>USB PD Reject</i> Message in response to the <i>USB PD DR_Swap</i> Message to become a UFP_U
Prefer UFP_U	- Issue a <i>USB PD DR_Swap</i> Message to become a UFP_U USB Mode response – Transmit a <i>USB PD Accept</i> Message DisplayPort Alt Mode response – Issue a <i>USB PD Hard Reset</i> - Issue a <i>USB PD Reject</i> Message in response to the <i>USB PD DR_Swap</i> Message to become a DFP_U Note: <i>If the port assignment results in a DFP_U role, the port is responsible for discovering and initiating Alternate Mode entry.</i>
DFP_U Only	- Do not issue a <i>USB PD DR_Swap</i> Message - Issue a <i>USB PD Reject</i> Message in response to all <i>USB PD DR_Swap</i> Message requests
UFP_U Only	- Do not issue a <i>USB PD DR_Swap</i> Message - Issue a <i>USB PD Reject</i> Message in response to all <i>USB PD DR_Swap</i> Message requests
No Preference	- Do not issue a <i>USB PD DR_Swap</i> Message USB Mode response – Transmit a <i>USB PD Accept</i> Message DisplayPort Alt Mode response – Issue a <i>USB PD Hard Reset</i>

The DFP_U shall discover and initiate DisplayPort Alt Mode entry after providing some time for Data Role swapping.

DisplayPort Alt Mode entry should **not** occur while a *USB PD Wait* Message is being issued in response to a *USB PD DR_Swap* Message. The DFP_U/UFP_U should retry a *USB PD DR_Swap* Message if a *USB PD Wait* Message is received. The *USB PD DR_Swap* Message retry should continue until either a *USB PD Accept* Message or *Reject* Message is received. If a *USB PD Wait* Message is received in response to a *USB PD DR_Swap* Message, and then a *USB PD DR_Swap* Message is received, the port should accept the *USB PD DR_Swap* Message and abandon its own *USB PD DR_Swap* Message request.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.1.3 USB PD V_{CONN} Policies

The DFP_U shall enter DisplayPort Alt Mode if the *USB PD Discover Identity* Command request returns a passive cable –or– has no response. If the *USB PD Discover Identity* Command response indicates that the cable is active, the DFP_U shall **not** enter DisplayPort Alt Mode if the active cable does not support DisplayPort Alt Mode. The V_{CONN} Source may disable sourcing V_{CONN} per *USB-C*.

Table 5-3 lists the *USB PD* V_{CONN} policy definitions.

Table 5-3: USB PD V_{CONN} Policies

Policy	Data Role	Definition
Potential V _{CONN} Source	DFP_U	- Issue a <i>USB PD VCONN_Swap</i> Message to become the V_{CONN} Source before discovering the cable and entering and configuring DisplayPort Alt Mode - Transmit a <i>USB PD Accept</i> Message in response to a <i>USB PD VCONN_Swap</i> Message if currently in the logical role of V_{CONN} Source - Transmit a <i>USB PD Accept</i> Message or <i>Wait</i> Message in response to a <i>USB PD VCONN_Swap</i> Message if not in the logical role of sourcing V_{CONN}
	UFP_U	- Transmit a <i>USB PD Accept</i> Message in response to a <i>USB PD VCONN_Swap</i> Message if currently in the logical role of sourcing V_{CONN} - Transmit a <i>USB PD Accept</i> Message or <i>Wait</i> Message in response to a <i>USB PD VCONN_Swap</i> Message if not in the logical role of sourcing V_{CONN}

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 5-3: USB PD V_{CONN} Policies (Continued)

Policy	Data Role	Definition
Potential Captive V _{CONN} Source	DFP_U	- Does not need to issue a USB PD VCONN_Swap Message to become the V_{CONN} Source before entering DisplayPort Alt Mode because the captive cable performance is known by design - Transmit a USB PD Accept Message or Not_Supported Message in response to a USB PD VCONN_Swap Message if currently in the logical role of sourcing V_{CONN} and responding with USB PD 3.0 - Transmit a USB PD Accept Message in response to a USB PD VCONN_Swap Message if currently in the logical role of sourcing V_{CONN} and responding with USB PD 2.0 - Transmit a USB PD Accept Message, Wait Message, Not_Supported Message, or Reject Message in response to a USB PD VCONN_Swap Message if not in the logical role of sourcing V_{CONN} and responding with USB PD 3.0 - Transmit a USB PD Accept Message, Wait Message, or Reject Message in response to a USB PD VCONN_Swap Message if not in the logical role of sourcing V_{CONN} and responding with USB PD 2.0
	UFP_U	- Transmit a USB PD Accept Message or Not_Supported Message in response to a USB PD VCONN_Swap Message if currently in the logical role of sourcing V_{CONN} and responding with USB PD 3.0 - Transmit a USB PD Accept Message in response to a USB PD VCONN_Swap Message if currently in the logical role of sourcing V_{CONN} and responding with USB PD 2.0 - Transmit a USB PD Accept Message, Wait Message, Not_Supported Message, or Reject Message in response to a USB PD VCONN_Swap Message if not in the logical role of sourcing V_{CONN} and responding with USB PD 3.0 - Transmit a USB PD Accept Message, Wait Message, or Reject Message in response to a USB PD VCONN_Swap Message if not in the logical role of sourcing V_{CONN} and responding with USB PD 2.0
Never V _{CONN} Source ^a	—	- Do not issue a USB PD VCONN_Swap Message to become the V_{CONN} Source - Transmit a USB PD Not_Supported Message in response to a USB PD VCONN_Swap Message if responding with USB PD 3.0 - Transmit a USB PD Accept Message in response to a USB PD VCONN_Swap Message if responding with USB PD 3.0 - Device Types 9 and 12, which make an initial connection as a DP Source device/DFP/V_{CONN} Source shall: - Transmit a USB PD Not_Supported Message in response to a USB PD VCONN_Swap Message if responding with USB PD 3.0 - Transmit a USB PD Accept Message in response to a USB PD VCONN_Swap Message if responding with USB PD 2.0

*a. The port does **not** support providing V_{CONN}.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.1.4 DP Source Device on a USB-C Receptacle

A *DP* Source device connected to a *USB-C* receptacle shall implement Device Type 1, 2, 3, –or– 4 (see Table 5-1 for definitions). The *USB PD* Data Role, Data Role Policy, and V_{CONN} Policy, and *USB-C* Power Role shall match the device type.

5.1.5 DP Sink Device on a USB-C-powered USB-C Receptacle

A *DP* Sink device is *USB-C* powered if it is receiving its operating power from the *USB-C* connector.

A *DP* Sink device connected to a *USB-C* receptacle shall implement Device Type 5, 6, –or– 7 (see Table 5-1 for definitions). The *USB PD* Data Role, Data Role Policy, and V_{CONN} Policy, and *USB-C* Power Role shall match the device type.

5.1.6 DP Sink Device on an Unrestricted Externally Powered USB-C Receptacle

A *DP* Sink device is unrestricted externally powered if the device is powered from an external power source (such as wall power) and not from the *USB-C* receptacle. An externally powered *DP* Sink device connected to a *USB-C* receptacle shall implement Device Type 8 (see Table 5-1 for definition). The *USB PD* Data Role, Data Role Policy, and V_{CONN} Policy, and *USB-C* Power Role shall match the device type.

5.1.7 DP Source Device on a USB-C Plug

A *DP* Source device that incorporates a *USB-C* plug shall implement Device Type 9 (see Table 5-1 for definition). The *USB PD* Data Role, Data Role Policy, and V_{CONN} Policy, and *USB-C* Power Role shall match the device type.

5.1.8 DP Sink Device on a USB-C Plug

A *DP* Sink device that incorporates a *USB-C* plug shall implement Device Type 10, 11, –or– 12 (see Table 5-1 for definitions). The *USB PD* Data Role, Data Role Policy, and V_{CONN} Policy, and *USB-C* Power Role shall match the device type.

5.1.9 USB-C-to-DP Receptacle Adapter, Cable Adapter, or Protocol Adapter

A protocol adapter is externally powered if the adapter is powered from an external power source (such as a battery or wall power) and **not** from the *USB-C* receptacle. A USB-C-to-DP receptacle adapter, cable adapter, or protocol adapter shall implement Device Type 13 –or– 14 (see Table 5-1 for definitions). The *USB PD* Data Role, Data Role Policy, V_{CONN} Policy, and *USB-C* Power Role shall match the device type.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.2 DisplayPort Alt Mode Discovery, Entry, and Exit

Discovery of support for *DisplayPort* and entry into/exit from DisplayPort Alt Mode shall be performed using *USB PD* Messages as described in the Discovery Process specified in *USB PD* and *USB-C*. These Messages shall be augmented for the specific case of *DisplayPort* as defined in this section.

The 16-bit *USB* Standard ID assigned by the USB-IF to *DisplayPort* (FF01h) is indicated in this section by the DP_SID.

Only a DFP_U that supports *DisplayPort* may behave as an Initiator for the *USB PD* Messages related to DisplayPort Alt Mode, apart from the *USB PD Attention* Command request, which may be issued only by the UFP_U.

5.2.1 Structured VDM Header

The Structured Vendor Defined Message (VDM) Header, as specified in *USB PD*, shall be used for the DP-specific commands (i.e., *DisplayPort Status Update* and *DisplayPort Configure* in conjunction with the DP_SID). *Table 5-4* is based on *USB PD*, *Table 6-25*, and includes updates related to the additional DP-specific commands.

Table 5-4: Structured VDM Header

Bit(s)	Description	Values
4:0	Command	0h = RESERVED, shall not be used. 1h = <i>USB PD Discover Identity</i> . 2h = <i>USB PD Discover SVIDs</i> . 3h = <i>USB PD Discover Modes</i> . 4h = <i>Enter Mode</i> . 5h = <i>Exit Mode</i> . 6h = <i>USB PD Attention</i> . 7h – Fh = RESERVED, shall not be used. 10h = <i>DisplayPort Status Update</i> . 11h = <i>DisplayPort Configure</i> . 12h – 1Fh = RESERVED for DP_SID use.
5	RESERVED	RESERVED (always 0).
7:6	Command Type Initiator shall clear to 00b.	00b = REQ (Request from Initiator Port). 01b = ACK (<i>USB PD</i> Responder ACK response). 10b = NAK (<i>USB PD</i> Responder NAK response). 11b = BUSY (<i>USB PD</i> Responder BUSY response).
10:8	Object Position	For <i>Enter Mode</i> Command requests/responses, <i>Exit Mode</i> Command requests/responses, and <i>USB PD Attention</i> Command requests: 000b = RESERVED. 001b – 110b = Index into the list of Vendor Defined Objects (VDOs) to identify the needed Mode VDO. 111b = Exit all Active Modes (equivalent of a power-on reset). Shall only be used with an <i>Exit Mode</i> Command request. For <i>USB PD Discover Identity</i> , <i>Discover SVIDs</i> , and <i>Discover Modes</i> Command requests/responses: 000b. 001b – 111b = RESERVED.
12:11	RESERVED	RESERVED (always 00b).
14:13	Structured VDM Version	Version number of the Structured VDM (not the <i>USB PD</i> version number). 00b = Version 1.0, shall not be used. 01b = Version 2.0. All other values are RESERVED.
15	VDM Type	1 = Structured VDM.
31:16	Standard or Vendor ID (SVID)	Base SID (for a <i>USB PD Discover SVIDs</i> Command request) or DP_SID, a 16-bit unsigned integer, assigned by the USB-IF.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

The UFP_U may support SVIDs other than the DP_SID, in which case the UFP_U may present them in an arbitrary order in its response. The DFP_U shall search all the SVIDs returned so as to discover the DP_SID.

The UFP_U may support a *DP* Source device, a *DP* Sink device, or both a *DP* Source and Sink device. An example of supporting both is a USB-C-to-DP adapter cable, which shall be bidirectional. When transmitting the Responder *USB PD Discover Modes* VDM, the UFP_U shall indicate its *DP* capabilities in the VDO, as defined in Table 5-5.

Future versions of this Standard may describe other modes associated with the DP_SID. Such modes shall be identified by having a non-zero value in bits 31:24 of the VDO. The DFP_U shall examine the list of modes returned until it finds 0s in bits 31:24 of the VDO and a non-zero value in bits 23:0 of the VDO (i.e., DP Capabilities). The DFP_U and UFP_U shall use the corresponding offset (indexed from 1) as the Object Position in the following commands:

- **Enter Mode**
- **DisplayPort Status Update**
- **DisplayPort Configure**
- *USB PD Attention*
- **Exit Mode**

Note: *Any value in the Object Position other than 0 or 7 indicates that the command shall be interpreted in the context of a Mode identified by the combination of the SVID and Object Position. If the value in the Object Position is **not** valid for DisplayPort Alt Mode or some other Mode, *USB PD* mandates that the Responder responds with a NAK.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 5-5: DP Capabilities (VDO in the Responder USB PD *Discover Modes* VDM)

Bit(s)	Description	Values ^a
1:0	Port Capability	00b = RESERVED. 01b = <i>DP</i> Sink device-capable (including <i>DP</i> Branch device). 10b = <i>DP</i> Source device-capable (including <i>DP</i> Branch device). 11b = Both <i>DP</i> Source and Sink device-capable.
5:2	Signaling for Transport of DisplayPort Protocol	XXX1b = Supports <i>DP</i> bit rates and electrical settings (shall always be set apart from diagnostic purposes). XX1Xb = RESERVED. X1XXb = RESERVED. 1XXXb = RESERVED.
6	Receptacle Indication	0 = <i>DP</i> interface is presented on a <i>USB-C</i> plug. 1 = <i>DP</i> interface is presented on a <i>USB-C</i> receptacle.
7	USB 2.0 Signaling Not Used	0 = <i>USB 2.0</i> may be needed on A6 – A7 –or– B6 – B7 while in DisplayPort Configuration. 1 = <i>USB 2.0</i> is not needed on A6 – A7 –or– B6 – B7 while in DisplayPort Configuration.
15:8	DP Source Device Pin Assignments Supported (reported by a <i>DP</i> Source device receptacle or <i>DP</i> Sink device (direct-attach) plug)	00000000b = <i>DP</i> Source device pin assignments are not supported. XXXXXXXX1b = RESERVED. XXXXXXXX1Xb = RESERVED. XXXXX1XXb = Pin Assignment C is supported. ^b XXXX1XXXb = Pin Assignment D is supported. ^{c d} XXX1XXXXb = Pin Assignment E is supported. ^e XX1XXXXXb = RESERVED. X1XXXXXXb = RESERVED. 1XXXXXXXb = RESERVED.
23:16	DP Sink Device Pin Assignments Supported (reported by a <i>DP</i> Sink device receptacle or <i>DP</i> Source device (direct-attach) plug)	00000000b = <i>DP</i> Sink device pin assignments are not supported. XXXXXXXX1b = RESERVED. XXXXXXXX1Xb = RESERVED. XXXXX1XXb = Pin Assignment C is supported. ^f XXXX1XXXb = Pin Assignment D is supported. ^{c g} XXX1XXXXb = Pin Assignment E is supported. ^h XX1XXXXXb = RESERVED. X1XXXXXXb = RESERVED. 1XXXXXXXb = RESERVED.
31:24	RESERVED	RESERVED (always 00h).

a. “X” value indicates “Don’t Care.”

b. See [Table 3-10](#) and [Table 3-11](#) for *DP* Source device Pin Assignment C.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

- c. A system supporting **DisplayPort** on a **USB-C** connector that supports **Multi-function Mode** (i.e., supports the simultaneous **USB SuperSpeed** and **DisplayPort**) shall offer both the multi-function pin assignment and associated **DP-only** pin assignment. The associated **DP-only** pin assignment is identified by the previous letter (e.g., Pin Assignment D for multi-function (**USB-C** and **DisplayPort**) and Pin Assignment C for **DisplayPort** only). The system shall offer both pin assignments, even if the system supports only one or two **DP** lanes.
- d. See [Table 3-12](#) and [Table 3-13](#) for **DP** Source device Pin Assignment D.
- e. See [Table 3-14](#) and [Table 3-15](#) for **DP** Source device Pin Assignment E.
- f. See [Table 3-16](#) and [Table 3-17](#) for **DP** Sink device Pin Assignment C.
- g. See [Table 3-18](#) and [Table 3-19](#) for **DP** Sink device Pin Assignment D.
- h. See [Table 3-20](#) and [Table 3-21](#) for **DP** Sink device Pin Assignment E.

The DFP_U shall ensure that the UFP_U's capabilities are consistent with its own capabilities, taking care to note that a UFP_U that uses a **USB-C** plug describes the pin assignments supported for the corresponding receptacle (i.e., a **DP** Sink device will describe the **DP** Source device pin assignments to which it connects), whereas a UFP_U that uses a **USB-C** receptacle describes its pin assignments directly (i.e., a **DP** Sink device will describe its own **DP** Sink device pin assignments). In general, the DFP_U will select a pin assignment for its **USB-C** receptacle (C, D, –or– E) and configure the Port Partner with the same pin assignment (same identifying letter).

An adapter may need more than the USB-C-specified level of power consumption on V_{CONN} to fully power its **DP** functionality. The DFP_U shall ensure that the V_{CONN} power defined in this Standard is available before issuing an **Enter Mode** Command request addressed to the DP_SID. The adapter shall **not** exceed the USB-C-specified level of power consumption on V_{CONN} before receiving an **Enter Mode** Command request addressed to the DP_SID.

5.2.2 DisplayPort Enter Mode Command

The **Enter Mode/Exit Mode** Command pair addressed to the DP_SID behaves as brackets for the various other DisplayPort Alt Mode-specific commands, which may only be issued while DisplayPort Alt Mode is active.

After receiving the **USB PD** Responder ACK response to the **Enter Mode** Command request, the DFP_U proceeds to the next step toward configuring the port by transmitting a **DisplayPort Status Update** Command to the UFP_U. If the UFP_U responds to the **Enter Mode** VDM with a **USB PD** Responder NAK response, DisplayPort Alt Mode is **not** entered and the UFP_U shall expose a **USB BB** device. The UFP_U is **not** allowed to transmit a **USB PD** Responder BUSY response.

After DisplayPort Alt Mode is entered, the DFP_U and UFP_U shall remain in that mode until one of the following three events occur:

- **Exit Mode** Command request is successful
- **USB PD Hard Reset**
- Device is disconnected

An adapter may consume power from V_{CONN} up to the limit defined in [Section 6.1](#) after receiving an **Enter Mode** Command request addressed to the DP_SID and entering DisplayPort Alt Mode.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A UFP_U may transmit a *USB PD Attention* Command request after entering DisplayPort Alt Mode, and shall do so as soon as possible after its status changes. When operating in *USB PD 2.0* Mode, *USB PD 2.0, Section 6.5.16*, imposes restrictions on the rate at which *USB PD Attention* Command requests are allowed to be generated. When operating in *USB PD 3.0* Mode, *USB PD* does not place restrictions on the rate at which *USB PD Attention* Command requests are allowed to be generated. However, the DFP_U might be using an internal interface (such as one compliant with *USB-C*) that has limited bandwidth and buffering. Consequently, the DFP_U might be unable to accept more than two closely spaced *USB PD Attention* Command requests. The UFP_U shall ensure a 10-ms gap before transmitting a third *USB PD Attention* Command request after transmitting two *USB PD Attention* Command requests spaced less than 10ms apart.

5.2.3 DisplayPort Status Update Command

The DFP_U may transmit a *DisplayPort Status Update* Command at any time, and will generally do so immediately after receiving the *USB PD* Responder ACK response to a *DisplayPort Enter Mode* Command request. The DFP_U provides its latest *DisplayPort Status Update* in the VDO. The UFP_U responds by transmitting its *DisplayPort Status Update* in the VDO. The UFP_U is **not** allowed to respond with a *USB PD* Responder NAK response. There is no mandate for the UFP_U to carry out any specific behavior based on the information that it receives concerning the DFP_U's status, and any use that it makes of this information is implementation-specific. However, this information may be useful for diagnostic purposes.

A DFP_U or UFP_U that integrates a *DP* Source or Sink device shall permanently set the appropriate connected status flag to 1. A DFP_U or UFP_U that integrates both a *DP* Source and Sink device may do one of the following:

- Permanently set both of the appropriate connected status flags to 1, –or–
- Indicate its current connected state by setting either the *DP Source/Sink Device Connected* status flag to 1

To resolve the *DP* signal direction when two systems that are capable of behaving as both *DP* Source and Sink devices are connected to one another, a vendor-dependent method of disabling one or the other function shall be provided. After one of the two functions is disabled, the respective system shall issue an updated *DisplayPort Status Update* Command with only the *DP Source/Sink Device Connected* bit set to 1, as appropriate.

Table 5-6 lists the **DisplayPort Status Update** as transmitted in the VDO.

Table 5-6: DisplayPort Status Update

Bit(s)	Description	Values
1:0	DP Source/Sink Device Connected	00b = Neither a DP Source device nor DP Sink device is connected, –or– the adapter is disabled. 01b = DP Source device is connected. 10b = DP Sink device is connected. ^a 11b = Both a DP Source and Sink device are connected.
2 ^b	Power Low	0 = Adapter is functioning normally or is disabled. 1 = Adapter has detected low power and disabled DP support.
3 ^b	Enabled	0 = Adapter DP functionality is disabled. 1 = Adapter DP functionality is enabled and operational.
4 ^b	Multi-function Preferred	0 = No preference for multi-function. 1 = Multi-function is preferred.
5 ^b	DisplayPort/USB Configuration Request	0 = Request change to DisplayPort Configuration (if currently in USB Configuration). 1 = Request change to USB Configuration (if currently in DisplayPort Configuration).
6 ^b	Exit DisplayPort Alt Mode Request	0 = Maintain the current mode. 1 = Request exit from DisplayPort Alt Mode (if currently in DisplayPort Alt Mode).
7 ^c	HPD State	0 = HPD_Low. 1 = HPD_High. ^d
8 ^c	IRQ_HPDP	0 = IRQ_HPDP has not been issued since the last status Message. 1 = IRQ_HPDP. ^e
31:9	RESERVED	RESERVED (always 0000000h).

- An adapter that contains a **DP** Branch device with its DPRX connected to the **USB-C** plug shall report that a **DP** Sink device is connected, regardless of whether the **DP** Branch device has a **DP** Source device connected.
- These status bits apply only to DisplayPort Status that is transmitted by a UFP_U to a DFP_U. When transmitting DisplayPort Status, the DFP_U shall clear these bits to 0. When receiving DisplayPort Status, the UFP_U shall ignore these bits.
- These status bits apply only to DisplayPort Status that is transmitted by a **DP** Sink device to a **DP** Source device. When transmitting DisplayPort Status, the **DP** Source device shall clear these bits to 0. When receiving DisplayPort Status, the **DP** Sink device shall ignore these bits.
- HPD state is reported after glitch filtering, IRQ_HPDP filtering (HPD state is maintained high when reporting an IRQ_HPDP), and de-bouncing, when applied.
- IRQ_HPDP may be generated only when the HPD state is HPD_High.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.2.4 DisplayPort Configure Command

The change to use DisplayPort Alt Mode is controlled by using the **DisplayPort Configure** Command with the VDO containing the details of the requested configuration. A DFP_U may transmit a **DisplayPort Configure** Command at any time while in DisplayPort Alt Mode. Before issuing the command, the DFP_U shall place the **USB-C** pins that are to be reconfigured to DisplayPort Configuration into the Safe state, as specified in **USB-C** (i.e., ensure that there is no **USB** on these pins). The UFP_U may enable DisplayPort Alt Mode immediately after receiving this command. The UFP_U shall respond to this command with a **USB PD** Responder ACK response after ensuring that the reconfigured pins are connected to **DisplayPort**.

Note: *The UFP_U may directly connect the reconfigured pins to the appropriate **DP** functionality (e.g., to pins on a **DP** connector, or to a **DP** protocol converter). In this case, a re-connection is **not** needed and the **USB PD** Responder ACK response may be immediately issued. After receiving the **USB PD** Responder ACK response to this command, the DFP_U shall change the reconfigured pins on its **USB-C** receptacle from the Safe state to **DisplayPort**, and then connect the SBU pins. If the DFP_U is a **DP** Source device, the HPD state shall be indicated to the DPTX. If the DFP_U is a **DP** Sink device, the HPD state shall be transmitted within a **DisplayPort Status Update** Command.*

The DFP_U may change the number of **DP** lanes that are connected as a result of reconfiguration. In this case, as well as placing any additional **USB-C** pins into the Safe state before transmitting the **DisplayPort Configure** Command, the DFP_U shall ensure that the HPD state is taken low and then isolate the SBU pins. The DFP_U shall not re-connect the SBU pins until after receiving the **USB PD** Responder ACK response and then re-connecting the reconfigured high-speed pins. In addition, when the DFP_U is a **DP** Source device, the DFP_U shall ensure that HPD_Low is maintained for a minimum of 3ms to ensure that its DPTX (e.g., GPU) sees an HPD disconnect event (and **not** an IRQ_HPDP).

The DFP_U may change to USB Configuration by using the **DisplayPort Configure** Command with the Select Configuration field programmed to **USB**. Before issuing the command, the DFP_U shall ensure that the HPD state is low, and then isolate the **USB-C** pins that were previously reconfigured to DisplayPort Configuration, including the SBU pins. When the DFP_U is a **DP** Source device, the DFP_U shall ensure that HPD_Low is maintained for a minimum of 3ms to ensure that its DPTX (e.g., GPU) sees an HPD disconnect event (and not an IRQ_HPDP). The UFP_U may start to use **USB** immediately after receiving this command. The UFP_U shall respond to this command with a **USB PD** Responder ACK response after connecting the reconfigured pins to **USB**, where **USB** functionality on these pins is supported. After receiving the **USB PD** Responder ACK response to this command, the DFP_U shall change the previously reconfigured pins on the **USB-C** receptacle to **USB**.

The DFP_U shall take into account the UFP_U's capabilities, together with the limitations of the cable connecting the two systems (in the case of a USB-C-to-USB-C cable), when determining the configuration of **DisplayPort** on the interface and select a configuration that is mutually supported, as per [Table 5-7](#).

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

The following considerations determine the mandates listed in [Table 5-7](#):

- DFP_U is always receptacle-based, is always capable of supporting Pin Assignments C and E, and is possibly capable of supporting Pin Assignment D.
- UFP_U that is receptacle-based is always capable of supporting Pin Assignments C and E, and is possibly capable of supporting Pin Assignment D.
- UFP_U that is implemented using a [USB-C](#) plug and terminates in a [DP](#) (or mDP) plug or receptacle with just wires or a signal-conditioning repeater only offers Pin Assignment E support.
- UFP_U that is implemented using a [USB-C](#) plug and contains a [DP](#) Branch device (e.g., a protocol converter) offers Pin Assignment C and possibly Pin Assignment D support.
- If the connection is made using a USB-C-to-USB-C cable, the DFP_U receives an indication of the (receptacle-based) UFP_U supporting Pin Assignments C and E and possibly supporting Pin Assignment D. However, Pin Assignment E is incompatible with the pin connections inside a USB-C-to-USB-C cable assembly; therefore, the DFP_U selects Pin Assignment C or possibly Pin Assignment D.
- DFP_U never selects Pin Assignment E when Pin Assignment C and possibly Pin Assignment D are offered by the UFP_U.

[Table 5-7](#) lists the various pin assignments that are offered by the various possible configurations, and the DFP_U-selected pin assignment in each case.

Table 5-7: DFP_U Pin Assignment Selection Mandates^a

DFP_U Pin Assignments Supported	Cable/Direct-attach	Peer Receptacle/Plug	UFP_U Assignment(s) Supported (Capabilities Response to USB PD <i>Discover Modes</i>)	Peer Multi-function Preferred	Configured Pin Assignment
C, D, E	USB-C-to-USB-C Cable	Receptacle	C, D, E	N	C or D
C, D, E			C, D, E	Y	C or D
C, E			C, D, E	N	C
C, E			C, D, E	Y	C
C, D, E			C, E	N/A	C
C, E			C, E	N/A	C
C, D, E	Direct-attach	Plug	C, D	N	C or D
C, D, E			C, D	Y	C or D
C, E			C, D	N	C
C, E			C, D	Y	C
C, D, E			C	N/A	C
C, E			C	N/A	C
C, D, E	USB-C-to-DP Cable/Receptacle		E	N/A	E
C, E			E	N/A	E

a. See [Table 3-10](#) through [Table 3-21](#) for DP Source and Sink device pin assignments.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table 5-8 lists the DisplayPort Configurations. The DFP_U shall select one of the supported bit rates and configurations and one pin assignment, unless the DFP_U is changing back to USB Configuration, in which case the pin assignment shall be cleared to 0. The Configuration shall indicate the UFP_U's *DisplayPort* role (either *DP* Source or Sink device), independently of whether the UFP_U is implemented with a receptacle or plug. The UFP shall select the corresponding *DP* Sink device pin assignment if it is a *DP* Sink device implemented on a receptacle –or– *DP* Source device implemented on a plug, and shall select the corresponding *DP* Source device pin assignment if it is a *DP* Source device implemented on a receptacle –or– *DP* Sink device implemented on a plug.

Notes: *Setting of appropriate multiplexers to support DisplayPort is associated with the DisplayPort Configure Command.*

It is not necessary for the DFP_U to exit DisplayPort Alt Mode to change the DisplayPort Configuration settings on the interface.

Table 5-8: DisplayPort Configurations

Bit(s)	Description	Values
1:0	Select Configuration	00b = Set configuration for <i>USB</i> . ^a 01b = Set configuration for UFP_U as a <i>DP</i> Source device. ^b 10b = Set configuration for UFP_U as a <i>DP</i> Sink device. ^b 11b = RESERVED.
5:2	Signaling for Transport of DisplayPort Protocol	0h = Bit rate is unspecified (used only when the <i>Select Configuration</i> field is programmed for USB Configuration). 1h = Select <i>DP</i> bit rates and electrical settings. All other values are RESERVED.
7:6	RESERVED	RESERVED (always 00b).
15:8	Configure UFP_U Pin Assignment	00h = De-select pin assignment. 04h = Select Pin Assignment C. ^c 08h = Select Pin Assignment D. ^d 10h = Select Pin Assignment E. ^e All other values are RESERVED.
31:16	RESERVED	RESERVED (always 0000h).

- a. *Selects USB Configuration.*
- b. *Selects DisplayPort Configuration.*
- c. *See Table 3-10 and Table 3-11 for DP Source device Pin Assignment C.*
See Table 3-16 and Table 3-17 for DP Sink device Pin Assignment C.
- d. *See Table 3-12 and Table 3-13 for DP Source device Pin Assignment D.*
See Table 3-18 and Table 3-19 for DP Sink device Pin Assignment D.
- e. *See Table 3-14 and Table 3-15 for DP Source device Pin Assignment E.*
See Table 3-20 and Table 3-21 for DP Sink device Pin Assignment E.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A UFP_U supporting DisplayPort Alt Mode that supports Multi-function Mode (i.e., supports the simultaneous [USB SuperSpeed](#) and [DisplayPort](#)) may be instructed by the DFP_U to configure the related DisplayPort-only pin assignment (i.e., Pin Assignment C or E; see [Table 5-7](#)). If the UFP_U supports only one or two [DP](#) lanes, the UFP_U shall ensure, when instructed to do so, that the channels normally associated with [USB SuperSpeed](#) that are not being used in this pin assignment are in the Safe state (as specified in [USB-C](#)).

The UFP_U shall respond with a [USB PD](#) Responder NAK response if the UFP_U is unable to change to the requested Configuration.

Note: This [USB PD](#) Responder NAK response indicates an error in the DFP_U.

5.2.5 USB PD Attention Command Request

A UFP_U uses the [USB PD Attention](#) VDM to signal to the DFP_U that there has been some change in the UFP_U's status since last reported. The UFP_U shall report its updated status in the VDO following the header, using the format defined in [Table 5-6](#). See [Section 5.2.3](#) for full details of the use of status reporting.

5.2.6 DisplayPort Exit Mode Command

A UFP_U uses the **Exit Mode** Command request to instruct the UFP_U to exit DisplayPort Alt Mode. The DFP_U shall issue an **Exit Mode** Command request only when the port is configured to be in USB Configuration. If the UFP_U receives an **Exit Mode** Command request when in DisplayPort Configuration, the UFP_U shall change to USB Configuration and then continue with the other actions that the UFP_U performs after receiving an **Exit Mode** Command request.

Note: Receipt of an **Exit Mode** Command request while not configured in USB Configuration indicates an error in the DFP_U.

The UFP_U that receives the **Exit Mode** VDM shall return to V_{CONN} default power consumption, and exit the mode, thus returning to normal [USB](#) operation. The UFP_U shall then transmit [USB PD](#) Responder ACK response. The UFP_U is **not** allowed to transmit a [USB PD](#) Responder NAK or Responder BUSY response.

After the DFP_U receives the [USB PD](#) Responder ACK response, the DFP_U exits into the state that the DFP_U enters on a new connection.

Note: The **Exit Mode** Command request may be addressed directly to the *DP_SID* plus the Object Position that identifies the mode, –or– to the *DP_SID* plus Object Position 111b (Exit All Modes).

5.3 Active Cable Support

All DFP_Us shall support active cables and use DisplayPort Alt Mode commands, as defined in [Section 4.2.1.3](#).

The following provides an overview of the general DisplayPort Alt Mode Discovery and Entry Processes:

- 1 [USB-C](#) CC Connection State Machines resolve power source/power sink and the initial data roles (DFP_U/UEP_U).
- 2 Initial V_{BUS} and V_{CONN} power is supplied.
- 3 The DFP_U shall attempt to discover the presence of an active cable before establishing a [USB PD](#) Explicit Power Contract by transmitting a [USB PD Discover Identity](#) Command request addressed to SOP'.
 - a **Active cable is discovered, –or– no response to SOP' USB PD Discover Identity** – DFP_U shall continue to step 4 ([USB PD](#) Explicit Power Contract).
 - b **Passive cable is discovered** – DFP_U shall continue as described in [Section 5.4](#).
- 4 DFP_U shall negotiate a [USB PD](#) Explicit Power Contract.
 - a **USB PD DR_Swap Message is needed** – Swap occurs, and then the new DFP_U shall start over at step 3.

Note: *The DFP_U and UEP_U may complete a [USB PD PR_Swap](#) Message anytime after the [USB PD](#) Explicit Power Contract is established.*

- 5 DFP_U shall transmit a [USB PD Discover Identity](#) Command request addressed to SOP'.
 - a **No response is received, USB PD Responder NAK response is received, –or– USB PD Responder ACK response is received that indicates modal operation is not supported** – DFP_U shall terminate the DisplayPort Alt Mode Discovery and Entry Processes.
 - b **USB PD Responder ACK response is received that indicates modal operation is supported** – DFP_U shall continue to step 6.
- 6 If a cable response to the [USB PD Discover Identity](#) Command request addressed to SOP' did **not** occur in step 3, the request shall be repeated and the actions below shall be taken.

Note: *See [USB PD 3.0](#) (or higher) regarding the maximum number of times that an SOP' [USB PD Discover Identity](#) can be transmitted, and the maximum time allocated for entering Alternate Modes.*

- a **No response** – DFP_U shall continue as described in [Section 5.4](#).
- b **USB PD Discover Identity VDO Product Type reports as a passive cable** – DFP_U shall continue as described in [Section 5.4](#).
- c **USB PD Discover Identity VDO Product Type reports as an active cable** – DFP_U shall continue to step 7.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Notes: *If a new **USB PD Explicit Power Contract** is needed as a result of discovering cable properties (e.g., increased current or no power), the DFP_U negotiates the Contract and continues the DisplayPort Alt Mode Discovery and Entry Processes.*

*If the cable plug implements **USB PD 2.0**, the SSTX1, SSTX2, SSRX1, and SSRX2 Directionality Support bits in the **USB PD Discover Identity VDO** shall be ignored, and use of DisplayPort Alt Mode will **not** be dependent on the setting of these bits.*

- 7 **USB PD Discover SVIDs** Command request addressed to SOP.
 - a **UFP_U does not support the DisplayPort SVID** – Terminate the DisplayPort Alt Mode Discovery and Entry Processes (this is no change from the passive mode cable case).
- 8 **USB PD Discover Modes** Command request addressed to SOP.
 - a **UFP_U-reported DP Capabilities do not match the DFP_U-supported capabilities** – Terminate the DisplayPort Alt Mode Discovery and Entry Processes (this is no change from the passive mode cable case).
- 9 **USB PD Discover SVIDs** Command request addressed to SOP'.
 - a **Active cable does not respond –or– responds with a NAK** – Terminate the DisplayPort Alt Mode Discovery and Entry Processes.
 - b **Active cable responds, but does not support the DisplayPort SVID** – Terminate the DisplayPort Alt Mode Discovery and Entry Processes.
- 10 **USB PD Discover Modes** Command request addressed to SOP'.
 - a **Active cable-reported DP capabilities do not match DFP_U-supported capabilities** – Terminate the DisplayPort Alt Mode process.
- 11 **Enter Mode** Command request addressed to SOP'.
- 12 **Optional – DisplayPort Status Update** Command request addressed to SOP'.
- 13 **Active cable VDO reports SOP" as being supported** – **Enter Mode** Command request addressed to SOP". Active cable **USB PD Discover Identity VDO** is reported.
- 14 **Optional, active cable VDO reports SOP" as being supported** – **DisplayPort Status Update** Command request addressed to SOP".
- 15 **Enter Mode** Command request addressed to SOP.
- 16 **DisplayPort Status Update** VDO is exchanged between the DFP_U and UFP_U.
- 17 Place **USB** SuperSpeed in the Safe state, as appropriate.
- 18 **DisplayPort Configure** Command request addressed to SOP'.
- 19 **Active cable VDO reports SOP" as being supported** – **DisplayPort Configure** Command request addressed to SOP".
- 20 **DisplayPort Configure** Command request addressed to SOP.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

The DFP_U shall use the following sequence of operation when exiting a DisplayPort Alt Mode connection by way of an active cable:

- 1 **DisplayPort Configure** Command request addressed to SOP to stop SOP from engaging in *DisplayPort*.
- 2 **DisplayPort Configure** Command request addressed to SOP' to stop SOP from engaging in *DisplayPort*.
- 3 **Active cable VDO reports SOP'' as being supported – DisplayPort Configure** Command request addressed to SOP'' to stop SOP from engaging in *DisplayPort*.
- 4 Remove *USB* SuperSpeed from the Safe state, as appropriate.
- 5 **Exit Mode** Command request addressed to SOP.
- 6 **Active cable VDO reports SOP'' as being supported – Exit Mode** Command request addressed to SOP''.
- 7 **Exit Mode** Command request addressed to SOP'.

The DFP_U shall use the following sequence of operation to change the configuration from Multi-function Mode (one or two *DP* lanes plus two *USB* SuperSpeed lanes) to four *DP* lanes:

- 1 Place *USB* SuperSpeed in the Safe state.
- 2 **DisplayPort Configure** Command request addressed to SOP'.
- 3 **Active cable VDO reports SOP'' as being supported – DisplayPort Configure** Command request addressed to SOP''.
- 4 **DisplayPort Configure** Command request addressed to SOP.

The DFP_U shall use the following sequence of operation to change the configuration from four *DP* lanes to Multi-function Mode (one or two *DP* lanes plus two *USB* SuperSpeed lanes):

Note: *USB SuperSpeed is already in the Safe state.*

- 1 **DisplayPort Configure** Command request addressed to SOP'. Note that the two unaffected *DP* lanes continue operation.
- 2 **Active cable VDO reports SOP'' as being supported – DisplayPort Configure** Command request addressed to SOP''. Note that the two unaffected *DP* lanes continue operation.
- 3 **DisplayPort Configure** Command request addressed to SOP.
- 4 Remove *USB* SuperSpeed from the Safe state.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.3.1 Active USB-C-to-USB-C Cable V_{CONN} Source

If a DFP_U V_{CONN} Source that provides power to an active USB-C-to-USB-C cable ceases to provide V_{CONN} (without performing a *USB PD VCONN_Swap* Message), the DFP_U V_{CONN} Source shall:

- 1 After re-applying V_{CONN} , issue an SOP'/SOP" *DisplayPort Enter Mode* Command request to place the active cable in DisplayPort Alt Mode.
- 2 To re-establish cable and UFP_U operation in DisplayPort Configuration:
 - Always transmit an SOP *DisplayPort Configure* Command
 - Transmit an SOP' *DisplayPort Configure* Command if the cable is an active cable
 - Also transmit an SOP" *DisplayPort Configure* Command if SOP" is present on the active cable

If a UFP_U V_{CONN} Source that provides power to an active USB-C-to-USB-C cable ceases to provide V_{CONN} (without performing a *USB PD VCONN_Swap* Message), the UFP_U V_{CONN} Source shall:

- 1 After re-applying V_{CONN} , issue a *USB PD VCONN_Swap* Message to pass responsibility for sourcing V_{CONN} to the DFP_U.
- 2 Issue a *USB PD Attention* Command request to place the DFP_U in USB Configuration.
- 3 Issue a *USB PD Attention* Command request to place the DFP_U and the active cable in DisplayPort Configuration.
- 4 **UFP_U V_{CONN} Source does not receive the expected *Enter Mode* and/or *DisplayPort Configure* Command requests** – Issue a *USB PD Hard Reset* to force the DFP_U to exit DisplayPort Alt Mode and re-establish *DP* operation.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

5.4 VDM Flows

Figure 5-1 illustrates the initial actions that occur when a new connection is detected. These include:

- UFP_U shall present a *USB 2.0 USB BB* if the tAMETimeout timer (see *USB-C, Section 5.1.3*) times out before DisplayPort Alt Mode is entered;
- Determining the cable or adapter properties by using a *USB PD Discover Identity* Command request addressed to SOP';
- **Negotiating a USB PD Explicit Power Contract** – Mandatory, but for simple DisplayPort Alt Mode devices, a Contract to not supply/consume power is sufficient; and
- **Establishing the port's data role** – **Optional**, but if used shall be performed before VDMs related to Alternate Modes are transmitted, or after successfully exiting DisplayPort Alt Mode. A DFP_U/UFP_U *USB PD DR_Swap* Message, if performed, shall be completed before entering DisplayPort Alt Mode.

The result is to move to either the DFP_U or UFP_U Message flow, depending on the port's role.

Note: *Implementers should take care to ensure that sufficient time is allowed to discover the connected display during system boot and resume from sleep.*

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

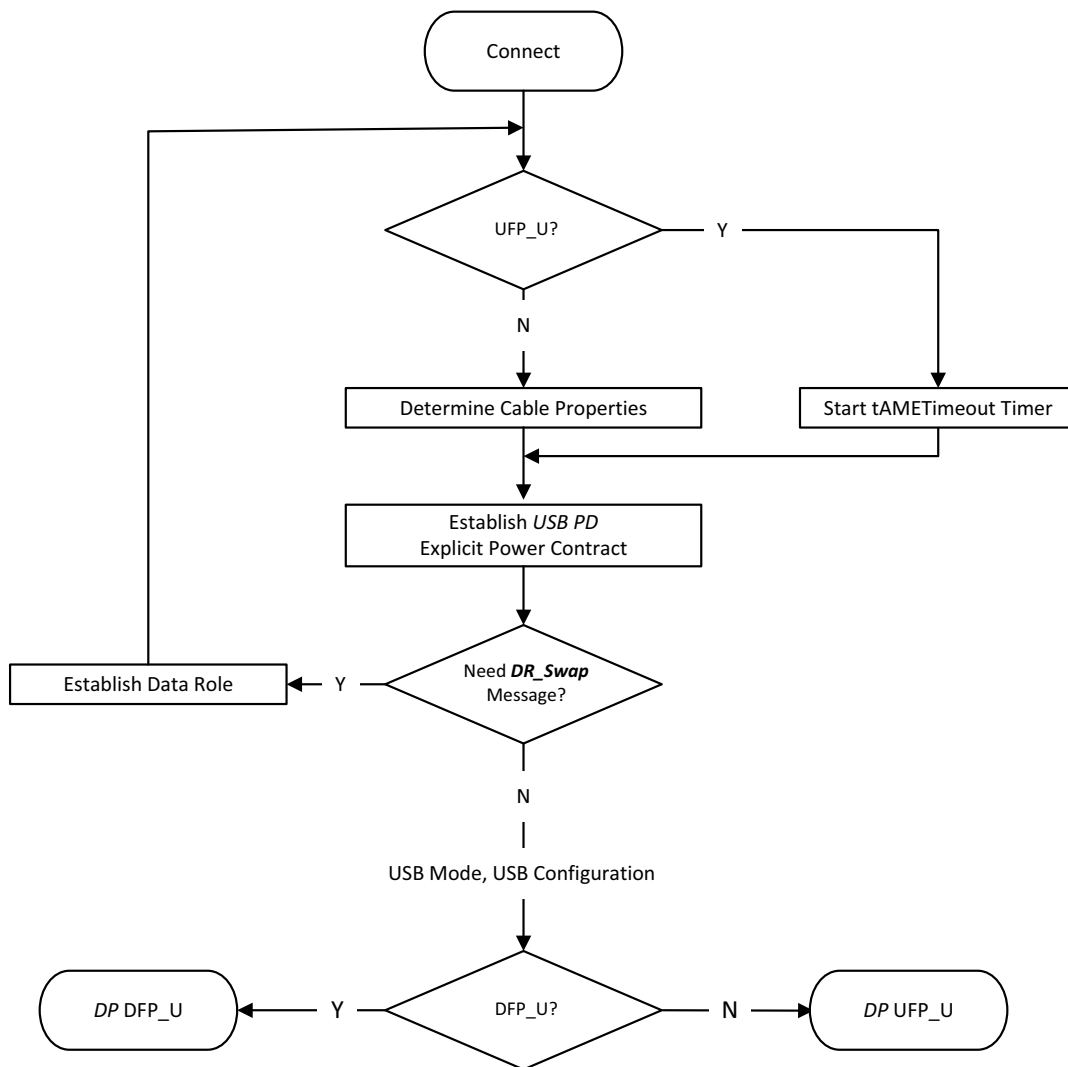


Figure 5-1: Connect Initial Actions

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-2 illustrates the following, up to the point that the port is ready to be configured to use DisplayPort Configuration:

- Message flow at the DFP_U for DisplayPort Alt Mode Discovery and Entry Processes
- Generation of **DisplayPort Status Update** Commands
- Receipt of **USB PD Attention** Command requests

While waiting to be configured to use **DisplayPort**, the port is in a DisplayPort Alt Mode/USB Configuration Wait state. While waiting in this state, if the DFP_U is a **DP** Sink device, the **DP** Sink device may detect a change in HPD state or an IRQ_HPD. In this case, the **DP** Sink device shall transmit a **DisplayPort Status Update** Command request and await a **DisplayPort Status Update** Command response (not shown in Figure 5-2).

When the DFP_U and UFP_U are attached by way of a USB-C-to-USB-C cable, the DFP_U shall ensure that the DFP_U, cable (where present), and UFP_U capabilities are compatible with one another (i.e., that the connection is neither a DP Source-to-DP Source device nor DP Sink-to-DP Sink device, and that there is a signal configuration supported by the **DP** Source device, **DP** Sink device, and cable). If the DFP_U, cable, and UFP_U are incompatible, the DFP_U shall remain in USB Mode.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

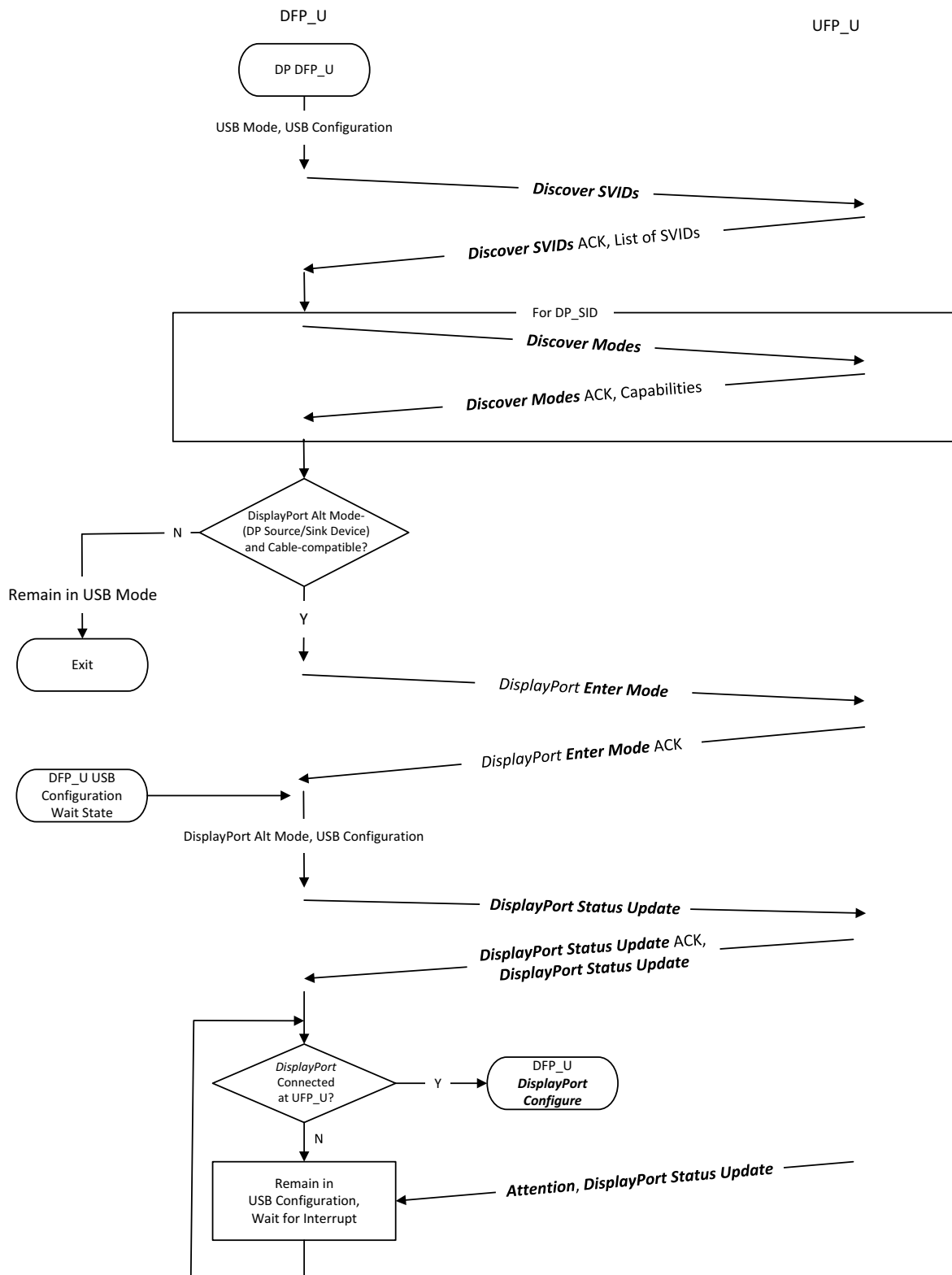


Figure 5-2: DFP_U DisplayPort Alt Mode Discovery and Entry Processes

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-3 illustrates the Message flow at the DFP_U for configuration, up to the point that the USB SuperSpeed and SBU connections start to use DisplayPort Configuration.

Note: If the UFP_U requests Multi-function Mode and the DFP_U supports Multi-function Mode, it is a matter of DFP_U implementation policy whether to select Multi-function Mode or remain in DisplayPort Alt Mode.

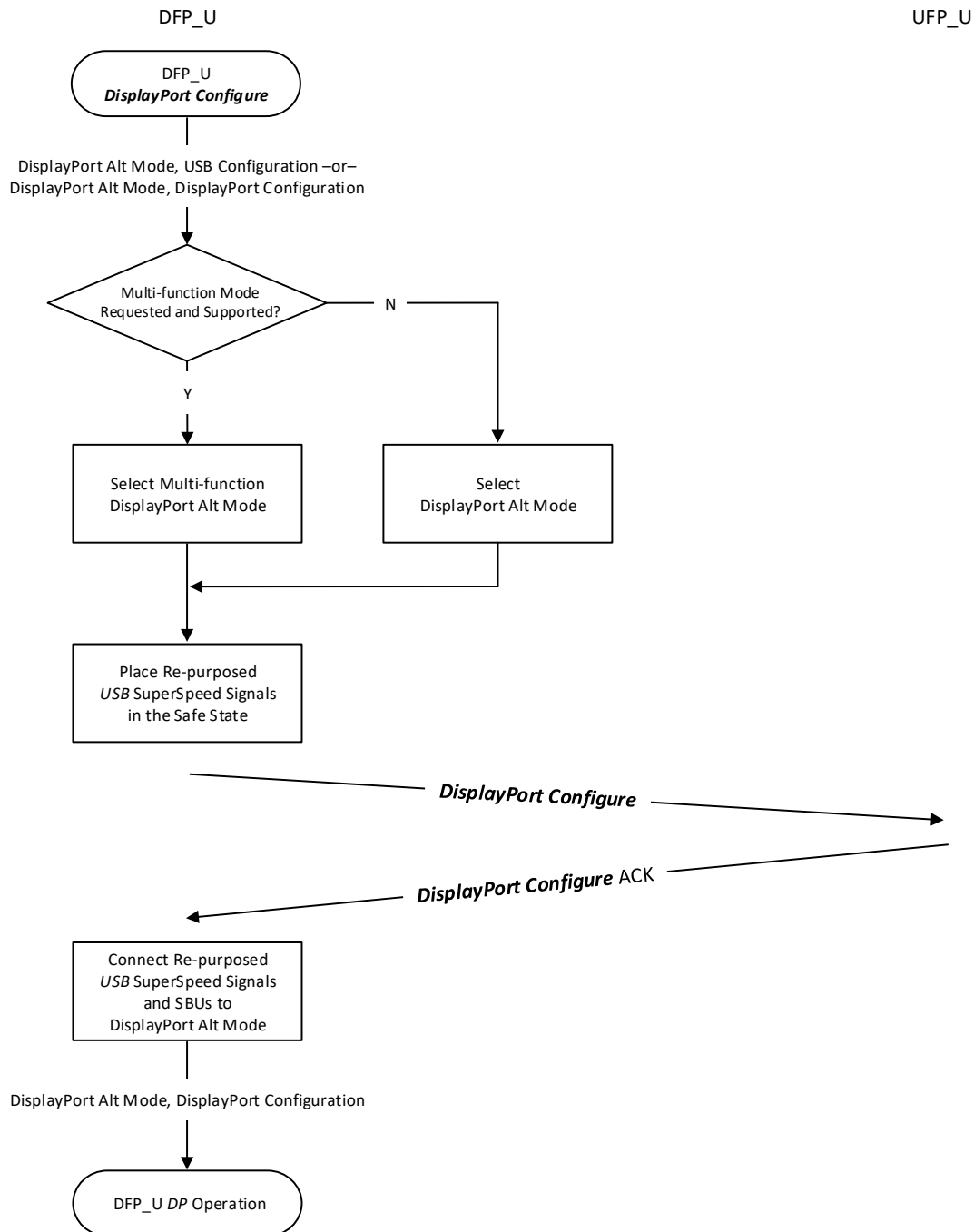


Figure 5-3: DFP_U DisplayPort Configuration

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-4 illustrates the Message flow when the DFP_U is operating in DisplayPort Configuration and receives a request for reconfiguration by a DisplayPort Status transmitted as the VDO in a **USB PD Attention** Command request received from the UFP_U. The UFP_U may transmit a DisplayPort Status that requests either of the following:

- Different configuration, –or–
- Change to USB Configuration instead of DisplayPort Configuration

If the DisplayPort Status indicates that the UFP_U no longer has a connected display, the DFP_U shall initiate a transition to USB Configuration. In addition, the DFP_U may make a policy change (e.g., in preparation to exiting DisplayPort Alt Mode before using a **USB PD DR_Swap** Message).

If the DFP_U is a **DP** Sink device, it may detect a change in HPD state or an IRQ_HPDP. In this case, the **DP** Sink device shall transmit a **DisplayPort Status Update** Command and await a **DisplayPort Status Update** Command response (not shown in Figure 5-4).

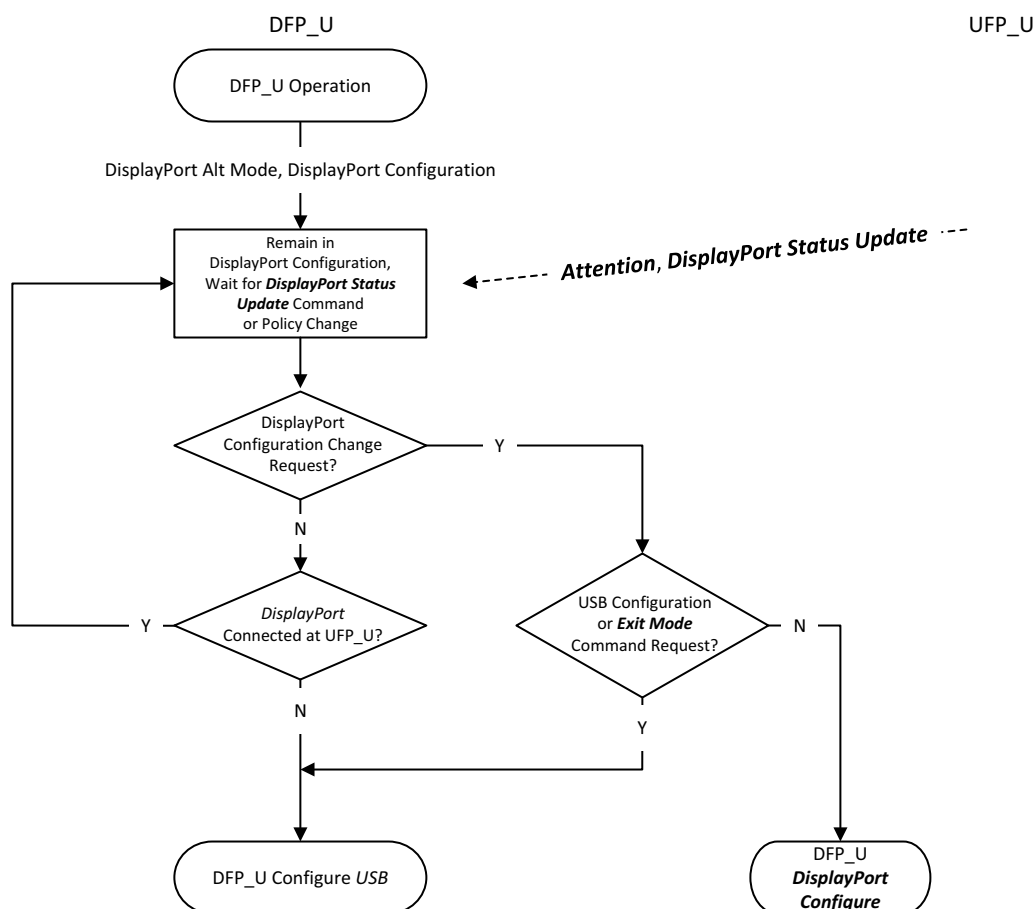


Figure 5-4: DFP_U DisplayPort Operation and Reconfiguration

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-5 illustrates the sequence at the DFP_U for reconfiguring the port to *USB* operation. This may be performed as a result of one of the following:

- Configuration request to reconfigure from DisplayPort Configuration
- Result of an **Exit Mode** Command request, which may be completed only when in USB Configuration

Before transmitting the **DisplayPort Configure** Command specifying USB Configuration, the DFP_U places the previously reconfigured pins into the USB Safe state and then isolates the SBU pins, as defined in Section 5.2.6.

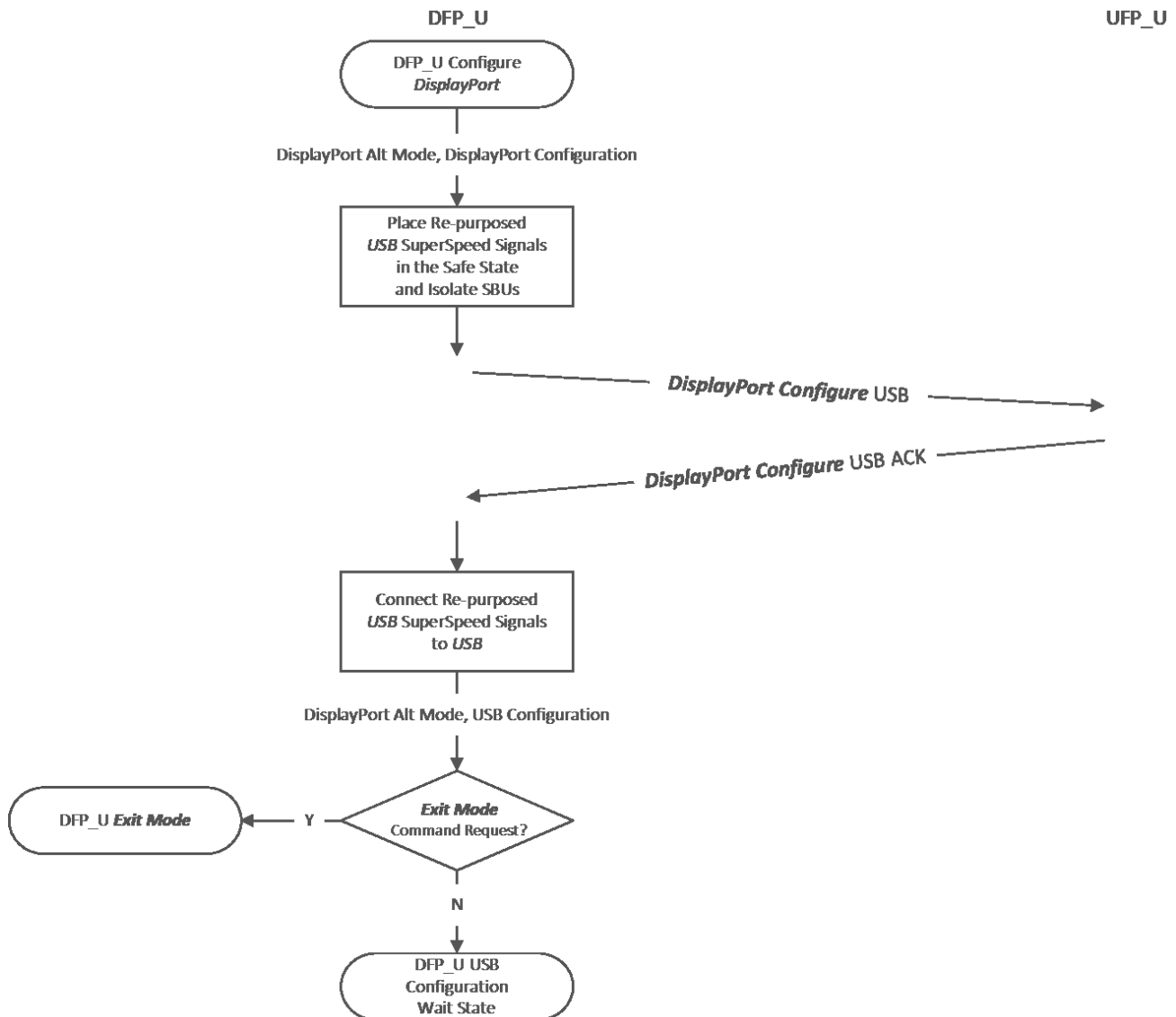


Figure 5-5: DFP_U Reconfigure to USB Configuration

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-6 illustrates the **Exit Mode** sequence at the DFP_U. This sequence is entered only when the port configuration has been set or reset to USB Configuration.

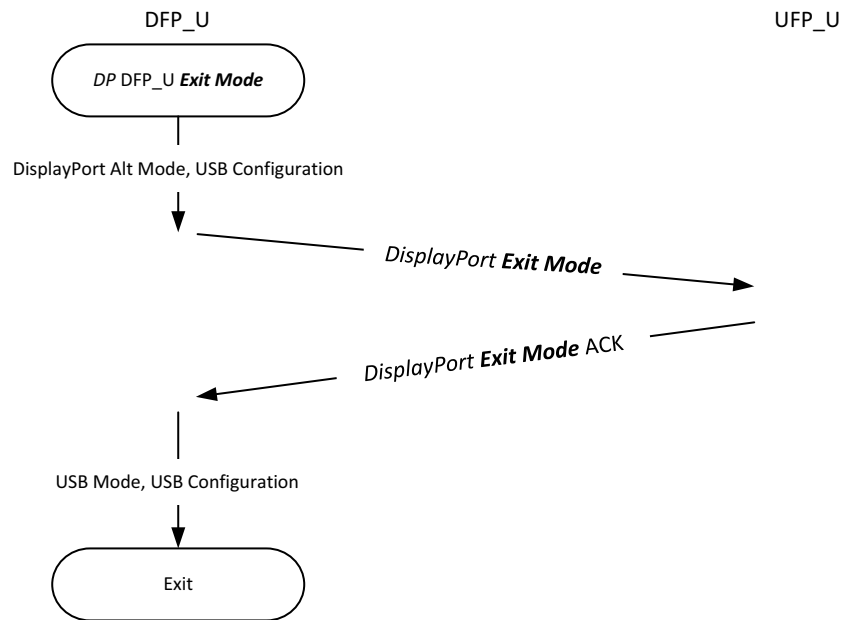


Figure 5-6: DFP_U Exit Mode Process

Figure 5-7 illustrates the Message flow for the UFP_U DisplayPort Alt Mode Discovery and Entry Processes, up to the point that the port enters a Wait state ready to be configured to use DisplayPort Configuration.

USB-C requires that the UFP_U present a *USB BB* device class if the UFP_U fails to receive and act on an **Enter Mode** Command request within the tAMETimeout timer period (see *USB-C*, Section 5.1.3). In this case, the UFP_U shall await further commands from the DFP_U (e.g., the UFP_U may receive a *DisplayPort Enter Mode* Command request even though it has already presented *USB BB*).

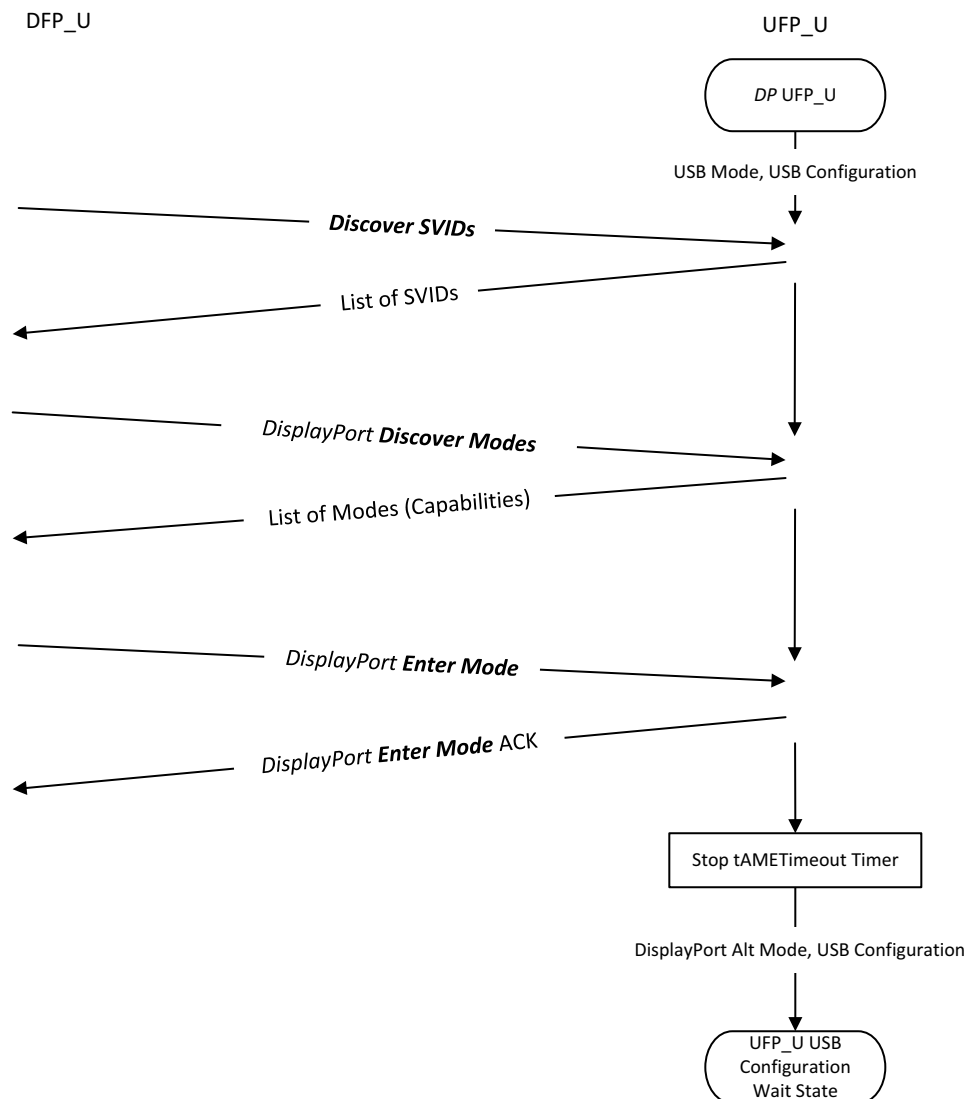


Figure 5-7: UFP_U DisplayPort Alt Mode Discovery and Entry Processes

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-8 illustrates the Message flow for the UFP_U while in the USB Configuration Wait state. In this state, the UFP_U may receive a command to do any one of the following:

- Report status
- Configure the UFP_U to DisplayPort Configuration
- Exit DisplayPort Alt Mode

During this state, the UFP_U may detect a change in connection status on its *DP* side (e.g., a *DP* Source or Sink device is attached, or a change in HPD state or an IRQ_HPDP) and consequently generate a *USB PD Attention* Command request toward the DFP_U. When entering the USB Configuration Wait state after receiving a *DisplayPort Enter Mode* Command request, the UFP_U shall wait to receive a DisplayPort Status command from the DFP_U, and then provide its status in its response. The UFP_U shall not generate a *USB PD Attention* Command request until a subsequent state change.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

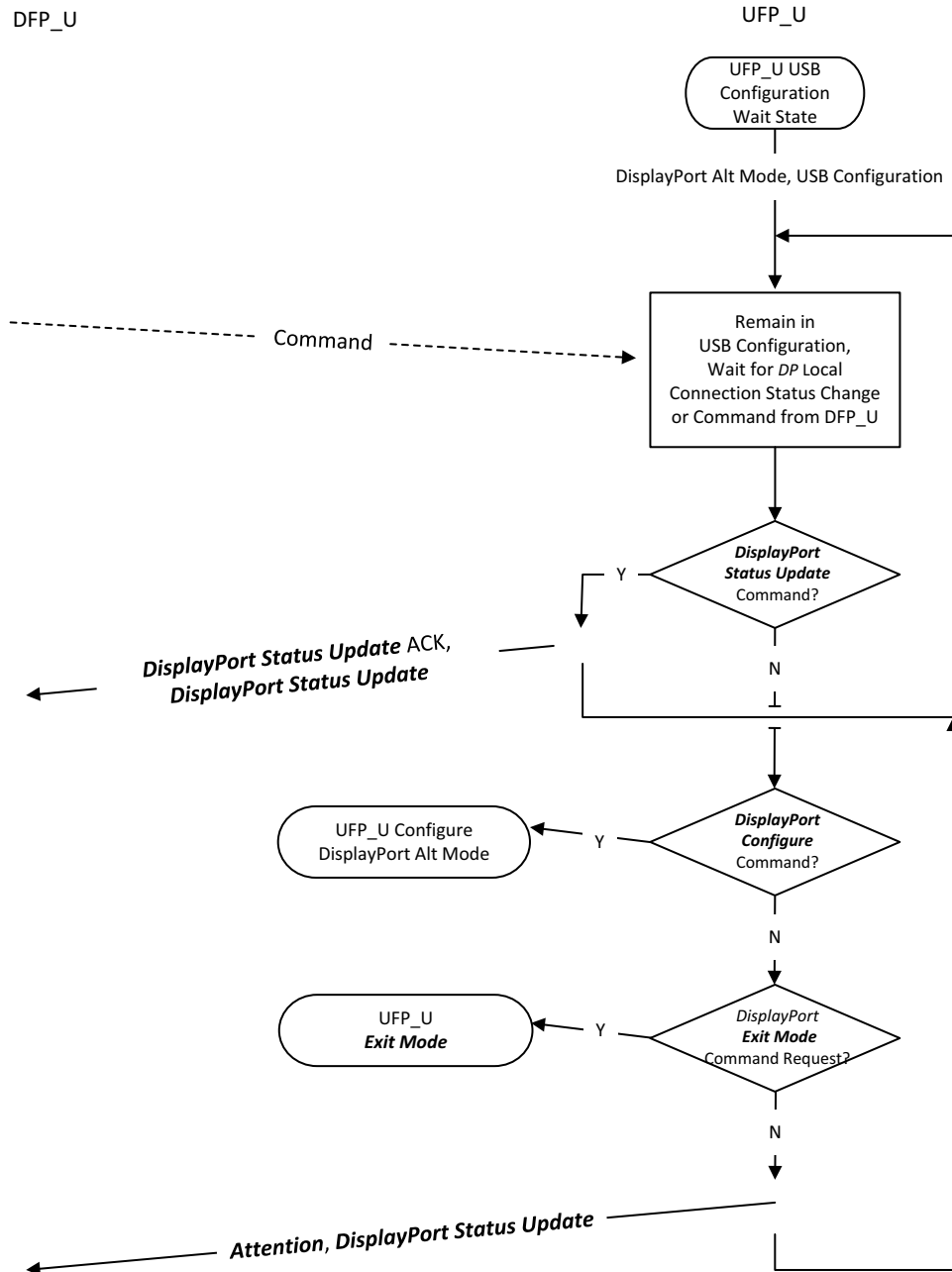


Figure 5-8: UFP_U USB Configuration Wait State

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-9 illustrates the Message flow at the UFP_U for configuration, up to the point that the *USB SuperSpeed* and SBU connections start to use DisplayPort Configuration.

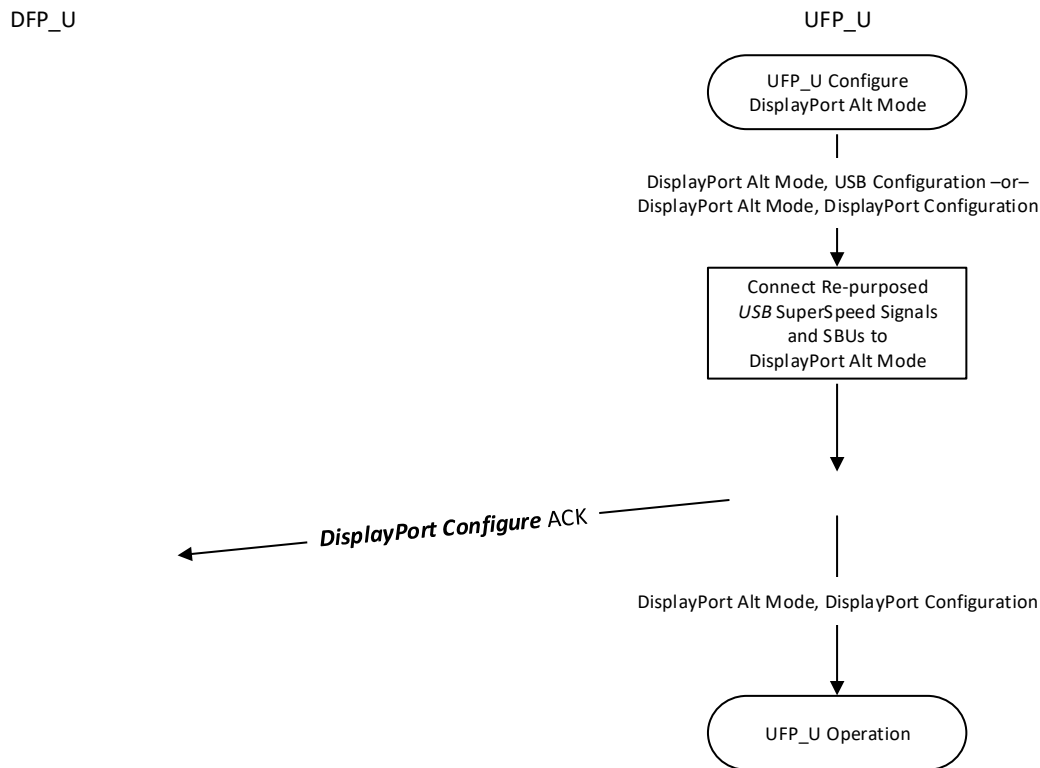


Figure 5-9: UFP_U DisplayPort Configuration

Figure 5-10 illustrates the Message flow for UFP_U operation while in DisplayPort Configuration. If the UFP_U detects a change in the connection status on its *DP* side (e.g., a *DP* Source or Sink device is disconnected, or a change in HPD state or an IRQ_HPDP), the UFP_U transmits a *USB PD Attention* Command request to the DFP_U. Simultaneously, the UFP_U waits for a command from the DFP_U. The UFP_U may receive one of the following:

- **DisplayPort Status Update** Command
- **DisplayPort Configure** Command to change to using USB Configuration instead of DisplayPort Configuration

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

DFP_U

UFP_U

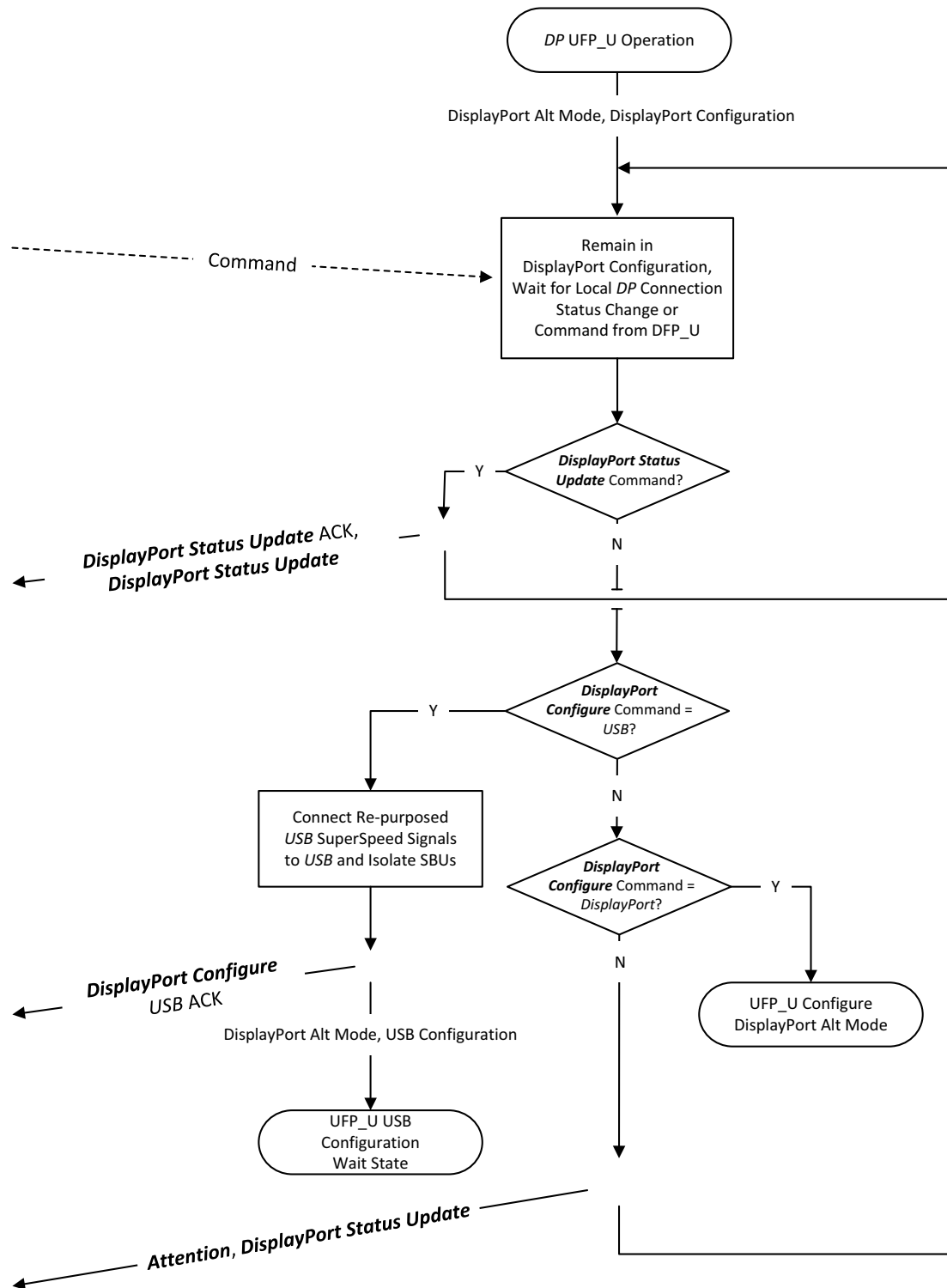


Figure 5-10: UFP_U DP Operation

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Figure 5-11 illustrates the **Exit Mode** sequence at the UFP_U. The UFP_U shall acknowledge the command, but take no further action. The UFP_U shall **not** transmit **USB PD Attention** Command requests after acknowledging the **Exit Mode** Command request. The UFP_U shall **not** re-start its tAMETimeout (**USB BB** connect) timer (i.e., **USB BB** is never displayed after the connection enters DisplayPort Alt Mode).

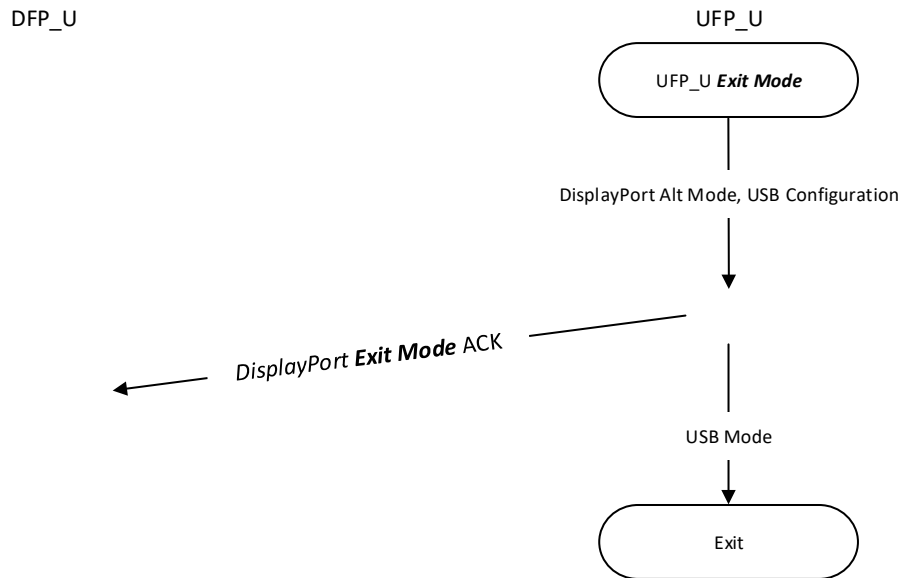


Figure 5-11: UFP_U Exit Mode Process

A port may be disconnected or subject to a **USB PD Hard Reset** at any time. The port resets the DisplayPort Alt Mode. Figure 5-12 illustrates the sequence of actions on a **USB PD Hard Reset** or disconnect at both the DFP_U and UFP_U. The UFP_U shall reduce its power consumption to no more than the default V_{CONN} . No Alternate Mode messaging occurs. After a **USB PD Hard Reset**, the port behaves as if a new connection was detected.

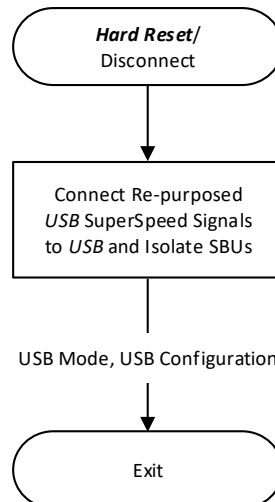


Figure 5-12: USB PD Hard Reset and Disconnect

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

6 Miscellaneous

This section describes *DP* Source and Sink device power management, low-power operation, and equivalent *USB* functionality.

6.1 V_{CONN} and V_{BUS}

As specified in *USB-C*, V_{CONN} shall be provided by the V_{CONN} Source at a voltage between 3.0 and 5.5V. Because *DP* adapters may need more power than that specified in *USB-C*, a V_{CONN} Source supporting *DisplayPort* on a *USB-C* receptacle shall support 1.5W at any voltage within this range when operating in DisplayPort Alt Mode.

A DP Source device/DFP_U that supports DisplayPort Alt Mode may be either a *USB* Power Provider or Sink with Accessory Support that has detected a V_{CONN}-powered accessory, as specified in *USB-C*. Assumptions about the presence of V_{BUS} may not be made because a Sink with Accessory Support operating as a DFP_U does **not** provide V_{BUS}.

A Sink with Accessory Support that supports *DisplayPort* on a *USB-C* receptacle shall **not** depend on power being available on V_{BUS}.

If a UFP_U implemented with a *USB-C* plug contains *USB* or other functionality (e.g., the UFP_U is a Multi-function captive cable dock), the *USB* or other functionality shall be powered from V_{BUS} and not from V_{CONN}. If V_{BUS} is unavailable, the *USB* or other functionality may be disabled. DisplayPort Alt Mode functionality shall continue to function, provided that V_{CONN} is available within the voltage range described above.

If the UFP_U implemented with a *USB-C* plug presents a *DP* or mDP receptacle connector, DP_PWR on the connector shall be powered from V_{BUS}, if available. If V_{BUS} is unavailable, DP_PWR may remain unpowered.

6.1.1 Active Cable Behavior

V_{CONN} shall be continuously provided to an active cable while the cable is in DisplayPort Alt Mode. V_{CONN} shall be provided to the active cable in all power and suspend states while in DisplayPort Alt Mode.

6.1.2 V_{CONN} Overcurrent Protection

The *USB PD* V_{CONN} Over-Current Protection (OCP) threshold in a V_{CONN} Source should be set at least 5% higher than the maximum-mandated operating current, I_{VCONN_max}, as follows:

$$I_{VCONN_max} = 1.5/V_{VCONN_min}$$

where:

- V_{VCONN_min} = Lowest voltage at which the V_{CONN} Source shall supply V_{CONN}

The V_{CONN} Source OCP should **not** trip on deviations that are greater than I_{VCONN_min} +5% with a duration of less than 1ms.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

6.2 Low-power Operation

6.2.1 Display Sleep

If the DFP_U is also a *DP* Source device, the DFP_U is in control of both the display power state and the DisplayPort Alt Mode setting. When the DFP_U is a *DP* Source device and puts the display to sleep (i.e., by writing to the SET_POWER & SET_DP_PWR_VOLTAGE register at DPCD Address 00600h within the *DP* Sink device), the *DP* Source device shall **not** exit DisplayPort Alt Mode (i.e., shall **not** issue an **Exit Mode** Command Request), and both the DFP_U and UFP_U shall maintain *USB PD* operation.

If the DFP_U is also a *DP* Sink device (e.g., a display), the *DP* Source device is in control of the display power state; however, the DFP_U on the connected port is in control of the DisplayPort Alt Mode setting. When the *DP* Source device (UFP_U) puts the display to sleep (i.e., by writing to DPCD Address 00600h within the *DP* Sink device), the *DP* Source device shall not request an exit from DisplayPort Alt Mode (i.e., shall **not** request the DFP_U to issue an **Exit Mode** Command Request), and both the DFP_U and UFP_U shall maintain *USB PD* operation. The V_{CONN} Source shall **not** reduce current if it is not also the *DP* Source device that initiated Sleep entry.

During the *DP* low-power state, the DFP_U/DP Source device may reduce the provided V_{CONN} power to an active cable from 1.5W to 150mW if the DFP_U/DP Source device grants an extended timeout. The DFP_U/DP Source device shall follow the active cable sleep/wake flow and process defined in *DP Standard*. For a passive cable that reports “ V_{CONN} required” in its *USB PD Discover Identity* Command response, the DFP_U/DP Source device shall provide V_{CONN} power per *USB-C*.

6.2.2 USB Suspend General Properties (Informative)

The system (DFP_U) may place the *USB* port into the USB Suspend state at any time. Typically, the system will do this when it is placed in a low-power state. The DFP_U maintains *USB* connectivity information on the port during the USB Suspend state (i.e., detects physical disconnect and connect events).

If the USB Suspend Supported bit is set to 1 in the *USB PD* Explicit Power Contract, the UFP_U shall reduce its V_{BUS} power consumption to no more than 25mW (intended to be sufficient to maintain *USB PD* if powered from V_{BUS}) and *USB-C* requires V_{CONN} to support only 7.5mA (i.e., 35-mW maximum).

If the USB Suspend Supported bit is **not** set to 1 in the *USB PD* Explicit Power Contract, V_{BUS} power and V_{CONN} (if needed to power an adapter or for an active cable) are maintained.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

6.2.3 USB-C DFP_U

DP functionality shall be maintained during the USB Suspend state. The DFP_U shall ensure that the USB Suspend Supported bit is cleared to 0 in the *USB PD* Explicit Power Contract. The DFP_U shall not exit DisplayPort Alt Mode simply because *USB* has been placed into its Suspend state. Note that *DP* connectivity information is maintained during the USB Suspend state.

During the system low-power state, the *DP* subsystem may put the display to sleep; however, the *DP* subsystem may also maintain *DP* connectivity information (e.g., so that a new *DP* connection may be treated as a wake event). The following cases describe DFP_U *DP* Source and Sink device behavior in the system low-power state.

- If the DFP_U is the *DP* Source device (HPD receiver)
 - **Maintaining the DP connection in the system low-power state: DP Source device needs to maintain HPD connectivity information** – Before placing the system in a low-power state, the DFP_U/DP Source device shall follow *DP Standard* low-power mandates and the DFP_U/DP Source device shall maintain DisplayPort Alt Mode. The DFP_U/DP Source device shall also maintain support for the full level of V_{CONN} power if the cable is an active cable or is being used to power the adapter; thus, the DFP_U/DP Source device shall ensure that the USB Suspend Supported bit is cleared to 0 in the *USB PD* Explicit Power Contract.
 - **Dropping the DP connection in the system low-power state: DP Source device does not need to maintain HPD connectivity information** – Before placing the system in the low-power state, the DFP_U/DP Source device shall exit DisplayPort Alt Mode. The DFP_U/DP Source device shall then place the *USB* connection in the USB Suspend state and reduce its V_{CONN} support to 7.5mA. In this case, exit from the system low-power state behaves like a new *DP* connection discovery.
- If the DFP_U is the *DP* Sink device (HPD transmitter)

The DFP_U/DP Sink device (Display) **cannot** negotiate *DisplayPort* low-power state. The DFP_U/DP Sink device may exit DisplayPort Alt Mode to reduce power consumption in the low-power state. The DFP_U/DP Sink device reduces the available V_{CONN} current in the low-power state according to *USB-C* V_{CONN} Source requirements.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

6.2.4 USB-C UFP_U that Supports a V_{CONN}-powered Adapter

The UFP_U does **not** provide V_{BUS}, and there is no *USB* data connectivity. There is therefore no concept of *USB* Suspend.

The following cases describe UFP_U *DP* Source and Sink device behavior in the system low-power state.

- If the UFP_U is the *DP* Source device (HPD receiver)
 - **Maintaining the DP connection in the system low-power state: DP Source device needs to maintain HPD connectivity information** – Before placing the system in the low-power state, the UFP_U/DP Source device shall follow *DP Standard* low-power mandates and the UFP_U/DP Source device shall maintain DisplayPort Alt Mode. The UFP_U/DP Source device shall also maintain support for the full level of V_{CONN} power to power the adapter; thus, the UFP_U/DP Source device shall ensure that the USB Suspend Supported bit is cleared to 0 in the *USB PD* Explicit Power Contract.
 - **Dropping the DP connection in the system low-power state: DP Source device does not need to maintain HPD connectivity information** – Before placing the system in the low-power state, the UFP_U/DP Source device shall exit DisplayPort Alt Mode. The UFP_U/DP Source device shall then place the system in the low-power state and reduce its V_{CONN} support to 7.5mA. In this case, exit from the system low-power state behaves like a new *DP* connection discovery.
- If the UFP_U is the *DP* Sink device (HPD transmitter)

The UFP_U/DP Sink device (Display) **cannot** negotiate *DisplayPort* low-power state. The UFP_U/DP Sink device may exit DisplayPort Alt Mode to reduce power consumption in the low-power state. The UFP_U/DP Sink device reduces the available V_{CONN} current in the low-power state according to *USB-C* V_{CONN} Source requirements.

6.3 Equivalent USB Functionality

USB-C strongly encourages devices that support Alternate Modes to provide equivalent *USB* functionality, where such exists, for best user experience (see *USB-C, Section 5.1*). An example of equivalent *USB* functionality for *DisplayPort* is the support of *USB AV* for the uncompressed transport of video with a resolution of 1080p60 24-bpp RGB.

6.4 USB BB URL

A device that is mandated to present a *USB BB* device shall include within the string indexed by **iAlternateModeString[n]** for DisplayPort Alt Mode either a manufacturer-specific permanently available URL, –or– the following VESA URL:

vesa.org/dp-usb-type-c

An example string is as follows:

DP-to-VGA adapter. For further assistance, see vesa.org/dp-usb-type-c.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A DisplayPort Alt Mode Examples (Informative)

This appendix provides two examples of using DisplayPort Alt Mode.

Examples are provided to illustrate how DisplayPort Alt Mode, negotiated using *USB PD*, is used to provide *DP* functionality beyond the basic functionality specified in *USB-C*. The examples are **not** intended to be complete designs.

A.1 DisplayPort Alt Mode Examples

There are many use scenarios for including *DisplayPort* on a *USB-C* connector. In all cases, the system with a *USB-C* connector may be a *USB* host or device, and, independently, may behave as a *DP* Source or Sink device.

Some possible examples include:

- *DisplayPort* over a USB-C-to-USB-C cable;
- *DisplayPort* over a USB-C-to-DP plug or receptacle cable (either direction);
- *DisplayPort* over a *USB-C* to other video format connector cable (e.g., *HDMI* or VGA). In these cases, the adapter would also contain a DP-to-other video format protocol converter;
- *DisplayPort* and *USB 3.2* to a dock (*DisplayPort* is limited to two lanes in this example because two of the *USB* SuperSpeed data signal pairs are used).

These particular examples are intended only to illustrate the overall Discovery Process and DisplayPort Alt Mode entry and exit flow by way of *USB PD* VDMs. Note that additional messaging is used to support DisplayPort Alt Mode-specific Configuration, Status, and Control.

A.2 USB-C USB Host-to-DP Video Adapter Example

This example describes a *USB* host with a *USB-C* receptacle that may also behave as a *DP* Source device, and connects to an adapter that has a *USB-C* plug and *DP* plug to adapt to a *DP* monitor.

Note that a *USB* device may support the same *DP* functionality.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A.2.1 DP Video Adapter Assumptions

The *DP* video adapter is a cable with a *USB-C* plug on one end and a *DP* connector on the other. It is an active adapter that contains circuitry (e.g., in the plug overmold, lump in the cable, etc.) that communicates with the *USB* host to expose its capabilities and enable its functionality.

The *DP* video adapter does **not** provide power to charge the *USB* host because *DisplayPort* does not support the provision of power over the cable.

The *USB* host in this example has the capability to source native *DP* video (i.e., behave as a *DP* Source device).

The *USB* host and *DP* video adapter both use the specific *USB* SID associated with *DisplayPort* (DP_SID) to identify themselves. The DP_SID is used firstly to indicate DisplayPort Alt Mode support, and secondly to distinguish commands that are specific to DisplayPort Alt Mode support.

The *USB* host is permitted to use the *DP* video adapter only if it identifies the adapter as a *DP* video adapter. Failure to meet this results in a connection to a *USB BB* device that will provide error information.

The *DP* video adapter is powered by V_{CONN} for the switch, CC logic, *USB PD* engine, and *DP* functions; however, *USB 2.0*-related functions must be powered by V_{BUS} .

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A.2.2 Block Diagrams

Figure A-1 illustrates a block diagram of the *USB* host system with its multiplexer. Figure A-2 illustrates a block diagram of the *DP* video adapter cable assembly.

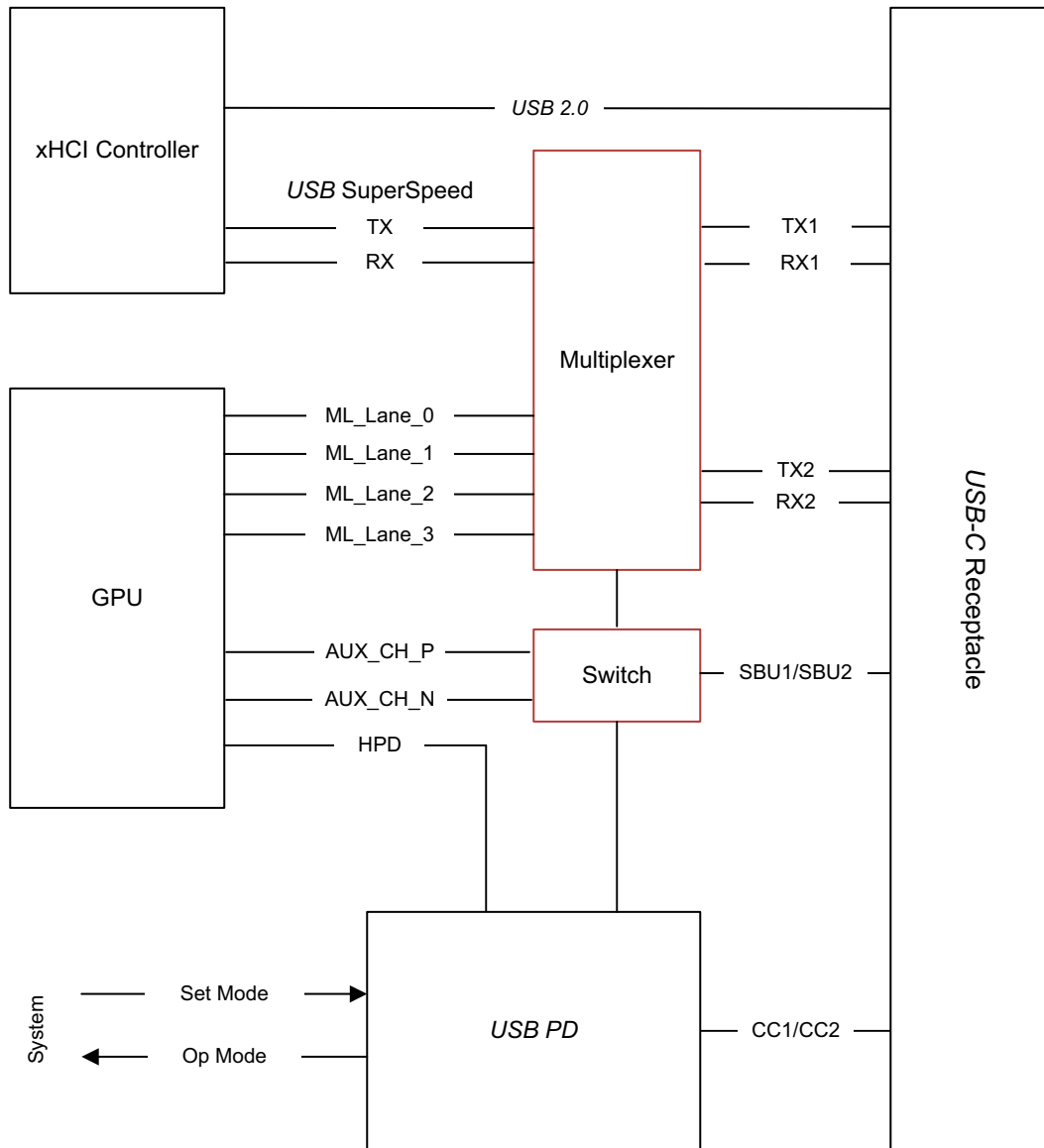


Figure A-1: USB Host System Block Diagram

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

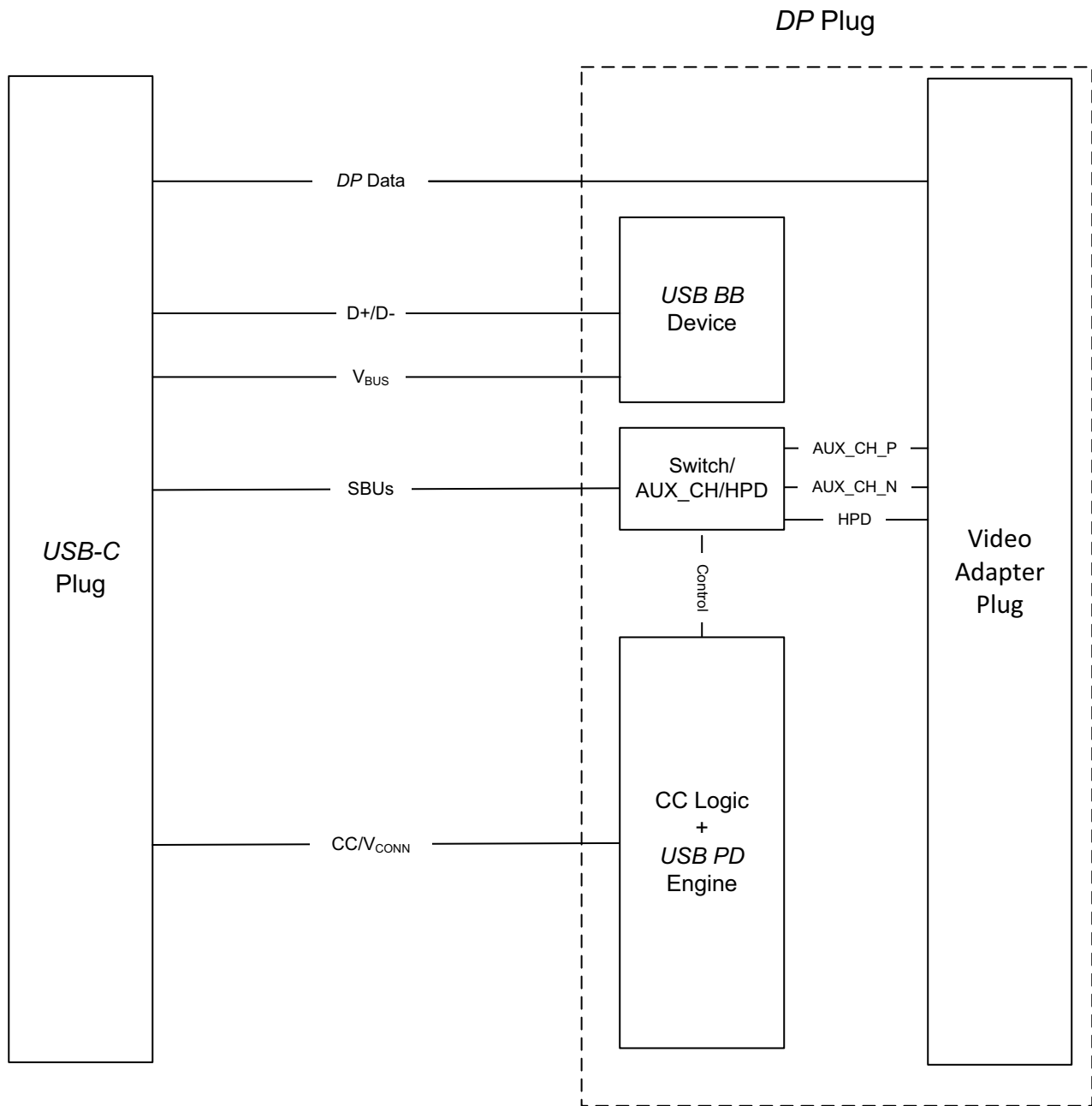


Figure A-2: DP Video Adapter Cable Block Diagram

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

The **USB** host has a multiplexer that may steer one of the following:

- **USB** SuperSpeed signals
- All four **DP** signal lanes
- Two **DP** signal lanes together with the **USB** SuperSpeed signals to the **USB-C** receptacle

The multiplexer, or a combination of the multiplexer and **DP** port output, manages the **USB-C** plug flipping. When the **USB** host system instructs the **USB PD** block to reconfigure the port's pins to output **DisplayPort**, the multiplexer is activated when the attached system identifies itself as a **DP** video adapter cable in the Discovery Process and responds to the **Enter Mode** Command request with an ACK. If the **DP** video adapter responds with a NAK any time during the Discovery Process, the **USB** host shall remain in the default **USB 2.0/USB 3.2** Configuration. The **USB** host sees that the operational mode has changed, based on the vendor-specific OpMode signal.

The **DP** video adapter cable assembly provides a **USB PD** interface to communicate with the **USB** host. By default, the cable assembly also provides a **USB 2.0** device interface. **USB** hosts that do **not** support **DP** video adapters use the **USB 2.0** device interface to do the following:

- Detect whether a **DP** video adapter is connected
- Provide users with information about the adapter and the **USB** host system's inability to use the adapter

The **DP** video adapter also contains circuitry to isolate the SBU connections until DisplayPort Alt Mode is established. The **DP** video adapter is powered by V_{CONN} for the switch, CC logic, **USB PD** engine, and **DP** functions; however the **USB 2.0**-related functions must be powered by V_{BUS} . The same video adapter may be plugged into a **USB** host or device (with dual-role discovery; i.e., either may behave as a **DP** Source device). In the latter case, V_{BUS} is unavailable and therefore is not used by the adapter.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A.2.3

Solution

The *USB* host system and the *DP* video adapter use *USB PD* DP-defined Messages to communicate and take advantage of advanced features.

The flow of Messages from the point after the *DP* video adapter is plugged into the *USB-C* receptacle, plug orientation has been determined, and the *USB* host/device relationship established is as follows:

- *USB* host system does **not** support *USB PD*
 - After the tAMETimeout timer times out, the *USB* host will see the *USB 2.0* device interface with *USB BB* and report to the user that an unsupported device has been connected, identifying the type of device from the *USB* descriptor information presented in *USB BB*.
- *USB* host system supports *USB PD* and advanced functions, but does not support *DP* video adapters
 - *USB* host may look for other advanced functions, but does support the *DP* video adapter even though its presence is discovered, because the video adapter responds with the DP_SID that the *USB* host does **not** recognize as identifying something that it supports.
 - Because *DP* video adapters are **not** supported, after the tAMETimeout timer times out the *USB* host will see the *USB 2.0* device interface with *USB BB* and report to the user that an unsupported device has been connected, identifying the type of device from the *USB* descriptor information presented in *USB BB*.
- *USB* host system supports *USB PD* and the *DP* video adapter
 - *USB* host looks for advanced functions and discovers a *DP* video adapter by way of responses to *USB PD* Messages, using the DP_SID.
 - Because a usable *DP* video adapter is discovered, the *USB* host enters DisplayPort Alt Mode and then reconfigures its *USB-C* receptacle to output *DP* signals on the connector. The video adapter cancels its tAMETimeout timer when entering DisplayPort Alt Mode.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

This example uses the basic extensions when connected that do the following:

Base Set VDMs used

- *USB PD Discover SVIDs*
- *USB PD Discover Modes* (DP_SID)
- *Enter Mode* (DP_SID, Alternate Mode offset)
- *Exit Mode* (DP_SID, Alternate Mode offset)

Process Outline

The *USB* host functions in its role as a DFP_U.

- 1 DFP_U detects the presence of a *USB-C* device (CC pins) and connector orientation.
- 2 DFP_U applies 5V to V_{BUS} and V_{CONN}.
- 3 DFP_U uses *USB PD* to make a *USB PD* Explicit Power Contract with the UFP_U.
- 4 DFP_U runs the *USB PD Discover SVIDs* Process.
 - a Transmits a *USB PD Discover SVIDs* Command request.
 - b Receives a *USB PD Discover SVIDs* Command response with a list of SVIDs for which the UFP_U supports modes.
- 5 DFP_U runs the *USB PD Discover Modes* Process.
 - a Transmits the *USB PD Discover Modes* VDM for each of the SVIDs previously discovered that it supports, including the DP_SID.
 - b Receives *USB PD Discover Modes* Command response with a list of modes associated with the DP_SID.
 - c **If a DP video adapter is not discovered** – Remains in USB Mode, discovers the *USB BB* device after the tAMETimeout timer times out, and then informs the user – done.
 - d Else.
- 6 DFP_U runs the *Enter Mode* Process.
 - a Transmits the *Enter Mode* VDM with the DP_SID and offset to mode.
 - b Receives a response:
 - 1 **USB PD Responder ACK response** – DFP_U is now attached to the *DP* video adapter, running in DisplayPort Alt Mode and USB Configuration.
 - 2 **USB PD Responder NAK response** – DFP_U returns to default *USB 3.2* Configuration.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

- 7 DFP_U discovers adapter status, and then waits until the adapter reports that it has a connected display.
- 8 DFP_U runs the **DisplayPort Configure** Process.
 - a Places signals to be reconfigured in the Safe state.
 - b Transmits a **DisplayPort Configure** Command to the adapter.
 - c Receives a **USB PD** Responder ACK response.
 - d Configures signals to use **DisplayPort**.
- 9 If the DFP_U receives a **DisplayPort Status Update** Command from the **USB-C** device that needs a configuration change, the DFP_U re-runs the **DisplayPort Configure** Process (possibly reconfiguring the interface back to USB Configuration).
- 10 DFP_U continues to run the mode until:
 - a Explicitly exited by an **Exit Mode** VDM.
 - b **USB PD Hard Reset**.
 - c Power is removed from the **USB** host.
 - d Adapter is disconnected.

The **USB** host and the **DP** video adapter will use the base set VDMs without change, together with the DP_SID-specific VDMs.

The **DP** video adapter shall follow and/or **not** interfere with normal **USB PD** mandates.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A.3 USB Dock with DisplayPort Alt Mode Example

In this example, a device uses its *USB-C* connector as a multi-function cradle dock for a *USB* host.

Note: *There are additional mechanical mandates that are **not** covered in this example.*

A.3.1 USB Dock with DisplayPort Alt Mode Assumptions

A multi-function cradle or captive cable dock has the following functions:

- Provides mechanical alignment and attachment in addition to those provided by the *USB-C* connector
- Uses *USB PD* to charge/power the *USB* host
- Sources at least two native *DP* video lanes on a *DP* receptacle
- Contains a *USB* hub that exposes three *USB* ports and attached internal *USB* devices:
 - *USB* Ethernet device (RJ-45 connector), and
 - *USB* audio device (3.5-mm phone jack)

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A.3.2 Block Diagram

Figure A-3 illustrates a block diagram of a *USB* dock that also supports a *DP* Source device.

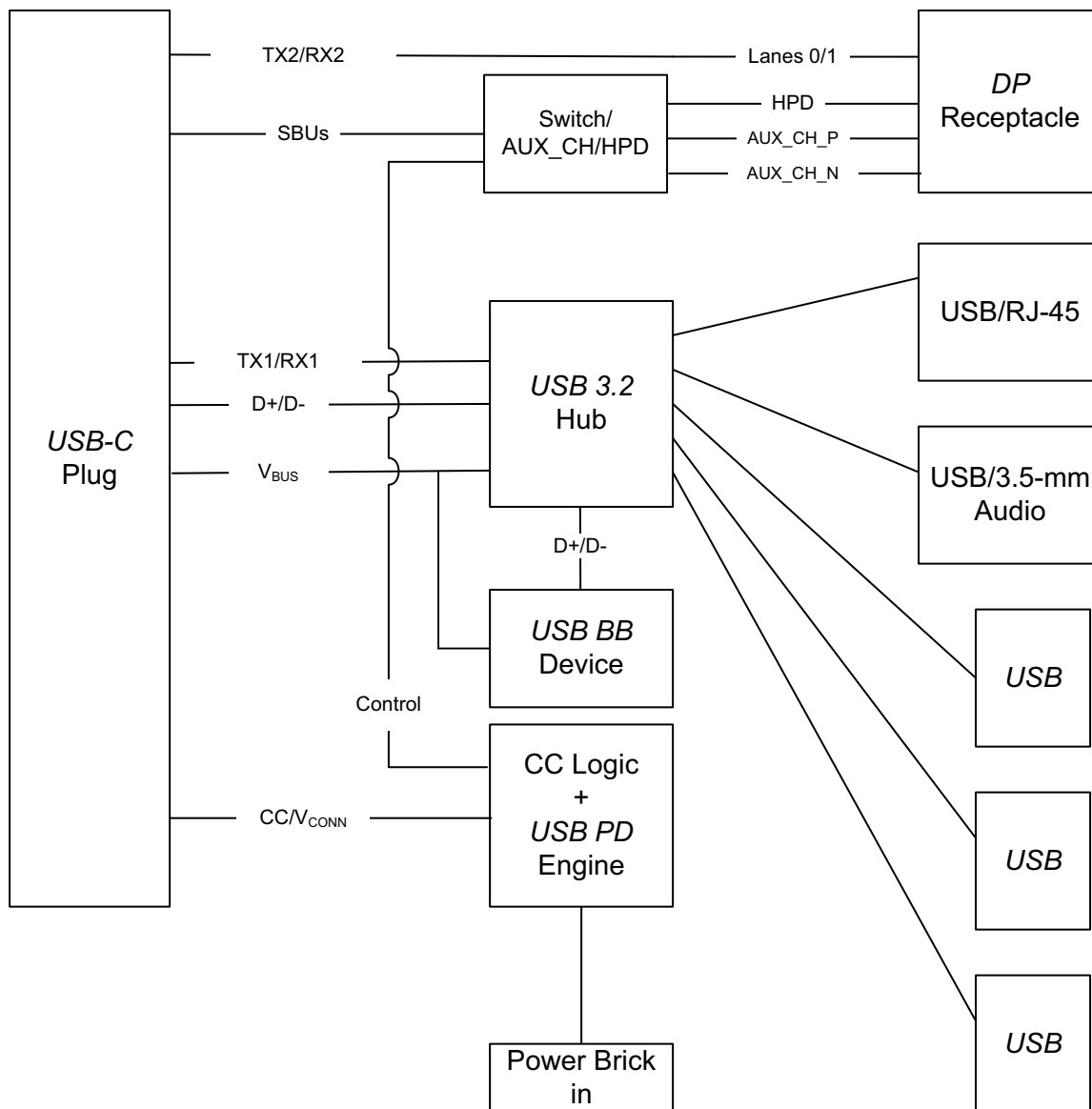


Figure A-3: Example Block Diagram of USB Dock with DisplayPort Alt Mode

The dock communicates to the *USB* host that it might behave as a two-lane *DP* Source device and participates in the negotiation process to place the *USB* host in the Alternate Mode that supports the dock. Because *USB PD* is used, it may also be used to negotiate power for the dock and *USB* host. In this example, the *USB* SuperSpeed signals allow the dock to work as a *USB* hub with a *USB* host that does **not** support *USB PD* or *DisplayPort*.

A more-complex dock may be designed that multiplexes TX1/RX1 to be either the *USB* SuperSpeed channel or to support an additional two *DP* lanes. Such a dock is beyond the scope of this example, but is not precluded by this Standard's support for Alternate Modes.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

A.3.3 Solution

The *USB* host system and *USB* dock with DisplayPort Alt Mode use *USB PD* VDMs to communicate and take advantage of advanced features.

The flow of Messages from the point where the *USB* dock with DisplayPort Alt Mode is plugged into the *USB-C* receptacle, plug orientation has been determined, and *USB* host/device relationship established is as follows:

- *USB* host system does **not** support *USB PD*
 - *USB* host will discover the *USB* hub and operate as it would when connected to a *USB* hub.
 - After the tAMETimeout timer times out, the *USB* host will see the *USB 2.0* device interface with *USB BB* and report to the user that an unsupported *USB-C* device has been connected, identifying the type of device from the *USB* descriptor information presented in *USB BB*.
- *USB* host system supports *USB PD* and advanced functions, but does **not** support the *USB* dock with DisplayPort Alt Mode
 - *USB* host will discover the *USB* hub and operate as it would for a *USB* hub.
 - *USB* host may look for other advanced functions, but does support the *USB* dock with DisplayPort Alt Mode.
 - Because the *USB* dock with DisplayPort Alt Mode is **not** supported, after the tAMETimeout timer times out the *USB* host will see the *USB 2.0* device interface with *USB BB* and report to the user that an unsupported *USB-C* device has been connected, identifying the type of device from the *USB* descriptor information presented in *USB BB*.
- *USB* host system supports the *USB* dock with DisplayPort Alt Mode
 - *USB* host looks for advanced functions by discovering whether the device reports SVIDs for advanced functions. The *USB* host then discovers a *USB* dock with DisplayPort Alt Mode by way of responses to *USB PD* Messages, using the DP_SID.
 - Because a usable *USB* dock with DisplayPort Alt Mode is discovered, the *USB* host enters DisplayPort Alt Mode, signals the dock to enter DisplayPort Alt Mode, and then reconfigures the *USB-C* receptacle to output *DP* signals on TX2/RX2 on the connector. The *USB* dock with DisplayPort Alt Mode cancels its tAMETimeout timer when entering DisplayPort Alt Mode. The *USB* host transmits *USB* SuperSpeed signals on TX1/RX1 and operates *USB* through the hub in the dock as normal.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

This example uses the basic extensions when docked that do the following:

Base Set VDMs used

- *USB PD Discover SVIDs*
- *USB PD Discover Modes* (DP_SID)
- *Enter Mode* (DP_SID, Alternate Mode offset)
- *Exit Mode* (DP_SID, Alternate Mode offset)

Process Outline

The *USB* host functions in its role as a DFP_U.

- 1 DFP_U detects presence of a *USB-C* device (CC pins) and connector orientation.
- 2 DFP_U applies 5V to V_{BUS} and V_{CONN}.
- 3 DFP_U discovers a *USB* hub in the dock with DisplayPort Alt Mode, and then enumerates the hub and attached *USB* devices (if any).
- 4 DFP_U makes a *USB PD* Explicit Power Contract with the dock device.
 - a DFP_U runs the *USB PD Discover SVIDs* Process.
 - b Transmits a *USB PD Discover SVIDs* Command request.
- 5 Receives a *USB PD Discover SVIDs* Command response with list of SVIDs for which the dock device has modes.
- 6 DFP_U runs the *USB PD Discover Modes* Process.
 - a Transmits the *USB PD Discover Modes* VDM for each of the SVIDs previously discovered that it supports, including the DP_SID.
 - b Receives a *USB PD Discover Modes* Command response with a list of modes associated with the DP_SID.
 - c **If a USB dock with DisplayPort Alt Mode is not discovered** – Remains in USB Mode, discovers the *USB BB* device after the tAMETimeout timer times out, and then informs the user – done.
 - d Else.
- 7 DFP_U runs the *Enter Mode* Process.
 - a Transmits the *Enter Mode* VDM with the DP_SID and offset to mode.
 - b Receives a response:
 - 1 **USB PD Responder ACK response** – DFP_U is now attached to the *USB* dock with DisplayPort Alt Mode, running in DisplayPort Alt Mode and USB Configuration.
 - 2 **USB PD Responder NAK response** – DFP_U returns to the default *USB 3.2* Configuration.
- 8 DFP_U discovers adapter status, and then waits until the adapter reports that it has a connected display.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

- 9 DFP_U runs the **DisplayPort Configure** Process.
 - a Places signals to be reconfigured to the Safe state.
 - b Transmits a **DisplayPort Configure** Command to the adapter.
 - c Receives a **USB PD** Responder ACK response.
 - d Configures signals to use **DisplayPort**; the **USB** host is now supporting native **DisplayPort** on the dock's **DP** connector in combination with **USB** SuperSpeed.
- 10 If the DFP_U receives a **DisplayPort Status Update** Command from the **USB-C** device that needs a configuration change, the **USB** host re-runs the **DisplayPort Configure** Process (possibly reconfiguring the interface back to USB Configuration).
- 11 DFP_U remains in this mode until:
 - a **USB** host is removed from the dock.
 - b Explicitly exited by an **Exit Alternate Mode** VDM.
 - c **USB PD Hard Reset**.
 - d Power is removed from the **USB** host.

The **USB** dock with DisplayPort Alt Mode shall follow and/or **not** interfere with normal **USB PD** mandates.

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

B Main Contributor History (Previous Versions)

Table B-1: Main Contributor History (Previous Versions)

Company	Name	Contribution	Version
Advanced-Connectek, Inc.	Steve Sedio	Technical Content	1.0b
Advanced Micro Devices, Inc.	Jim Hunkins	Technical Content	1.0, 1.0b
Allion Labs, Inc.	John Lin	Technical Content	1.0b
	Brian Shih	Technical Content	1.0b
Amphenol AssembleTech	Jesse Jaramillo	Technical Content	1.0b
Analogix Semiconductor, Inc.	Mehran Badii	Technical Content	1.0, 1.0b
	Greg Stewart	Technical Content	1.0, 1.0b
Apple, Inc.	Sree Anantharanan	Technical Content	1.0, 1.0b
	Brian Back	Technical Content	1.0b
	Bill Cornelius	Technical Content	1.0, 1.0b
	Will Ferry	Technical Content	1.0
	Derek Iwamoto	Technical Content	1.0
	Souvik Mukherjee	Technical Content	1.0
	Anil Pannikkat	Technical Content	1.0
	Bob Ridenour	Technical Content	1.0
	Jennifer Tsai	Technical Content	1.0b
	Colin Whitby-Strevens	Subgroup Document Editor	1.0, 1.0a, 1.0b
Avatar Tech Pubs	Trish McDermott	Technical Writer	1.0, 1.0a, 1.0b
BizLink Technology, Inc.	Morphy Hsieh	Technical Content	1.0b
Broadcom Corporation	Chris Pasqualino	Technical Content	1.0
Chrontel, Inc.	David Tsai	Technical Content	1.0, 1.0b
Dai-Ichi Seiko (formerly I-PEX)	Greg Young	Technical Content	1.0, 1.0b
Diodes	Tat Choi	Technical Content	1.0b
DisplayLink Corporation	Tony Lambord	Technical Content	1.0b
Ellisys Corporation	Abel Astley	Technical Content	1.0b
	Rick Bogart	Technical Content	1.0b
Foxconn Interconnect Technology, Ltd. (formerly Foxconn Electronics)	Glenn Moore	Technical Content	1.0, 1.0b
	Steve Sedio	Technical Content	1.0
Google, Inc.	David Schneider	Technical Content	1.0
GoPro, Inc.	Bill Orner	Technical Content	1.0b

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table B-1: Main Contributor History (Previous Versions) (Continued)

Company	Name	Contribution	Version
Intel Corporation	Simon Ellis	Subgroup Leader	1.0, 1.0a
	Mike Hamann	Technical Content	1.0, 1.0b
	Robert “Jamie” Johnston	Technical Content	1.0, 1.0b
	Ziv Kabiry	Technical Content	1.0b
	Christine Krause	Technical Content	1.0, 1.0b
	Sara Shachor	Technical Content	1.0b
	Gal Yedidia	Technical Content	1.0b
JAE Electronics, Inc.	Mark Saubert	Technical Content	1.0, 1.0b
	Toshio Shimoyama	Technical Content	1.0
Keysight Technologies	Brian Fetz	Technical Content	1.0b
LUXSHARE-ICT, Inc.	Juan Calderon	Technical Content	1.0b
	Scott Shuey	Technical Content	1.0b
MegaChips Technology America	Alan Kobayashi	DisplayPort Task Group Chair	1.0, 1.0a, 1.0b
		Technical Content	1.0, 1.0a, 1.0b
Microchip Technology	Mark Bohm	Technical Content	1.0b
NVIDIA Corporation	David Stears	Technical Content	1.0, 1.0b
	Bill Simms	Technical Content	1.0b
NXP Semiconductors (formerly Philips Semiconductor)	Vijendra Kuroodi	Technical Content	1.0b
Parade Technologies, Ltd.	Craig Wiley	Subgroup Co-leader, Technical Content	1.0, 1.0a, 1.0b
Qualcomm, Inc.	Alan Chan	Technical Content	1.0b
	Chris Wiesner	Technical Content	1.0b
Realtek Semiconductor Corporation	Jay Lin	Technical Content	1.0b
ROHM Semiconductor	Vijendra Kuroodi	Technical Content	1.0
Semtech Corporation	Chi Tai Hong	Technical Content	1.0
Siliconworks Co., Ltd.	Ben Huh	Technical Content	1.0b
Synaptics, Inc.	Jeff Lukanc	Technical Content	1.0b
Synopsys	Moren Christiansen	Technical Content	1.0b
TE Connectivity, Ltd.	Jim McGrath	Technical Content	1.0
	Scott Shuey	Technical Content	1.0
Tektronix, Inc.	Yogesh Pai	Technical Content	1.0b
	Medhu Acharya	Technical Content	1.0b
Teledyne LeCroy	Mike Micheletti	Technical Content	1.0b

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED

Table B-1: Main Contributor History (Previous Versions) (Continued)

Company	Name	Contribution	Version
Texas Instruments, Inc.	Mike Campbell	Technical Content	1.0b
	Richard Hubbard	Technical Content	1.0
	Yoon Lee	Technical Content	1.0, 1.0b
VESA – CPM	Jim Choate	Technical Content	1.0b
VIA Technologies, Inc.	Jay Tseng	Technical Content	1.0b
VTM Group	Bob Crepps	Technical Content	1.0, 1.0b

DISTRIBUTION TO NON-MEMBERS IS PROHIBITED